Transforming Luxuries into Necessities: How Inequality Affects Growth?*

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Abstract

Products often begin as luxuries and become necessities as they diffuse down the income distribution. How does income inequality shape this transformation and long-run growth? Using barcode-level data, we document this diffusion and show that greater inequality reduces overall innovation while concentrating it in richer-consumer products. We develop an endogenous growth model with hierarchical demand and firm dynamics, where innovation raises productivity within existing products and creates new products at the frontier. Rising inequality reallocates innovation away from necessities, creating an endogenous bottleneck for long-run growth. Calibrated to U.S. data, the model implies sizable welfare gains from redistribution.

Keywords: Inequality, Innovation, Non-homothetic preference, Growth.
JEL codes: H54, N45,O11, O16, O38, O53, O57.

*Researchers own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business. The conclusions drawn from the NielsenIQ data are those of the researchers and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein. We thank Benjamin Jones, Reto Foellmi, David Hemous, and seminar participants at HKU, PKU, BUAA, CUFE, and SWUFE for their helpful comments. We thank Yucheng Yang for sharing the GS1 Global Data. We thank Ruichen Ma and Ziyue Wang for their excellent research assistance. Financial support from the Natural Science Foundation of China (Grant No. 72322007;72503230) is gratefully acknowledged by Mingzhi Xu.

1 Introduction

Rising income inequality is not only about who gets what, but also shapes what gets invented, who benefits from new technologies, and how much growth innovation delivers (Jones and Kim, 2018; Aghion et al., 2019a; De Loecker et al., 2020). New products often begin as high-end goods targeted at affluent consumers and only gradually become widely used necessities as costs fall and markets expand (Matsuyama, 2002; Foellmi and Zweimüller, 2006). This transformation means that products at different stages of their life cycle serve very different segments of the income distribution. Consider a cutting-edge consumer product consumed primarily by wealthy households and a mature household product consumed by almost everyone. As inequality rises, the purchasing power of their respective customers changes differently, potentially creating very different incentives for the producers of these two products to innovate. This naturally raises several questions: How does rising inequality reshape the allocation of innovation across firms and sectors? What are the aggregate consequences for economic growth? What are the effects of redistribution through taxes and subsidies on economic growth and welfare?

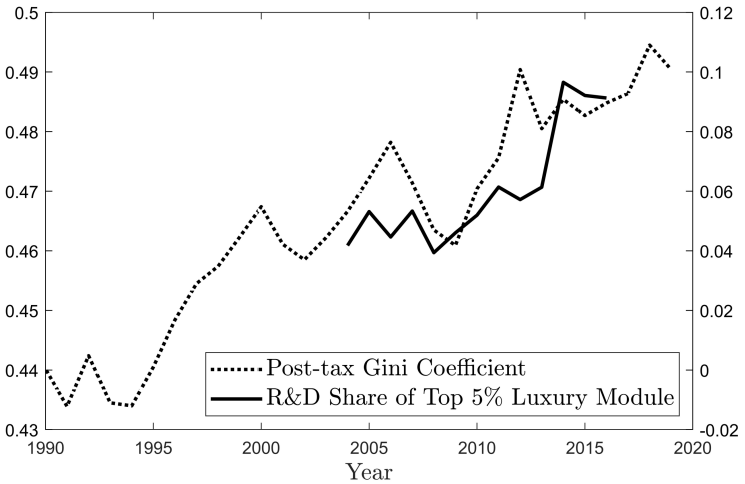
This paper attempts to answer these questions by combining empirical, theoretical, and quantitative analyses using an endogenous growth model. We show that inequality distorts the direction of innovation by shifting R&D toward premium, rich-serving products and away from productivity improvements in basic goods. This change in the sectoral distribution of R&D weakens innovation-led growth. We call this mechanism the R&D reallocation channel of inequality.

The R&D reallocation channel emerges through the interaction of hierarchical demand and the product life cycle. New varieties are introduced at high prices and initially sold to affluent consumers. As incumbent innovation and entry raise productivity and intensify competition, prices fall, markets expand, and products gradually diffuse to lower-income households. Higher inequality weakens demand and profits in mass-market necessity sectors while strengthening the relative profitability of premium sectors. R&D consequently shifts toward premium products and away from productivity improvements in necessities. Slower productivity growth and price declines in necessity sectors delay the release of labor and household purchasing power toward emerging sectors, thereby constraining the expansion of new product categories and slowing aggregate growth.

This inequality-driven reallocation in R&D is visible in the U.S. aggregate time series. Figure 1 plots the post-tax Gini coefficient for the United States, obtained from the World Inequality Database, alongside the share of R&D allocated to luxury-oriented product segments. Firm-level R&D data from BvD Orbis are aggregated to the product-module, which is a fine-grained NielsenIQ classification that groups closely comparable products. We define luxury-oriented modules as the top 5% of product modules ranked by the average income of their consumers. The figure shows that, as income inequality rises, R&D expenditure becomes increasingly concentrated in these luxury-oriented product segments.

To study this mechanism, we build a matched dataset linking household purchases and incomes to firm outcomes, product ownership, and patenting over the product life cycle. The dataset

Figure 1: Income Inequality and the R&D in Luxuries



Notes. The dotted line plots the U.S. post-tax Gini coefficient (left axis), from the World Inequality Database. The solid line plots the share of total R&D attributable to the luxury-oriented modules (right axis). We define luxury-oriented modules as the top 5% of product modules ranked by the average income of their consumers.

combines nationally representative household purchases with firm-level financial and patent data, allowing us to track how demand, innovation, and market structure evolve within narrowly defined product markets. Our empirical analysis delivers four stylized facts that characterize a common product life cycle and its link to innovation. First, products enter as high-priced varieties consumed primarily by high-income households and gradually diffuse to lower-income consumers as prices fall. Second, innovation is concentrated where early demand is strongest: firms serving higher-income consumers engage more intensively in patenting. Third, diffusion is accompanied by entry and declining market concentration, consistent with intensifying competition. Fourth, greater exposure to inequality is associated with weaker aggregate innovation, particularly in segments serving low-income consumers, and with a higher concentration of innovative activity.

Motivated by these facts, we build an endogenous growth model with heterogeneous households, hierarchical (non-homothetic) demand, and multiple margins of innovation. Households allocate spending across a continuum of goods ranked from necessities to higher-end products, as in Matsuyama (2002) and Foellmi and Zweimüller (2006). On the supply side, potential entrants choose whether to enter and how much to invest to raise their productivity (effectively reduce costs) when they become incumbents, while new varieties expand the frontier. Entry and cost-reducing innovation generate a natural product life cycle: goods arrive as high-priced luxuries, prices fall as productivity rises and competition intensifies, and adoption spreads down the income distribution. Entry increases and concentration declines along the same path, matching the diffusion and market-structure patterns in the data. The model delivers a tight link from inequality to the direction of innovation and long-run growth. Along the balanced growth path, luxury sectors sustain higher prices and greater innovation intensity, consistent with the cross-sector evidence. When income concentrates at the top, the mass market for necessities shrinks relative to premium demand, weakening incentives to innovate in broadly consumed goods. As a result, innovative effort shifts away

from necessities and slows productivity growth in these sectors. Because productivity improvements in necessities release both labor and household purchasing power toward emerging products, slower innovation in these sectors constrains the expansion of the product frontier. Necessities therefore become the bottleneck for long-run growth.

Quantitatively, we calibrate the model to match the core module-level moments that define the luxury-to-necessity transition: the post-entry decline in prices, the gradual shift of adoption toward lower-income households, and the accompanying evolution of entry and concentration within a market. The calibrated economy also matches standard aggregates such as R&D intensity, firm-size concentration, and long-run growth. With the model disciplined by these facts, we use it to evaluate mechanisms and policy-relevant counterfactuals. A first decomposition shows that price declines and diffusion down the income distribution are driven in roughly equal parts by incumbents' cost-reducing R&D and by entry. Comparative statics that vary income inequality imply quantitatively important effects on growth, with magnitudes comparable to those observed in the historical U.S. experience. Finally, a standard progressive-tax schedule that restores post-tax inequality to its 1970s level delivers large welfare gains: around 99% of households benefit, with average welfare rising by about 44.58%. Therefore, rising inequality does more than shift demand: it reallocates R&D toward premium goods and away from productivity improvements in necessities, whose technological progress is critical for product diffusion and aggregate growth. Policies that strengthen the mass market for basics can therefore redirect innovation toward broad growth and welfare.

Our paper makes three main contributions to the literature. First, we provide a comprehensive empirical characterization of the "from luxury to necessity" flying-geese pattern. While this pattern has been noted in the literature, existing studies typically rely on aggregate evidence or focus on specific margins. By contrast, we use comprehensive micro-level data to trace how consumption goods diffuse across consumers over the product life cycle, and to jointly document the evolution of innovation activity and firm dynamics along this process.

Our second contribution is to develop a unified theoretical framework that integrates multiple margins of innovation—process innovation by incumbents, innovation-driven entry that intensifies market competition, and the creation of new product varieties—and embeds them in a setting with non-homothetic demand. While each of these dimensions has been studied in isolation, relatively few papers incorporate all three within a unified framework. Their interaction generates a rich product life cycle in which prices decline, adoption expands toward lower-income consumers, entry rises, and market concentration falls as products mature. The framework therefore connects household demand, firm dynamics, and long-run product expansion in a way that can be disciplined by the micro-level evidence documented in this paper.

Our third contribution is to identify a novel R&D reallocation channel through which inequality affects long-run growth. Existing hierarchical-demand models primarily emphasize how price and market-size effects shape the overall incentive to innovate. In our framework, inequality also changes the allocation of R&D resources across sectors and innovation activities. By shifting innovative

effort away from necessities, greater inequality slows the release of labor and household purchasing power toward emerging products, thereby reducing overall economic growth.

Related Literature. Our theoretical model is most closely related to the literature on innovation under hierarchical demand. In these models, economic growth transforms goods that were once luxuries into necessities, generating the "flying-geese" pattern studied by Matsuyama (2002), Foellmi and Zweimüller (2006), Foellmi et al. (2014), and Oberfield (2023). In Matsuyama (2002) and Oberfield (2023), technological progress arises from learning by doing, leaving no role for the allocation of R&D resources. Foellmi and Zweimüller (2006) introduce purposeful innovation but restrict it to the creation of new goods. Foellmi et al. (2014) additionally allow for a one-time process innovation within each sector and show how its profitability depends on the price and market-size effects of inequality.

The precise distinction of our framework is the endogenous allocation of scarce R&D resources across sectors and innovation activities. Incumbent firms continuously invest in productivity improvement, potential entrants invest in entering existing product markets, and inventors develop new product categories. Inequality therefore affects not only the profitability of innovation through the conventional price and market-size effects, but also how R&D resources are divided among these activities and across the product ladder. When inequality redirects R&D away from necessity sectors, technological progress slows in the sectors that constitute the bottleneck for product expansion, thereby reducing long-run growth. Endogenous entry and competition also allow the model to account for the joint evolution of prices, adoption, entry, and market concentration over the product life cycle. The framework remains tractable and is well suited for quantitative analysis disciplined by these micro-level patterns.

Our model identifies productivity growth in necessity sectors as an *endogenous bottleneck* to long-run growth. Existing studies emphasize supply-side bottlenecks arising from weak links, critical inputs, or complementarities in production and innovation (Jones, 2011; Aghion et al., 2017; Acemoglu et al., 2024; Jones and Tonetti, 2025; Jones and Hall, 2026). In our model, by contrast, the bottleneck emerges from the hierarchy of demand. Because necessities are purchased by all households, productivity growth in these sectors releases purchasing power throughout the income distribution and resources for new-product creation, thereby expanding demand for higher-ranked goods. Slow productivity growth in necessities thus constrains both demand expansion and resource reallocation. The bottleneck is endogenous because it is determined by the evolving consumption hierarchy rather than by a fixed technological structure.

This demand-side perspective also connects our paper to the literature on non-homothetic demand and the direction of technological progress. Existing studies show that rising average income redirects demand and innovation toward more income-elastic sectors (Bohr et al., 2023; Comin et al., 2025). We identify a complementary distributional channel: holding average income fixed, changes in the income distribution reallocate effective market size and R&D across consumption tiers. By shifting innovation from widely consumed necessities toward luxuries, greater inequality

tightens the endogenous growth bottleneck and reduces long-run growth.

Our paper also contributes to the broader literature on inequality and growth. While evidence on the reduced-form relationship remains mixed (Persson and Tabellini, 1994; Forbes, 2000; Barro, 2000; Banerjee and Duflo, 2003; Baselgia and Foellmi, 2023), our mechanism operates through the direction of innovation. Inequality changes the composition of demand and, in turn, the relative profitability of cost reductions across sectors. In our model, a rise in inequality shifts innovative effort toward premium, rich-serving goods and away from cost reductions in necessities, where diffusion and scale generate broad productivity gains, lowering the aggregate return to innovation and slowing growth. A related literature studies innovation as a driver of inequality (Hémous and Olsen, 2022; Aghion et al., 2019a), which emphasizes how technological change reshapes the income distribution. Our analysis highlights the reverse feedback: the income distribution shapes innovation incentives by determining which markets are large enough to reward cost-reducing R&D and broad diffusion.

Finally, our paper builds on the endogenous-growth literature with firm dynamics. Pioneering contributions model productivity growth through innovations (Klette and Kortum, 2004; Lentz and Mortensen, 2008; Acemoglu et al., 2018) in the tradition of Romer (1990), Aghion and Howitt (1992), and Grossman and Helpman (1991). Subsequent work allows incumbents to invest in internal R&D alongside external innovation (Akcigit and Kerr, 2018; Peters, 2020; Akcigit and Ates, 2023). We extend this line by adding hierarchical demand and household heterogeneity, which makes market size by income group a determinant of entry, innovation incentives, and the sectoral allocation of R&D.

2 Empirical Analysis

We start from what households buy and then trace each product back to its supplier and forward to the supplier's innovation and financial outcomes. This linkage lets us follow a product's life cycle as it moves from high-income to broader consumers, and relate that diffusion to firms' innovative activity. In this section we first describe the data, and then lay out the key diffusion facts. We relegate additional descriptions and robustness check to Appendix A.

2.1 Data Description

The analysis draws on four primary sources. First, we use the NielsenIQ Homescan Consumer Panel, collected by NielsenIQ (U.S.) and made available through the Kilts Center for Marketing at the University of Chicago Booth School of Business. Second, we use GS1 Company Database to link the sample products obtained through NielsenIQ Homescan Consumer Panel to firms. Third, we incorporate the PatentsView dataset, compiled by the U.S. Patent and Trademark Office (USPTO). Finally, to obtain comprehensive firm-level financial information, we draw on the Bureau van Dijk (BvD) Orbis database.

NielsenIQ Homescan Consumer Panel. We use the NielsenIQ Homescan Consumer Panel (NHCP) to obtain detailed information on household demographics, geographic location, and purchasing behavior. The panel, maintained and updated annually by NielsenIQ, covers a geographically and demographically representative sample of roughly 40,000 to 60,000 U.S. households. NielsenIQ implements quarterly coverage monitoring and weekly reviews of sample representativeness to ensure data quality. Participating households record purchases using in-home scanners, providing information on the date of purchase, Universal Product Code (UPC), number of units, and total expenditure. To address potential selection bias, NielsenIQ constructs projection factors—sampling weights that scale to the U.S. household population—so that the dataset is projectable to the national level. In what follows, we introduce several key terms used throughout the analysis, define their empirical counterparts in the data, and explain how they map into variables in the model.

(i) Product Modules. Unless otherwise noted, we define a market in this paper as the set of transactions within a given product module. Each product module encompasses a variety of products that are defined at the most detailed level within the NielsenIQ data, with each item being distinctly identified by a 12-digit UPC paired with version codes. NielsenIQ organizes roughly 1,000 modules into a hierarchy of about 100 product groups and 10 departments (Figure A.1), and modules within the same group contain close substitutes such as the brands in a representative shampoo module (Figure A.2). For ease of exposition, we use the terms "product" and "UPC" interchangeably to refer to this identifier throughout the analysis. We exclude so-called "Magnet" products that lack standard UPCs—items such as in-store baked goods—because only a subset of households report these purchases, and coverage is unavailable prior to 2007.

(ii) Firms. We link each UPC to its owning company using the GS1 Company Database. GS1 is a global not-for-profit standards organization that assigns barcodes to manufacturers worldwide. The database provides company names and GS1 prefixes—unique numerical codes embedded in the initial digits of each barcode—that connect UPCs to their parent firms (Hottman et al., 2016).[1] For instance, *Procter & Gamble* appears in our data as a single firm encompassing multiple brands, including *Crest* in oral hygiene and *Pantene and Head & Shoulders* in hair care. Restricting attention to GS1 companies with sales recorded in the NielsenIQ data during our sample period (2004–2016) yields 44,030 distinct firms.

To broaden the description of these firms and facilitate the estimation of firm-level productivity, we incorporate financial data from the Bureau van Dijk (BvD) Orbis database. Orbis provides comprehensive information on both listed and unlisted firms worldwide, including operating revenues, tangible fixed assets, employment, employee costs, and investment. We

[1]For 16.53% of UPCs, no firm code could be matched between the GS1 data and the UPC list. These UPCs are excluded from the analysis. With 2.36 million products spanning categories such as food, non-food groceries, health and beauty aids, and general merchandise, the NHCP database accounts for about 66% of total products from U.S. GS1 firms, thereby ensuring the representativeness of our dataset.

match GS1 company names to Orbis entries using a string-matching algorithm supplemented by manual validation, achieving a success rate of 86.4 percent. Matched firms span a broad range of NAICS four-digit industries, with consumer-focused sectors such as grocery wholesaling and food manufacturing dominating (Table A.1), and are geographically concentrated in major economic centers including California, New York, Florida, and Texas (Panel (a) of Figure A.4).

(iii) Consumer Groups. We classify consumers into groups based on annual household income, as reported in the NielsenIQ panel. Each participating household is asked to record its combined total income, which NielsenIQ assigns to a predefined income range. A practical challenge arises because the top income category in the demographic survey changes over time, whereas the lower income ranges remain stable. For instance, in 2004–2005 the highest category was defined as "$100,000 or more." Beginning in 2006, NielsenIQ subdivided this range into four brackets: $100,000–124,999, $125,000–149,999, $150,000–199,999, and $200,000 or more. Following NielsenIQ's convention from 2010 onward, we collapse these upper ranges into a single top-income group to maintain consistency over time. Unless otherwise noted, we define "rich" or "high-income" consumers as those in the highest income category—roughly the top decile of the distribution—and "poor" or "low-income" consumers as those with income below $20,000, approximately the bottom decile. This harmonized definition ensures comparability across survey years while capturing meaningful variation at both ends of the income distribution.

(iv) Revenues and Prices. Firm revenue, used as a proxy for firm size, is computed as the sum of projection factor–weighted annual household expenditures on a firm's products. Product revenue is defined analogously, aggregating expenditures at the product level. Product prices are then calculated as the ratio of annual product revenue to projection factor–weighted units sold. To account for inflation, we express all prices in real terms using the U.S. Consumer Price Index (CPI) from the World Bank. Consumer expenditures exhibit substantial spatial and income-group variation, with higher spending concentrated in coastal states and steady growth over time for both poor and rich consumers (Figure A.3).

PatentsView Database. We measure firm-level innovation using detailed patent data from the PatentsView database, an official platform maintained by the U.S. Patent and Trademark Office (USPTO). PatentsView provides disambiguated assignee identifiers, allowing consistent linkage of patents to firms. Following the standard approach in the innovation literature, our baseline measure of innovation output is the number of patent applications filed by a firm in a given year that are ultimately granted (He and Tian, 2013). This measure captures the timing of inventive activity at the application stage while restricting attention to patents that pass the threshold of being granted, thus reflecting substantive technological advancement. To complement this measure, we also construct claim-weighted patent counts as an indicator of patent quality, since patents with more claims confer broader and more expansive protection (Aghion et al., 2019a).

To link patents to firms, we employ a multi-step string-matching procedure based on the Levenshtein edit distance, implemented in Python's *FuzzyWuzzy* package, with a high similarity threshold of 0.9 and cross-validation using phonetic and n-gram algorithms, supplemented by manual verification. The resulting matches reproduce the right-skewed distribution of patenting activity in the USPTO universe (Panels (a) and (b) of Figure A.5), with the matched sample exhibiting higher median and upper-quartile counts due to the consolidation of assignee name variants (Panel (c) of Figure A.5). Patent activity is concentrated in innovation hubs including California and Massachusetts (Panel (b) of Figure A.4), and the most patent-intensive firms include Microsoft, General Electric, and Apple (Panel (d) of Figure A.5).

2.2 Motivational Facts

We first document empirical facts on the diffusion of new products across consumers with different income levels. We begin by describing how newly introduced products spread within the market, and then investigate potential explanations for these diffusion patterns. Specifically, we consider how innovation by incumbent firms, the entry of new competitors, and changes in the income distribution shape the process through which products move from high- to low-income consumers. Finally, we shed light on how shifts in the income distribution influence the innovation distribution.

Fact I: Product Diffusion Along the Income Distribution. We measure product diffusion at the *module* level rather than at the UPC level. A module groups closely related UPCs that share similar physical attributes and serve the same consumption purpose, such as sparkling water, yogurt, or shampoo. Each module contains multiple UPCs that vary by brand, package size, or other variants. This aggregation reduces mechanical UPC-level churn while preserving substantial within-category variation, making it well suited for tracing diffusion patterns along the income distribution.[2]

We trace products' spread across the distribution – whether adoption begins with high-income households and diffuses downward, or vice versa. We examine this pattern by estimating

$$Outcome_{fmt} = \sum_{y=1}^{12} \mu_y \times D_{fmy} + \eta_{fm} + \eta_{ft} + \eta_{mt} + \epsilon_{fmt}. \tag{1}$$

[2]This aggregation is motivated by three considerations. First, UPCs are a noisy proxy for innovation, since barcode changes often reflect packaging revisions, bundle adjustments, or temporary promotions rather than genuine product development, mechanically inflating product entry and exit. Aggregating to the module level filters out this noise and yields a more stable measure of product availability and adoption. Second, although prices and promotions are set at the UPC level, firms typically manage innovation and product upgrading at a broader level, so new lines or quality improvements usually affect clusters of related UPCs rather than a single barcode (Braguinsky et al., 2021). Third, our interest is in diffusion across income groups. The relevant margin is whether a product type becomes accessible to lower-income households, not which specific UPC they choose within that category. Module-level adoption therefore better captures the shift from non-consumption to consumption, and more broadly, the movement of a good from luxury to necessity. This view is consistent with Ma et al. (2025), who show that UPC-level purchases are highly volatile and often distorted by minor stock-keeping changes.

The outcome variable, $Outcome_{fmt}$, captures the diffusion of module m sold by firm f in year t. We measure $Outcome_{fmt}$ in two complementary ways: (i) the income position of the module's buyers, computed as the projection-factor-weighted mean income percentile, and (ii) the module's expenditure share accounted for by lower-income households (income below \$20,000).[3] The key regressor, D_{fmy}, is a dummy variable indicating event time relative to entry, which is equal to 1 for the y-th sales period after firm f's initial sale of products in module m. The specification includes a rich set of fixed effects. Firm–module fixed effects (η_{fm}) absorb persistent differences in product attributes and firm positioning within a module. Firm–year fixed effects (η_{ft}) absorb time-varying firm-wide shocks, such as changes in brand reputation or marketing intensity. Module–year fixed effects (η_{mt}) absorb common shocks within a module-year, including shifts in tastes or technology. Standard errors are clustered at the firm level to allow arbitrary correlation within firms over time.

Figure 2 shows how adoption of a module shifts across the income distribution as it ages. Each point is an event-time estimate of years since entry, controlling for firm–module, firm–year, and module–year fixed effects. The x-axis is periods since the firm first sells in the module. The y-axis reports diffusion: the average buyer's income quantile in Panel (a) and the low-income expenditure share in Panel (b). Adoption clearly moves down the income distribution over time. In Panel (a), the average buyer's income rank falls by about five percentiles within twelve years of entry. Panel (b) shows a corresponding rise in the low-income expenditure share, indicating gradual penetration into poorer segments. A natural mechanism is falling prices as technology improves and competition intensifies (Jaravel, 2019; Bohr et al., 2023). We assess this channel by re-estimating the baseline specification with the module's annual price as the dependent variable.

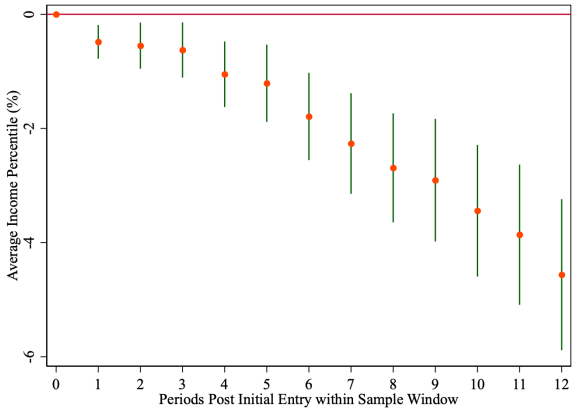
As shown in Panel (c), average prices experience a decline of roughly 21 percent over the 12 years following entry. Panel (d) shows a similar downward price path in both the entrant and incumbent subsamples. Firms observed in all 13 sample years from 2004 to 2016 are classified as incumbents, while all others are classified as entrants. This pattern suggests that the decline in prices reflects both within-firm adjustment and changes in the composition of sellers over the product life cycle, which in turn motivates the intensive- and extensive-margin decompositions in Facts II and III studied next. Taken together, the diffusion pattern shown in Figure 2 is consistent with a reservation-price mechanism (Simonovska, 2015; Bertoletti et al., 2018; Hu et al., 2025): lower-income households have lower willingness-to-pay thresholds, so declining prices naturally broaden product adoption among poorer consumers.[4]

Appendix robustness checks confirm the stability of these patterns. The diffusion result holds at the finer UPC level (Figure A.6), on a balanced panel of firm–module pairs (Figure A.7), and on a sample of modules entering after 2004 (Figure A.8). The price decline is robust to the inverse

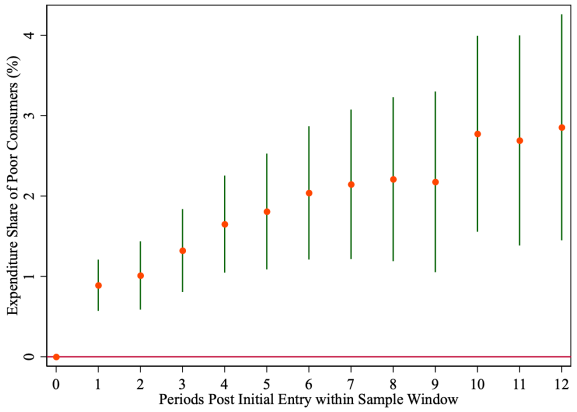
[3] We weight households using NielsenIQ's *projection factor* and compute income percentiles by year. This reweighting aligns the income distribution in the NielsenIQ sample with that of the U.S. population.

[4] Differences in demand elasticities and endogenous markups may also shape adoption patterns (Atkeson and Burstein, 2008), but their net effect on prices over the life cycle is theoretically ambiguous. Our reduced-form, module-level estimates therefore capture the average effect of these forces. The combination of steadily declining prices and rising participation by lower-income households suggests that, in this setting, cost reductions and competitive pressures outweigh any tendency for markups to rise.

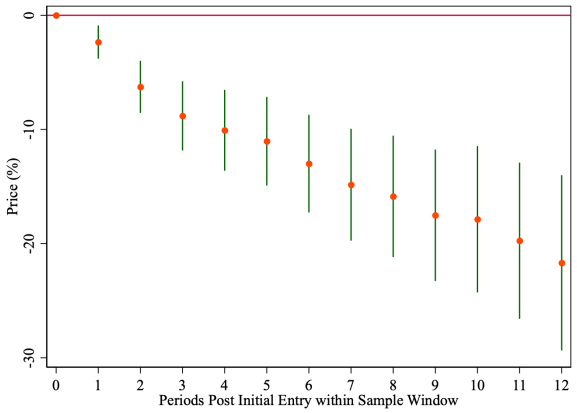
Figure 2: Product Diffusion Across the Income Distribution



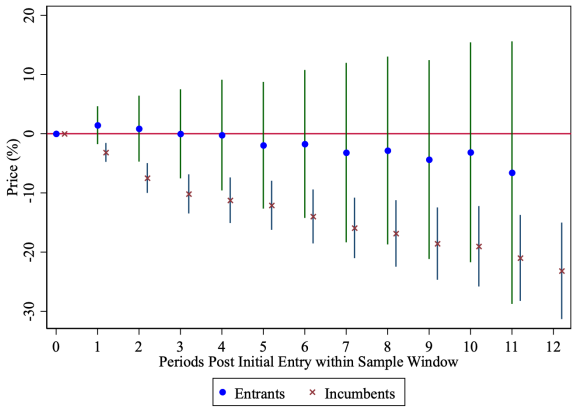
(a) Decline in Buyers' Income



(b) Growing Share of Low-Income Buyers



(c) Falling Prices Over Product Age



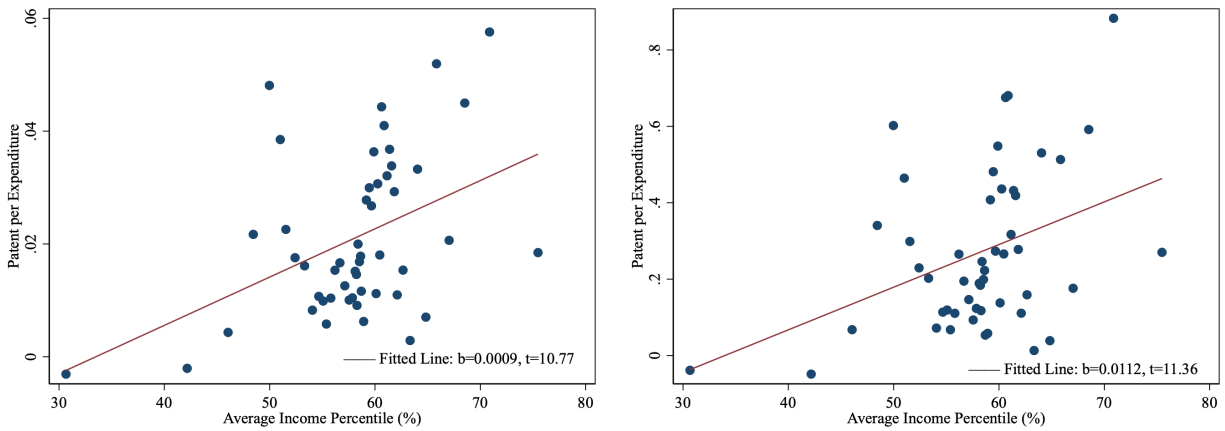
(d) Falling Prices: Entrants vs. Incumbents

Notes. Each point plots the estimated coefficient from a regression of product-level outcomes on the cumulative periods relative to initial entry within the sample window. The x-axis measures periods after initial sales records, and the y-axis shows the estimated change in the corresponding outcome relative to the initial year. All regressions include firm–product, firm–year, and product–year fixed effects, with standard errors clustered at the firm level. In Panel (a), the dependent variable is the average income quantile of a product's buyers, showing whether its customer base shifts toward lower-income consumers as it matures. In Panel (b), the dependent variable is the expenditure share of the bottom income tercile, reflecting the growing importance of poorer consumers in total spending. Panel (c) uses the annual price to track how average prices evolve after initial year. Panel (d) further illustrates that the decline in prices is driven concurrently by both new entrants and established firms. To facilitate interpretation, we report the percentage change of price in Panels (c) and (d), computed by dividing the estimated marginal effect by the sample mean price. Solid dots represent point estimates, and vertical bars denote 95% confidence intervals based on robust standard errors.

hyperbolic sine transformation, size-adjusted unit prices, and a split by innovative versus non-innovative firms (Figure A.9), and firm-level TFP rises steadily over the product life cycle (Figure A.10), ruling out product obsolescence as an alternative explanation. To address left truncation in the baseline event-time measure, we further match each module to its technology birth year from Kalyani et al. (2025) (see Table A.2, Figure A.11 for more details) and re-estimate the Fact I diffusion dynamics (Figure A.12) using years since technology birth. All results remain qualitatively unchanged.

Fact II: Innovation by Incumbents. Fact I shows that, as products age, prices fall and adoption shifts toward lower-income households. A natural mechanism is cost-reducing innovation by incumbent firms. We therefore examine how innovative activity varies with the income profile of a firm's customer base. Figure 3 plots firm-level innovation intensity against the average income percentile of the firm's customer base, computed as the expenditure-weighted average of household income percentiles among the firm's consumers. Innovation intensity is measured as patents per million dollars of consumer expenditure on the firm's products. Each panel displays a binned scatter plot, pooling all firm-year observations from 2004 to 2016, absorbing year fixed effects and controlling for the log number of household groups and the log number of product modules, with observations weighted by firm expenditure. These controls and the expenditure weighting are designed to mitigate the concern that a raw correlation between demand orientation and innovation may primarily reflect firm composition, size, marketing, or sector differences.

Figure 3: Innovation Intensity by Customer Income Segment, 2004–2016



(a) Patent Count-based Innovation Intensity (b) Patent Quality-based Innovation Intensity

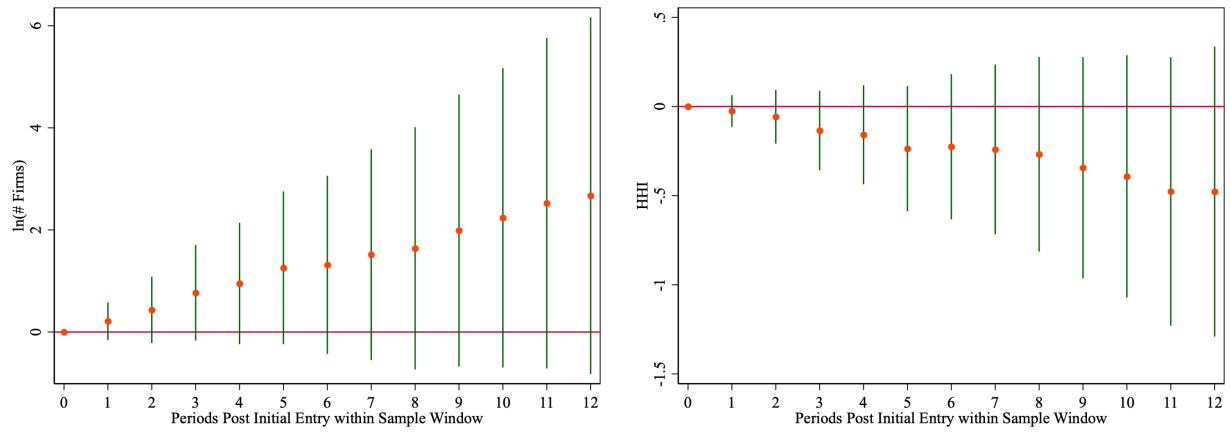
Notes. Each panel displays a binned scatter plot of firm-level innovation intensity against the average income percentile of the firm's customer base, with 50 quantiles. The average income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. All firm-year observations from 2004 to 2016 are pooled, with year fixed effects absorbed and controls for the log number of household groups and the log number of product modules. Observations are weighted by firm expenditure. In Panel (a), innovation intensity is defined as patents per million dollars of consumer expenditure on the firm's products, with a fitted slope of 0.0009 ($t = 10.77$). In Panel (b), innovation intensity is defined as claim-weighted patents per million dollars of consumer expenditure, with a fitted slope of 0.0112 ($t = 11.36$). Innovation intensity variables are winsorized at the 1% and 99% percentiles within each year.

The relationship is positive: firms serving higher-income customers exhibit higher innovation intensity. Patent counts mask heterogeneity in patent value. As a quality adjustment, Panel (b) uses claim-weighted patent counts following Aghion et al. (2019a), and the positive gradient persists. Altogether, this concentration of innovation at the top of the income distribution is consistent with incumbents pushing the frontier in high-income segments, with subsequent diffusion as costs fall.

Appendix Figures A.14–A.16 probe the robustness of this pattern. Figure A.14 restricts to single-product firms—ensuring that firm-level patents map to the same product line as the customer income profile—and further to a balanced panel, removing entry and exit margins. Figure A.15 replaces the time-varying income percentile with either the firm-level sample mean or the entry-year value, fixing each firm's demand orientation throughout the analysis. Figure A.16 plots rich- and poor-market innovation intensity by year, showing that the gap widens over time as rich-market firms innovate more and poor-market firms less. The positive gradient persists in every case, confirming that the association is not driven by patent-product mismatch in diversified firms, firm turnover, time-varying customer composition, or the choice of estimation specification.

Fact III: Entry of New Competitors. Declining prices may reflect not only cost-reducing innovation by incumbents (Fact II) but also intensifying competition as new firms enter. To assess this market-structure channel, we estimate a module-year analogue of (1) in which the dependent variable is the number of active firms in module m and year t. The specification includes module fixed effects to absorb time-invariant differences across product categories and year fixed effects to absorb aggregate shocks; standard errors are clustered at the module level.

Figure 4: Entry and Declining Concentration over the Product Life Cycle



(a) Number of Active Firms

(b) Market Concentration (HHI)

Notes. Each panel reports event-time estimates from regressions of module-year market-structure outcomes on periods since the module's introduction. The x-axis measures years relative to the entry year, and the y-axis shows the estimated change in the outcome relative to the initial year. Panel (a) uses the log number of active firms in module m and year t. Panel (b) uses the Herfindahl–Hirschman Index, $HHI_{mt} = \sum_{f \in m} s_{ft}^2$, where s_{ft} is firm f's expenditure share within module m in year t. All regressions include module fixed effects and year fixed effects, with standard errors clustered at the module level. Solid dots represent point estimates, and vertical bars denote 95% confidence intervals based on robust standard errors.

Figure 4 shows rapid entry following a module's introduction. Panel (a) indicates that within five periods the number of active firms nearly doubles, pointing to substantial entry by new competitors. Panel (b) shows a parallel decline in concentration. The Herfindahl–Hirschman Index, $HHI_{mt} = \sum_{f \in m} s_{ft}^2$ falls steadily over event time, where s_{ft} is firm f's revenue share within module m in year t. Because the figure is designed to summarize the underlying pattern rather than to estimate a fully saturated specification with detailed fixed effects and controls, the year-by-year coefficients are naturally measured less precisely. Even so, the overall life-cycle pattern is clear: market participation broadens and concentration erodes as products mature, consistent with stronger competitive pressure that reinforces the price declines and downstream diffusion documented in Fact I.

Appendix robustness checks confirm the stability of these patterns. The de-concentration result holds under alternative concentration measures, including the CR4 and CR20 (Figure A.17), and on a sample of modules entering after 2004 to avoid left censoring (Figure A.18). Across both exercises, the core pattern of rising firm participation and falling concentration over the product life cycle remains unchanged.

Fact IV: Inequality and the Distribution of Innovation. If diffusion is powered by innovation and competition (Facts I–III), then changes in the income distribution should matter. We therefore study how income inequality affects firm innovation. A direct regression of patenting on inequality is hard to interpret because inequality and innovation may co-move: innovative firms can expand in particular markets, while changes in inequality can reshape the profitability of further innovation. To reduce this concern, we construct a Bartik-style, firm-specific exposure to inequality shocks using a predicted (and more plausibly exogenous) component of state-level inequality, denoted $Shock_{ft}^{ineq}$. Appendix A.8 provides details on measurement and the estimating equation.

$$Shock_{ft}^{ineq} = \sum_{s} \lambda_{sft_0} \widehat{Gini}_{st}, \qquad \widehat{Gini}_{st} \equiv Gini_{s,2004} \times \frac{1}{N-1} \sum_{j \neq s} \frac{Gini_{jt}}{Gini_{j,2004}} \tag{2}$$

where f indexes firms, t indexes years, and s indexes U.S. states. The weights λ_{sft_0} are predetermined baseline expenditure shares: $\lambda_{sft_0} \equiv E_{sft_0} / \sum_s E_{sft_0}$, where E_{sft_0} is consumer expenditure on firm f in state s in the firm's entry year t_0. These shares are complete—they sum to unity for every firm—which eliminates incomplete-share bias. Following Boyce et al. (2016), $Gini_{st}$ is the state-year Gini index constructed from the NielsenIQ. The predicted series $\widehat{Gini}_{st}$ follows Doerr et al. (2024): it interacts each state's initial inequality, $Gini_{s,2004}$, with the national evolution of inequality, measured by the average proportional change in the Gini across other states. Using $\widehat{Gini}_{st}$ rather than realized $Gini_{st}$ downweights state-specific movements that may be mechanically correlated with local firm outcomes while preserving the common shift in the income distribution.

Our outcome is firm-year patenting intensity, measured as granted patent applications per thousand dollars of consumer expenditure, using PatentsView data from the U.S. Patent and Trademark Office. We also use a claim-weighted patent count that weights each patent by its number of claims

to capture technological scope (Aghion et al., 2019b). Expenditure scaling purges firm-size effects from the innovation measure. Following Miguel and Kremer (2004); Xu (2022); Franklin et al. (2024), we adopt a reduced-form shift-share specification, entering $Shock_{ft}^{ineq}$ directly as the regressor. We control for two classes of covariates. The first class consists of shift-share exposure-level controls aggregated with the same entry-year state shares: predicted log average household income, predicted log per-capita GDP, and predicted log population, each constructed via the same leave-one-out procedure. The second class consists of firm-level covariates: the log of the number of product modules supplied (product scope), the firm's expenditure share (market power), and the log of firm age. The estimating sample requires $t > t_0$, ensuring that the initial sales shares are measured strictly before the outcome period (Borusyak et al., 2025). Standard errors are clustered at the firm level.

Table 1 reports the main results based on the Gini index. Greater exposure to inequality is associated with weaker firm innovation. In Panel A, Column (1) shows a negative and statistically significant coefficient of -0.0085. Columns (2)–(5) indicate that this result is robust across alternative samples: firms entering in 2004, firms selling to both low- and high-income consumers, small firms below the median expenditure, and firms observed in at least four years. The effect also varies systematically across firms. In Columns (6) and (7), we split firms by their average buyer income percentile at entry into *poor-market* and *rich-market* firms. The negative effect is concentrated among poor-market firms, while the estimate for rich-market firms is small and statistically insignificant. In economic terms, a one-standard-deviation increase in inequality exposure (0.018) reduces patenting intensity by 0.153 patents per million dollars of expenditure, or about 3.6 fewer patent applications per firm-year given the average annual expenditure of $23.4 million. Panel B further shows that these findings are not driven by low-value patents, as the estimates remain very similar when patent counts are weighted by claims. Taken together, the evidence suggests that rising inequality not only reshapes the distribution of innovation across firms, but also depresses aggregate patenting, with the strongest effect falling on mass-market firms.

Table A.3 reports balance tests following Borusyak et al. (2022): 21 of 24 pre-period characteristics are uncorrelated with the inequality shock, with only the employment rate significant at the 1% level and two variables marginally significant, supporting the shift-orthogonality assumption. Table A.4 shows that augmenting the specification with controls mapping onto these three imbalanced variables leaves the negative and significant effect unchanged. Figure A.20 shows that the predicted Gini shock ranges from 0.30 to 0.45 with substantial cross-state dispersion, and Table A.5 documents effective sample sizes of 253.5 at the state-year level and 21.2 at the state level, with the five largest state-year weights collectively accounting for only 5.5 percent of total exposure—together satisfying the concentration requirement for reliable shift-share inference. Tables A.6 and A.7 show that the result survives excluding the six largest expenditure states (40 percent of total) and the top one percent of high-exposure firms, respectively, confirming that no subset of states or firms drives the finding. Table A.8 reports exposure-robust estimates following Borusyak et al. (2022) and Adão et al. (2019), confirming that the results are not an artifact of firm-level cluster-

ing. Figures A.21 and A.22 address measurement error: the NHCP-based Gini is positively and significantly correlated with the U.S. Census Bureau measure, and the Gini, Theil, and Generalized Entropy indices are virtually collinear. Tables A.9 and A.10 confirm that the full-sample coefficient remains negative and significant and the poor-market concentration persists under the Theil and GE indices, respectively, indicating that the results are not an artifact of a single inequality metric.

Summary of Motivational Facts. Our evidence reveals a systematic pattern in how consumer goods diffuse from higher-income to lower-income households. Over a module's life cycle, the expenditure share of poorer consumers rises, while both the average income of buyers and product prices decline (Fact I). Two forces appear to sustain this diffusion. First, demand from high-income consumers in the early stage of a product's life cycle is associated with stronger incumbent innovation, consistent with subsequent cost reductions over time (Fact II). Second, as the market expands, entry rises and concentration falls, pointing to stronger competition and further pressure on markups (Fact III). In turn, greater exposure to inequality is associated with weaker patenting, with the decline concentrated among firms that rely more heavily on low-income demand (Fact IV).

These patterns suggest that the distribution of income plays a central role in shaping how innovation begins and diffuses across consumers. When purchasing power is concentrated at the top, firms can cover fixed innovation costs by selling to affluent buyers first. High-end demand then finances experimentation and early cost reductions; as prices fall, the market expands and mass-market demand sustains further innovation and entry. Inequality thus shapes where innovation starts and how quickly its gains diffuse. To discipline these empirical regularities and assess how the income distribution shapes firm dynamics and long-run growth, the model below formalizes the underlying mechanism.

3 Model

In this section, we build a dynamic general-equilibrium model that links heterogeneous households on a quality ladder to innovating, heterogeneous firms with endogenous product variety.

3.1 Households

The economy is populated by a continuum of households with identical preferences but heterogeneous incomes. Each household is endowed with labor l, drawn from a continuous and differentiable distribution $\mathcal{G}(l)$ over $[\underline{l}, \infty)$ with mean normalized to one, $\mathbb{E}[l] = 1$. This distribution also represents the income distribution in our model. Preferences are defined over a continuum of vertically differentiated final goods, capturing a hierarchical structure of needs.

Following Matsuyama (2002) and Foellmi and Zweimüller (2006), we assume that a household

Table 1: Inequality Exposure and Firm Innovation

	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel A. Dependent variable: Patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0085**	-0.0118***	-0.0085**	-0.0111**	-0.0081**	-0.0129**	-0.0016
	(0.0037)	(0.0043)	(0.0037)	(0.0051)	(0.0037)	(0.0057)	(0.0044)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	243,889	184,352	243,485	110,313	236,300	118,388	125,498
Adjusted R^2	0.546	0.548	0.547	0.531	0.538	0.520	0.571
	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel B. Dependent variable: Claim-weighted patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0752**	-0.0953***	-0.0753**	-0.1118***	-0.0707**	-0.1075**	-0.0226
	(0.0306)	(0.0354)	(0.0307)	(0.0420)	(0.0308)	(0.0463)	(0.0382)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	243,889	184,352	243,485	110,313	236,300	118,388	125,498
Adjusted R^2	0.548	0.554	0.548	0.524	0.541	0.529	0.565

Notes. This table reports firm-year regressions of innovation intensity on exposure to income-inequality shocks. In Panel A, the dependent variable is the number of ultimately granted patent applications filed by firm f in year t (PatentsView/USPTO), scaled by the firm's total consumer expenditure in thousands of dollars. In Panel B, the dependent variable is the claim-weighted analogue, where each patent is weighted by its number of claims to capture technological scope. The main regressor, $Shock_{ft}^{ineq}$, is the firm-specific inequality exposure: a weighted average of predicted state-year Gini coefficients, with weights given by the firm's complete initial sales shares across U.S. states in its entry year t_0. Predicted state-level inequality is constructed via the leave-one-out national-trend instrument variable procedure. The set of controls (denoted "Controls" above) comprises two classes. The first class consists of shift-share exposure-level controls aggregated with the same entry-year state shares: predicted log average household income, predicted log per-capita GDP, and predicted log population, each constructed via the same leave-one-out procedure as predicted inequality. These absorb firm exposure to first-moment and scale characteristics of state-level economic conditions that may co-move with inequality. The second class consists of firm-level covariates: the log of the number of product modules supplied (product scope), the firm's expenditure share (market-power proxy), and the log of firm age (years since first entry). Columns (1)–(5) report baseline and robustness samples. Column (1) is the full sample. Column (2) restricts to firms whose entry year is 2004. Column (3) restricts to firms with positive sales to both low- and high-income consumers during the sample period (low- and high-income households are defined in the NielsenIQ data description). Column (4) restricts to firms with below-median total consumer expenditure over the sample period (small firms). Column (5) restricts to firms observed in at least four sample years (surviving firms), thereby excluding short-lived entrants. Columns (6)–(7) split firms by sales orientation: *Poor Market* (*Rich Market*) indicates firms whose average buyer income percentile at entry is below (above) the sample median, where the buyer income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. Because this orientation is fixed at the firm's entry year, the split is predetermined with respect to subsequent innovation and inequality shocks, ruling out reverse causality from a firm's innovative activity to its customer composition. All specifications include firm fixed effects and year fixed effects. The estimating sample requires $t > t_0$, ensuring that the initial sales shares used to construct exposures are measured strictly before the outcome period. All continuous variables are winsorized at the 1st and 99th percentiles within each year. Standard errors (in parentheses) are clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

with endowment l derives utility from effective consumption:

$$U_l(0) = \int_0^\infty e^{-\rho t} \frac{C_l(t)^{1-\sigma}}{1-\sigma} dt$$

where $\rho > 0$ is the subjective discount rate, $1/\sigma$ is the intertemporal elasticity, and

$$C_l(t) = \int_0^{N(t)} i^{-\gamma} C_l(i,t) di + \min_{\{i \in (0, N_t]\}} \{(C_l(i,t))\} \cdot X_l(t), \tag{3}$$

where $C_l(i,t) \in \{0,1\}$ denotes a binary decision to consume variety i, $N(t)$ is the measure of available differentiated goods, and $X_l(t)$ is a continuously consumed luxury service. Assume that $\gamma \in [0,1)$. The index $i \in (0, N(t)]$ orders goods by consumption priority: lower-i goods are more basic as they carry a higher utility weight.[5] The luxury service $X_l(t)$ yields utility only if all manufactured varieties are consumed, i.e., if $\min_{i \in (0,N(t)]} C_l(i,t) = 1$[6].

Let $P(i,t)$ denote the price of manufactured variety i and $P^X(t)$ the price of the luxury service $X_l(t)$. Labor is supplied inelastically at the prevailing wage $W(t)$. Following Foellmi and Zweimüller (2006), we assume further that each household has the same income composition (identical labor and profit shares) and has no initial saving at $t = 0$. For a household with labor endowment l, the present-value budget constraint is given by

$$\int_0^\infty e^{-R(t,0)} \left[\int_0^{N(t)} P(i,t) C_l(i,t) di + P^X(t) X_l(t) \right] dt \leq \int_0^\infty W(t) l \cdot e^{-R(t,0)} dt + \mathbb{V}(0) l,$$

where $R(t,0) \equiv \int_0^t r(s)\, ds$ is the cumulative interest factor and $\mathbb{V}(0)$ is the aggregate market value of all firms at $t = 0$ defined in equation (B.23) in the appendix.[7] For tractability, we assume that each household owns l shares of all firms as the expected labor endowment l is equal to one. Thus, a household with a larger l has a higher income level.

Given the binary nature of consumption choices, households follow a reservation-price strategy. For each differentiated good i, a household with labor endowment l purchases the good if and only if its market price does not exceed its reservation price. Solving the household's utility-maximization

[5] Because consumption of each manufactured variety is binary, the conventional classification of goods into "luxuries" and "necessities" based on income elasticities is not well defined: the income elasticity of demand is either zero or infinite. Following Matsuyama (2002), we instead adopt an expenditure-share–based definition. A good is defined as a necessity if its expenditure share rises when household income falls, and as a luxury if its expenditure share rises when household income increases.

[6] Our specification of the luxury service is similar to the treatment of leisure in Matsuyama (2002). In his model, sufficiently wealthy households reduce labor supply once they have consumed all discrete goods, deriving utility from leisure at the margin. In contrast, in our model such households allocate resources to a continuously consumed luxury service once they have consumed all manufactured varieties. Moreover, inequality affects aggregate labor supply in Matsuyama (2002), while aggregate labor supply remains constant in our setting. As luxury services are hard to measure in the data, our analysis focuses on manufactured goods.

[7] The present-value budget constraint implies that households are allowed to save at the interest rate $r(t)$. However, we will show in the proof of Lemma B.8 in the Appendix that, along the BGP, net savings are zero for every households.

problem yields the consumption rule:

$$C_l(i,t) = \begin{cases} 1, & P(i,t) \leq \kappa_l(i,t), \\ 0, & P(i,t) > \kappa_l(i,t), \end{cases} \qquad \text{with} \quad \kappa_l(i,t) = \frac{i^{-\gamma} e^{-\rho t}}{\mu_l C_l(t)^{\sigma}} \cdot e^{R(t,0)}, \tag{4}$$

where $\kappa_l(i,t)$ denotes the reservation price of good i for a household of type l at time t, μ_l represents the marginal utility of wealth (the Lagrange multiplier associated with the household's budget constraint). In the balanced growth equilibrium (BGE) formally defined later, $P(i,t)$ is increasing in i and $\kappa_l(i,t)$ is decreasing in i, which leads to a a hierarchical consumption structure: goods are purchased in an increasing order of i, and households with higher income consume more varieties. Furthermore, only sufficiently rich households can consume all $N(t)$ varieties and allocate their remaining expenditure to luxury services. Lemma B.7 in Appendix B.3.1 characterizes this consumption pattern.

3.2 Production

On the supply side, the economy consists of three types of firms: competitive producers of final goods, monopolistically competitive producers of intermediate goods, and competitive luxury service providers.

Final Goods Producers. While much of the related literature typically assumes a single homogeneous final good, our framework introduces a continuum of final goods $i \in (0, N(t)]$, each linked to its own range of intermediates. This structure captures heterogeneity in sectoral specialization and allows the scope of product variety to evolve endogenously over time. For each final-good variety i, a unit mass of identical perfectly competitive firms produces output using a measure $F(i,t)$ of specialized intermediate inputs supplied by firms indexed by $j \in [0, F(i,t)]$. The representative final-good producer operates with the decreasing-returns-to-scale technology:

$$Y(i,t) = \frac{1}{1-\beta} \int_0^{F(i,t)} M_i(j,t)^{1-\beta} dj,$$

where $M_i(j,t)$ denotes the quantity of intermediate input j used in the production of good i, and $\beta \in (0,1)$ governs the elasticity of substitution across intermediates[8]. A larger $F(i,t)$ reflects a broader range of specialized suppliers within sector i, raising productivity through the variety-expansion channel.

The representative final-good producer chooses the set of inputs $M_i(j,t)$ to maximize profits, taking prices as given:

$$\Pi(i,t) \equiv \max_{\{M_i(j,t)\}} \left\{ \frac{P(i,t)}{1-\beta} \int_0^{F(i,t)} M_i(j,t)^{1-\beta} dj - \int_0^{F(i,t)} Q_i(j,t) M_i(j,t) dj \right\}, \tag{5}$$

[8]Here we assume a decreasing-returns-to-scale production technology. In Appendix C.2, we consider an alternative specification with constant returns to scale, and show that all the theoretical results continue to hold.

where $P(i,t)$ is the price of final good i, and $Q_i(j,t)$ is the price of intermediate good j. The first-order condition of this problem yields the demand for intermediate inputs:

$$M_i(j,t) = \left[\frac{P(i,t)}{Q_i(j,t)}\right]^{\frac{1}{\beta}}. \tag{6}$$

We define sector i as the collection of intermediate producers whose outputs are exclusively used in the production of final good i. This formulation highlights two structural forces that shape modern production networks: the deepening of intra-sectoral specialization and the endogenous expansion of input variety through innovation and market entry.

Intermediate Goods Producers. Within each sector i, a continuum of $F(i,t)$ monopolistically competitive firms produces differentiated intermediate goods. Each firm operates a linear production technology:

$$M_i(j,t) = \left(\tilde{A}_i(j,t)^{\theta}\bar{A}(t)^{1-\theta}\right) L_i(j,t), \tag{7}$$

where $L_i(j,t)$ denotes labor employed by firm j in sector i. Firm productivity has two components: an idiosyncratic term $\tilde{A}_i(j,t)$, and an aggregate productivity term $\bar{A}(t)$, which will be formally defined below. The Cobb–Douglas combination of these components captures the interaction between firm-specific efficiency and economy-wide technological progress, with $\theta \in (0,1)$ determining their relative importance. Each intermediate producer chooses its labor hired to maximize profits:

$$\max_{L_i(j,t)} \left\{Q_i(j,t)M_i(j,t) - W(t)L_i(j,t)\right\}, \tag{8}$$

subject to equation (6) and equation (7), where $W(t)$ is the wage rate.

The first-order condition of (8) yields firm profits:

$$\Pi_i(j,t) = \beta(1-\beta)^{\frac{1-\beta}{\beta}}\left[\frac{P(i,t)^{\frac{1}{1-\beta}}}{W(t)}\bar{A}(t)^{(1-\theta)}\tilde{A}_i(j,t)^{\theta}\right]^{\frac{1-\beta}{\beta}}. \tag{9}$$

equation (9) shows that profits rise with both the sectoral price $P(i,t)$ and the firm's own productivity $\tilde{A}_i(j,t)$. To aggregate across firms, we define sector-level productivity as

$$B(i,t) \equiv \bar{A}(t)^{(1-\theta)\frac{1-\beta}{\beta}} \int_0^{F(i,t)} \tilde{A}_i(j,t)^{\theta\frac{1-\beta}{\beta}} dj \tag{10}$$

The term $B(i,t)$ summarizes the effective productivity of sector i, which increases with both firm-level productivity and the number of intermediate producers. The positive relationship between $B(i,t)$ and $F(i,t)$ arises from decreasing marginal returns to each intermediate input in final-goods production, implying that a richer network of suppliers raises sectoral efficiency. Given this

representation, sectoral output supply and labor demand can be expressed as:

$$Y(i,t) = (1-\beta)^{\frac{1-2\beta}{\beta}} \left[\frac{W(t)}{P(i,t)}\right]^{-\frac{1-\beta}{\beta}} B(i,t), \tag{11}$$

$$L(i,t) = \left[\frac{W(t)}{(1-\beta)P(i,t)}\right]^{-\frac{1}{\beta}} B(i,t), \tag{12}$$

both are proportional to sectoral productivity $B(i,t)$. The novel feature here lies in the structure of the relative price $P(i,t)$. Rather than serving as a mere equilibrium outcome, $P(i,t)$ encapsulates consumers' willingness to pay along the hierarchy of needs, linking demand heterogeneity directly to production outcomes.

Following the standard expanding-variety framework of Romer (1990), we interpret $N(t)$—the number of available final-good varieties—as a proxy for aggregate productivity, and simply set $\bar{A}(t) = N(t)$. This formulation captures the idea that technological progress operates primarily through the expansion of product varieties. Alternative measures—such as the economy-wide average firm productivity would yield similar implications. The analysis is reported in the Appendix.

Luxury Service Providers. Finally, we introduce the production of luxury services. These services are supplied by representative competitive firms operating a linear technology:

$$X(t) = \bar{A}(t)^{1-\gamma} L^X(t), \ \gamma \in (0,1),$$

where $L^X(t)$ denotes labor employed in the luxury service sector. The exponent $1-\gamma$ captures the degree to which productivity improvements in goods-producing sectors spill over to luxury services. This specification ensures the existence of BGP with a stationary relative price of luxury services $P^X(t)$ and keeps the utility contributions from goods and services comparable along the growth path. Firms in this sector operate under perfect competition and do not engage in innovation on their own. Because technological progress in the luxury service sector is not our focus, we simply assume productivity growth in luxury service sector arises entirely through spillovers from the goods-producing sectors, as aggregate productivity $\bar{A}(t)$ diffuses throughout the economy.

3.3 Innovation

Innovation in the economy takes three distinct forms. First, incumbent intermediate-good producers engage in R&D to improve their own productivity—what we refer to as *incumbents' innovation*. Second, potential entrants invest in R&D to introduce new intermediate varieties, a process we label *firm entry*. Finally, a separate group of inventors develops entirely new final goods, thereby creating new consumption categories and expanding the economy's sectoral frontier—a process we refer to as *new-good invention*.

Incumbents' Innovation We start with incumbents' R&D decisions. An incumbent that invests in R&D faces a Poisson arrival rate of successful innovation, denoted by z. Each success raises the firm's productivity by a factor $\lambda > 1$. For firm j in sector i, the innovation arrival rate is

$$z_i(j,t) = \psi \left(\frac{\tilde{A}_i(j,t)}{\bar{A}(t)} \right)^{-\iota} \left(\bar{A}(t) L_i^I(j,t) \right)^{\alpha}, \tag{13}$$

where $L_i^I(j,t)$ is labor allocated to R&D. The technology is concave in R&D effort, with $\alpha \in (0,1)$ governing the elasticity of innovation with respect to R&D labor and $\psi > 0$ indexing baseline R&D efficiency. As in standard endogenous-growth models, aggregate productivity $\bar{A}(t)$ enters R&D as an external knowledge stock, so that a given amount of research labor is more productive in a more advanced economy. The term $\left(\tilde{A}_i(j,t)/\bar{A}(t)\right)^{-\iota}$ captures diminishing returns to pushing the frontier: conditional on research effort, relatively more productive firms face a lower marginal effectiveness of R&D, which prevents systematically faster growth for already-leading incumbents. We impose the following parameter restriction:

Assumption 1 *The parameter ι satisfies $\iota = \alpha\theta\frac{1-\beta}{\beta}$.*

Assumption 1 imposes a standard scaling restriction that keeps the incumbent's R&D problem homogeneous (see, e.g., Akcigit and Kerr (2018); Peters (2020)). With this restriction, a firm's productivity affects incentives only through a normalization, so $\tilde{A}_i(j,t)$ need not be tracked as an additional state variable. The optimal R&D policy can therefore be written in terms of normalized sectoral and aggregate objects, which delivers a tractable characterization of innovation behavior.

Assume that all incumbent firms face an exogenous exit risk, modeled as a Poisson process with constant rate δ. Then the value function of firm j with productivity $\tilde{A}_i(j,t)$, denoted $\tilde{V}_i(\tilde{A}_i(j,t),t)$, satisfies the following Hamilton–Jacobi–Bellman (HJB) equation along the BGP:

$$\begin{aligned} r\tilde{V}_i(\tilde{A}_i(j,t),t) - \dot{\tilde{V}}_i(\tilde{A}_i(j,t),t) \;=\; & \max_{L_i^I(j,t)} \Pi_i(j,t) - W(t)\cdot L_i^I(j,t) \\ & + z_i(j,t)\cdot\left[\tilde{V}_i(\lambda\tilde{A}_i(j,t),t) - \tilde{V}_i(\tilde{A}_i(j,t),t)\right] \\ & - \delta\tilde{V}_i(\tilde{A}_i(j,t),t), \end{aligned} \tag{14}$$

where r is the equilibrium interest rate along the BGP. The first line on the right-hand side represents the firm's operating profit net of R&D costs, while the second and the third capture the expected gain from successful innovation and the loss from exogenous exit, respectively. In Lemma B.1 of Appendix B.2.2, we provide a characterization of the value function

$$\tilde{V}_i(A,t) = A^{\theta\frac{1-\beta}{\beta}} V_i(t), \tag{15}$$

and show that the R&D intensity $z_i(j,t)$ is identical across firms j in the same sector i and depends solely on sector-level fundamentals. We simply write it as $z_i(t)$. This property substantially simplifies the characterization of equilibrium innovation dynamics.

Firm Entry As a second type of innovation, potential entrepreneurs in each sector i conduct R&D aimed at creating new firms. Entry is stochastic and follows a Poisson process, with the arrival rate of successful innovation given by

$$e(i,t) = \psi_e \bar{A}(t) L_i^E(t), \tag{16}$$

where $L_i^E(t)$ denotes the amount of labor devoted to R&D for new firm creation in sector i, and $\psi_e > 0$ measures the efficiency of the entry process. Upon a successful innovation, a new firm is created and assigned a productivity level A drawn from the existing distribution of incumbent firms within the same sector. The optimization problem of potential entrants in sector i is given by

$$\max_{L_i^E(t)} \left\{ e(i,t) \cdot \mathbb{E}_t[A^{\theta\frac{1-\beta}{\beta}} V_i(t)] - W(t) \cdot L_i^E(t) \right\}, \tag{17}$$

where $\mathbb{E}_t[A^{\theta\frac{1-\beta}{\beta}} V_i(t)]$ represents the expected value of a newly created firm conditional on successful entry. The number of firms in sector i evolves according to

$$\dot{F}(i,t) = e(i,t) - \delta F(i,t), \tag{18}$$

where $e(i,t)$ captures the inflow of new entrants and $\delta F(i,t)$ represents the exit of incumbent firms due to exogenous shocks. More technical details on firm entry are provided in Appendix B.2.

Given this structure, sectoral productivity is determined jointly by incumbent innovation and firm dynamics. As defined in equation (10), sectoral productivity $B(i,t)$ depends on aggregate productivity $\bar{A}(t)$, the number of firms $F(i,t)$, and firm-specific productivity $\tilde{A}_i(j,t)$. As shown in Appendix B.3.3, the evolution of $B(i,t)$ follows

$$\frac{\dot{B}(i,t)}{B(i,t)} = \underbrace{(1-\theta)\frac{1-\beta}{\beta}\frac{\dot{\bar{A}}(t)}{\bar{A}(t)}}_{\text{technology spillovers}} + \underbrace{z_i(t) \cdot \left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right)}_{\text{incumbents' innovation}} + \underbrace{\frac{e(i,t)}{F(i,t)}}_{\text{firm dynamics}} - \delta\,. \tag{19}$$

equation (19) indicates that sectoral productivity growth is shaped by three key forces: aggregate technological spillovers, innovation by incumbent firms, and the entry of new firms. While new entrants do not immediately raise average productivity—since their productivity is drawn from the existing distribution—they expand the range of intermediate inputs available for production. This expansion along the extensive margin increases the diversity of inputs, improves resource allocation within the sector, and ultimately contributes to higher overall productivity growth.

New-Good Invention New-good invention creates entirely new categories of final consumption goods, expanding the consumption frontier by generating new sectors. It is carried out by a continuum of inventors embedded in households and distributed according to the income distribution $\mathcal{G}(l)$, so that their consumption is fully internalized in aggregate household consumption. Inventors undertake R&D by hiring labor, and successful invention follows a Poisson process with an arrival

rate linear in invention labor and aggregate productivity. The law of motion for the number of final goods is given by

$$\dot{N}(t) = \phi \bar{A}(t) L^N(t), \tag{20}$$

where $L^N(t)$ denotes the total labor input devoted to invention activities, and $\phi > 0$ measures the efficiency of the invention process. Each successful invention gives rise to a new sector producing a distinct final good. Upon invention, each inventor is endowed with ownership of a unit mass of final-goods firms in the newly established sector. The initial intermediate entrants in that sector begin with a baseline productivity level equal to a fraction ζ of aggregate productivity $\bar{A}(t)$, i.e. $\tilde{A}_{N(t)}(j,t) = \zeta \bar{A}(t)$ for all $j \in [0, F(N(t),t)]$. In the equilibrium, ζ is assumed to be relatively small to reflect that newly created sectors typically start with lower productivity. The precise restriction on ζ is discussed in Appendix B.5.

The inventor's optimization problem can therefore be written as

$$\max_{L^N(t)} \{V^F(N(t),t)\dot{N}(t) - W(t)L^N(t)\}, \tag{21}$$

where $V^F(N(t),t)$ denotes the value of a final-goods firm in a newly established sector, corresponding to the present discounted value of all future profits as defined in equation (5). This formulation highlights that inventors' incentives to create new consumption goods depend on the profitability of final-goods producers, which in turn reflects household demand for newly invented varieties. Because demand declines with the index $N(t)$, aggregate income constrains the market size for high-$N(t)$ goods. Consequently, inventors have limited incentive to develop excessively luxurious varieties, as the expected firm value $V^F(N(t),t)$ becomes too low to justify invention costs. As aggregate productivity $\bar{A}(t)$ rises, however, higher household income expands effective demand, making the invention of more luxurious goods profitable.

Given $N(t) = \bar{A}(t)$, equation (20) implies that

$$g_n \equiv \frac{\dot{N(t)}}{N(t)} = \phi L^N(t).$$

As in Romer (1990), the engine of economic growth is driven by the growth of $N(t)$, which is determined by the labor $L^N(t)$ for the final product innovation. As $L^N(t)$ is endogenously determined by the labor market clearing condition, a key feature of our model is that hierarchical consumer demand and labor allocation in other sectors of the economy play a central role in economic growth.

4 Balanced Growth Equilibrium

In this section, we characterize the balanced growth equilibrium (BGE) and study properties of such an equilibrium.

4.1 Characterizing the BGE

We characterize the equilibrium along a BGP. When doing so, we suppress the time index t whenever no confusion arises. Define aggregate final consumption index as

$$C(t) = \int_{\underline{l}}^{\infty} C_l(t) d\mathcal{G}(l).$$

We normalize the price index P^C of this aggregate final consumption bundle to 1. Define aggregate output as

$$Y(t) = \int_0^{N(t)} P(i,t)Y(i,t)di + X(t)P^X(t),$$

where X denotes aggregate luxury services and satisfies

$$X(t) = \int_{\underline{l}}^{\infty} X_l(t) d\mathcal{G}(l)$$

in equilibrium. In the proof of Proposition 1, we show that $Y(t) = C(t)$ holds in a BGE.

Definition 1 *A BGE consists of paths of prices* $\{W, r, P^X, \{P(i)\}_i, \{Q_i(j)\}_{i,j}\}$*, quantities* $\{N, \{F(i)\}_i, \{Y(i)\}_i, \{M_i(j)\}_{i,j}, \{C_l(i)\}_{l,i}, \{X_l\}_l, X, C, Y\}$*, and labor choices* $\{\{L_i(j)\}_{i,j}, \{L_i^I(j)\}_{i,j}, \{L_i^E\}_i, L^N, L^X\}$ *such that: (i) households maximize lifetime utility subject to their budget constraints taking interest rates and prices as given; (ii) competitive final-good producers in each sector i chooses intermediate goods to maximize their profits; (ii) monopolistically competitive intermediate goods firms for each sector i chooses labor to maximize profits, taking downward-sloping demand curve as given; (iii) incumbent intermediate goods firms hire labor to conduct innovations to maximize firm value; (iv) entrants and new-good innovators choose labor to solve problems* (17) *and* (21) *taking wages as given; (v) the mass of intermediate goods firms $F(i)$ satisfies equation* (18)*; (vi) competitive luxury services providers choose labor to maximize profits. (vii) markets for final goods, intermediate goods, luxury services, and labor all clear; and (viii) along the BGE, r and P^X are constant,* $\{C, X, Y, W, N\}$*, and* $\{C_l\}_l$ *grow at constant rates.*

In Appendix B, we show that, along the BGP, Y, C, X, C_l, and W all grow at the same rate $(1-\gamma)g_n$, and $X_l = 0$ for households with $l < \bar{l}$, but X_l grows at rate $(1-\gamma)g_n$ for households with $l \geq \bar{l}$, where $\bar{l}$ is endogenously determined in Lemma B.5. In addition, at any point in time t^*, as a new sector $N(t^*)$ is created, the number of firms in that sector, $F(N(t^*), t)$, changes with t due to entry and exit and approaches a constant in the long run. As other new products are being invented over time, product $N(t^*)$ moves down the product ladder. As aggregate labor supply is fixed at 1 and sectors expand at rate g_n, employment and entrant R&D labor input in each sector decline at rate g_n.

A convenient way to express the BGP in stationary form is to normalize the product space from

$(0, N(t)]$ to $(0, 1]$. Accordingly, define the normalized product index

$$\tilde{i}(t) \equiv \frac{i}{N(t)}.$$

This index orders goods by their relative position along the product ladder and plays a central role in characterizing sectoral outcomes along the BGP. In Lemma B.10, we show that two sectors with the same $\tilde{i}$ at different points in time will share identical (possibly detrended) values for output, innovation rates, firm numbers, and other sector-specific variables.

Based on this property, each sectoral variable can be rewritten as the product of a common trend and a stationary profile in $\tilde{i}$. Throughout, detrended variables are written in lowercase. As an illustration, let $V_i(t)$ denote the productivity-adjusted firm value defined in equation (15), and define its detrended counterpart as $v(\tilde{i}) \equiv e^{-g_v t} V_i(t)$, where g_v is the constant growth rate of $V_i(t)$. Once $v(\tilde{i})$ is determined, all sectoral supply-side variables—including innovation intensity and productivity—are uniquely pinned down, so that $v(\tilde{i})$ summarizes the relevant supply-side information.

The same approach can be applied to the demand side. Define $\mathcal{L}(\tilde{i}(t))$ as the lowest labor endowment level among households that purchase product $\tilde{i}$ at time t:

$$\mathcal{L}(\tilde{i}(t)) = \inf\{l : C_l(i, t) = 1\}.$$

The household with $l = \mathcal{L}(\tilde{i}(t))$ is the marginal buyer such that any household with $l' > l$ will also buy product $\tilde{i}$ at time t. The function $\mathcal{L}(\tilde{i})$ provides a compact summary of the demand for good $\tilde{i}$. The detailed characterization of $\mathcal{L}(\tilde{i})$ can be found in Appendix B.3.1.

In Proposition 1, we show that the BGE can be characterized by a joint system for $v(\tilde{i})$ and $\mathcal{L}(\tilde{i})$, together with a small number of endogenous scalars including the growth rate g_n of N.

Proposition 1 *The BGE can be characterized by two functions,* $\{v(\tilde{i}), \mathcal{L}(\tilde{i})\}$, $\tilde{i} \in (0, 1]$, *and four endogenous variables,* $\{w, \eta, y, g_n\}$, *which are jointly determined by the following system:*

(i) The function $v(\tilde{i})$ *solves*

$$v'(\tilde{i}) = \theta \frac{1-\beta}{\beta \tilde{i}} \cdot \left[\left(\frac{v(\tilde{i})}{h(g_n)} \right)^{\frac{\alpha}{1-\alpha}} - 1 \right] v(\tilde{i}), \tag{22}$$

subject to the boundary condition

$$v(1) = \frac{w}{\psi_e \zeta^{\theta \frac{1-\beta}{\beta}}}.$$

Here, $h(\cdot)$ *is given by*

$$h(g_n) = \frac{w \left(\theta \frac{1-\beta}{\beta} \right)^{\frac{1-\alpha}{\alpha}}}{\left[\left(\lambda^{\theta \frac{1-\beta}{\beta}} - 1 \right) \psi \right]^{\frac{1}{\alpha}} \alpha} g_n^{\frac{1-\alpha}{\alpha}}.$$

(ii) There exists a constant $\eta > 0$ denoting the share of product varieties universally consumed across all households. If $\tilde{i} \geq \eta$, $\mathcal{L}(\tilde{i})$ satisfies

$$\mathcal{L}'(\tilde{i}) = \frac{w^{1-\beta}}{y\beta^{\beta}(1-\beta)^{1-\beta}} \left\{ [\rho + \delta + ((\sigma-1)(1-\gamma)+1)\, g_n]\, v(\tilde{i}) + w \left[\frac{w}{\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) v(\tilde{i}) \cdot \psi\alpha} \right]^{\frac{1}{\alpha-1}} \right\}^{\beta}, \tag{23}$$

subject to the boundary conditions $\mathcal{L}(\eta) = \underline{l}$. For $\tilde{i} \in (0, \eta)$, $\mathcal{L}(\tilde{i}) = \underline{l}$.

(iii) The four variables are determined by four conditions detailed in Appendix B.4: w from the labor market clearing condition; η from the resource constraint; y from the price normalization; g_n from the entry condition of final producers.

The proof is shown in Appendix B.4. This proposition characterizes the equilibrium through two ODEs. Equation (22) is derived from the firm entry condition together with the dynamics of sector-level productivity. Its boundary condition reflects the assumption that the productivity of any newly established sector equals a fraction ζ of the aggregate productivity. The function $h(\cdot)$ corresponds to $\lim_{\tilde{i}\to 0} v(\tilde{i})$ for a given growth rate g_n. Equation (23), in turn, is obtained from the incumbent firms' HJB equation combined with the demand curve summarized by $\mathcal{L}(\tilde{i})$. The boundary condition shows that the poorest household with endowment $\underline{l}$ buys all products $\tilde{i} \in (0, \eta]$. This means that sectors in $(0, \eta]$ produce basic products that are consumed by all consumers. Figure B.1 in Appendix B.1 illustrates our model structure and equilibrium feature.

4.2 Sectoral Dynamics along the BGP

Beyond the technical characterization in Proposition 1, our model admits tractable qualitative properties of sectoral dynamics. In particular, as one moves up the product ladder (i.e., to higher $\tilde{i}$), the equilibrium exhibits monotone patterns in productivity, prices, firm value, and innovation incentives. As shown in Appendix B.3.3, we define detrended variables:

$$z(\tilde{i}) = z_i(t),\ b(\tilde{i}) = e^{-\frac{1-\beta}{\beta} g_n t} B(i,t),\ p(\tilde{i}) = e^{\gamma g_n t} P(i,t).$$

Proposition 2 presents the result with its proof given in Appendix B.5.

Proposition 2 *Along the BGP, for $\tilde{i} \in (0, 1]$, detrended sectoral productivity $b(\tilde{i})$ is strictly decreasing in $\tilde{i}$, whereas the detrended price $p(\tilde{i})$, productivity-adjusted firm value $v(\tilde{i})$, and innovation intensity $z(\tilde{i})$ are strictly increasing. Moreover,*

$$p(\tilde{i}) = y\mathcal{L}'(\tilde{i}) \quad \text{for} \quad \tilde{i} \in [\eta, 1). \tag{24}$$

Proposition 2 has four implications. First, the detrended productivity $b(\tilde{i})$ is lower in more luxurious sectors (higher $\tilde{i}$). Equivalently, within any given sector i, the productivity $B(i,t)$ rises over time as the sector matures, reflecting both incumbent innovation and entry-induced reallocation.

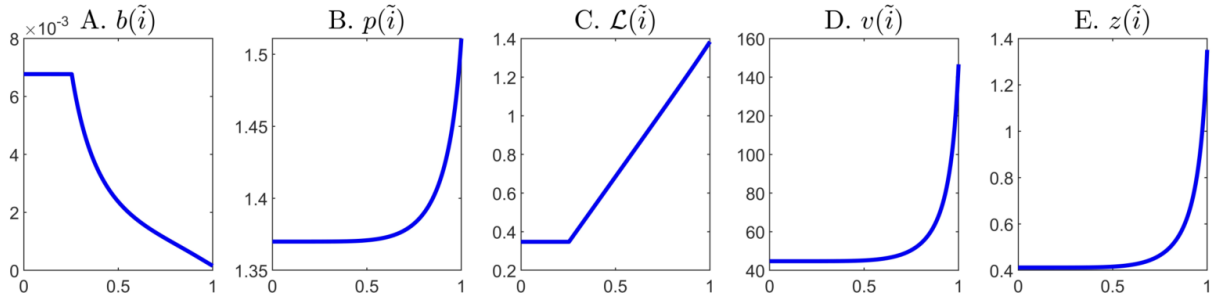
Second, the detrended price is higher for a more luxurious product. Intuitively, new sectors for more luxurious products begin with lower productivity and therefore higher marginal costs. With constant markups for both intermediate- and final-good producers, these higher costs translate one-for-one into higher prices. This result implies that, for any fixed sector i, the product price $P(i,t)$ declines over time at an asymptotic rate γg_n. As product ages, incumbents conduct innovations to improve productivity so that the price falls over time. The trend growth reflects aggregate productivity growth. Equation (24) shows that, in detrended terms, the price of product $\tilde{i} \in [\eta, 1)$ is equal to the total income that the marginal buyer is willing to spend for the product with an additional level of luxury.

Third, the productivity-adjusted firm value $v(\tilde{i})$ increases with $\tilde{i}$. This object measures the value of a marginal improvement in productivity. Because luxury sectors sustain higher prices, an additional unit of output is worth more, raising the payoff to productivity gains and hence $v(\tilde{i})$.

Finally, the innovation intensity $z(\tilde{i})$ rises with the luxury level. Intuitively, a higher productivity-adjusted value strengthens the incentive to invest in R&D, increasing the Poisson arrival rate of innovation. These monotone sectoral patterns yield a set of testable predictions, and they line up with the empirical regularities documented in Section 2.2.

Figure 5 illustrates Proposition 2 based on the calibrated parameter values in Section 5. Notice that some lines become almost flat when $\tilde{i}$ is below $\eta = 0.253$. This is because the market size does not expand as all households consumes products in $(0, \eta]$. In Panel C, $\mathcal{L}(\tilde{i})$ is exactly flat at $\underline{l}$ on $(0, \eta)$ and is increasing and convex on $[\eta, 1)$.

Figure 5: Sectoral Dynamics



Notes. The figure illustrates several detrended sectoral variables under the parameter listed in Table 2, including: detrended productivity $b(\tilde{i})$ in Panel A, detrended product prices $p(\tilde{i})$ in Panel B, the marginal buyer $\mathcal{L}(\tilde{i})$ in Panel C, productivity-adjusted firm value $v(\tilde{i})$ in Panel D, and innovation intensity $z(\tilde{i})$ in Panel E.

4.3 Innovation Allocation and the Growth Bottleneck

Changes in the income distribution affect not only who consumes each product, but also which products firms find most profitable to improve. By changing the composition of demand along the product ladder, they generate heterogeneous R&D responses across sectors. We first characterize

how sectoral innovation changes whenever a shift in the income distribution alters the aggregate growth rate, holding all other parameters fixed.

Proposition 3 *Consider two income distributions such that $g_n^1 > g_n^2$ in the BGE. Then the logarithmic change in sector-level innovation intensity, $\log z^1(\tilde{i}) - \log z^2(\tilde{i})$, is positive for all $\tilde{i}$ and strictly decreasing in $\tilde{i}$.*

Proposition 3 delivers two key implications, and its proof is given in Appendix B.6. First, aggregate growth and innovation move together: any change in income distribution that lowers (raises) the long-run growth rate also lowers (raises) innovation intensity in every sector. Second, the response is systematically tilted toward the lower end of the product ladder. When growth falls, innovation contracts most in necessity sectors; when growth rises, innovation expands most in those same sectors. Intuitively, a widening in inequality shifts purchasing power away from low-income households, tightening demand where necessities are priced. Because the willingness to pay for necessity goods is anchored by the marginal low-income buyer, a rise in inequality depresses necessity-sector prices and profits both in the current period and in the expected future, reducing the return to cost-reducing improvements and weakening R&D incentives precisely where diffusion generates the largest welfare gains. By contrast, luxury sectors face a different trade-off. Higher inequality can raise their current price by expanding rich-household demand, but it also lowers the expected future price once the market reaches lower-income households. These two forces offset each other, so innovation in luxury sectors responds much less. Thus, inequality shocks do not simply rescale innovation uniformly—they reshape its sectoral allocation.

Having characterized the heterogeneous response of sectoral innovation, we next examine how innovation in necessity sectors relates to the long-run rate of product-category expansion.

Proposition 4 *The BGE growth rate of $N(t)$ is determined by the innovation intensity in the most necessity sectors:*

$$g_n = \frac{1}{\theta}\frac{\beta}{1-\beta}\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right)\lim_{\tilde{i}\to 0} z(\tilde{i}).$$

The proof of Proposition 4 is given in Appendix B.7. Together with Proposition 2, which establishes that $z(\tilde{i})$ is strictly increasing in $\tilde{i}$, Proposition 4 implies that $\lim_{\tilde{i}\to 0} z(\tilde{i})$ is a lower bound for $z(\tilde{i})$ for all $0 < \tilde{i} \leq 1$, and hence the infimum of the innovation schedule. Aggregate growth is therefore pinned down by the sector with the slowest technological progress, echoing the bottleneck logic emphasized by Acemoglu et al. (2024); Jones and Tonetti (2025). In our framework, the most necessary sectors endogenously become the bottleneck.

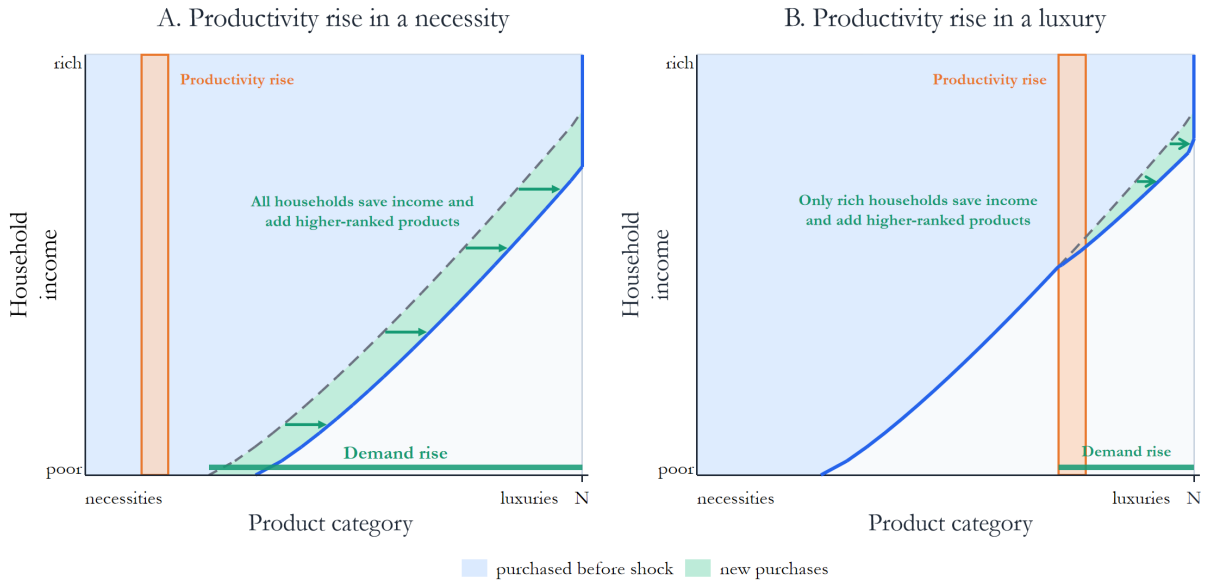
Why does innovation in necessity sectors ultimately determine the long-run rate of product-category expansion? Expanding the product space requires both purchasing power and labor to be continuously released from existing sectors toward newly created ones. Productivity growth in necessity sectors facilitates this process by lowering the expenditure and labor required to satisfy

existing needs. Slower innovation in necessities therefore delays the release of purchasing power and labor toward new products, constraining long-run product expansion.

Figure 6 illustrates this upward demand externality through a partial-equilibrium experiment. Starting from a given equilibrium, we introduce a positive productivity shock in one sector while holding wages and household income fixed. A productivity improvement in a necessity lowers its price for a broad set of households, releasing purchasing power and increasing demand for a wide range of products higher up the consumption hierarchy. In contrast, a productivity improvement in a luxury benefits only the households already consuming it and therefore increases demand only for a narrower range of even more luxurious products. Thus, productivity growth in necessity sectors generates a broader upward demand externality. Appendix B.7 formally characterizes this effect.

Productivity growth also generates a parallel labor-saving externality. By reducing the labor required to satisfy existing needs, productivity improvements in necessity sectors release labor toward newly created products.

Figure 6: Upward Demand Spillover of Productivity Growth



Notes. The horizontal axis ranks products from necessities to luxuries, while the vertical axis ranks households from poor to rich. The light-blue area indicates household–product pairs purchased before the shock, and the dashed green curve represents the consumption frontier after the shock. The orange band marks the product experiencing the productivity improvement, and the green segment along the horizontal axis indicates the range of products whose aggregate demand increases. Panel A considers a productivity improvement in a necessity purchased by all households, and Panel B considers an improvement in a luxury purchased only by richer households.

This mechanism parallels a familiar insight from the structural transformation literature (Johnston and Mellor, 1961; Gollin et al., 2007; Restuccia et al., 2008). These studies emphasize that sluggish productivity growth in agriculture delays the reallocation of resources toward higher-productivity activities and thereby depresses aggregate performance. In our setting, necessity sectors play an analogous role, with slow innovation delaying the release of both purchasing power and labor toward new products.

4.4 Inequality and Growth: An Analytical Example

In this subsection, we examine analytically how income inequality affects innovation and long-run growth. To obtain a tractable characterization, we impose the following parameter restrictions:

$$\sigma = 1, \qquad \alpha = \frac{1}{2}, \qquad \theta\frac{1-\beta}{\beta} = 1. \tag{25}$$

Under these restrictions, the differential equations determining the productivity-adjusted firm value $v(\tilde{i})$ and incumbent innovation intensity $z(\tilde{i})$ admit closed-form solutions:

$$v(\tilde{i}) = \frac{2wg_n}{(\lambda-1)^2\psi^2}\frac{1}{1-(1-g_n/\bar{g})\,\tilde{i}}, \qquad z(\tilde{i}) = \frac{g_n}{\lambda-1}\frac{1}{1-(1-g_n/\bar{g})\,\tilde{i}},$$

where $\bar{g} \equiv \frac{(\lambda-1)^2\psi^2}{2\psi_e\zeta}$[9]. We further assume that the discount rate ρ is sufficiently small, under which equilibrium uniqueness can be established analytically.[10]

To compare inequality across different income distributions, we follow Bergemann and Bonatti (2024) and define that $\mathcal{G}^2$ is a *strict mean-preserving spread* of $\mathcal{G}^1$ if the two distributions have the same mean and satisfy

$$\int_0^x \left[\mathcal{G}^2(l) - \mathcal{G}^1(l)\right] dl > 0$$

for every x in their supports. Intuitively, relative to $\mathcal{G}^1$, $\mathcal{G}^2$ places more weight on the tails of the income distribution and less weight on intermediate income levels, while leaving mean income unchanged. Thus, $\mathcal{G}^2$ represents an unambiguously more unequal income distribution.

Proposition 5 *Under the parameter restrictions in equation (25) and for a sufficiently small discount rate ρ, if $\mathcal{G}^2$ is a strict mean-preserving spread of $\mathcal{G}^1$, then $g_n^2 < g_n^1$.*

The proof of Proposition 5 is provided in Appendix B.8. The proposition establishes a strong comparative-static result: under the stated conditions, any strict mean-preserving spread of the income distribution reduces long-run growth. Combined with Propositions 3 and 4, it highlights the mechanism linking inequality to the endogenous growth bottleneck: greater inequality reallocates R&D away from necessity sectors, slowing innovation precisely where technological progress is most important for long-run product expansion.

To isolate this R&D reallocation channel from changes in aggregate innovation resources, Appendix C.1 introduces separate markets for skilled and unskilled labor, with skilled labor used in innovation and unskilled labor used in production. Because the aggregate supply of skilled labor is fixed, so is the total amount of R&D resources. We show that, when the initial level of inequality is sufficiently high, a further increase in inequality reallocates R&D away from necessity sectors and lowers long-run growth. Thus, the result operates through the allocation, rather than the total amount, of R&D resources. If we shut down incumbent R&D and firm entry, skilled labor is used

[9] The derivation are provided in Appendix B.8.

[10] For larger values of ρ, our numerical exercises continue to yield a unique equilibrium, but establishing uniqueness analytically becomes considerably more difficult. We therefore impose this restriction to keep the analysis tractable.

exclusively for the invention of new products, as in Foellmi and Zweimüller (2006), and the rate of product expansion is independent of the income distribution. Once these additional innovation margins are introduced, however, inequality changes how the same fixed stock of R&D resources is allocated across sectors and innovation activities. This endogenous R&D allocation margin constitutes the key distinction between our framework and the existing hierarchical-demand literature.

5 Quantitative Analysis

In our model, rising inequality tilts innovation away from widely consumed goods, delays product diffusion, and slows growth. In this section, we calibrate our model to quantify these forces and to evaluate three counterfactual questions: what drives product diffusion, how large the effect of inequality on growth is, and how alternative income tax schedules affect welfare.

5.1 Model Calibration

Parameter values and their identification are summarized in Table 2. The calibration separates parameters that can be pinned down directly from standard references or simple moment mappings (external calibration) from those that require solving the model and matching a small set of informative moments (internal calibration). The internal targets are chosen to discipline three objects that are central for our mechanism: aggregate growth, firms' innovation effort and size dispersion, and the speed of diffusion and entry after a product's introduction.

Externally Calibrated Parameters. We set preferences to conventional values: the discount rate is $\rho = 0.02$ and the inverse intertemporal elasticity of substitution is $\sigma = 2$, following Acemoglu et al. (2018). The elasticity of innovation with respect to R&D is set to $\alpha = 0.5$, consistent with estimates in Hall and Van Reenen (2000), Blundell et al. (2002), and Acemoglu et al. (2018). Firm exit is fixed at $\delta = 0.099$, targeting establishment exit rates in the U.S. Business Dynamics Statistics (BDS) over 2004–2016. The parameter β is disciplined by operating profitability. In the model, the average profit-to-sales ratio of intermediate producers equals $1-(1-\beta)^2$, which maps directly into the combined pre-tax profit and R&D expenditure rate reported in Akcigit and Kerr (2018). This yields $\beta = 0.056$. The hierarchy-preference parameter γ is disciplined by the long-run relation between aggregate growth and the trend in sectoral prices. The proof of Proposition 1 implies

$$g_p \;=\; -\frac{\gamma}{1-\gamma}\, g_y, \tag{26}$$

where g_y is the asymptotic growth rate of per-capita output and g_p is the asymptotic growth rate of sector-level prices $P(i,t)$. We target $g_y = 1.6\%$, the average U.S. per-capita GDP growth rate over 2010–2019, and set $g_p = -0.86\%$, the average price trend for necessity goods in our sample.[11]

[11] We use the 2010s for the aggregate growth target to avoid the extreme fluctuations around the 2008–2009 financial crisis. Using 2004–2016 excluding 2008–2009 yields very similar results. We define necessity goods as those whose base-year (2004) target-consumer income lies in the bottom 5% of the income distribution; in the model, the

equation (26) then implies $\gamma = 0.343$.

Finally, we discipline the baseline income distribution using the World Inequality Database (WID) (Piketty et al., 2018). To obtain a transparent, one-parameter summary of U.S. inequality, we assume incomes follow a Pareto distribution and choose the shape parameter ϵ to match the average U.S. post-tax Gini coefficient in the 2010s, 0.484. This mapping implies $\epsilon = 1.533$.[12] Together with the external targets for U.S. aggregate growth and price trends in the 2010s, this pins down a baseline economy that matches the key moments used in our calibration and provides a coherent starting point for the counterfactual exercises.

Table 2: Calibration of Structural Parameters

Parameter	Interpretation	Sources	Value
	External Calibrated Parameters		
ρ	Discount rate	Acemoglu et al. (2018)	0.02
σ	Inverse of Intertemporal elasticity of substitution	Acemoglu et al. (2018)	2
β	Firm profitability	Akcigit and Kerr (2018)	0.056
δ	Firm exit rate	U.S. Business Dynamics Statistics	0.099
α	Elasticity of innovation to R&D	Hall and Van Reenen (2000)	0.50
γ	Degree of hierarchy preference	equation (26)	0.343
ϵ	Pareto exponent in income distribution	World Inequality Database	1.533
	Internal Calibrated Parameters		
ϕ	Productivity of inventors	GDP per-capita growth rate	2.855
ψ	Productivity of incumbents' innovation	Firm-level R&D-to-sales ratio	0.216
ψ_e	Productivity of firm entry	Normalization	1
λ	Step size of a successful innovation	Firm size dispersion	1.053
ζ	Productivity of newly-established sector	Firm number growth rate	0.575
θ	Productivity spillover on production	Product diffusion across households	0.374

Notes. "External" parameters are pinned down either by standard values in the literature or by direct mappings that do not require solving the full equilibrium (e.g., β from the implied profit-share identity). "Internal" parameters are chosen jointly so that the solved BGE matches the listed moments. Moments based on post-entry dynamics are computed as module-by-entry-year averages over the first twelve years after a module is introduced. The income distribution is parameterized as Pareto; the WID post-tax Gini in the 2010s identifies ϵ one-to-one within this family.

Internally Calibrated Parameters. The remaining parameters are disciplined by matching model moments to their empirical counterparts. For transparency, each parameter is tied to a single primary target.

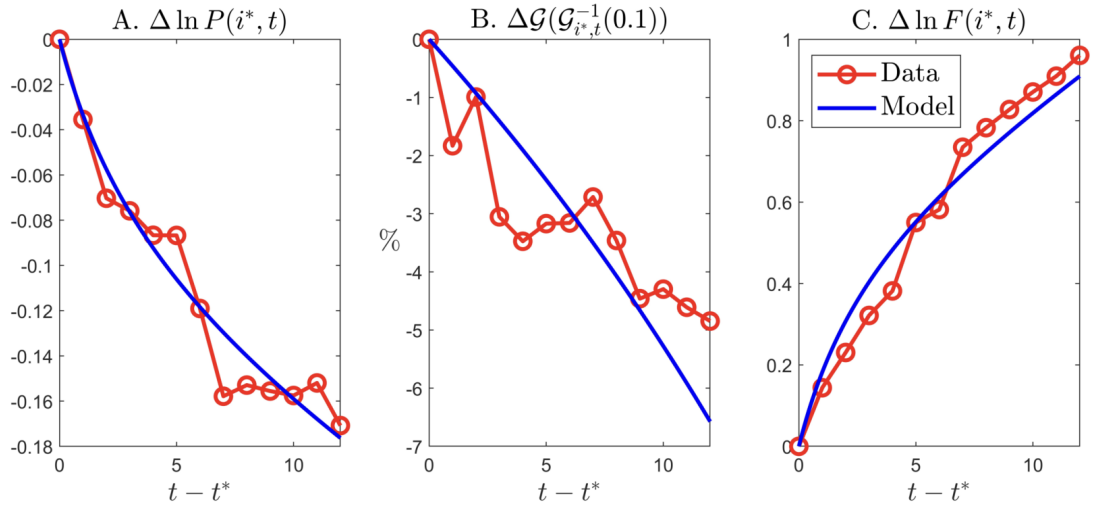
The parameter ϕ governs the efficiency of new final-good invention and is the main determinant of the long-run rate of variety creation. We choose $\phi = 2.987$ to match per-capita GDP growth of 1.6% in the 2010s. The parameter ψ governs incumbents' innovation efficiency. We choose $\psi = 0.216$ to match the average firm-level R&D-to-sales ratio of 4.1% reported in Akcigit and Kerr (2018). We normalize the entry-efficiency parameter to $\psi_e = 1$, which fixes the level of the mass of firms without affecting growth rates or the cross-sector slopes that are central to the paper. In the model, scaling ψ_e simply rescales the number of operating firms and the level of

asymptotic sectoral price trend is governed by the price trend for necessities. Results are not sensitive to γ; doubling or halving γ leaves the main quantitative conclusions largely unchanged (see Appendix Section E.2).

[12] Data are from the World Inequality Database (WID): https://wid.world/country/usa/.

output, leaving balanced-growth rates and the monotone patterns across $\tilde{i}$ unchanged. This scale invariance makes ψ_e a natural normalization. The parameter ζ governs the initial productivity level in a newly introduced sector. A higher ζ makes early entry more attractive but leaves less room for subsequent expansion as the sector matures. We choose $\zeta = 0.575$ to match the average growth in the number of active firms over the first twelve years after a product's introduction. The innovation step size λ governs the thickness of the right tail of the firm-size distribution. We pin down λ using the module-level top-10-percent output share, defined as the fraction of total output produced by the largest 10 percent of firms within a module. We set $\lambda = 1.053$ to match the empirical top-10-percent share of 83.40% among necessity goods in our sample, close to the value reported in Akcigit and Kerr (2018). The details of solving firm distribution dynamics are provided in Appendix E.1. The parameter θ governs the extent to which aggregate productivity improvements translate into firm-level productivity, and therefore how quickly products become affordable to lower-income households. We discipline θ using the speed at which the income position of the marginal consumer falls after entry. Empirically, we proxy the marginal consumer by bottom-decile (D1) purchasers within a module and target the average decline in their economy-wide income quantile over the first twelve years after entry. This yields $\theta = 0.374$.

Figure 7: Model fit



Notes. The figure compares the model's predictions (blue) to the data (red) for the evolution of a product module after it is introduced. Each empirical series is constructed as a module-by-entry-year average. Panel A plots the cumulative log change in the module's average price relative to the entry year. Panel B plots the change in the economy-wide income quantile of the module's bottom-decile buyers (a proxy for the marginal buyer cutoff). Panel C plots the log change in the number of active firms in the module.

Model Fit A natural test of the calibrated economy is whether it reproduces the empirical signature of diffusion: new products enter as high-end purchases and then become progressively cheaper and more widely consumed. Figure 7 compares the model with the data by tracing product-level dynamics following entry. The blue line plots the model-implied trajectory of any fixed product i^* after its invention at some time t^*, while the red line shows the corresponding trajectory in the

data. All empirical series are computed as module-by-entry-year averages. For any variable X of interest, we define $\Delta X(t) \equiv X(t) - X(t^*)$ as its cumulative changes after entry. Panel A reports log price dynamics, $\Delta \ln P(i^*, t)$. Panel B tracks the change in the overall income percentile of the 10th-percentile buyer—our proxy for the marginal buyer. Specifically, it plots $\Delta \mathcal{G}(\mathcal{G}_{i^*,t}^{-1}(0.1))$, where $\mathcal{G}_{i^*,t}$ denotes the income distribution of buyers of product i^* at time t. Panel C reports the log change in the number of active firms relative to the entry date.

The model matches the three diffusion patterns emphasized in Section 2.2. Prices fall after entry and the pace of decline decelerates with maturity: over the first twelve years, the average price decline is 17.1% in the data and 17.5% in the model, with similar early-versus-late dynamics (years 1–6: 1.98% vs. 1.96% per year; years 6–12: 0.86% vs. 0.93%). Although price dynamics are not included among the target moments, the model replicates them quite well. The income position of marginal buyers shifts downward in both the data and the model, indicating diffusion toward lower-income households. Firm participation rises rapidly after entry and then gradually slows; the model implies firm-number growth of 10.26% per year in years 1–6 and 5.29% in years 7–12, close to the empirical 9.70% and 6.31%. Overall, the calibrated economy captures the joint post-entry dynamics of prices, diffusion, and market structure, providing a disciplined baseline for the counterfactuals below.

5.2 Mechanism of the "Luxury-to-Necessity" Transition

Having disciplined the model to match post-entry dynamics, we use it to unpack the diffusion mechanism—why prices fall and the marginal buyer moves down the income distribution. The strategy is a clean channel decomposition within the calibrated economy. We take a single sector i and feed it alternative paths for its sectoral productivity $B(i,t)$ that selectively shut down incumbent cost-reducing innovation and entry/exit in equation (19). All other sectors remain on the BGP, so aggregate objects such as $\bar{A}(t)$, $W(t)$, and the common spillover term are held fixed. Because sector i is measure zero, this perturbation generates no general-equilibrium feedback; any change in diffusion reflects only the sector's own productivity evolution.

Inserting $P(i,t) = \kappa_l(i,t)$ in equation (11), diffusion is pinned down by the marginal-buyer condition,

$$1 - \mathcal{G}(\ell(i,t)) = (1-\beta)^{\frac{1-2\beta}{\beta}} W(t)^{-\frac{1-\beta}{\beta}} \kappa_l(i,t)^{\frac{1-\beta}{\beta}} B(i,t), \tag{27}$$

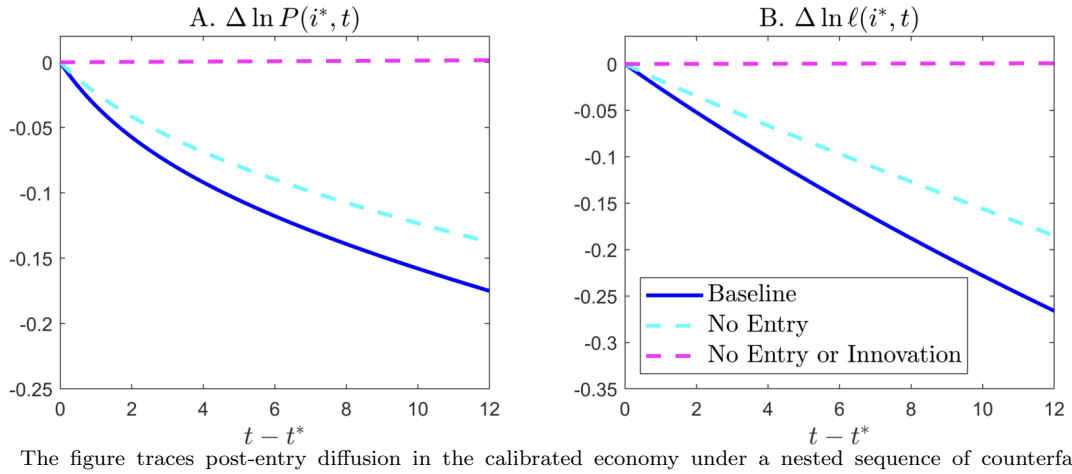
where $\ell(i,t)$ is the labor endowment of the marginal buyer, defined in equation (B.19), and $\kappa_l(i,t)$ is that buyer's reservation price. With aggregates held fixed, higher $B(i,t)$ lowers the effective price, expands coverage, and pushes the marginal buyer toward lower incomes (a lower $\ell(i,t)$).

Equation (19) implies that sectoral productivity growth can be decomposed into three terms:

$$\underbrace{S(t) = (1-\theta)\frac{1-\beta}{\beta}\frac{\dot{\bar{A}}(t)}{\bar{A}(t)}}_{\text{Common Spillovers}}, \qquad \underbrace{I_i(t) = z_i(t)\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right)}_{\text{Incumbent R\&D}}, \qquad \underbrace{E_i(t) = \frac{e(i,t)}{F(i,t)} - \delta}_{\text{Firm Entry–Exit}}.$$

The spillover term $S(t)$ is common across sectors. The term $I_i(t)$ captures incumbents' within-firm cost reductions from R&D, while $E_i(t)$ captures entry- and exit-driven changes in average sectoral efficiency. Our counterfactuals hold the common spillover component $S(t)$ fixed and shut down the other two channels sequentially.

Figure 8: Decomposition of Product Diffusion



Notes. The figure traces post-entry diffusion in the calibrated economy under a nested sequence of counterfactuals that selectively remove the sector-specific forces behind productivity growth. Panel A reports the log change in product price; Panel B reports the the log change in endowment level of the marginal consumers. "Baseline" incorporates all three components in equation (19). "No Entry" sets $E_i(t) = 0$, shutting down firm dynamics. "No Entry or Innovation" additionally sets $I_i(t) = 0$, leaving only the common spillover component.

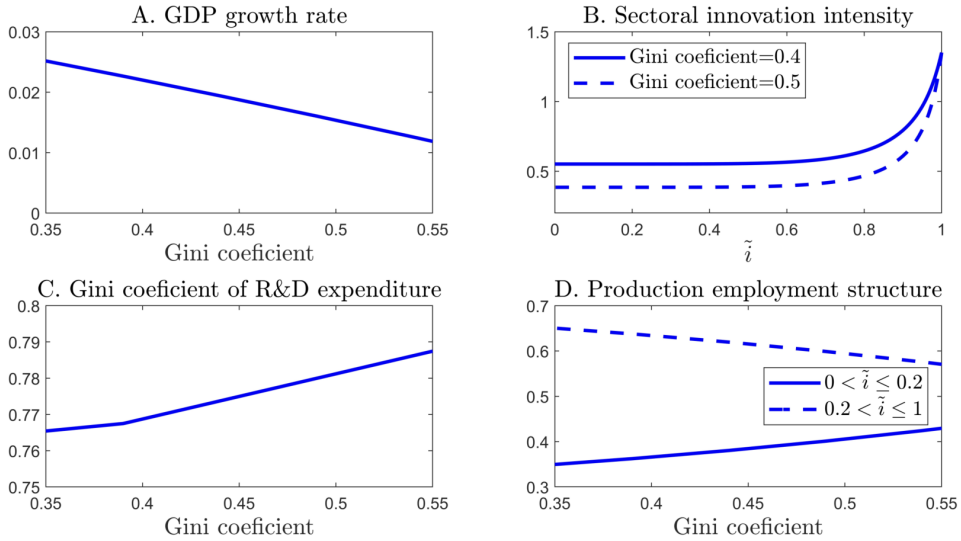
Figure 8 summarizes the implied price paths and diffusion patterns. We trace a fixed product i^* after its invention at some time t^*. Panel A shows that the steady post-entry price decline in the baseline is largely accounted for by endogenous sectoral forces: once firm dynamics are removed, the decline becomes much more muted; once both firm dynamics and incumbents' innovation are removed, prices are essentially flat. Panel B shows the corresponding movement of the marginal buyer's income (essentially $\ell(i, t)$ determined in equation (27)). In the baseline, the income of the marginal buyer falls substantially, indicating diffusion toward lower-income households. Shutting down firm dynamics markedly slows this movement, and shutting down both entry and innovation nearly eliminates it. Put differently, without incumbents' innovation and the extensive-margin adjustment of firms, a new product remains disproportionately confined to its initial, higher-income customers and fails to complete the luxury-to-necessity transition.

5.3 Income Inequality, R&D Allocation and Growth

We next study how income inequality affects economic growth through a set of comparative statics exercises in our model. Around the benchmark calibration, we vary the Pareto exponent to generate different degrees of inequality, while adjusting the minimum income threshold so that mean income remains fixed. This exercise isolates the role of redistribution in shaping demand composition and innovation incentives, holding aggregate resources constant.

Figure 9 summarizes the key implications. Panel A shows that higher inequality is associated

Figure 9: Income Inequality, Innovation, and Long-Run Growth



Notes. The figure illustrates how changes in income inequality shift innovation incentives and long-run growth in the model. Inequality is generated by varying the Pareto exponent ϵ while adjusting the minimum income threshold to keep mean income constant. Panel A plots the implied output growth rate as inequality varies. Panel B compares the sectoral innovation schedule across the product ladder under two inequality levels. Panels C and D plot, respectively, the Gini coefficient of incumbent R&D expenditure and the production labor in necessity and non-necessity sectors.

with slower output growth. Quantitatively, a 0.01 increase in the Gini coefficient reduces the long-run GDP growth rate by about 0.06 percentage points. The key intuition underlying this growth effect is captured by Proposition 5. They are illustrated by Panel B, which compares the sectoral innovation schedules $z(\tilde{i})$ under two levels of inequality (Gini coefficients of 0.4 and 0.5). As the income gap widens, moving from the solid line to the dashed line, innovation declines in all sectors, but the shift is not parallel: the contraction is largest in necessity sectors and becomes progressively more muted toward luxury sectors. This is the key reallocation force in the model. By eroding the purchasing power of the marginal low-income buyers, higher inequality compresses necessity-sector prices and profits and weakens incentives for cost-reducing innovation precisely where diffusion is the broadest, which leads to slower growth.

Panel C provides another perspective on this mechanism: inequality increases the concentration of R&D across firms, as measured by the Gini coefficient of firm-level R&D expenditure. Greater concentration, in turn, reduces the efficiency of aggregate R&D by shifting innovative effort toward luxuries, even though innovation in necessities generates higher social returns. Through this reallocation of innovative effort, inequality imposes an additional drag on long-run growth. This mechanism is robust to alternative specifications of the labor market. For instance, in Appendix Section C.1, we consider an economy in which skilled labor is employed in R&D and unskilled labor in production. Even when aggregate R&D labor is held fixed, greater inequality continues to increase the concentration of R&D and reduce long-run growth.

Panel D illustrates how labor becomes trapped within necessity sectors by showing the production employment structure across necessity and non-necessity sectors. We approximate necessity

sectors as those with $0 < \tilde{i} \leq 0.2$. Specifically, Panel D plots production labor employed in necessity sectors as a share of total production labor across all sectors. As productivity growth in these sectors slows, lower production efficiency draws more labor into necessity-good production, thereby raising their employment share. This distortion hinders the spread of productivity improvements and delays the shift of labor toward more advanced sectors, thereby dampening variety expansion and long-run growth.

5.4 Welfare Effects of Redistribution

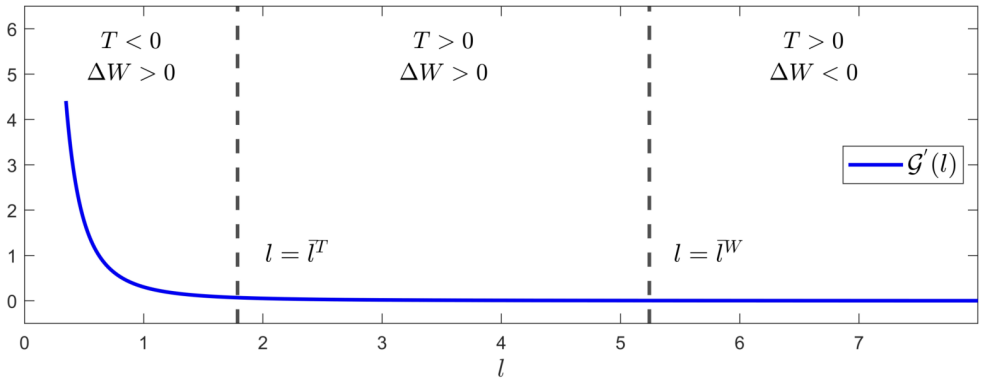
The previous subsection shows that inequality shapes both disposable income and the incentives that drive innovation and growth. Redistribution is therefore a natural policy lever in our setting. We therefore use the calibrated economy to ask how a more progressive tax-and-transfer system reshapes welfare across households.

Following the public-finance literature, we begin by representing redistribution with a parsimonious reduced-form schedule and later discuss alternative forms of redistribution. Specifically, we adopt a standard reduced-form schedule widely used in the public-finance literature,

$$T(l) = l - \frac{l^{1-\upsilon}}{\int l^{1-\upsilon} d\mathcal{G}(l)}, \tag{28}$$

where $\upsilon \in [0, 1]$ governs progressivity. The schedule is budget balanced by construction, i.e., $\int T(l) d\mathcal{G}(l) = 0$ for any υ. Since income in the model is proportional to the endowment l, $T(l)$ can be interpreted as a tax on endowments (equivalently, on pre-tax income). This functional form is widely used as a tractable representation of tax-and-transfer systems (see, e.g., Benabou (2002); Heathcote et al. (2017); Oberfield (2023)). As shown in Appendix Section E.3, if the pre-tax endowment distribution is Pareto with exponent ϵ, then the after-tax distribution remains Pareto with an adjusted exponent $\epsilon/(1-\upsilon)$, so higher υ maps into lower inequality in a transparent way.

Figure 10: Welfare Effects of Redistribution



Notes. The figure illustrates the distributional logic for $\upsilon = 0.16$. Households with $l < \bar{l}^T$ receive net transfers ($T(l) < 0$) and gain from both the level and growth channels. Households with $\bar{l}^T < l < \bar{l}^W$ pay positive net taxes but still gain because the growth effect dominates. For households with $l > \bar{l}^W$, the level loss from taxation outweighs the growth gain, so $\Delta W < 0$. The cutoff $\bar{l}^T$ solves $T(\bar{l}^T) = 0$, and $\bar{l}^W$ solves $\Delta W(\bar{l}^W) = 0$.

Redistribution matters for welfare through two channels. The *level effect* reallocates contemporaneous consumption: transfers relax the budgets of low-income households and tighten those of high-income households. The *growth effect* operates through the balanced-growth rate g_c: by compressing inequality, a more progressive schedule shifts innovation toward necessity sectors, speeds diffusion, and raises long-run growth. The growth effect benefits all households, while the level effect changes sign across the distribution. Figure 10 illustrates the logic by partitioning households into three regions. For low-income households with $l < \bar{l}^T$, taxes are negative ($T(l) < 0$), so both the level effect and the growth effect raise welfare. For middle-income households with $\bar{l}^T < l < \bar{l}^W$, taxes are positive ($T(l) > 0$) but the burden is modest under a progressive schedule; here the positive growth effect dominates, so welfare still rises on net. For high-income households with $l > \bar{l}^W$, the tax burden is largest; the negative level effect outweighs the growth effect, and welfare falls.

Table 3 reports the consequences across three alternative degrees of progressivity υ. Welfare is measured using a *compensating endowment variation*: the percentage increase in endowment in the pre-policy equilibrium that makes a household indifferent between the pre-policy allocation and the allocation under the progressive schedule.

Table 3: Welfare Effects of Redistribution under Alternative Tax Progressivity

	Progressivity parameter υ		
Outcome	0.16	0.10	0.05
Inequality (post-tax)			
Pre-policy Gini	0.484	0.484	0.484
Post-policy Gini	0.376	0.416	0.449
Net tax rates (percent of income)			
5th percentile household	-29.62	-18.34	-9.18
Median (50th percentile)	-21.13	-13.49	-6.92
95th percentile household	5.01	2.34	0.82
Growth effects (per-capita GDP growth rate, %)			
Pre-policy growth	1.64	1.64	1.64
Post-policy growth	2.35	2.10	1.88
Welfare effects (compensating endowment change, %)			
Avg. welfare change	44.58	28.56	14.59
Share with gains ($l < \bar{l}^W$)	98.44	98.79	99.04
Avg. welfare change for gainers	45.21	28.93	14.59
Share with losses ($l > \bar{l}^W$)	1.56	1.21	0.96
Avg. welfare change for losers	-10.32	-8.26	-5.33

Notes. The table reports outcomes in the calibrated economy under the reduced-form tax-and-transfer schedule in equation (28). A higher υ implies a more progressive schedule and therefore a larger reduction in post-tax inequality. "Net tax rate" is $T(l)/l$ evaluated at the indicated income percentiles; negative values indicate net transfers. Welfare changes are measured as compensating endowment changes: the percentage increase in pre-policy endowment that makes a household indifferent between the pre-policy equilibrium and the post-policy equilibrium. The cutoff $\bar{l}^W$ denotes the income threshold at which the welfare change is zero; households with $l < \bar{l}^W$ gain and those with $l > \bar{l}^W$ lose.

The first column presents our benchmark policy, which increases progressivity to $\upsilon = 0.16$. This policy reduces the post-tax Gini coefficient from 0.484 to 0.376, approximately restoring inequality

to its level in the 1970s. Under this policy, transfers are substantial for households at the bottom and middle of the distribution, whereas positive net taxes are concentrated among top earners. Specifically, households at the bottom 5th percentile receive a net transfer rate of 29.62%, while those at the top 5th percentile face a net tax rate of 5.01%. In welfare terms, nearly the entire population (approximately 98.44%) experiences sizable gains, with an average welfare improvement of 44.58%. By contrast, welfare losses are concentrated in a small upper tail of the distribution (approximately 1.56%), where the average loss is -10.32%. Overall, the average welfare improvement is 36.15%. The remaining columns consider lower degrees of progressivity ($\upsilon = 0.10$ and $\upsilon = 0.05$). As expected, both the extent of redistribution and the growth response become weaker, but the qualitative pattern remains unchanged: welfare gains accrue to nearly all households, while losses are concentrated at the very top of the distribution. These results suggest that even modest increases in progressivity can generate sizable welfare gains in the calibrated economy. The underlying mechanism is closely tied to why inequality matters in our framework: progressive redistribution strengthens demand at the necessity margin, thereby redirecting innovation toward sectors with broader diffusion and higher growth returns.

Alternative redistribution schemes. In the preceding experiments, high-income households experience welfare losses because the tax burden outweighs the associated growth benefits. We therefore examine two alternative redistribution schemes to see whether there exist other redistribution rules that can improve the welfare of the rich.

First, because the tax rate under the benchmark transfer rule in equation (28) increases with income, households in the extreme upper tail face disproportionately large welfare losses. To mitigate these losses, we introduce a two-stage schedule. For households with $l < l_{\text{cap}}$, the schedule follows a similar rule in equation (28) with progressivity parameter υ; and households with $l \geq l_{\text{cap}}$ instead face a constant tax rate $\bar{\tau}$. Further details are provided in Appendix Section E.4. Capping the tax rate substantially reduces welfare losses at the top of the distribution, although it also attenuates the growth response.

Second, we consider a locally redistribution scheme designed to illustrate that welfare losses among high-income households are not an inevitable consequence of redistribution. In principle, the growth effect induced by redistribution can be sufficiently large to outweigh the direct tax burden borne by high-income households, allowing even the rich to experience welfare gains. Under this scheme, all households with $l \geq l_{\text{cap}}$ pay a low constant tax rate τ_R, and the resulting revenue is transferred uniformly to a group of households below l_{cap}. Further details are provided in Appendix Section E.5. Our numerical results show that this constructed redistribution scheme increases welfare for all households and therefore constitutes a Pareto improvement.

6 Conclusion

This paper links inequality to growth through a simple but underappreciated margin: the speed at which innovations diffuse from luxuries to necessities. Using micro evidence that connects households' purchases to firms' innovative activity and market outcomes, we document a systematic life-cycle pattern within product markets. As products mature, prices fall, adoption shifts toward lower-income consumers, and market structure becomes less concentrated. Yet innovation is disproportionately oriented toward high-income demand, and greater exposure to rising inequality is associated with weaker innovation among firms serving the mass market.

To interpret these facts, we develop a tractable endogenous-growth model with hierarchical demand, incumbent productivity-improvement R&D, and endogenous entry. These ingredients generate a tight link between the income distribution and the composition of innovative effort. Our model shows that inequality is not only a distributional outcome, but also reallocates market size across segments and the private returns to innovation. The policy implication is therefore sharper than "redistribute to raise welfare". By sustaining the mass market for widely consumed goods, progressive tax-and-transfer policies can tilt innovation incentives back toward cost reductions that diffuse broadly, speeding the luxury-to-necessity transition. In this sense, reducing post-tax inequality can deliver a double dividend: it improves equity today while fostering innovation that raises broad-based living standards over time.

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Online Appendix

Contents of the Appendix

Appendix A provides additional data details and empirical facts.

Appendix B presents proofs in the baseline model.

Appendix C checks the robustness of the theoretical results.

Appendix D describes the numerical algorithm used in computation.

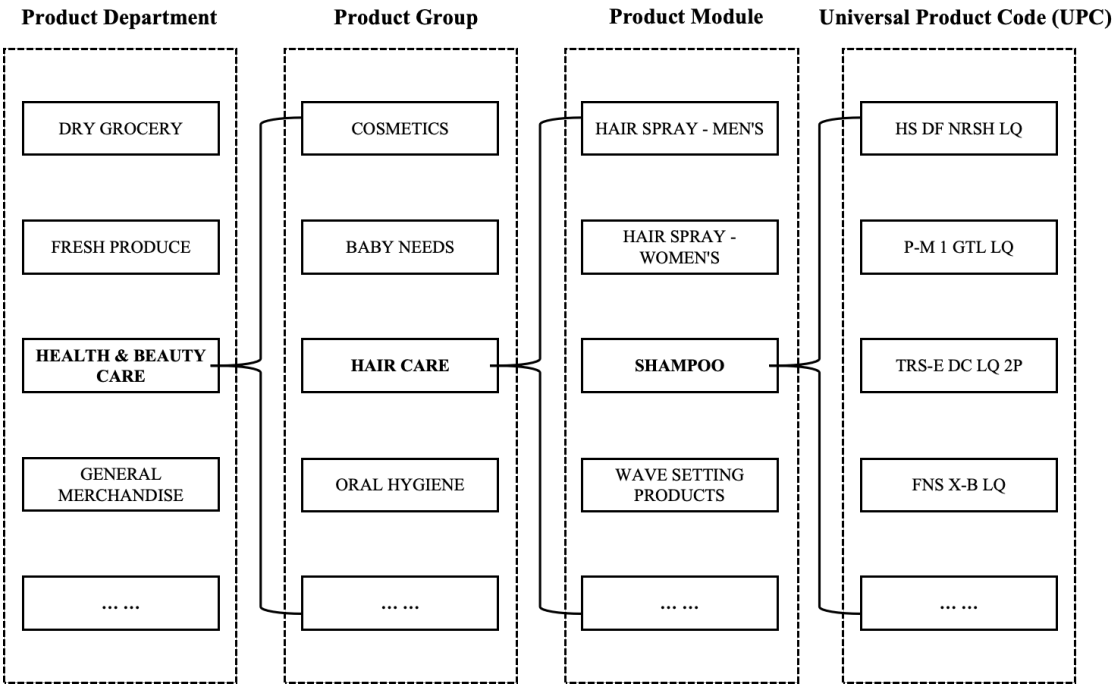
Appendix E reports robustness checks for the quantitative analysis.

A Appendix: Data and Empirical Facts

A.1 NielsenIQ Homescan Consumer Panel

To ensure data quality, NielsenIQ implements multiple validation checks, including quarterly coverage monitoring and weekly reviews of sample representativeness. NielsenIQ organizes roughly 1,000 product modules into a hierarchical structure of about 100 product groups and 10 product departments. Figure A.1 illustrates this hierarchy. At the broadest level, products are assigned to departments such as Health and Beauty Care, which contain narrower product groups such as Hair Care. Each group is further divided into product modules that capture relatively close substitutes, such as Shampoo. At the most granular level, products are distinguished by their UPCs, which uniquely identify individual items by brand and package size.

Figure A.1: NielsenIQ Product Hierarchy from Departments to UPCs



Notes. This figure illustrates the product hierarchy defined by NielsenIQ, from broad departments to specific Universal Product Codes (UPCs).

Figure A.2 illustrates a representative shampoo module. Within this product module, well-known brands including Pantene, Head and Shoulders, and Paul Mitchell offer products that are close substitutes for consumers. Although package sizes vary across UPCs, these shampoos exhibit a high degree of similarity in design and function. The high comparability of UPCs within a module provides a well-suited empirical setting for the product-module-level analyses that follow.

Figure A.3 documents the spatial and distributional patterns of consumer expenditures. Higher spending is concentrated in coastal states, while lower spending is more common in inland regions. Expenditures by poor consumers are markedly lower than those by rich consumers, yet both groups

Figure A.2: UPCs of Different Brands within Shampoo Market



(a) Pantene Pro-V, 12.6 OZ



(b) Matrix, 13.5 OZ



(c) Head & Shoulders, 13.5 OZ



(d) Paul Mitchell, 10.14 OZ



(e) Garnier Fructis, 13.0 OZ

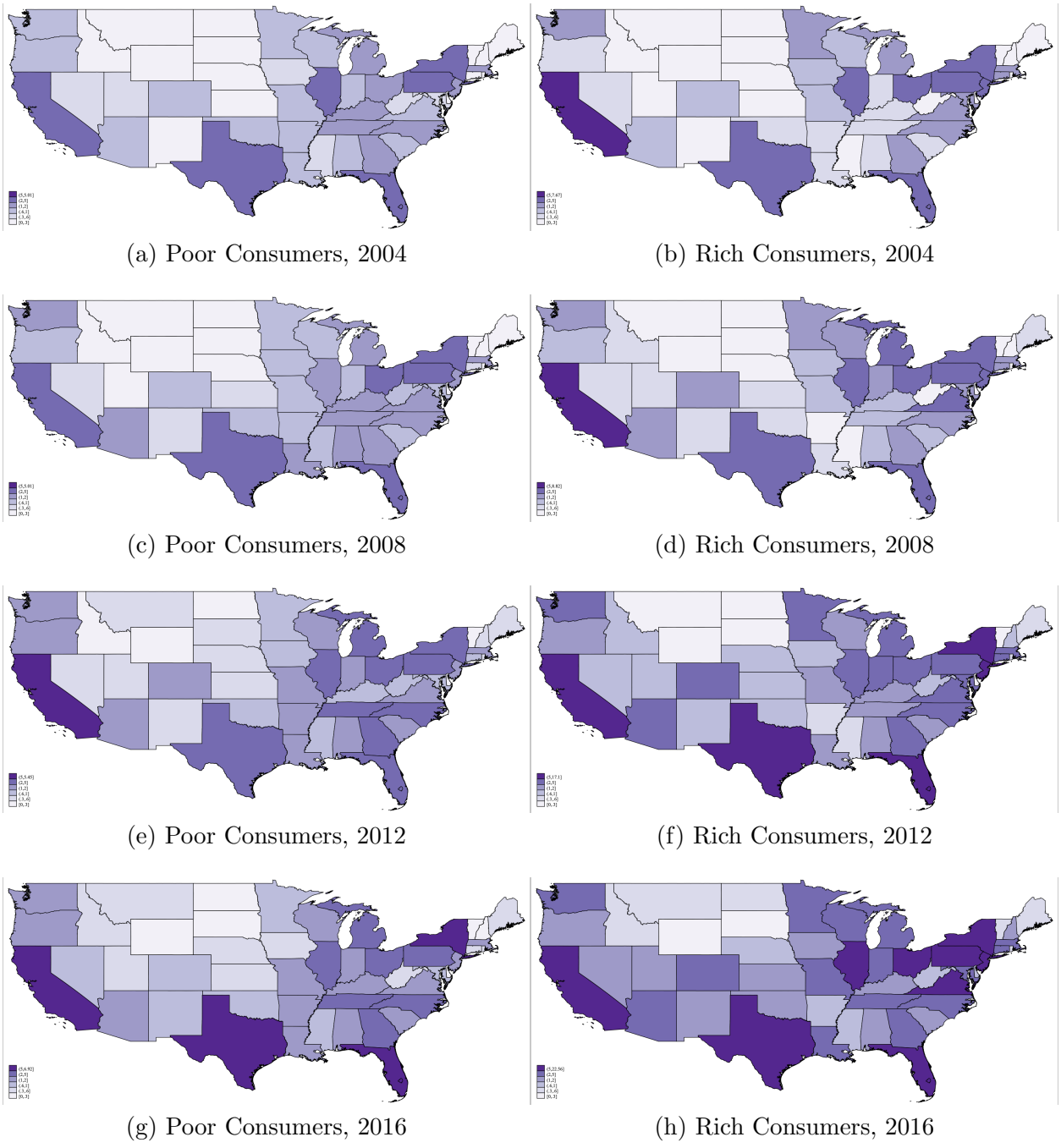


(f) TIGI, 6.76 OZ

Notes. This figure illustrates several representative Universal Product Codes (UPCs) within the product module Shampoo. Pictures of UPCs can be downloaded from https://www.upcitemdb.com/. Despite nuances in package size, UPCs of different product brands show large similarity in product design and utility.

exhibit steady growth over time. In 2016, for example, state-level expenditures ranged from roughly $50 million to over $6 billion among poor consumers, and from about $400 million to more than $22 billion among rich consumers. This substantial variation across consumer groups, regions, and time provides a rich setting for analyzing how local consumption patterns shape economic outcomes.

Figure A.3: Geographic Distribution of Consumer Expenditures by Income Group, 2004–2016 (Billions of USD)



(a) Poor Consumers, 2004

(b) Rich Consumers, 2004

(c) Poor Consumers, 2008

(d) Rich Consumers, 2008

(e) Poor Consumers, 2012

(f) Rich Consumers, 2012

(g) Poor Consumers, 2016

(h) Rich Consumers, 2016

Notes. The figure maps projection factor–weighted consumer expenditures (in billions of U.S. dollars) across states for 2004, 2008, 2012, and 2016. Panels (a), (c), (e), and (g) display expenditures of poor consumers (household income below $20,000, roughly the bottom decile). Panels (b), (d), (f), and (h) display expenditures of rich consumers (household income in the top income bracket, roughly the top decile). Expenditures are consistently higher among rich consumers and more concentrated in coastal states, while poorer consumers account for lower levels of spending, especially in inland regions.

A.2 BvD Orbis Database

To measure production activities and firm performance, we incorporate financial data from the Bureau van Dijk (BvD) Orbis database. Orbis provides comprehensive financial information on both listed and unlisted firms worldwide. We match GS1 company names to Orbis entries using a string-matching algorithm supplemented by manual validation, achieving a success rate of 86.4 percent. This high match rate confirms broad coverage of the GS1 sample within Orbis and is consistent with recent work such as Bajgar et al. (2020).

In Orbis, gross output is measured as operating revenues. Capital is proxied by tangible fixed assets, and labor by the number of employees. Orbis also reports the cost of employees for each firm. Intermediate inputs are calculated as the difference between turnover and value added (Bajgar et al., 2020). Although material costs constitute a substantial component of intermediate inputs, this variable is missing for most sample firms. Firm-level investment is reported directly in Orbis.

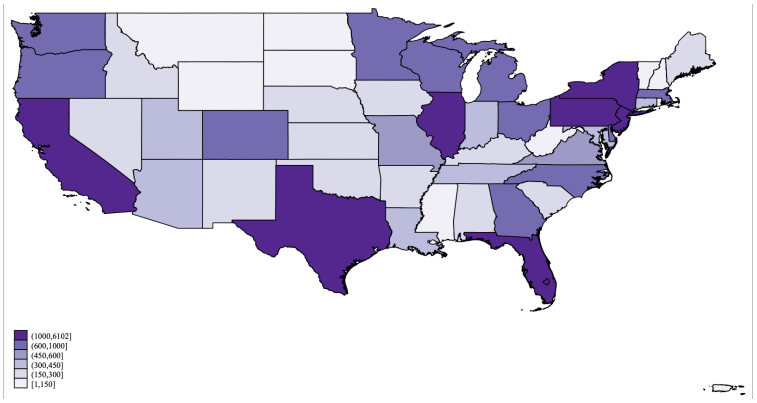
Table A.1 reports the distribution of matched firms across NAICS four-digit industries (2022 classification). For brevity, we report only industries with more than 100 firms. Panel (a) of Figure A.4 maps the geographic distribution of matched firms across U.S. states, showing concentrations in major economic centers including California, New York, Florida, and Texas. Together, the breadth of industry and geographic coverage indicates that the matched sample is broadly representative of the U.S. economy.

Table A.1: Industry Distribution of Sample Firms (Top NAICS Four-Digit Sectors)

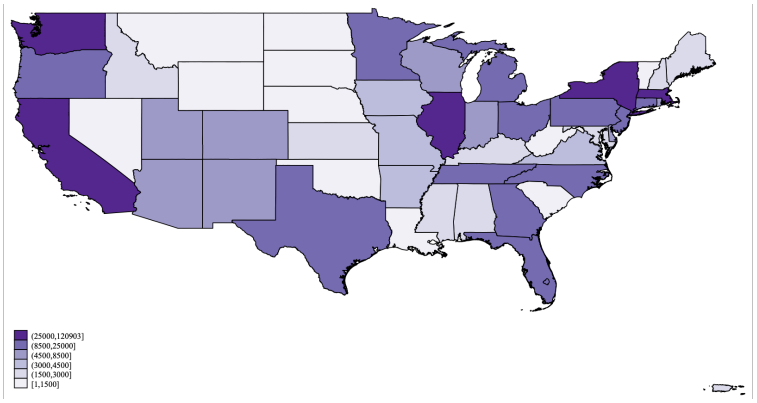
NAICS	Industry Description	Firm Count
4244	Grocery and Related Product Merchant Wholesalers	3504
5614	Business Support Services	2137
3121	Beverage Manufacturing	1975
3119	Other Food Manufacturing	1502
3118	Bakeries and Tortilla Manufacturing	1363
7225	Restaurants and Other Eating Places	999
4452	Specialty Food Retailers	919
3114	Fruit and Vegetable Preserving and Specialty Food Manufacturing	669
4249	Miscellaneous Nondurable Goods Merchant Wholesalers	614
4599	Other Miscellaneous Retailers	545
3399	Other Miscellaneous Manufacturing	481
4248	Beer, Wine, and Distilled Alcoholic Beverage Merchant Wholesalers	469
4239	Miscellaneous Durable Goods Merchant Wholesalers	466
3256	Soap, Cleaning Compound, and Toilet Preparation Manufacturing	464
3115	Dairy Product Manufacturing	441
5416	Management, Scientific, and Technical Consulting Services	410
4451	Grocery and Convenience Retailers	409
3254	Pharmaceutical and Medicine Manufacturing	393
3116	Animal Slaughtering and Processing	392
3113	Sugar and Confectionery Product Manufacturing	339
4242	Drugs and Druggists' Sundries Merchant Wholesalers	310
1113	Fruit and Tree Nut Farming	268
3261	Plastics Product Manufacturing	251
1119	Other Crop Farming	222
3112	Grain and Oilseed Milling	221
1121	Cattle Ranching and Farming	205
4232	Furniture and Home Furnishing Merchant Wholesalers	179
4234	Professional and Commercial Equipment and Supplies Merchant Wholesalers	178
5511	Management of Companies and Enterprises	177
4238	Machinery, Equipment, and Supplies Merchant Wholesalers	168
3222	Converted Paper Product Manufacturing	158
4243	Apparel, Piece Goods, and Notions Merchant Wholesalers	154
5131	Newspaper, Periodical, Book, and Directory Publishers	146
5121	Motion Picture and Video Industries	146
5239	Other Financial Investment Activities	146
4453	Beer, Wine, and Liquor Retailers	141
4885	Freight Transportation Arrangement	138
4561	Health and Personal Care Retailers	134
1112	Vegetable and Melon Farming	130
3111	Animal Food Manufacturing	129
4594	Office Supplies, Stationery, and Gift Retailers	129
3391	Medical Equipment and Supplies Manufacturing	128
4236	Household Appliances and Electrical and Electronic Goods Merchant Wholesalers	119
4591	Sporting Goods, Hobby, and Musical Instrument Retailers	116
1151	Support Activities for Crop Production	115
4581	Clothing and Clothing Accessories Retailers	113
4492	Electronics and Appliance Retailers	112
1114	Greenhouse, Nursery, and Floriculture Production	108
4491	Furniture and Home Furnishings Retailers	104
3152	Cut and Sew Apparel Manufacturing	103

Notes. The table reports the distribution of sample firms across the top NAICS four-digit industries (2022 classification). To conserve space, only industries with more than 100 firms are shown. Consumer-oriented industries such as grocery wholesaling, food manufacturing, and beverage production dominate, reflecting the prevalence of firms producing everyday goods.

Figure A.4: Geographic Distribution of BvD–Orbis Firms and Innovation in the United States, 2004-2016



(a) BvD Orbis Matched Sample Firm Count



(b) Total Patent Count Filed by BvD Orbis Matched Sample Firms

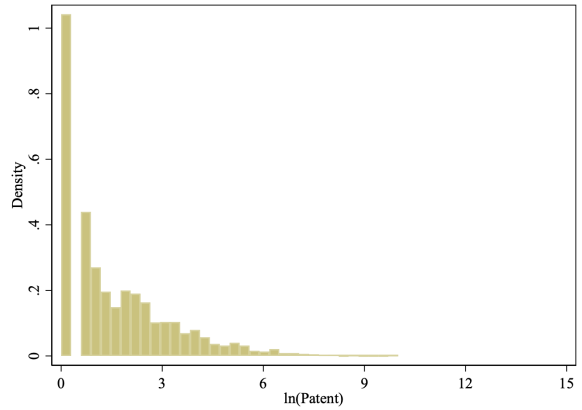
Notes. Panel (a) shows the spatial distribution of BvD Orbis matched firm count in our sample based on their headquarters' location as reported in the BvD Orbis database. The majority of firms are concentrated in economically active states such as California, New York, Florida, and Texas, while relatively few are located in sparsely populated states. For clarity of presentation, Hawaii, Alaska, and Guam are omitted; together they account for only 0.51% of the sample. Panel (b) maps the distribution of patents filed by firms in our Bvd Orbis matched sample across U.S. states, based on the location of firm headquarters. Patent activity is concentrated in economically dynamic states such as California, New York, Massachusetts, and Texas. For clarity of presentation, Hawaii, Alaska, and Guam are omitted, as together they account for only 0.33% of total patents.

A.3 Patent–Firm Matching in GS1 Data

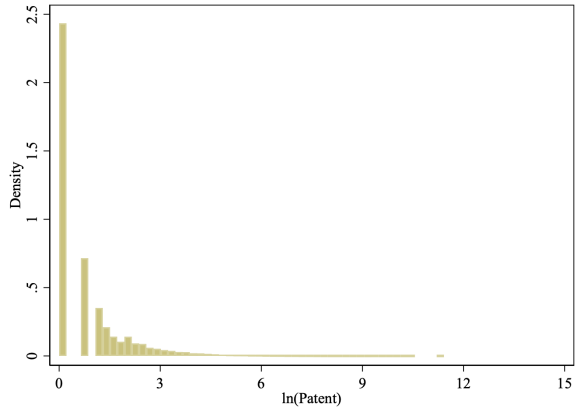
Matching Procedure. A crucial step in linking innovation outcomes to firms in the GS1 data is to match patent assignee names with firm names. Using Python's *FuzzyWuzzy* package, we implement an approximate matching algorithm based on the Levenshtein edit distance. This algorithm measures the similarity between two strings by calculating the minimum number of single-character edits—insertions, deletions, or substitutions—required to transform one string into the other (Levenshtein, 1966). Compared with exact string matching, this approach successfully links many pairs of slightly different names that refer to the same entity. For example, the algorithm reconciles variants such as "Honeywell International," "Honeywelll International Inc.," and "Honeywell International, Inc." with the GS1 entry "Honeywell International Inc." To reduce the risk of false positives, we apply a high similarity threshold of 0.9 and cross-validate matches using alternative approaches, including vector decomposition and phonetic algorithms. No single matching algorithm is universally superior, and each has strengths and limitations. To mitigate sensitivity to the choice of method, we combine several widely used techniques from the matching literature (Raffo and Lhuillery, 2009). We employ *bigram*, *scotoken*, and *token_soundex* within Stata's *matchit* package to filter implausible pairs, and we complement the edit-distance approach with the 2-gram algorithm available in Stata's *reclink2* package, which performs well with large string permutations in alphabetic languages (Thoma et al., 2010). The automated procedure is further supplemented with manual verification to ensure the accuracy of the final matches.

Validation. The matching procedure delivers a close correspondence between the sample and the USPTO universe. Panel (b) of Figure A.4 shows that patents in the sample are concentrated in the same innovation hubs highlighted in the literature, particularly California and Massachusetts, where clusters of firms drive technological progress. Panels (a) and (b) of Figure A.5 confirm that both the matched sample and the USPTO universe display strikingly similar right-skewed distributions, indicating that the overall shape of patenting activity is preserved. Panel (c) of Figure A.5 shows that the matched sample exhibits higher median and upper-quartile patent counts than the USPTO universe. This upward shift is consistent with the consolidation of assignee names with minor spelling differences into single firms, which raises measured counts without altering the underlying distribution. Panel (d) of Figure A.5 lists the most patent-intensive firms, led by Microsoft, General Electric, and Apple. The presence of these global technology leaders at the top of the distribution confirms that the matching procedure reproduces the broad structure of the USPTO data while capturing the firms most central to U.S. innovation.

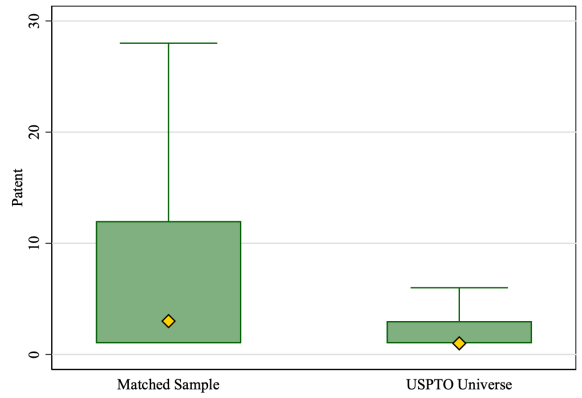
Figure A.5: Validation of Patent Match, 2004–2016



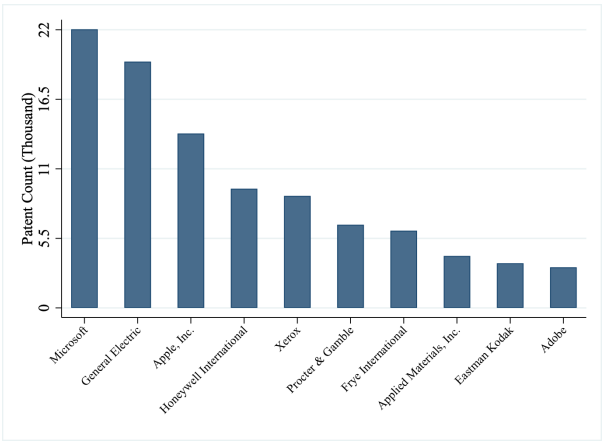
(a) Patent Count Distribution: Matched Sample



(b) Patent Count Distribution: USPTO Universe



(c) Box Plots: Matched Sample vs. USPTO Universe



(d) Top Ten Patent-Holding Firms

Notes. This figure validates the quality of the patent matching procedure. Panels (a) and (b) compare the distribution of firm-level patent counts (log scale) between the matched sample and the full universe of USPTO assignees. Both distributions are right-skewed with broadly similar shapes, indicating that the matched sample captures the overall structure of patenting activity. Panel (c) presents box plots of firm-level patent counts for the matched sample and the USPTO universe. Diamonds indicate medians, and extreme values are excluded for clarity. Panel (d) lists the ten firms in the matched sample with the highest number of granted patents over the sample period. Patent counts are aggregated at the firm level after consolidating assignee names. High-technology firms including Microsoft, General Electric, and Apple dominate the ranking, reflecting the concentration of patenting activity among global leaders.

A.4 Robustness of Fact I: Product Diffusion Along the Income Distribution

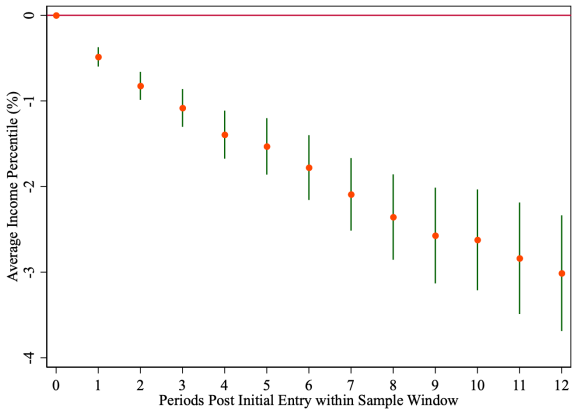
This subsection reports a series of robustness checks for the Fact I product diffusion pattern. We re-estimate the event-time specification at the finer UPC level, on a balanced panel of firm–module pairs, and on a sample of modules that enter after 2004 to avoid left censoring. We also verify the price result using alternative price measures—the inverse hyperbolic sine transformation, size-adjusted unit prices, and a split by innovative versus non-innovative firms—and document firm-level TFP growth as direct evidence on the technological mechanism underlying the price decline. Across all these exercises, the core diffusion pattern remains stable.

Figure A.6 replicates the Fact I diffusion patterns at the UPC level. The analysis moves from product modules to individual UPCs, restricting to UPCs with no version changes during the sample period and adding UPC fixed effects to absorb time-invariant UPC attributes. The same patterns emerge. Following entry, the average buyer income percentile declines (Panel a), the low-income expenditure share rises (Panel b), and real prices fall (Panel c). The price decline is present among both entrants and incumbents (Panel d). That the module-level results carry through to the finer UPC unit confirms that the diffusion pattern is not an artifact of compositional changes across UPCs within modules or of version churn in barcode identifiers.

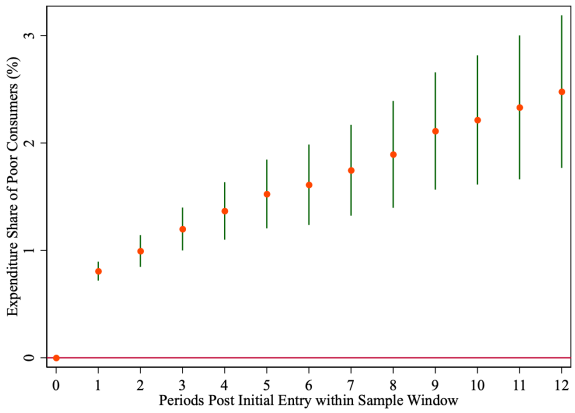
Figure A.7 re-estimates the diffusion pattern on a balanced panel of firm–module pairs observed in at least ten years. This restriction removes entry and exit margins from the variation, so the event-time path is identified from within-spell changes rather than from compositional shifts as firms and products enter and leave the sample. Firm–module, firm–year, and module–year fixed effects further ensure that the coefficients reflect deviations within a given firm–module spell, net of firm-wide shocks and module-level common trends. The same diffusion pattern holds. As a module ages, the average buyer income percentile falls, the low-income expenditure share rises, and real prices decline. That the pattern survives in a sample stripped of turnover confirms that the luxury-to-necessity diffusion reflects within-product aging rather than the changing composition of products over the life cycle.

Figure A.8 re-estimates the diffusion pattern on a sample of modules that enter after 2004, the first year of the sample, so that event time zero corresponds to the year in which the firm first sells the module rather than a left-censored starting point. This restriction addresses the concern that modules already present in 2004 have unknown pre-sample ages, which could attenuate or distort the estimated life-cycle pattern. While this approach removes left-truncated modules entirely, a complementary strategy that retains the full sample is to recover each module's true technology birth year from external patent and text data, as we do in the next subsection A.5. The diffusion results are unchanged. Following true entry, the average buyer income percentile declines (Panel a), the low-income expenditure share rises (Panel b), and real prices fall (Panel c). Panel (d) shows that the price decline occurs among both entrants and incumbents, indicating that the cost reductions underlying diffusion are not confined to either new or established firms. That the pattern holds when product age is measured without left censoring confirms that the luxury-to-necessity diffusion reflects the actual aging of products from their market introduction.

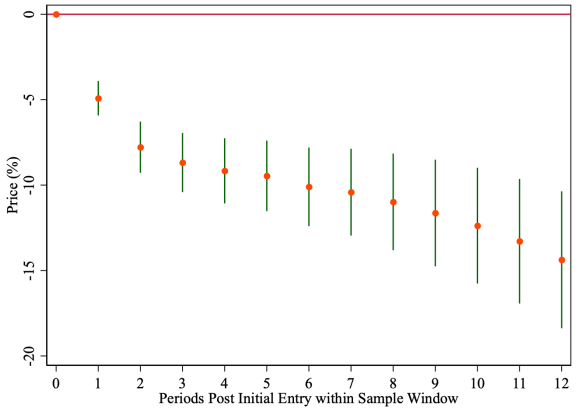
Figure A.6: Product Diffusion Across the Income Distribution: UPC-Level Robustness



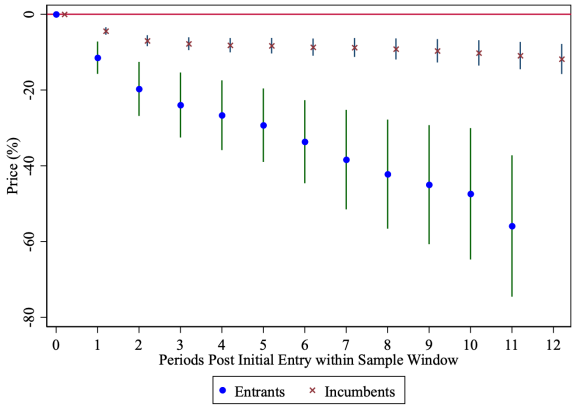
(a) Decline in Buyers' Income



(b) Growing Share of Low-Income Buyers



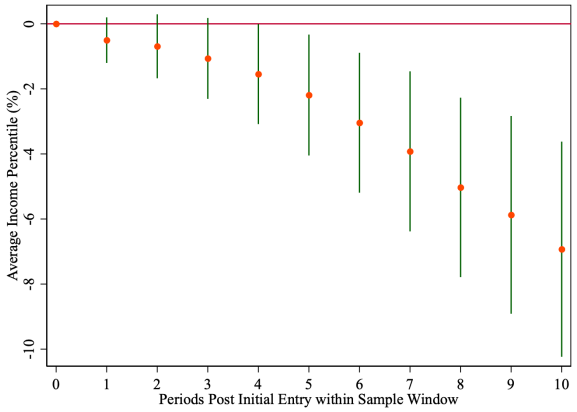
(c) Falling Prices Over Product Age



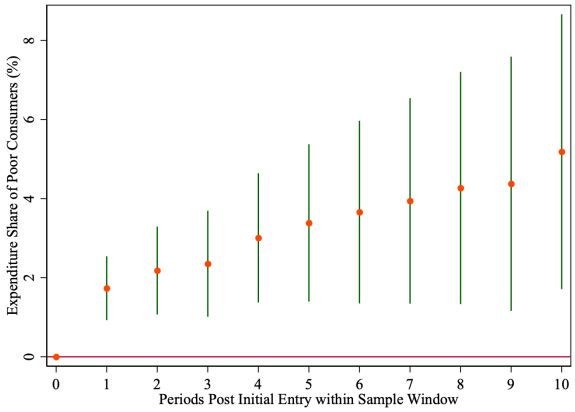
(d) Falling Prices: Entrants vs. Incumbents

Notes. Each point plots the estimated coefficient from a regression of UPC-level outcomes on the cumulative periods relative to initial entry within the sample window. The x-axis measures periods after initial sales records, and the y-axis shows the estimated change in the corresponding outcome relative to the initial year. All regressions include firm–product, firm–year, product–year, and UPC fixed effects, with standard errors clustered at the firm level. The sample excludes UPCs that experience version changes during the sample period. In Panel (a), the dependent variable is the average income quantile of a UPC's buyers, showing whether its customer base shifts toward lower-income consumers as it matures. In Panel (b), the dependent variable is the expenditure share of the bottom income tercile, reflecting the growing importance of poorer consumers in total spending. Panel (c) uses the annual price to track how average prices evolve after initial year. Panel (d) further illustrates that the decline in prices is driven concurrently by both new entrants and established firms. To facilitate interpretation, we report the percentage change of price in Panels (c) and (d), computed by dividing the estimated marginal effect by the sample mean price. Solid dots represent point estimates, and vertical bars denote 95% confidence intervals based on robust standard errors.

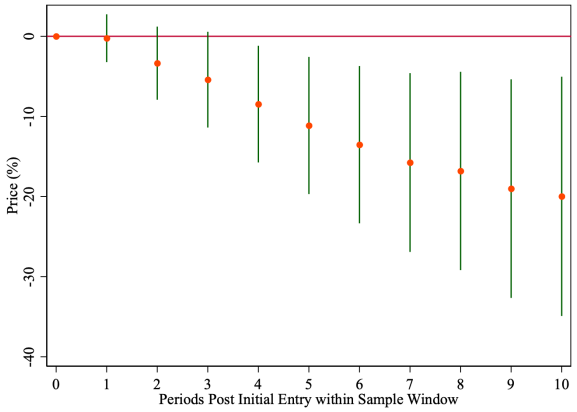
Figure A.7: Product Diffusion Across the Income Distribution: Balanced-Panel Robustness



(a) Decline in Buyers' Income



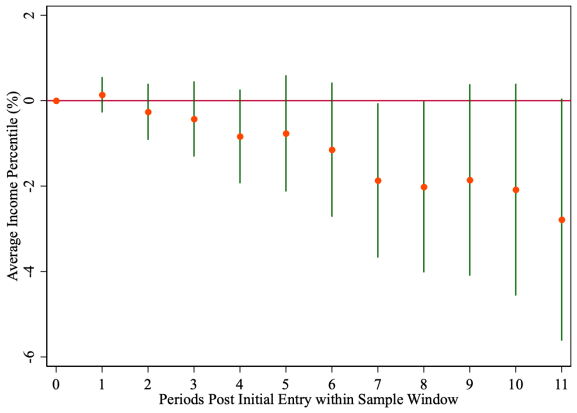
(b) Growing Share of Low-Income Buyers



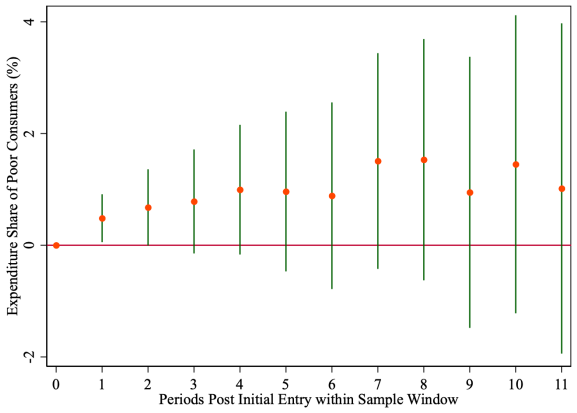
(c) Falling Prices Over Product Age

Notes. Each point plots the estimated coefficient from a regression of product-level outcomes on the cumulative periods relative to initial entry within the sample window. The sample restricts to firm–module pairs observed in at least ten years, removing entry and exit margins from the variation. The x-axis measures periods after initial sales records, and the y-axis shows the estimated change in the corresponding outcome relative to the initial year. All regressions include firm–product, firm–year, and product–year fixed effects, with standard errors clustered at the firm level. In Panel (a), the dependent variable is the average income quantile of a product's buyers, showing whether its customer base shifts toward lower-income consumers as it matures. In Panel (b), the dependent variable is the expenditure share of the bottom income tercile, reflecting the growing importance of poorer consumers in total spending. Panel (c) uses the annual price to track how average prices evolve after initial year. To facilitate interpretation, we report the percentage change of price in Panel (c), computed by dividing the estimated marginal effect by the sample mean price. Solid dots represent point estimates, and vertical bars denote 95% confidence intervals based on robust standard errors.

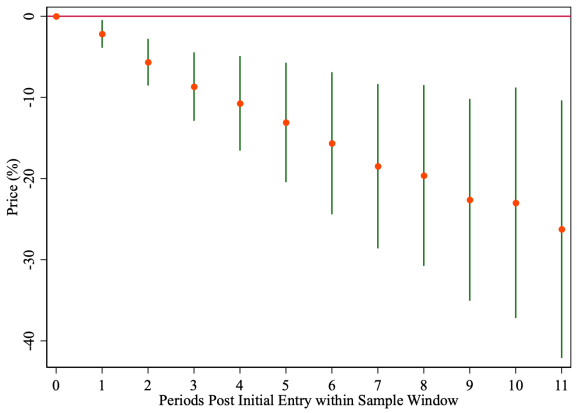
Figure A.8: Product Diffusion Across the Income Distribution: New-Module Robustness



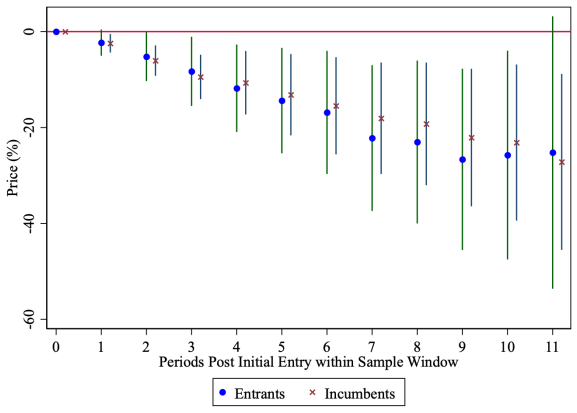
(a) Decline in Buyers' Income



(b) Growing Share of Low-Income Buyers



(c) Falling Prices Over Product Age



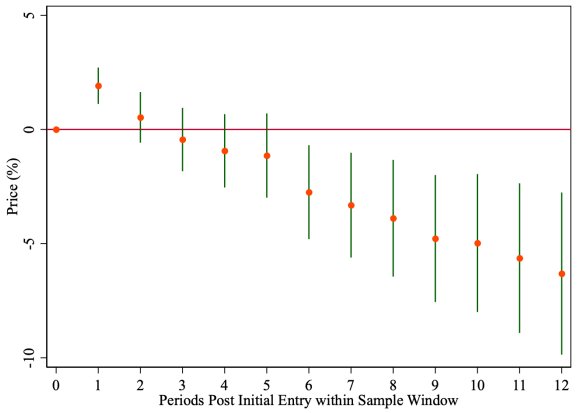
(d) Falling Prices: Entrants vs. Incumbents

Notes. Each point plots the estimated coefficient from a regression of product-level outcomes on the cumulative periods relative to initial entry within the sample window. The sample excludes firm–module pairs already present in 2004, the first year of the sample, so that event time zero corresponds to the year in which the firm first sells the module within the sample window. The x-axis measures periods after initial sales records, and the y-axis shows the estimated change in the corresponding outcome relative to the initial year. All regressions include firm–product, firm–year, and product–year fixed effects, with standard errors clustered at the firm level. In Panel (a), the dependent variable is the average income quantile of a product's buyers, showing whether its customer base shifts toward lower-income consumers as it matures. In Panel (b), the dependent variable is the expenditure share of the bottom income tercile, reflecting the growing importance of poorer consumers in total spending. Panel (c) uses the annual price to track how average prices evolve after initial year. Panel (d) further illustrates that the decline in prices is driven concurrently by both new entrants and established firms, where incumbents are firms observed in all 13 sample years. To facilitate interpretation, we report the percentage change of price in Panels (c) and (d), computed by dividing the estimated marginal effect by the sample mean price. Solid dots represent point estimates, and vertical bars denote 95% confidence intervals based on robust standard errors.

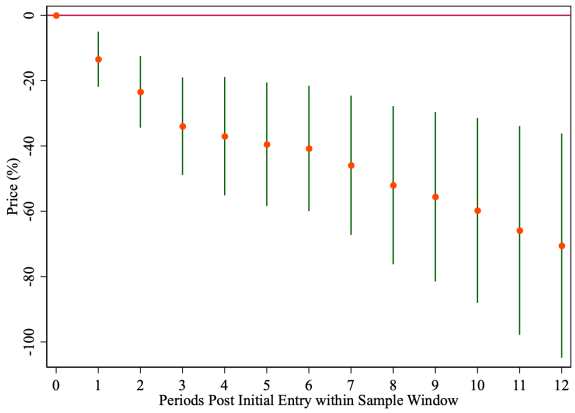
Figure A.9 reports three robustness checks on the baseline price result. Panel (a) uses the inverse hyperbolic sine (IHS) transformation of real price. The baseline specification uses the price level rather than log price because some observations are recorded as zero in the Nielsen data, typically reflecting promotional transactions such as buy-one-get-one-free offers or free samples where the item carries no incremental cost. The IHS transformation accommodates these zero-price observations while reducing the influence of extreme values (Doran et al., 2022). The price decline persists under this transformation, though the estimated magnitude is more modest than in the baseline because IHS transformation compresses large values. Panel (b) uses size-adjusted average price, where total quantity is computed across equivalent package sizes to address within-module variation in packaging. The downward price path remains clear. Panel (c) splits the sample by whether the firm filed at least one ultimately granted patent during the sample period. Both innovative and non-innovative firms exhibit declining prices as their products age, but the decline is steeper among innovative firms, consistent with cost-reducing innovation contributing to the price dynamics. Across all three alternatives, the central finding of falling prices over the product life cycle is unchanged.

Figure A.10 provides direct evidence on the technological mechanism underlying the price decline documented in Fact I, while also addressing an alternative interpretation of the diffusion pattern. One concern is that the downward shift in buyer income and falling prices could reflect product obsolescence rather than successful diffusion: if a product is gradually abandoned by consumers and displaced by newer alternatives, high-income households would exit first and prices would fall as demand collapses, mechanically generating the observed pattern. Under this obsolescence interpretation, firms would cease investing in these declining products, and productivity would stagnate or decline. The data show the opposite. Firm-level total factor productivity rises steadily as products age, with the pattern holding under both the Olley and Pakes and the Levinsohn and Petrin methodologies. This productivity growth is also consistent with the cost reductions that drive falling prices and the subsequent diffusion of products toward lower-income households. Together with the patent evidence in Fact II, the TFP results confirm that the luxury-to-necessity transition is accompanied by genuine technological progress rather than by the decline of obsolete products.

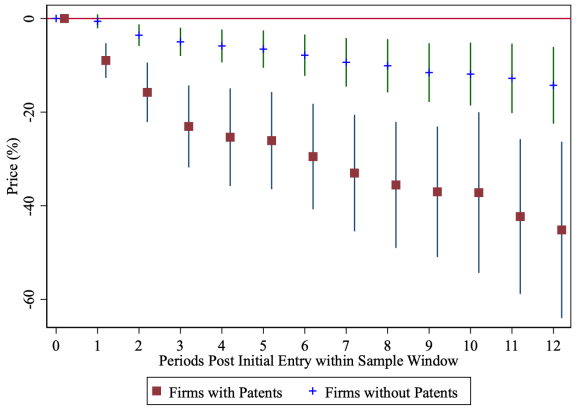
Figure A.9: Product Price Dynamics: Alternative Specifications



(a) Falling Prices: Inverse Hyperbolic Sine



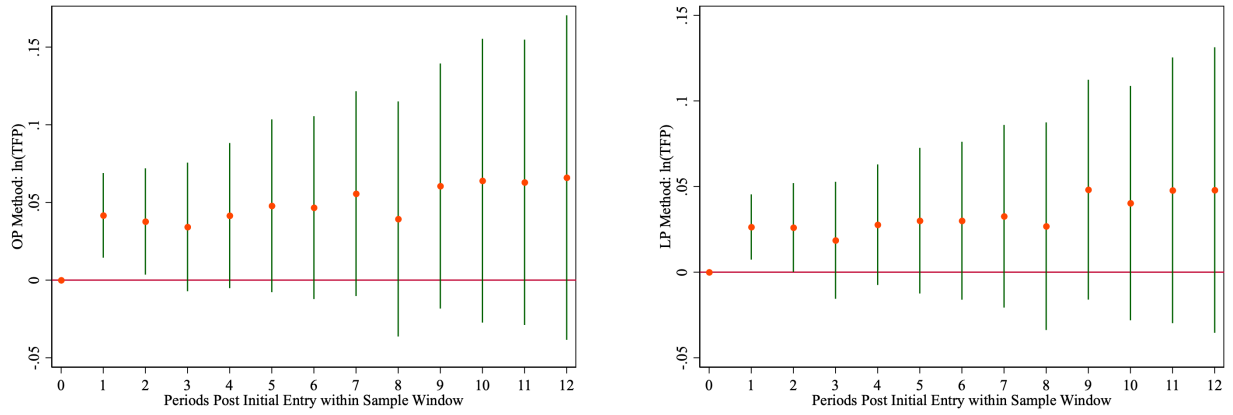
(b) Falling Prices: Size-Adjusted



(c) Falling Prices: Innovative vs. Non-Innovative Firms

Notes. Each point plots the estimated coefficient from a regression of product-level prices on the cumulative periods relative to initial entry within the sample window. The x-axis measures periods after initial sales records, and the y-axis shows the estimated percentage change in price relative to the initial year. All regressions include firm–product, firm–year, and product–year fixed effects, with standard errors clustered at the firm level. Prices are deflated using the U.S. CPI. Panel (a) uses the inverse hyperbolic sine transformation of real price, which accommodates zero-price observations and reduces the influence of extreme values. Panel (b) uses size-adjusted average price, where total quantity is computed across equivalent package sizes to address within-module variation in packaging. Panel (c) splits the sample by whether the firm filed at least one ultimately granted patent during the sample period: innovative firms (squares) and non-innovative firms (plus markers). To facilitate interpretation, coefficients are scaled to percentage changes by dividing by the sample mean price. Solid markers represent point estimates, and vertical bars denote 95% confidence intervals based on robust standard errors.

Figure A.10: Further Evidence on Firm Innovation: Growing TFP Pattern



(a) Olley and Pakes (OP) Methodology

(b) Levinsohn and Petrin (LP) Methodology

Notes. This figure provides additional evidence on firm innovation by documenting firm-level total factor productivity (TFP) growth over the product life cycle. Each point plots the estimated coefficient from a regression of firm-level log TFP on the cumulative periods relative to initial entry within the sample window. The x-axis measures periods after the initial year of market entry, and the y-axis shows the estimated change in log TFP relative to the initial year. All regressions include firm and year fixed effects, with standard errors clustered at the firm level. Firm-level TFP is obtained from the BvD Orbis database, matched to NielsenIQ firms via the GS1–BvD concordance. Panel (a) estimates TFP using the Olley and Pakes (OP) methodology. Panel (b) estimates TFP using the Levinsohn and Petrin (LP) methodology. Solid dots represent point estimates, and vertical bars denote 95% confidence intervals based on robust standard errors.

A.5 Robustness of Fact I: Correcting for Left Truncation with Technology Birth Years

The robustness checks above address left truncation in the event-time measure by dropping modules already present in 2004. This subsection takes a complementary approach that retains the full sample. The baseline event-time variable in Facts I and III, D_{fmy}, relies on the first observed in-sample sales record (2004–2016), so modules that debuted commercially prior to 2004 have unknown pre-sample ages, masking their early-stage diffusion dynamics. We address this bias by matching each module to its technology birth year recorded in Kalyani et al. (2025), constructing a corrected time metric—Years Since Technology Birth—that counts true elapsed years from the technology's original launch to each observation year. Kalyani et al. (2025) leverage comprehensive U.S. patent texts spanning 1976 to 2014, combined with Wikipedia and corporate earnings call transcripts, to systematically identify distinct novel technologies dating back to the 1970s. This metric largely recovers pre-2004 technology origins and allows the diffusion dynamics to be traced from the actual inception of the underlying technology rather than from the first in-sample sales record.

Product–Technology Matching Procedure. We construct product–technology mappings in two stages. In the first stage, we manually classify NielsenIQ product modules into a set of broad product categories. The purpose of this intermediate classification is to abstract from the detailed module-level nomenclature and place products into economically meaningful categories that can be compared consistently across modules. The set of broad categories is specified prior to the technology-matching exercise. Researchers conduct this classification manually, using the product module as the primary source of information and referring to its broader product-group and department classifications only when necessary to resolve ambiguous cases.

In the second stage, we use OpenAI GPT-5.5 to link these broad product categories to technologies. On the product side, the model receives the standardized broad-category name and its generic product definition. On the technology side, the model relies on the standardized technology names and descriptive titles in the technology dictionary of Kalyani et al. (2025). We restrict the candidate set to the 1,899 pre-defined technologies contained in this dictionary. For each broad product category, the model may identify zero, one, or multiple relevant technologies but cannot introduce technology identifiers outside the fixed dictionary.

The matching criterion is deliberately restrictive. A technology is considered relevant only when it has a direct relationship to the product category itself, a defining function of the product, or an essential component or material embodied in the product. For each proposed link, GPT-5.5 reports the corresponding technology, the nature of the relationship, a confidence assessment, and a short rationale identifying the specific component or function underlying the match. The same matching standard is used in subsequent reviews of ambiguous cases and cross-category consistency.

The operational instructions given to GPT-5.5 can be summarized as follows:

(i) Assign each broad product category zero, one, or multiple technologies from the fixed tech-

nology dictionary of Kalyani et al. (2025).

(ii) Treat the standardized broad product category and its generic definition as the product-side basis for the matching decision.

(iii) Report the exact technology name used in the dictionary. The accompanying technology description may inform interpretation but may not be substituted for the technology name or used to create a new technology category.

(iv) Accept a match only when the relationship is direct. A qualifying technology must characterize the product category itself, a defining product function, or an essential component or material.

(v) Exclude relationships arising only through upstream manufacturing, logistics, transportation, retail operations, advertising, inventory management, generic firm-level software or data infrastructure, isolated abbreviations, or incidental packaging, unless the product category itself concerns packaging or materials.

(vi) For each proposed match, report the technology name, relationship type, confidence assessment, and a specific explanation of the component, material, or function that establishes the direct relationship. Generic semantic similarity is insufficient for a valid match.

(vii) Retain multiple technologies when they capture distinct direct mechanisms embodied in the same product category, and return no match when no technology in the dictionary satisfies the direct-link criterion.

Once the broad-category–technology crosswalk is constructed, each NielsenIQ product module inherits the technology associated with the broad product category to which it was manually assigned.

Details on Matched Technologies. We impose several additional restrictions to obtain a accurate and interpretable set of product–technology links. First, we retain only broad-category–technology matches classified as *high confidence*. Second, when a broad product category is associated with multiple distinct qualifying technologies, we exclude the corresponding product modules because such cases do not permit an unambiguous assignment of the technology underlying the product's life cycle. Finally, we exclude the single matched technology dated 2003. All other technologies in the matched sample originated in or before 1997, and retaining this isolated post-1997 observation would create a highly asymmetric distribution of technology emergence dates and make its corresponding event-time measure difficult to compare with the rest of the sample.

These restrictions yield a final sample of 53 product modules for the technology life-cycle regressions. The retained broad-category–technology matches fall into three mutually exclusive and readily interpretable groups: exact or near-name correspondence between the product category

and the technology (Type I), direct component or material relationships (Type II), and direct functional relationships (Type III). Appendix Table A.2 reports the complete set of retained matches, including the manually assigned broad product category from NielsenIQ product module, the corresponding technology, the matching type, and the rationale for the broad-category–technology link.

Table A.2: Product–Technology Matches Based on Broad Product Categories

Broad Product Category	Technology	Type	Match Rationale
Absorbent hygiene products	Super absorbent	II	Absorbent hygiene products rely on superabsorbent materials to retain fluid and limit leakage, making the technology a central material embodied in the product category.
Aerosol fabric-care products	Aerosol particles	III	Aerosol fabric-care products disperse treatment agents through a spray, making aerosolization a direct feature of the product-delivery mechanism.
Aerosol food sprays	Aerosol particles	III	Aerosol food sprays disperse edible ingredients as fine droplets, making aerosolization a defining feature of the product's delivery mechanism.
Aerosol-generating smoking products	Aerosol particles	III	Aerosol-generating smoking products function by producing an inhalable aerosol, making aerosol particles a direct output of product use.
Aerosol household fragrance products	Aerosol particles	III	Aerosol household fragrance products disperse fragrance through small airborne droplets or particles, making aerosolization a direct product-delivery mechanism.
Aerosol personal-care products	Aerosol particles	III	Aerosol personal-care products apply active or fragrance ingredients through a spray, making aerosolization a direct delivery mechanism.
Aerosol pest-control products	Aerosol particles	III	Aerosol pest-control products disperse active ingredients through fine droplets or particles, making aerosolization intrinsic to the product's application mechanism.
Automotive lubricants	Liquid lubricant	III	Automotive lubricants reduce friction between moving components in liquid form, making liquid lubrication the defining function of the product category.
Borax-based laundry additives	Sodium tetraborate	II	Borax-based laundry additives are characterized by the use of borate compounds, of which sodium tetraborate is a defining material.
Cooling appliances	Removing heat	III	Cooling appliances operate by removing heat from an enclosed space and transferring it elsewhere; heat removal therefore represents their defining functional mechanism.
Dental floss products	Poly tetrafluoroethylene	II	Dental floss products may use PTFE filaments or tape, making poly tetrafluoroethylene a direct material component of the product.

Continued on next page

Table A.2: Product–Technology Matches Based on Broad Product Categories (Continued)

Broad Product Category	Technology	Type	Match Rationale
Disposable plastic drinkware accessories	Isotactic polypropylene	II	Disposable plastic drinkware accessories commonly use polypropylene-based structural materials, making isotactic polypropylene a direct material associated with the category.
Disposable plastic food-service products	Isotactic polypropylene	II	Disposable plastic food-service products commonly use rigid polypropylene materials, making isotactic polypropylene a direct structural material associated with the category.
Electric cooking appliances	Joule heating	III	Electric cooking appliances use resistive electrical heating to generate the heat required for cooking, steaming, or drying food.
Electric fabric-care appliances	Joule heating	III	Electric fabric-care appliances rely on resistive heating to generate heat or steam for treating fabrics.
Electric heating appliances	Joule heating	III	Electric heating appliances convert electrical energy into heat through resistance, making Joule heating their defining functional mechanism.
Household tissue products	Ply tissue	II	Household tissue products use layered tissue structures to provide strength and absorbency, making ply tissue a direct material characteristic of the product category.
Household water-filtration products	Carbon filter	II	Household water-filtration products use carbon filtration media to remove tastes, odors, and other substances from water, making carbon filters a direct functional component.
Inhaled respiratory products	Aerosol particles	III	Inhaled respiratory products can deliver active ingredients as aerosolized particles to the respiratory tract, making aerosolization a direct delivery mechanism.
Label-printing devices	Label printer	I	Label-printing devices and label printers refer to the same underlying product function and closely corresponding product class.
Lubricant products	Liquid lubricant	III	Lubricant products are designed to reduce friction between moving surfaces, and liquid lubrication directly captures this defining product function.
Packaged software products	Software distribution	III	Packaged software products reach users through the distribution of executable or installable software, making software distribution directly related to the product's delivery form.
Pressurized aerosol cleaning products	Aerosol particles	III	Pressurized aerosol cleaning products release their contents as a directed spray or particle stream, so aerosolization is intrinsic to the product's mode of use.
Printer and computer supplies	Toner cartridge	II	Printer and computer supplies include replaceable printing consumables, among which toner cartridges constitute a direct physical product component.

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Table A.2: Product–Technology Matches Based on Broad Product Categories (Continued)

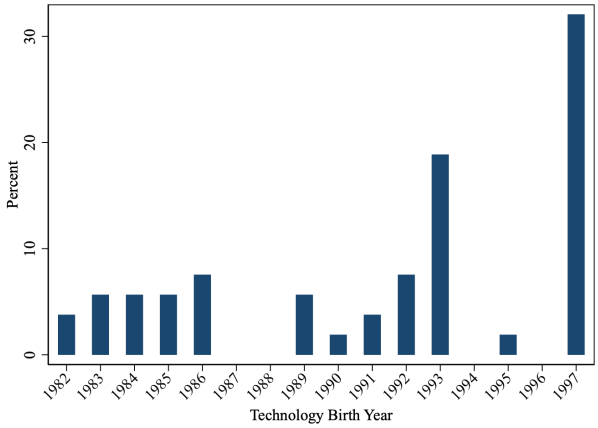
Broad Product Category	Technology	Type	Match Rationale
Printer cartridges	Toner cartridge	I	Printer cartridges and toner cartridges represent closely corresponding product categories, with the latter identifying a specific major class of replaceable printer cartridge.
Printing devices	Printing technologies	III	Printing devices are defined by transferring text or images onto physical media, making printing technology the product category's central function.
Protective coating products	Polyurethane polymer	II	Protective coating products may use polyurethane polymers as film-forming materials, creating a direct material relationship between the product category and the technology.
Starch-based products	Starch derivatives	II	Starch-based products obtain their functional properties from starch and modified starch materials, making starch derivatives a direct material relationship.
Synthetic cosmetic applicators	Polyurethane polymer	II	Synthetic cosmetic applicators commonly use polymer-based foam materials, making polyurethane polymers a direct material embodied in this product category.
Temperature-measurement devices	Temperature sensor	II	Temperature-measurement devices rely on temperature-sensing components to perform their defining measurement function.

Notes. This table reports the 30 broad product categories retained in the technology life-cycle analysis. Each broad product category is manually assigned by the researchers from the NielsenIQ product classification. The subsequent technology-matching procedure is conducted at the broad-product-category level using the fixed technology dictionary of Kalyani et al. (2025). The table reports the broad product category, the corresponding technology, the type of direct relationship, and the rationale for the category–technology match. The retained matches are classified into three mutually exclusive categories: exact or near-name correspondence (Type I), direct component or material relationships (Type II), and direct functional relationships (Type III).

Appendix Figure A.11 documents the distribution of technology birth years for all matched products, which span 1982 to 1997. While technologies emerge steadily across the 1980s and early 1990s, the distribution features a prominent concentration in 1997, with more than 30 percent of all matched technologies originating in this single year. Since 1997 lies only seven years prior to the start of our 2004–2016 sample window, the gap between true technology inception and our first observed sales record is relatively short for the vast majority of product modules. In other words, this compressed time lag largely mitigates the severity of left truncation and measurement error inherent in our baseline event-time metric.

Empirical Analysis. We replicate Fact I using the same three outcome variables: the average income percentile of consumers purchasing each product module, the expenditure share of low-income consumers, and product prices. The core regressors are event-time dummies indicating

Figure A.11: Birth Year Distribution of Matched Technologies

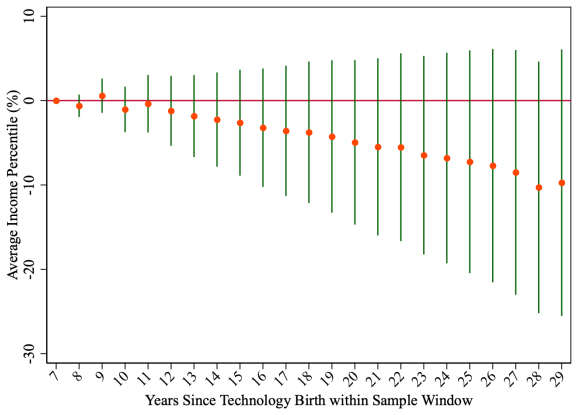


Notes. This histogram plots the fraction of product modules matched to each technology emergence year sourced from Kalyani et al. (2025). The horizontal x-axis records each technology's true birth year, ranging from 1982 to 1997. The vertical y-axis measures the fraction of all matched product modules falling into each birth-year bin.

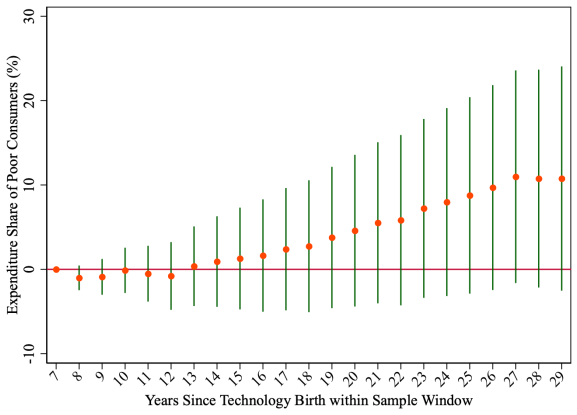
years relative to the technological birth of each module, with each dummy equal to one in the y-th year following module birth within the sample window. For modules without a matched technology birth date, we impute the birth year as the average technological birth year of all modules in the same product department. The specification includes three sets of fixed effects. Firm–module fixed effects absorb persistent differences in product attributes and firm positioning within a module. Firm–year fixed effects absorb time-varying firm-wide shocks such as changes in brand reputation or marketing intensity. Module–sales age fixed effects—where sales age is the cumulative number of years a firm has sold the module within the sample window—absorb module-specific lifecycle confounders that co-vary with market maturity, including shifting consumer tastes, growing brand recognition, expanding retail coverage, and iterative product upgrades. Standard errors are clustered at the firm level.

Appendix Figure A.12 replicates the Fact I patterns. Following module birth, the average buyer income percentile declines (Panel a), the low-income expenditure share rises (Panel b), and average prices fall (Panel c). Adoption thus moves down the income distribution as products age and prices decline. We also replicate Fact III using the same two outcome variables: the number of active firms in each product module and the module-market HHI. The specification includes module fixed effects, which absorb time-invariant differences across product categories, and year fixed effects, which absorb aggregate shocks. Standard errors are clustered at the module level. Appendix Figure A.13 replicates the Fact III patterns. Following module birth, the number of active firms rises (Panel a) and market concentration falls (Panel b).

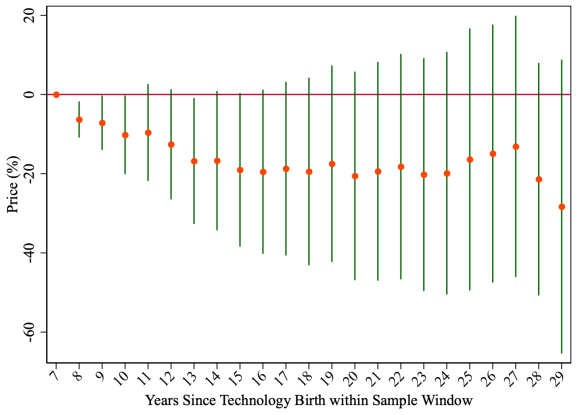
Figure A.12: Product Diffusion Across the Income Distribution Using Birth Year from Kalyani et al. (2025)



(a) Decline in Buyers' Income



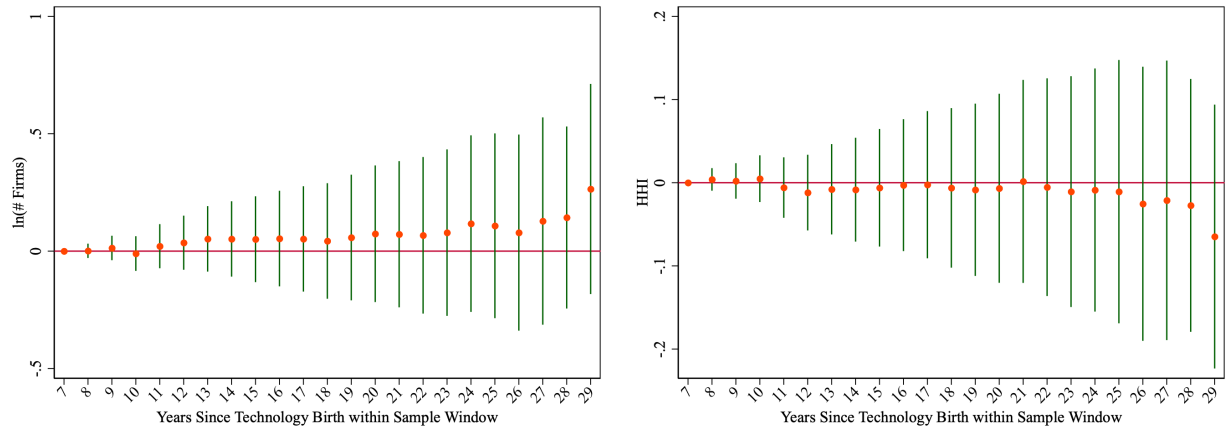
(b) Growing Share of Low-Income Buyers



(c) Falling Prices Over Product Age

Notes. Each point plots the estimated coefficient from a regression of product-level outcomes on the years relative to technology birth within the sample window. The x-axis measures cumulative years since technology birth, and the y-axis shows the estimated change in the corresponding outcome relative to the seventh year post initial entry. All regressions include firm–product, firm–year, and sales period-product fixed effects, with standard errors clustered at the firm level. In Panel (a), the dependent variable is the average income quantile of a product's buyers, showing whether its customer base shifts toward lower-income consumers as it matures. In Panel (b), the dependent variable is the expenditure share of the bottom income tercile, reflecting the growing importance of poorer consumers in total spending. Panel (c) uses the annual price to track how average prices evolve after initial year. To facilitate interpretation, we report the percentage change of price in Panel (c), computed by dividing the estimated marginal effect by the sample mean price. Solid dots represent point estimates, and vertical bars denote 95% confidence intervals based on robust standard errors.

Figure A.13: Entry and Declining Concentration over the Product Life Cycle Using Birth Year from Kalyani et al. (2025)



(a) Number of Active Firms

(b) Market Concentration (HHI)

Notes. Each panel reports event-time estimates from regressions of module-year market-structure outcomes on periods since the module's introduction. The x-axis measures years relative to the entry year. The y-axis shows the estimated change in the outcome relative to the seventh year post entry. Panel (a) uses the log number of active firms in module m and year t. Panel (b) uses the Herfindahl–Hirschman Index, $HHI_{mt} = \sum_{f \in m} s_{ft}^2$, where s_{ft} is firm f's revenue share within module m in year t. All regressions include module fixed effects and year fixed effects. Dots denote point estimates and whiskers denote 95% confidence intervals based on standard errors clustered at the module level.

A.6 Robustness of Fact II: Innovation by Incumbents

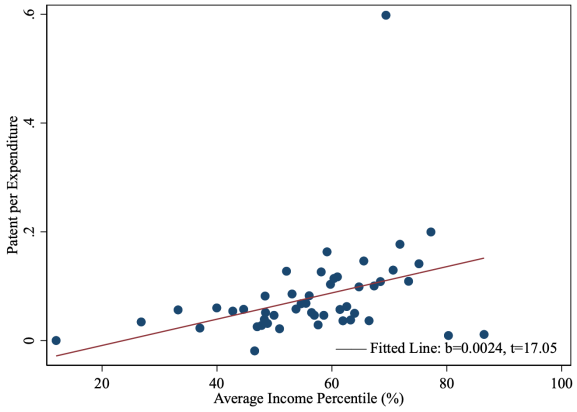
This subsection reports robustness checks for the Fact II pattern. The exercises restrict the sample to single-product firms and a balanced panel to address patent-product mismatch and firm turnover, replace the time-varying customer income percentile with time-invariant or entry-year measures, and plot the rich-poor innovation gap by year as a time-series complement to the cross-sectional gradient. The positive association between customer income and innovation intensity remains stable throughout.

Figure A.14 probes the robustness of this pattern to two concerns raised by the firm-level nature of the patent data. First, because patents are measured at the firm level, they need not map one-to-one to the specific product modules serving high- or low-income households for diversified firms. Restricting the sample to single-product firms—firms selling in exactly one product module—eliminates this concern, as the firm's patenting activity and its customer income profile pertain to the same product line. Second, the baseline pattern could in principle reflect firm turnover or compositional changes over the sample period rather than a stable relationship. The further balanced-panel restriction—keeping only single-product firms observed in all 13 years—removes entry and exit margins from the variation. Across both restrictions, the positive gradient between customer income percentile and innovation intensity persists, for both patent counts and claim-weighted patents. That the pattern survives these more demanding samples indicates that it is not driven by patent-product mismatch in diversified firms or by the changing composition of firms over time.

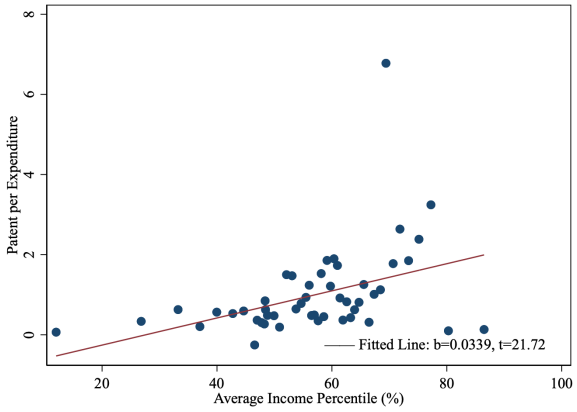
Figure A.15 additionally examines whether the baseline gradient is sensitive to how the firm's customer income profile is measured, using two alternative measures that differ in the source of identifying variation. The first measure, used in panels (a) and (b), is the firm-level mean of the income percentile averaged over all sample years. Because this measure is time-invariant within each firm, it ensures that each firm occupies a fixed position in the income-percentile distribution throughout the sample period, preventing firms from shifting between high- and low-income segments from year to year. The positive gradient persists under this measure, indicating that the pattern reflects stable differences in firm demand orientation rather than transitory fluctuations in customer composition. The second measure, used in panels (c) and (d), is the income percentile measured strictly in the firm's entry year. Because entry-year customer composition is fixed before the outcome period, it does not shift alongside subsequent changes in firm innovation. The gradient again remains positive. Together, these two alternatives—one fixing each firm's demand orientation to a stable within-sample average, the other fixing the demand measure at the start of the firm's tenure—show that the baseline correlation is not driven by time-varying customer composition or by within-firm co-movement between innovation and the income profile of a firm's consumers.

Figure A.16 complements the cross-sectional binned scatter plots with a time-series view, plotting the average innovation intensity of rich- and poor-market firms separately for each year from 2004 to 2016. Firms are split at the within-year median of the average buyer income percentile. In both panels, rich-market firms consistently exhibit higher innovation intensity than poor-market

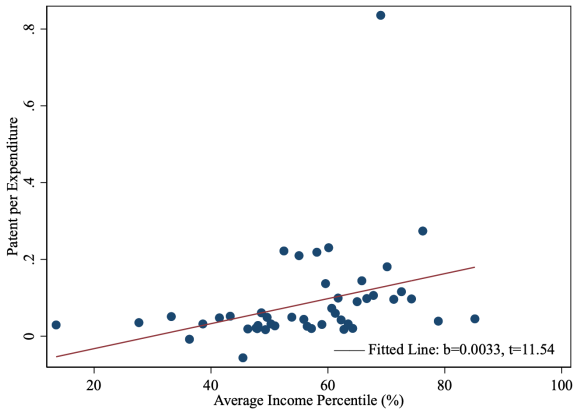
Figure A.14: Innovation Intensity by Customer Income Segment: Single-Product and Balanced-Panel Robustness



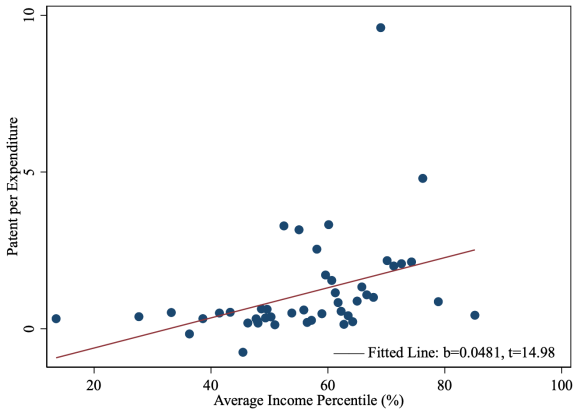
(a) Single-Product: Patent Count-based Innovation Intensity



(b) Single-Product: Patent Quality-based Innovation Intensity



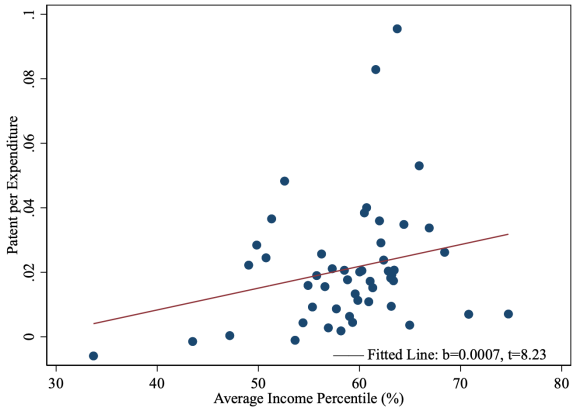
(c) Single-Product, Balanced Panel: Patent Count-based Innovation Intensity



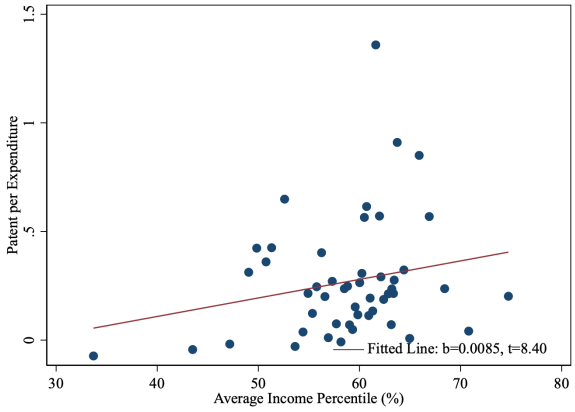
(d) Single-Product, Balanced Panel: Patent Quality-based Innovation Intensity

Notes. Each panel displays a binned scatter plot of firm-level innovation intensity against the average income percentile of the firm's customer base, with 50 quantiles. The average income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. All specifications absorb year fixed effects, control for the log number of household groups and the log number of product modules, and weight observations by firm expenditure. Panels (a) and (b) restrict the sample to single-product firms—firms selling in exactly one product module. Panels (c) and (d) further restrict to the balanced panel of single-product firms observed in all 13 sample years (2004–2016). In panels (a) and (c), innovation intensity is patents per million dollars of consumer expenditure, with fitted slopes of 0.0024 ($t = 17.05$) and 0.0033 ($t = 11.54$), respectively. In panels (b) and (d), innovation intensity is claim-weighted patents per million dollars of consumer expenditure, with fitted slopes of 0.0339 ($t = 21.72$) and 0.0481 ($t = 14.98$), respectively. Innovation intensity variables are winsorized at the 1% and 99% percentiles within each year.

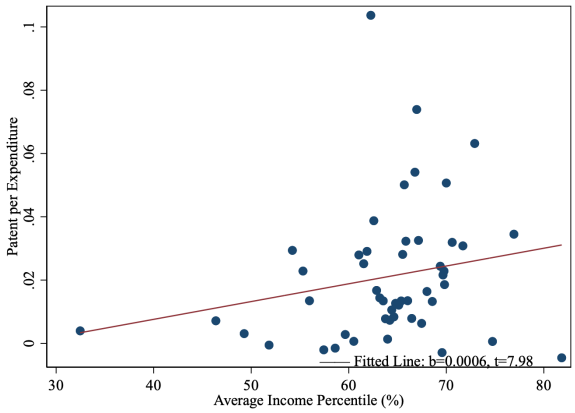
Figure A.15: Innovation Intensity by Customer Income Segment: Alternative Income-Percentile Measures



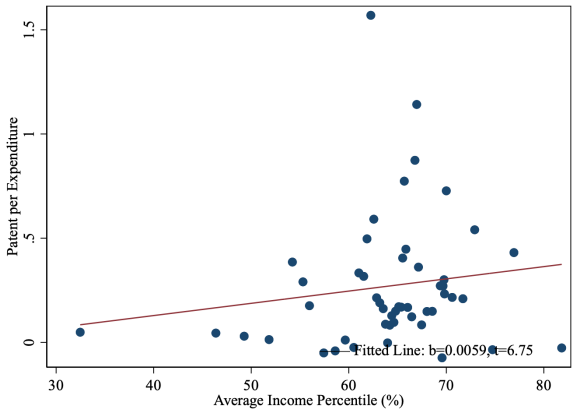
(a) Firm-Mean Income Percentile: Patent Count-based Innovation Intensity



(b) Firm-Mean Income Percentile: Patent Quality-based Innovation Intensity



(c) Entry-Year Income Percentile: Patent Count-based Innovation Intensity

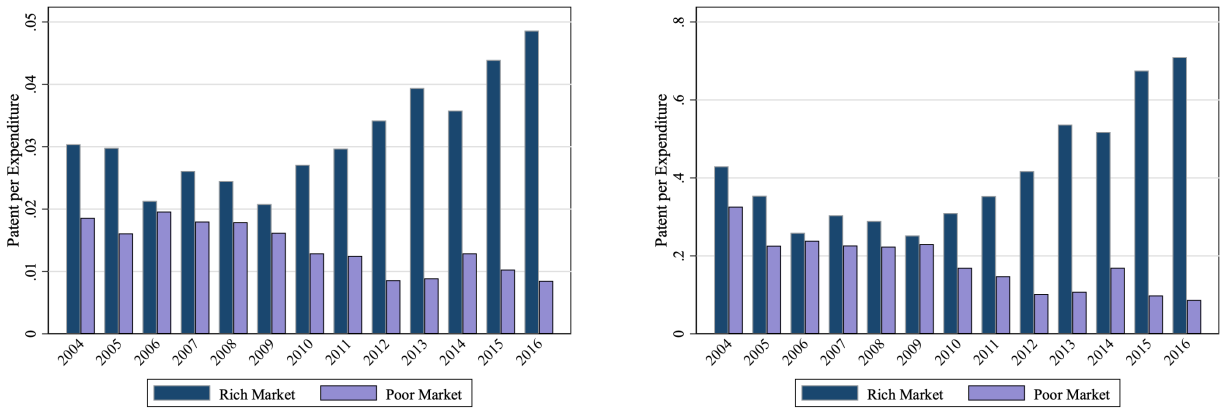


(d) Entry-Year Income Percentile: Patent Quality-based Innovation Intensity

Notes. Each panel displays a binned scatter plot of firm-level innovation intensity against the average income percentile of the firm's customer base, with 50 quantiles. All specifications absorb year fixed effects, control for the log number of household groups and the log number of product modules, and weight observations by firm expenditure. The baseline specification uses the time-varying average income percentile in each firm-year. Panels (a) and (b) instead use the firm-level mean of the income percentile averaged over all sample years, a time-invariant firm characteristic. Panels (c) and (d) use the income percentile measured in the firm's entry year, which is predetermined with respect to subsequent innovation. In panels (a) and (c), innovation intensity is patents per million dollars of consumer expenditure, with fitted slopes of 0.0007 ($t = 8.23$) and 0.0006 ($t = 7.98$), respectively. In panels (b) and (d), innovation intensity is claim-weighted patents per million dollars of consumer expenditure, with fitted slopes of 0.0085 ($t = 8.40$) and 0.0059 ($t = 6.75$), respectively. Innovation intensity variables are winsorized at the 1% and 99% percentiles within each year.

firms, and the gap widens over the sample period: innovation intensity among rich-market firms rises over time, while that among poor-market firms declines. The divergence is more pronounced in Panel (b), where patents are weighted by claims. This time-series pattern aligns with the cross-sectional gradient documented above, confirming that the positive association between customer income and innovation intensity is not specific to the binned-scatter specification or to pooling across years.

Figure A.16: Innovation Intensity by Customer Income Segment and Year, 2004–2016



(a) Patent Count-based Innovation Intensity

(b) Patent Quality-based Innovation Intensity

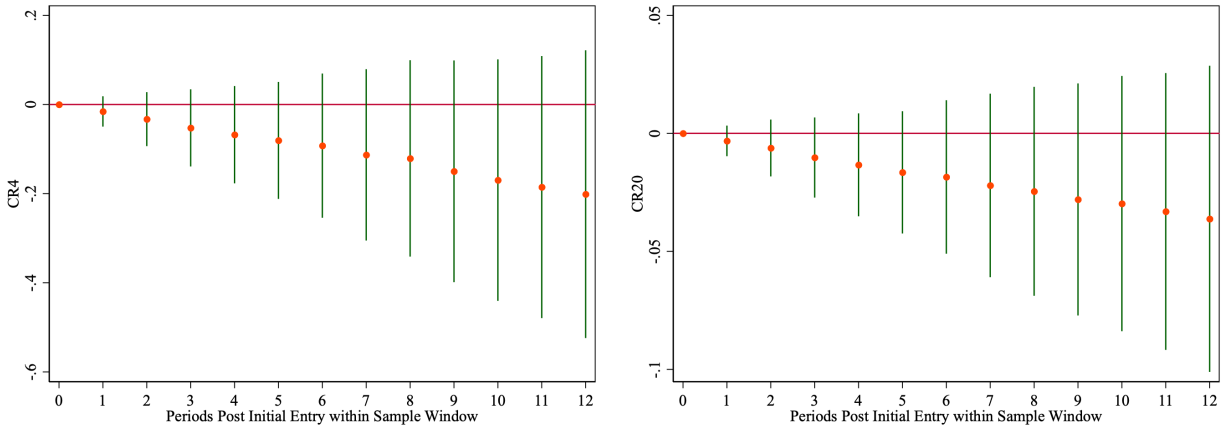
Notes. Each panel plots the expenditure-weighted average innovation intensity by year, separately for rich-market and poor-market firms. Firms are classified as rich-market (poor-market) if their average buyer income percentile in a given year is above (below) the within-year 50th percentile, where the buyer income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. In Panel (a), innovation intensity is defined as patents per million dollars of consumer expenditure on the firm's products. In Panel (b), innovation intensity is defined as claim-weighted patents per million dollars of consumer expenditure. Innovation intensity variables are winsorized at the 1% and 99% percentiles within each year.

A.7 Robustness of Fact III: Entry of New Competitors

This subsection reports robustness checks for the Fact III entry and de-concentration pattern. We verify the decline in market concentration using alternative concentration measures—the CR4 and CR20—in place of the baseline Herfindahl–Hirschman Index. We also re-estimate the firm-entry and concentration dynamics on a sample of modules that enter after 2004, so that event time zero corresponds to true module introduction rather than a left-censored starting point. Across both exercises, the core pattern of rising firm participation and falling concentration over the product life cycle remains stable.

Figure A.17 verifies the Fact III de-concentration pattern using alternative concentration measures. Panel (a) reports the CR4, the combined expenditure share of the four largest firms in each module-year, and Panel (b) reports the CR20, the corresponding share of the twenty largest firms (Amiti and Heise, 2025). Both measures decline steadily over the product life cycle, confirming that the fall in HHI documented in the baseline is not an artifact of the specific functional form of the Herfindahl index. The CR4 falls by roughly 0.25 within twelve years of entry, while the CR20 declines by a smaller magnitude, consistent with entry occurring primarily among smaller firms that gradually erode the dominance of the largest incumbents. Together with the baseline HHI result, these estimates establish that market participation broadens and concentration falls across multiple metrics as products mature.

Figure A.17: Entry and Declining Concentration over the Product Life Cycle: CR4 and CR20



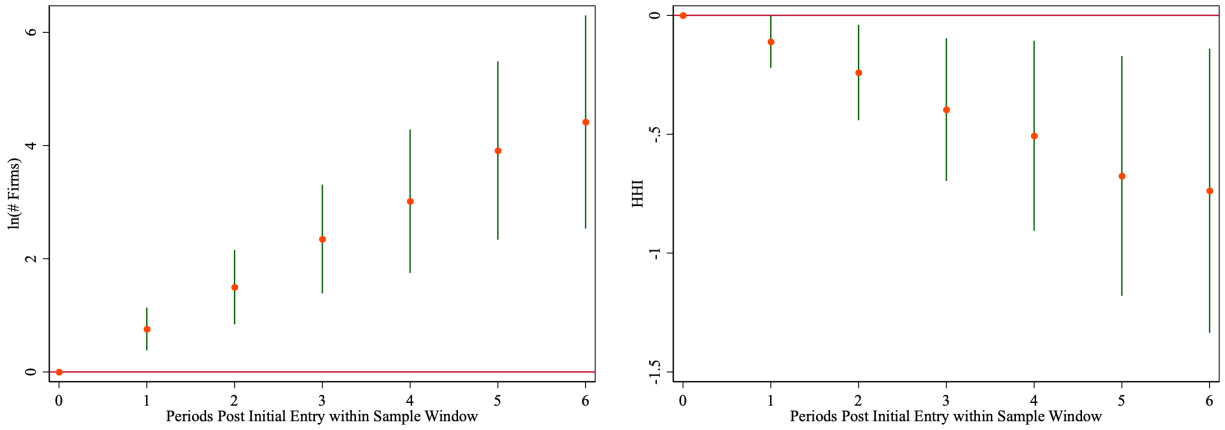
(a) Top-Four Concentration Ratio (CR4) (b) Top-Twenty Concentration Ratio (CR20)

Notes. Each panel reports event-time estimates from regressions of module-year market-concentration outcomes on periods since the module's introduction. The x-axis measures years relative to the entry year, and the y-axis shows the estimated change in the outcome relative to the initial year. Panel (a) uses the CR4, defined as the combined expenditure share of the four largest firms within module m and year t. Panel (b) uses the CR20, defined as the combined expenditure share of the twenty largest firms within module m and year t. All regressions include module fixed effects and year fixed effects, with standard errors clustered at the module level. Solid dots represent point estimates, and vertical bars denote 95% confidence intervals based on robust standard errors.

Figure A.18 re-estimates the Fact III market-structure dynamics on a sample of modules that enter after 2004, the first year of the sample, so that event time zero corresponds to the year in which

the module first appears in the sample rather than a left-censored starting point. This restriction addresses the concern that modules already present in 2004 have unknown pre-sample ages, which could attenuate or distort the estimated life-cycle pattern of entry and concentration. The same patterns hold. Following true entry, the number of active firms rises sharply (Panel a) and market concentration, measured by the HHI, falls steadily (Panel b). That the entry and de-concentration patterns survive when product age is measured without left censoring confirms that the broadening of market participation over the product life cycle reflects the actual maturation of modules from their market introduction rather than compositional changes among left-truncated observations.

Figure A.18: Entry and Declining Concentration over the Product Life Cycle: New-Module Robustness



(a) Number of Active Firms

(b) Market Concentration (HHI)

Notes. Each panel reports event-time estimates from regressions of module-year market-structure outcomes on periods since the module's introduction. The sample excludes modules already present in 2004, the first year of the sample, so that event time zero corresponds to the year in which the module first appears within the sample window. The x-axis measures years relative to the entry year, and the y-axis shows the estimated change in the outcome relative to the initial year. Panel (a) uses the log number of active firms in module m and year t. Panel (b) uses the Herfindahl–Hirschman Index, $HHI_{mt} = \sum_{f \in m} s_{ft}^2$, where s_{ft} is firm f's expenditure share within module m in year t. All regressions include module fixed effects and year fixed effects, with standard errors clustered at the module level. Solid dots represent point estimates, and vertical bars denote 95% confidence intervals based on robust standard errors.

A.8 Fact IV: Empirical Specification and Robustness

This subsection details the empirical specification and robustness checks for Fact IV. We first present the shift-share reduced-form specification, including the construction of the inequality shock, the control set, and sample restrictions. We then report the main estimates and their economic magnitude, followed by a sequence of robustness exercises: balance tests on pre-period characteristics, additional controls for imbalanced variables, sensitivity to excluding influential states and firms, exposure-robust standard errors, validation of the inequality measure against official data, and alternative inequality indices.

Baseline Specification. To identify how income inequality affects firm innovation, we estimate the following shift-share specification:

$$N_{ft}^{Pat} = \beta \underbrace{\sum_{s} \lambda_{sft_0} \widehat{Gini}_{st}}_{Shock_{ft}^{ineq}} + \boldsymbol{\delta}' \sum_{s} \lambda_{sft_0} \widehat{\mathbf{Z}}_{st} + \mathbf{X}_{ft}\boldsymbol{\gamma} + \mu_f + \mu_t + \epsilon_{ft}, \tag{A.1}$$

where f indexes firms, t indexes years, and s indexes U.S. states. The outcome N_{ft}^{Pat} measures innovation intensity as patents per thousand dollars of consumer expenditure. We employ two complementary measures: the count of patent applications filed by firm f in year t that are ultimately granted, and a claim-weighted patent count that weights each patent by its number of claims to capture technological scope (Aghion et al., 2019a). Expenditure scaling—in thousands of dollars for coefficient readability—purges firm-size effects from the innovation measure.

The initial share of firm f's sales in state s in its entry year t_0 is

$$\lambda_{sft_0} \equiv \frac{E_{sft_0}}{\sum_s E_{sft_0}},$$

where E denotes consumer expenditure. As is standard in the shift-share literature, our constructed shares are complete—they sum to unity for every firm—which eliminates the incomplete-share bias, whereby cross-sectional variation in total exposure $S_f = \sum_s \lambda_{sft_0}$ would load non-experimental variation into the instrument. State-year inequality is measured by the Gini index, following Boyce et al. (2016):

$$Gini_{st} = 1 + \frac{1}{n} - \frac{2}{n^2 \cdot \overline{Income}_{st}} \sum_{i \in \Omega_{st}} (n - i + 1)\, Income_{i,st}, \tag{A.2}$$

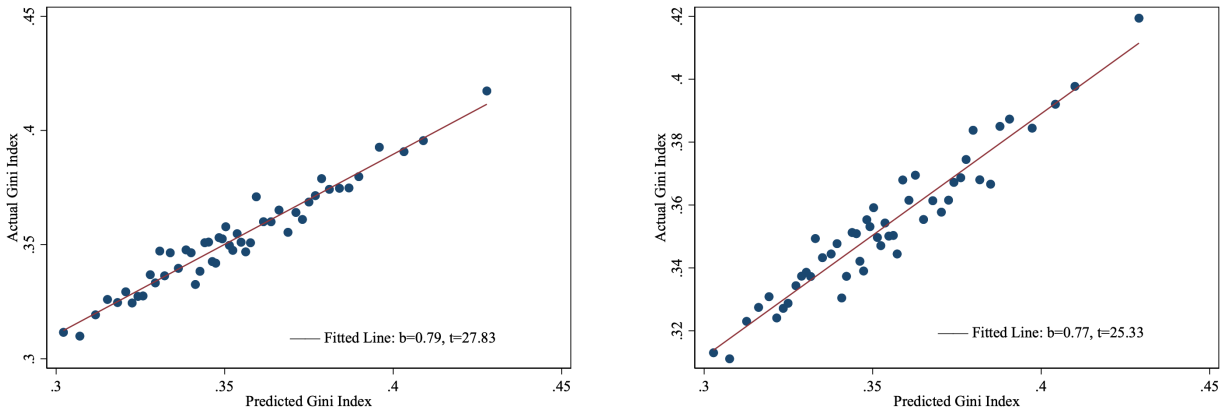
where Ω_{st} denotes the set of households in state s in year t, indexed in non-decreasing income order, $n = |\Omega_{st}|$ is the number of households, and $\overline{Income}_{st}$ is mean household income. The Gini coefficient lies in $[0, 1]$, with higher values denoting greater inequality. To address the potential endogeneity of observed state-level inequality, we adopt a instrumental-variable strategy in the spirit of Doerr et al. (2024). The instrument interacts each state's initial inequality level with the

leave-one-out national evolution of inequality:

$$\widehat{Gini}_{st} \equiv Gini_{s,2004} \times \frac{1}{N-1} \sum_{j \neq s}^{N} \frac{Gini_{jt}}{Gini_{j,2004}}. \tag{A.3}$$

Intuitively, $\widehat{Gini}_{st}$ captures the counterfactual path of inequality in state s had that state tracked the national trend, scaled by its initial inequality level. Because the national component excludes state s itself, the instrument purges variation originating from the state's own inequality dynamics. We use $\widehat{Gini}_{st}$ as an instrument for the observed state-level Gini index over 2004–2016. Figure A.19 provides a first-stage check: it plots actual against predicted state-level Gini, with observations binned into 50 quantiles. The relationship is positive, tightly clustered around the 45-degree line, and highly significant—the fitted slope is 0.79 ($t = 27.83$) in the full sample and 0.77 ($t = 25.33$) when the base year 2004 is excluded—confirming that the instrument captures meaningful variation in actual inequality.

Figure A.19: Actual versus Predicted State-Level Gini Index



(a) Full Sample (2004–2016)

(b) Excluding Base Year 2004

Notes. Each point represents a U.S. state-year observation, binned into 50 quantiles of the predicted Gini index. The solid line is the linear fit. Panel (a) uses all state-year observations from 2004 to 2016; panel (b) excludes the base year 2004. The fitted slope and t-statistic are reported in each panel. The positive and highly significant relationship confirms that the predicted Gini index is a strong predictor of actual state-level inequality.

Following Miguel and Kremer (2004); Xu (2022); Franklin et al. (2024), we adopt a reduced-form shift-share specification, entering the exposure-weighted predicted inequality $Shock_{ft}^{ineq}$ directly as the regressor. The coefficient of interest is β. Our identification follows the shift-based exogenous shift-share framework of Borusyak et al. (2022): the idealized experiment randomly assigns state-level inequality shifts independent of any determinant of firm innovation, and cross-firm variation in exposure to these common shifts identifies the causal effect. In our design, the exposure shares λ_{sft_0} are measured in each firm's entry year and held fixed thereafter, so the same predicted state-level inequality movement generates heterogeneous firm-level demand shocks $Shock_{ft}^{ineq}$—a firm with a

larger initial sales share in state s derives more of its revenue from that state's income distribution, so a given rise in inequality there reallocates spending across income tiers more forcefully for that firm. The shift-share instrument thus isolates only the variation in inequality mediated through these predetermined geographic exposures. In practice, however, even the leave-one-out predicted Gini—designed to approximate the idealized exogenous shift by purging each state's own inequality dynamics—is not randomly assigned across states, and two classes of confounders may still threaten the exclusion restriction: shift-level (state-year) variables that correlate with the predicted inequality shift and directly affect innovation, and unit-level (firm-year) variables that correlate with both the constructed exposure and the error term. We address both through controls.

The first class consists of shift-level confounders. As Borusyak et al. (2022) show, when a state-level observable both correlates with the inequality shift and can directly affect firm outcomes, the appropriate control is its shift-share aggregate—constructed with the same exposure shares λ_{sft_0} as the instrument—so that the instrument leverages only variation in predicted inequality orthogonal to these characteristics. The vector $\widehat{\mathbf{Z}}_{st}$ thus contains predicted state-level variables that are aggregated to the firm level via $\sum_s \lambda_{sft_0}\widehat{\mathbf{Z}}_{st}$. Following Hu et al. (2025), it comprises three elements in log form: predicted average household income, predicted per-capita GDP, and predicted population, each constructed using the leave-one-out national-trend procedure in Equation A.3. State-level inequality may co-move with the level of economic development—proxied by per-capita GDP—and with market size—proxied by population—both of which could independently shape firms' innovation incentives through demand scale, input-cost, or agglomeration channels. Absorbing firm exposure to these first-moment and scale characteristics isolates the effect of the second moment of the income distribution, namely inequality, from confounding state-level economic conditions.

The second class consists of unit-level confounders: firm-year observables that may correlate with both the error term and the constructed exposure. The vector $\mathbf{X}_{ft}$ includes the log of the number of product modules supplied, to capture product scope; the firm's national expenditure share, as a proxy for market power; and the log of firm age, measured as years since first entry during 2004–2016. These covariates absorb within-firm changes in scale and market position that may correlate with both innovation output and the geographic composition of demand.

All continuous variables are winsorized at the 1st and 99th percentiles within each year. This mitigates the influence of extreme observations arising from measurement error in patent counts or expenditure shares while preserving the cross-sectional and within-firm variation that identifies β. The estimating sample is restricted to firm-years satisfying $t > t_0$, where t_0 is the firm's entry year. This condition ensures that the initial sales shares λ_{sft_0} are measured strictly before the outcome period, eliminating mechanical feedback from a firm's subsequent innovation and growth to its geographic sales composition (Borusyak et al., 2025). The specification thus exploits only the predetermined geographic distribution of demand at entry, interacted with subsequent national trends in state-level inequality, to identify the causal effect of inequality on innovation. We include firm fixed effects μ_f, which absorb time-invariant firm heterogeneity, and year fixed effects μ_t, which absorb common aggregate shocks. Standard errors are clustered at the firm level.

Main Estimates. Table 1 reports the main results. Greater exposure to income inequality depresses firm innovation. In Panel A, Column (1) yields a negative and statistically significant coefficient on $Shock_{ft}^{ineq}$ of -0.0085. Columns (2)–(5) show that this result is similar across alternative samples. Column (2) restricts to firms entering in 2004, which aligns the measurement of exposure shares across a single entry cohort and removes cross-cohort differences in initial market conditions. Column (3) restricts to firms that sell to both low- and high-income consumers, which are precisely the firms through which the luxury-to-necessity diffusion channel operates. Column (4) restricts to small firms below the median of total expenditure, which often lack the geographic diversification or internal resource buffers that allow firms to absorb local demand shocks, so their innovation decisions are more sensitive to state-level inequality movements. Column (5) restricts to firms observed in at least four years, excluding short-lived entrants whose patenting may be driven by survival concerns rather than demand-side shocks, ensuring that the estimated innovation response comes from firms whose R&D decisions track consumer demand. In economic terms, a one-standard-deviation increase in inequality exposure (0.018) reduces patenting intensity by 0.153 patents per million dollars of expenditure. Given that the average firm in the sample generates \$23.4 million in annual consumer expenditure, this translates to a decline of approximately 3.6 patent applications per firm-year.

The effect is concentrated among firms that serve lower-income households. Columns (6) and (7) split the sample by the firm's sales orientation at entry. *Poor Market* (*Rich Market*) indicates firms whose average buyer income percentile at entry is below (above) the sample median, where the buyer income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. Because the sales orientation is measured in the entry year, this split is predetermined with respect to subsequent innovation and inequality shocks, which rules out reverse causality from innovative activity to customer composition. For poor-market firms, the coefficient is -0.0129 and significant at the 5% level. For rich-market firms, the estimate falls to -0.0016 and is statistically indistinguishable from zero. The poor-market coefficient is roughly eight times larger in magnitude. Panel B confirms that these patterns are not driven by low-value patents. When patents are weighted by the number of claims, the same robustness and heterogeneity patterns hold.

Falsification Tests. The shift-share identification relies on the quasi-random assignment of the state-level inequality shocks conditional on the included controls, an assumption whose plausibility can be assessed via balance tests. Following Borusyak et al. (2022, 2025), we implement this in two complementary ways. First, we conduct a unit-level balance test: each firm characteristic measured in the firm's entry year is aggregated to the state-year level via the *ssaggregate* procedure and regressed on the predicted state-year Gini index, with year fixed effects absorbed and standard errors clustered at the state level. Second, we conduct a shock-level balance test: each state characteristic drawn from the 2000 IPUMS 5% census (Ruggles et al., 2025)—a pre-period that predates the sample by at least four years—is regressed on the predicted Gini index over 2005–2016,

weighted by average exposure shares s_n, again with year fixed effects and state-level clustering. In both exercises the Gini index is standardized to mean zero and unit variance, so each coefficient measures the change in the corresponding characteristic per one-standard-deviation increase in predicted inequality.

Table A.3 reports the results. Panel A shows that the entry-year firm characteristics are well balanced: of the ten variables tested, including patenting activity, expenditure, product scope, and geographic and household reach, none is significantly related to the inequality shock at the 5 percent level, and only the log household count is marginally significant at the 10 percent level. Panel B reports the shock-level balance for fourteen state-level demographic and labor-market characteristics. Twelve of the fourteen are statistically insignificant, including the manufacturing, agriculture, and service employment shares, educational attainment, racial composition, and household structure. The employment rate is significant at the 1 percent level: states with higher predicted inequality have lower employment rates, with a one-standard-deviation increase in Gini associated with a 2.5 percentage point lower employment rate. The female share is marginally significant at the 10 percent level.

Overall, we fail to reject balance in 21 of the 24 pre-period characteristics at conventional significance levels, which supports the plausibility of the shift-orthogonality assumption. The one notable imbalance is the employment rate: a one-standard-deviation increase in predicted Gini is associated with a 2.5 percentage point lower employment rate (p=0.002). To address this directly, we augment the baseline specification with additional controls that map onto the imbalanced variables identified in the balance tests. Specifically, we include the exposure-weighted state employment rate and the exposure-weighted female share—both constructed from the 2000 IPUMS census using the same entry-year state shares that define the inequality shock, and each interacted with year fixed effects to allow their influence to vary over time—as well as the firm-level log household count. These three controls ensure that the employment rate imbalance and the two marginally significant variables are directly absorbed.

Table A.4 reports the results. The estimated effect of inequality exposure remains negative and statistically significant across the full sample and every robustness specification. The heterogeneity pattern is likewise unchanged: the effect is concentrated among firms serving poor-market consumers and is statistically indistinguishable from zero among rich-market firms. The stability of the estimates after directly absorbing these channels provides consistent support for the shift-orthogonality assumption underlying the identification strategy.

Properties of Inequality Shocks and Exposure Shares. The validity of a shift-share design rests not only on the exogeneity of the underlying shocks but also on their distributional properties and the concentration of exposure shares (Borusyak et al., 2022). Figure A.20 plots the distribution of the predicted state-level Gini index, $\widehat{Gini}_{st}$, which serves as the inequality shock in the shift-share construction. Panel (a) includes all state-year observations over 2004–2016, while Panel (b) excludes the base year 2004 from which the initial inequality level used to construct the leave-

Table A.3: Falsification Tests: Balance of Pre-Period Characteristics

Balance variable	Coef.	SE
Panel A. Unit-level balance (firm entry-year characteristics)		
Patent dummy	0.0011	(0.0009)
Patent per $k expenditure	0.0005	(0.0024)
Claim-weighted patent per $k expenditure	0.0337	(0.0551)
Log firm expenditure	0.0001	(0.0448)
Module diversification	-0.0051	(0.0054)
Log module count	0.0017	(0.0098)
State diversification	-0.0009	(0.0099)
Log state count	0.0176	(0.0302)
Log household count	0.0184*	(0.0094)
Household diversification	-0.0007	(0.0031)
No. of state-year observations	588	
Panel B. Shock-level balance (state characteristics, IPUMS 2000)		
Employment rate	-0.0248***	(0.0077)
Manufacturing employment share	0.0049	(0.0085)
Agriculture employment share	0.0010	(0.0023)
Service employment share	-0.0057	(0.0056)
Central city population share	0.0235	(0.0184)
Mean household size	-0.0561	(0.0411)
Living alone share	0.0043	(0.0041)
Less than high school share	-0.0060	(0.0073)
White share	0.0175	(0.0278)
Black share	0.0236	(0.0176)
Native American share	0.0020	(0.0021)
Asian share	-0.0046	(0.0032)
Female share	0.0035*	(0.0018)
Log mean age	0.0097	(0.0072)
No. of state-year observations	588	

Notes. This table reports falsification tests examining whether the inequality shock is orthogonal to pre-period characteristics, following the balance-test framework of Borusyak, Hull, and Jaravel (2022). Panel A reports unit-level balance. The firm-level variables are measured in each firm's entry year and drawn from the NielsenIQ Homescan Consumer Panel (NHCP) matched to PatentsView/USPTO. *Patent dummy* equals one if the firm filed at least one ultimately granted patent in its entry year. *Patent per $k expenditure* is the count of granted patents scaled by the firm's total consumer expenditure in thousands of dollars. *Claim-weighted patent per $k expenditure* weights each patent by its number of claims to capture technological scope, then scales by expenditure. *Log firm expenditure* is the natural log of the firm's total consumer expenditure. *Module diversification* is a dummy equal to one if the firm sells in more than one product module, and *log module count* is the log of the number of product modules in which the firm has positive sales. *State diversification* is a dummy equal to one if the firm sells in more than one U.S. state, and *log state count* is the log of the number of states in which the firm has positive sales. *Log household count* is the log of the number of distinct households purchasing the firm's products, and *household diversification* is a dummy equal to one if the firm sells to more than one household. Each firm characteristic is aggregated to the state-year level via the *ssaggregate* procedure, then regressed on the standardized state-year Gini index, weighted by average exposure shares s_n, with year fixed effects absorbed and standard errors clustered at the state level. Panel B reports shock-level balance. The state-level variables are drawn from the 2000 IPUMS 5% census and are computed for the population aged 16 and older unless otherwise noted. *Employment rate* is the share of the population aged 16+ that is employed. *Manufacturing employment share*, *agriculture employment share*, and *service employment share* are the shares of employed workers aged 16+ in manufacturing, agriculture, and services, respectively. *Central city population share* is the share of the population residing in a central city within a metropolitan area. *Mean household size* is the average number of persons per household, and *living alone share* is the share of households with exactly one member. *Less than high school share* is the share of the population aged 16+ with less than a high school diploma. *White share*, *black share*, *Native American share*, and *Asian share* are the shares of the population aged 16+ in each racial category. *Female share* is the share of women aged 16+. *Log mean age* is the log of average age among the population aged 16+. Each state characteristic is regressed on the standardized state-year Gini index over 2005–2016, weighted by s_n, with year fixed effects absorbed and standard errors clustered at the state level. In both panels, the Gini index is standardized to have mean zero and unit variance, so coefficients represent the change in the balance variable per one-standard-deviation increase in predicted inequality. Of the 24 balance variables tested, 21 show no statistically significant relationship with the inequality shock at conventional levels. The employment rate in Panel B is significant at the 1% level: states with higher predicted inequality have lower employment rates, a pattern we address by including predicted income and per-capita GDP as exposure-level controls in the main specification. The household count (Panel A) and female share (Panel B) are marginally significant at the 10 percent level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

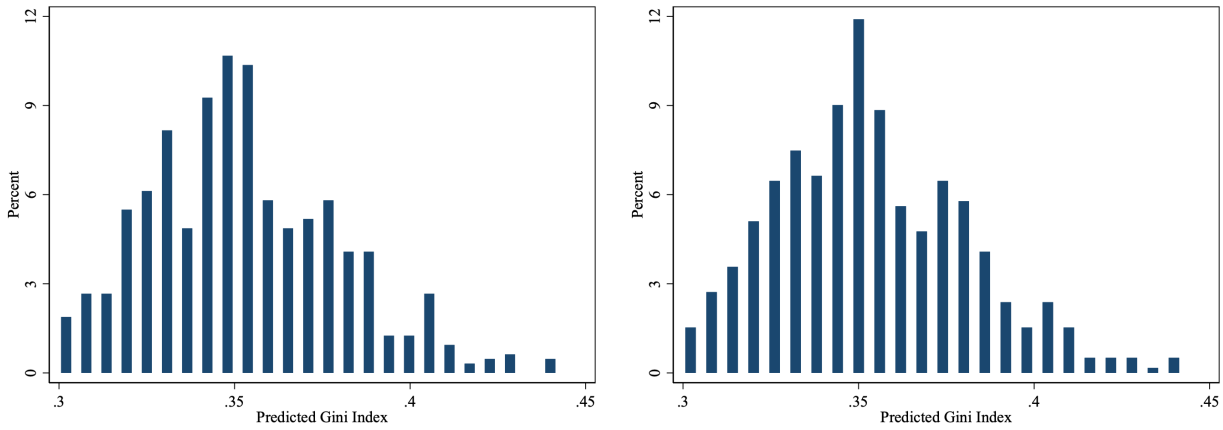
Table A.4: Inequality Exposure and Firm Innovation: Additional Controls

	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel A. Dependent variable: Patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0094**	-0.0134***	-0.0094**	-0.0082	-0.0089**	-0.0135**	-0.0028
	(0.0042)	(0.0049)	(0.0042)	(0.0059)	(0.0042)	(0.0065)	(0.0049)
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	243,889	184,352	243,485	110,313	236,300	118,388	125,498
Adjusted R^2	0.547	0.548	0.547	0.531	0.538	0.521	0.572
	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel B. Dependent variable: Claim-weighted patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0844**	-0.1096***	-0.0850**	-0.0939*	-0.0795**	-0.1168**	-0.0321
	(0.0346)	(0.0399)	(0.0347)	(0.0483)	(0.0349)	(0.0532)	(0.0417)
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	243,889	184,352	243,485	110,313	236,300	118,388	125,498
Adjusted R^2	0.548	0.555	0.548	0.524	0.541	0.530	0.566

Notes. This table reports firm-year regressions of innovation intensity on exposure to income-inequality shocks, augmenting the baseline specification of Table 1 with additional controls for the imbalanced pre-period characteristics identified in the falsification tests of Table A.3. In Panel A, the dependent variable is the number of ultimately granted patent applications filed by firm f in year t (PatentsView/USPTO), scaled by the firm's total consumer expenditure in thousands of dollars. In Panel B, the dependent variable is the claim-weighted analogue, where each patent is weighted by its number of claims to capture technological scope. The main regressor, $Shock_{ft}^{ineq}$, is the firm-specific inequality exposure: a weighted average of predicted state-year Gini coefficients, with weights given by the firm's complete initial sales shares across U.S. states in its entry year t_0. Predicted state-level inequality is constructed via the leave-one-out national-trend instrument variable procedure. The set of baseline controls (denoted "Baseline controls" above) comprises two classes. The first class consists of shift-share exposure-level controls aggregated with the same entry-year state shares: predicted log average household income, predicted log per-capita GDP, and predicted log population, each constructed via the same leave-one-out procedure as predicted inequality. These absorb firm exposure to first-moment and scale characteristics of state-level economic conditions that may co-move with inequality. The second class consists of firm-level covariates: the log of the number of product modules supplied (product scope), the firm's expenditure share (market-power proxy), and the log of firm age (years since first entry). The additional controls (denoted "Additional controls" above) comprise three variables that map onto the imbalanced covariates in Table A.3: the firm-level log household count, the exposure-weighted state employment rate, and the exposure-weighted state female share. The latter two are constructed from the 2000 IPUMS five-percent census using the same entry-year state shares that define the inequality shock, and each is interacted with year fixed effects to allow their influence to vary over time. Columns (1)–(5) report baseline and robustness samples. Column (1) is the full sample. Column (2) restricts to firms whose entry year is 2004. Column (3) restricts to firms with positive sales to both low- and high-income consumers during the sample period (low- and high-income households are defined in the NielsenIQ data description). Column (4) restricts to firms with below-median total consumer expenditure over the sample period (small firms). Column (5) restricts to firms observed in at least four sample years (surviving firms), thereby excluding short-lived entrants. Columns (6)–(7) split firms by sales orientation: *Poor Market* (*Rich Market*) indicates firms whose average buyer income percentile at entry is below (above) the sample median, where the buyer income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. Because this orientation is fixed at the firm's entry year, the split is predetermined with respect to subsequent innovation and inequality shocks, ruling out reverse causality from a firm's innovative activity to its customer composition. All specifications include firm fixed effects and year fixed effects. The estimating sample requires $t > t_0$, ensuring that the initial sales shares used to construct exposures are measured strictly before the outcome period. All continuous variables are winsorized at the 1st and 99th percentiles within each year. Standard errors (in parentheses) are clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

one-out instrument is drawn. In both panels, the predicted Gini index ranges from roughly 0.30 to 0.45. The dispersion in $\widehat{Gini}_{st}$ across states and years confirms that the inequality shock is sufficiently variable to support the shift-share design: with little cross-state variation in the shock, the large-sample approximation underlying the shift-share estimator would be unreliable in finite samples.

Figure A.20: Distribution of the Predicted Gini Shock



(a) Full Sample (2004–2016)

(b) Excluding Base Year 2004

Notes. This figure plots the distribution of the predicted state-level Gini index, $\widehat{Gini}_{st}$, which serves as the inequality shock in the shift-share design. Panel (a) includes all state-year observations over 2004–2016. Panel (b) excludes the base year 2004, from which the initial inequality level used to construct the leave-one-out instrument is drawn. Both histograms are expressed in percent of observations.

Table A.5 further characterizes the exposure shares that map state-level shocks into firm-level treatment. At the state-year level (Column 1), the effective sample size—the inverse of the Herfindahl index of the exposure weights, $1/\sum_k s_k^2$—is 253.5, indicating that the identifying variation is dispersed across a large number of effective state-year units. The five largest state-year weights range from 0.0113 to 0.0109, and collectively account for only 5.5 percent of total exposure. At the state level (Column 2), the effective sample size is 21.2 out of 49 states, with California (12.9 percent) carrying the largest weights. While state-level exposure is more concentrated, the effective number of states remains well above the level at which a single state could dominate the identifying variation. Together, the broad dispersion of both the inequality shocks and the exposure shares indicates that the shift-share variation satisfies the concentration requirement for reliable large-sample inference, and that the baseline estimates are unlikely to be driven by a handful of outlying states or years.

Sensitivity Analyses. The descriptive evidence above suggests that the shift-share variation is sufficiently dispersed, but a more direct test is to re-estimate the baseline after removing the observations with the greatest potential to dominate the results. We first exclude the six states with the largest total consumer expenditure—California, Texas, Florida, New York, Pennsylvania, and

Table A.5: Summary Statistics for Inequality Exposure Shares

	(1) State-year Exposure Weights	(2) State Exposure Weights
Mean	0.0017	0.0204
Standard Deviation	0.0020	0.0236
Interquartile Range	0.0016	0.0189
Effective Sample Size ($1/\sum s^2$)	253.5	21.2
1st Largest Weight	0.0113	0.1286
2nd largest Weight	0.0112	0.0824
3rd Largest Weight	0.0110	0.0707
4th Largest Weight	0.0109	0.0561
5th Largest Weight	0.0109	0.0451
Observation Count	588	49

Notes. This table summarizes the distribution of the exposure shares used to construct the firm-level inequality shock $Shock_{ft}^{ineq} = \sum_s \lambda_{sft_0} \widehat{Gini}_{st}$. Column (1) reports statistics for the state-year level exposure weights s_{st}, obtained by aggregating the firm-level entry-year shares λ_{sft_0} to the state-year level via the *ssaggregate* procedure. Column (2) reports statistics for the state-level exposure weights, obtained by summing s_{st} over all years within each state. The effective sample size is the inverse of the Herfindahl index of the exposure weights, $1/\sum_k s_k^2$, which measures the concentration of the shift-share variation: a smaller value indicates that a few states dominate the identifying variation. The sample includes 49 states (Alaska, Hawaii, and U.S. territories are excluded) over 2004–2016.

Ohio—which together account for about 40 percent of personal consumption expenditures across the 49-state sample. Table A.6 reports the results. The full-sample coefficient remains negative in both panels, though it declines in magnitude from -0.0085 to -0.0049 in Panel A ($p = 0.052$) and from -0.0752 to -0.0433 in Panel B ($p = 0.049$). The attenuation is expected: dropping states that carry the largest exposure shares mechanically reduces the cross-state dispersion in $\widehat{Gini}_{st}$ that identifies the effect. The heterogeneity pattern is unchanged, with the negative effect concentrated among poor-market firms and absent among rich-market firms. That the result survives the removal of states that collectively represent two-fifths of total expenditure confirms that it is not driven by the economic weight of a few large states.

We then exclude the top one percent of firms ranked by their average inequality exposure, corresponding to influential firms whose initial sales shares are most heavily concentrated in high-inequality states. Table A.7 shows that the full-sample coefficient remains negative at -0.0069 in Panel A ($p = 0.062$) and -0.0768 in Panel B ($p = 0.014$). The poor-market coefficient is -0.0102 in Panel A and -0.1082 in Panel B, both statistically significant, while the rich-market coefficient is indistinguishable from zero. The stability of the estimates after removing the firms with the greatest leverage confirms that the baseline effect reflects the average response of the broad cross-section of firms, rather than the behavior of a small number of highly exposed outliers. Taken together, these leave-out tests corroborate the concentration analysis above: the identifying variation is broad enough that no subset of states or firms is responsible for the main finding.

Alternative Clustering for Exposure-robust Standard Errors. The baseline inference clusters standard errors at the firm level, which assumes that the regression residuals are uncorrelated across firms conditional on the fixed effects and controls. In a shift-share design, however, this assumption may fail: firms with similar initial exposure shares mechanically receive correlated in-

Table A.6: Inequality Exposure and Firm Innovation: Excluding Top Six Expenditure States

	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel A. Dependent variable: Patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0049*	-0.0067**	-0.0050*	-0.0067*	-0.0045*	-0.0080**	0.0001
	(0.0025)	(0.0029)	(0.0025)	(0.0035)	(0.0025)	(0.0036)	(0.0035)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	208,961	154,770	208,585	93,586	201,946	101,002	107,956
Adjusted R^2	0.543	0.550	0.544	0.508	0.535	0.518	0.568
	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel B. Dependent variable: Claim-weighted patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0433**	-0.0520**	-0.0437**	-0.0647**	-0.0397*	-0.0740**	0.0068
	(0.0220)	(0.0249)	(0.0221)	(0.0305)	(0.0221)	(0.0313)	(0.0313)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	208,961	154,770	208,585	93,586	201,946	101,002	107,956
Adjusted R^2	0.549	0.562	0.550	0.509	0.542	0.530	0.569

Notes. This table reports firm-year regressions of innovation intensity on exposure to income-inequality shocks, excluding the six states with the largest total consumer expenditure: California, Texas, Florida, New York, Pennsylvania, and Ohio. These six states account for about 40 percent of total personal consumption expenditures across the 49-state sample. In Panel A, the dependent variable is the number of ultimately granted patent applications filed by firm f in year t (PatentsView/USPTO), scaled by the firm's total consumer expenditure in thousands of dollars. In Panel B, the dependent variable is the claim-weighted analogue, where each patent is weighted by its number of claims to capture technological scope. The main regressor, $Shock_{ft}^{ineq}$, is the firm-specific inequality exposure: a weighted average of predicted state-year Gini coefficients, with weights given by the firm's complete initial sales shares across U.S. states in its entry year t_0. Predicted state-level inequality is constructed via the leave-one-out national-trend instrumental-variable procedure. The set of controls (denoted "Controls" above) comprises two classes. The first class consists of shift-share exposure-level controls aggregated with the same entry-year state shares: predicted log average household income, predicted log per-capita GDP, and predicted log population, each constructed via the same leave-one-out procedure as predicted inequality. These absorb firm exposure to first-moment and scale characteristics of state-level economic conditions that may co-move with inequality. The second class consists of firm-level covariates: the log of the number of product modules supplied (product scope), the firm's expenditure share (market-power proxy), and the log of firm age (years since first entry). Columns (1)–(5) report baseline and robustness samples. Column (1) is the full sample after excluding the six states. Column (2) restricts to firms whose entry year is 2004. Column (3) restricts to firms with positive sales to both low- and high-income consumers during the sample period (low- and high-income households are defined in the NielsenIQ data description). Column (4) restricts to firms with below-median total consumer expenditure over the sample period (small firms). Column (5) restricts to firms observed in at least four sample years (surviving firms), thereby excluding short-lived entrants. Columns (6)–(7) split firms by sales orientation: *Poor Market* (*Rich Market*) indicates firms whose average buyer income percentile at entry is below (above) the sample median, where the buyer income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. Because this orientation is fixed at the firm's entry year, the split is predetermined with respect to subsequent innovation and inequality shocks, ruling out reverse causality from a firm's innovative activity to its customer composition. All specifications include firm fixed effects and year fixed effects. The estimating sample requires $t > t_0$, ensuring that the initial sales shares used to construct exposures are measured strictly before the outcome period. All continuous variables are winsorized at the 1st and 99th percentiles within each year. Standard errors (in parentheses) are clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.7: Inequality Exposure and Firm Innovation: Excluding Top One Percent High-Exposure Firms

	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel A. Dependent variable: Patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0069*	-0.0090**	-0.0069*	-0.0080*	-0.0066*	-0.0102*	-0.0019
	(0.0037)	(0.0040)	(0.0037)	(0.0048)	(0.0037)	(0.0056)	(0.0048)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	242,316	183,207	241,912	109,675	234,833	117,480	124,833
Adjusted R^2	0.548	0.549	0.548	0.531	0.539	0.521	0.572
	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel B. Dependent variable: Claim-weighted patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0768**	-0.0866**	-0.0769**	-0.1084***	-0.0737**	-0.1082**	-0.0328
	(0.0314)	(0.0340)	(0.0315)	(0.0399)	(0.0316)	(0.0459)	(0.0431)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	242,316	183,207	241,912	109,675	234,833	117,480	124,833
Adjusted R^2	0.549	0.556	0.549	0.525	0.542	0.530	0.566

Notes. This table reports firm-year regressions of innovation intensity on exposure to income-inequality shocks, excluding the top one percent of firms ranked by their average inequality exposure. These are the firms whose initial sales shares are most heavily concentrated in high-inequality states, and whose exposure to the inequality shock is consequently the largest. Excluding them addresses the concern that the baseline estimates may be driven by a small number of highly exposed firms rather than by the broad cross-section of firms. In Panel A, the dependent variable is the number of ultimately granted patent applications filed by firm f in year t (PatentsView/USPTO), scaled by the firm's total consumer expenditure in thousands of dollars. In Panel B, the dependent variable is the claim-weighted analogue, where each patent is weighted by its number of claims to capture technological scope. The main regressor, $Shock_{ft}^{ineq}$, is the firm-specific inequality exposure: a weighted average of predicted state-year Gini coefficients, with weights given by the firm's complete initial sales shares across U.S. states in its entry year t_0. Predicted state-level inequality is constructed via the leave-one-out national-trend instrumental-variable procedure. The set of controls (denoted "Controls" above) comprises two classes. The first class consists of shift-share exposure-level controls aggregated with the same entry-year state shares: predicted log average household income, predicted log per-capita GDP, and predicted log population, each constructed via the same leave-one-out procedure as predicted inequality. These absorb firm exposure to first-moment and scale characteristics of state-level economic conditions that may co-move with inequality. The second class consists of firm-level covariates: the log of the number of product modules supplied (product scope), the firm's expenditure share (market-power proxy), and the log of firm age (years since first entry). Columns (1)–(5) report baseline and robustness samples. Column (1) is the full sample after excluding the top one percent high-exposure firms. Column (2) restricts to firms whose entry year is 2004. Column (3) restricts to firms with positive sales to both low- and high-income consumers during the sample period (low- and high-income households are defined in the NielsenIQ data description). Column (4) restricts to firms with below-median total consumer expenditure over the sample period (small firms). Column (5) restricts to firms observed in at least four sample years (surviving firms), thereby excluding short-lived entrants. Columns (6)–(7) split firms by sales orientation: *Poor Market* (*Rich Market*) indicates firms whose average buyer income percentile at entry is below (above) the sample median, where the buyer income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. Because this orientation is fixed at the firm's entry year, the split is predetermined with respect to subsequent innovation and inequality shocks, ruling out reverse causality from a firm's innovative activity to its customer composition. All specifications include firm fixed effects and year fixed effects. The estimating sample requires $t > t_0$, ensuring that the initial sales shares used to construct exposures are measured strictly before the outcome period. All continuous variables are winsorized at the 1st and 99th percentiles within each year. Standard errors (in parentheses) are clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

equality shocks, and may also have correlated unobserved shocks because they operate in similar states. Adão et al. (2019) show via Monte Carlo simulation that this issue can be serious in practice, and Borusyak et al. (2022) propose an equivalent shift-level two-stage least squares regression that produces valid standard errors, because estimation occurs at the same level at which the shifts are assigned. We implement this approach using the *ssaggregate* procedure, which first residualizes the outcome and treatment on firm-level controls and then averages them across firms within each state-year cell with weights equal to the average exposure share s_n. The resulting 588 state-year observations are then estimated via two-stage least squares, in which the aggregated inequality exposure is instrumented by the predicted state-level Gini index, with state and year fixed effects absorbed and standard errors clustered at the state level.

Table A.8 reports the results. The full-sample coefficient is -0.0095 in Panel A ($p = 0.030$) and -0.0724 in Panel B ($p = 0.056$), close in magnitude to the reduced-form firm-level estimates of -0.0085 and -0.0752. The modest attenuation in statistical significance is expected: exposure-robust standard errors are deliberately more conservative, as they account for the mechanical correlation in residuals across firms with overlapping geographic exposure. The heterogeneity pattern is unchanged. The negative effect is concentrated among poor-market firms and absent among rich-market firms. The first-stage F statistics range from 51.268 to 346.462, all well above the conventional threshold of 10 or 15 for strong instruments, confirming that the predicted Gini index remains a strong instrument for the aggregated exposure after the shift-level transformation. That the qualitative conclusions survive this more demanding inference standard reinforces that the results are not an artifact of firm-level clustering.

Measurement Error and Alternative Inequality Measures. A practical concern is that state-year inequality computed from the NielsenIQ Homescan Consumer Panel (NHCP) may be subject to sampling and measurement error. We address this concern in three ways. First, we benchmark our NHCP-based Gini against the official U.S. Census Bureau (USCB) measure. Panels (a) and (b) of Figure A.21 plot the two series against each other at the state level for 2006 and 2016, using scatter points. The fitted slope is 0.29 ($t = 3.53$) in 2006 and 0.19 ($t = 1.95$) in 2016, confirming a positive and significant cross-state correlation between the NHCP and USCB Gini measures in both years. This alignment indicates that the NHCP-based inequality series captures meaningful cross-state variation despite differences in sampling frame, and that measurement error in the state-level Gini is unlikely to be severe enough to drive the results.

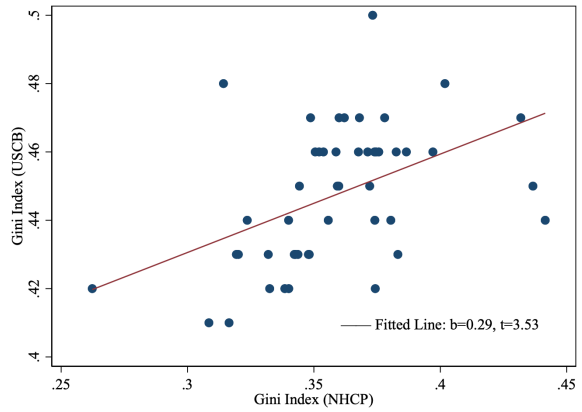
Second, we verify that the spatial and time-series patterns of our Gini measure aligns with well-documented facts. Panel (c) of Figure A.21 plots the cross-state distribution of the Gini index in 2004 and 2016. The distribution shifts markedly to the right over this period, indicating a broad-based increase in state-level income inequality that is consistent with the well-documented rise in U.S. inequality during the 2000s and 2010s. Panel (d) of Figure A.21 displays the geographic distribution of the average state-level Gini over 2004–2016. Louisiana, Alabama, and Mississippi exhibit the highest inequality, consistent with the well-documented pattern of elevated income

Table A.8: Inequality Exposure and Firm Innovation: Exposure-Robust Standard Errors

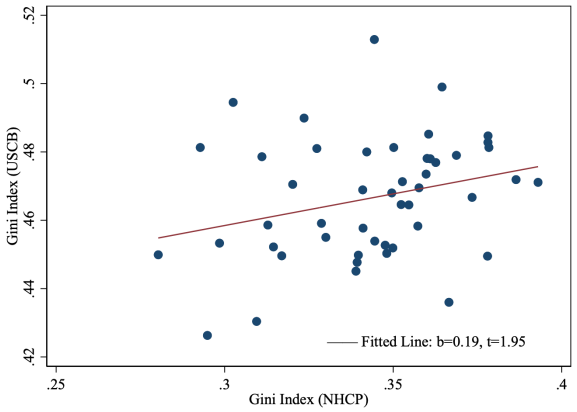
	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel A. Dependent variable: Patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0095**	-0.0105***	-0.0095**	-0.0091**	-0.0096**	-0.0117**	-0.0046
	(0.0042)	(0.0036)	(0.0043)	(0.0045)	(0.0038)	(0.0056)	(0.0060)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	89.731	346.462	90.041	131.216	100.919	121.022	51.268
Observations	588	588	588	588	588	588	588
	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel B. Dependent variable: Claim-weighted patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0724*	-0.0787***	-0.0728*	-0.0864**	-0.0783**	-0.0897*	-0.0371
	(0.0369)	(0.0295)	(0.0372)	(0.0416)	(0.0342)	(0.0500)	(0.0542)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F-stat	89.731	346.462	90.041	131.216	100.919	121.022	51.268
Observations	588	588	588	588	588	588	588

Notes. This table reports exposure-robust estimates following Borusyak et al. (2022). Firm-level variables are first aggregated to the state-year level using the *ssaggregate* procedure, with weights given by the average exposure share s_n across firms. Firm-level controls are partialled out prior to aggregation. The aggregated data are then estimated via two-stage least squares, in which the firm-level inequality exposure $Shock_{ft}^{ineq}$ is instrumented by the state-level predicted Gini index. All specifications absorb state fixed effects and year fixed effects, and standard errors are clustered at the state level. In Panel A, the dependent variable is the number of ultimately granted patent applications scaled by the firm's total consumer expenditure in thousands of dollars. In Panel B, the dependent variable is the claim-weighted analogue. We also report the Kleibergen-Paap rk Wald F statistic from the first stage, which measures instrument strength. Columns (1)–(5) report baseline and robustness samples. Column (1) is the full sample. Column (2) restricts to firms whose entry year is 2004. Column (3) restricts to firms with positive sales to both low- and high-income consumers during the sample period (low- and high-income households are defined in the NielsenIQ data description). Column (4) restricts to firms with below-median total consumer expenditure over the sample period (small firms). Column (5) restricts to firms observed in at least four sample years (surviving firms), thereby excluding short-lived entrants. Columns (6)–(7) split firms by sales orientation: *Poor Market* (*Rich Market*) indicates firms whose average buyer income percentile at entry is below (above) the sample median, where the buyer income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. Because this orientation is fixed at the firm's entry year, the split is predetermined with respect to subsequent innovation and inequality shocks, ruling out reverse causality from a firm's innovative activity to its customer composition. All columns require $t > t_0$, that is, the outcome year strictly follows the firm's entry year, ensuring that the initial sales shares used to construct exposures are measured strictly before the outcome period. The estimating sample comprises 588 state-year observations (49 states times 12 years, 2005–2016). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

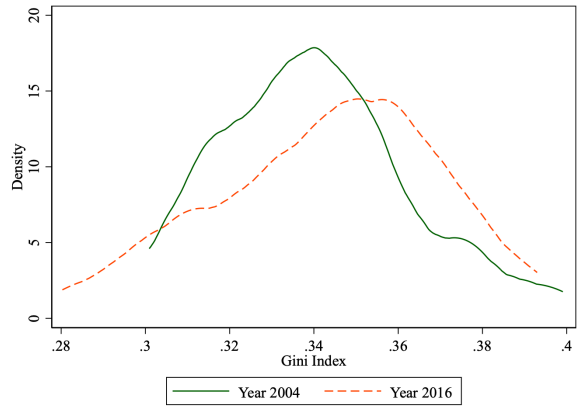
Figure A.21: Validation of Gini Index



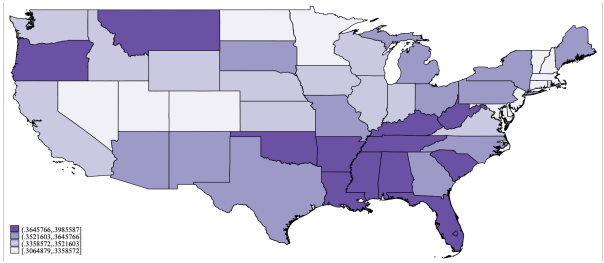
(a) Correlation between Estimated Gini from NHCP and Gini from USCB, 2006



(b) Correlation between Estimated Gini from NHCP and Gini from USCB, 2016



(c) Distribution and Evolution of Estimated Gini Index from NHCP



(d) Geographic Distribution of Estimated Mean Gini Index from NHCP, 2004–2016

Notes. Panels (a) and (b) validate the state-level Gini index computed from the NielsenIQ Homescan Consumer Panel (NHCP) against the official Gini index published by the U.S. Census Bureau (USCB). Each circle is a scatter point of U.S. states, with the NHCP-estimated Gini on the horizontal axis and the USCB Gini on the vertical axis. The solid line is the fitted regression line. The estimated slope is 0.29 ($t = 3.53$) in 2006 and 0.19 ($t = 1.95$) in 2016, confirming a positive and significant correlation between the two measures in both years. Panel (c) plots the kernel density distribution of the NHCP state-level Gini index for 2004 (solid line) and 2016 (dashed line), illustrating the rightward shift in the distribution over the sample period. Panel (d) displays a geographic map of the average state-level Gini index over 2004–2016, with darker shades denoting higher inequality.

inequality in the Deep South, while Mountain and Plains states are more equal. Both patterns provide external validity checks on our inequality series.

Third, we assess whether the results depend on the functional form of the inequality measure. We construct two alternative indices from the same NHCP household-income data. The first is the Theil index:

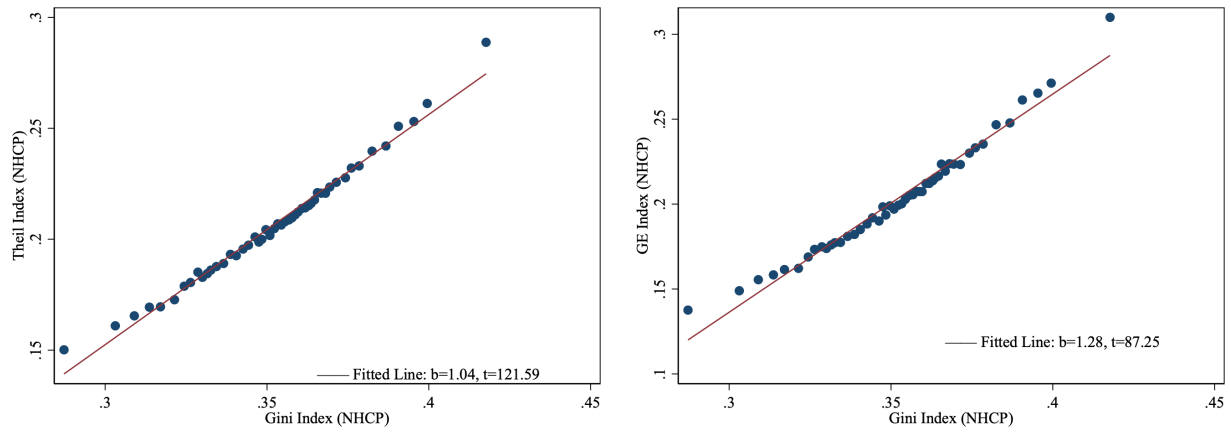
$$Theil_{st} = \sum_{i \in \Omega_s} \omega_i \cdot \frac{Income_i}{\overline{Income}_{st}} \cdot \ln\left(\frac{Income_i}{\overline{Income}_{st}}\right), \tag{A.4}$$

where ω_i is the population share of household i and $\overline{Income}_{st}$ is mean household income in state s and year t. Like the Gini, the Theil index is sensitive to changes throughout the income distribution. The second is the Generalized Entropy (GE) measure with parameter two, which places greater weight on the upper tail:

$$GE_{st} = \frac{1}{2} \sum_{i \in \Omega_s} \omega_i \cdot \left(\frac{Income_i}{\overline{Income}_{st}}\right)^2 - \frac{1}{2}. \tag{A.5}$$

Figure A.22 confirms that the three series are virtually collinear at the state-year level: after residualizing against year fixed effects and binning into 50 quantiles, the fitted slope of the Theil on the Gini is 1.04 ($t = 121.59$), and that of the GE on the Gini is 1.28 ($t = 87.25$). Because the three indices differ in their sensitivity to different parts of the income distribution, their near-perfect correlation indicates that our estimates are unlikely to be an artifact of a single inequality metric or of measurement error specific to the Gini functional form.

Figure A.22: Correlation between Inequality Indexes



(a) Theil and Gini Index

(b) GE and Gini Index

Notes. This figure validates that the Gini index captures similar dispersion patterns as alternative inequality measures. Both panels plot binned scatter points (50 quantiles) of state-year observations, after residualizing both variables against year fixed effects. All indexes are computed from the NielsenIQ Homescan Consumer Panel (NHCP). Panel (a) relates the Theil index to the Gini index; the fitted slope is 1.04 ($t = 121.59$). Panel (b) relates the generalized entropy (GE) index to the Gini index; the fitted slope is 1.28 ($t = 87.25$). The near-unit slopes and extremely high t-statistics confirm that the three inequality measures are virtually similar.

Table 1 establishes that greater inequality exposure reduces firm innovation, with the effect

concentrated among firms serving lower-income consumers. This pattern is robust to the choice of inequality measure. Appendix Tables A.9 and A.10 replicate the full set of specifications using the Theil index and the generalized entropy (GE) index with parameter two, respectively. Across both alternative metrics, the full-sample coefficient remains negative and significant, the effect is concentrated in poor-market firms, and rich-market firms show no measurable response. Because the Gini, Theil, and GE indices differ in their sensitivity to different parts of the income distribution, their consistent results indicate that the estimates are not an artifact of a single inequality metric.

Table A.9: Inequality Exposure and Firm Innovation: Theil Index

	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel A. Dependent variable: Patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0042**	-0.0051**	-0.0042**	-0.0060***	-0.0040**	-0.0064**	-0.0004
	(0.0017)	(0.0020)	(0.0017)	(0.0022)	(0.0017)	(0.0025)	(0.0023)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	243,889	184,352	243,485	110,313	236,300	118,388	125,498
Adjusted R^2	0.546	0.548	0.547	0.531	0.538	0.520	0.571
	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel B. Dependent variable: Claim-weighted patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0392***	-0.0450***	-0.0392***	-0.0640***	-0.0369**	-0.0565***	-0.0090
	(0.0146)	(0.0166)	(0.0146)	(0.0191)	(0.0147)	(0.0208)	(0.0204)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	243,889	184,352	243,485	110,313	236,300	118,388	125,498
Adjusted R^2	0.548	0.554	0.548	0.524	0.541	0.529	0.565

Notes. This table reports firm-year regressions of innovation intensity on exposure to income-inequality shocks, where inequality is measured by the Theil index rather than the Gini coefficient. In Panel A, the dependent variable is the number of ultimately granted patent applications filed by firm f in year t (PatentsView/USPTO), scaled by the firm's total consumer expenditure in thousands of dollars. In Panel B, the dependent variable is the claim-weighted analogue, where each patent is weighted by its number of claims to capture technological scope. The main regressor, $Shock_{ft}^{ineq}$, is the firm-specific inequality exposure: a weighted average of predicted state-year Theil indices, with weights given by the firm's complete initial sales shares across U.S. states in its entry year t_0. Predicted state-level inequality is constructed via the leave-one-out national-trend instrumental-variable procedure. The set of controls (denoted "Controls" above) comprises two classes. The first class consists of shift-share exposure-level controls aggregated with the same entry-year state shares: predicted log average household income, predicted log per-capita GDP, and predicted log population, each constructed via the same leave-one-out procedure as predicted inequality. These absorb firm exposure to first-moment and scale characteristics of state-level economic conditions that may co-move with inequality. The second class consists of firm-level covariates: the log of the number of product modules supplied (product scope), the firm's expenditure share (market-power proxy), and the log of firm age (years since first entry). Columns (1)–(5) report baseline and robustness samples. Column (1) is the full sample. Column (2) restricts to firms whose entry year is 2004. Column (3) restricts to firms with positive sales to both low- and high-income consumers during the sample period (low- and high-income households are defined in the NielsenIQ data description). Column (4) restricts to firms with below-median total consumer expenditure over the sample period (small firms). Column (5) restricts to firms observed in at least four sample years (surviving firms), thereby excluding short-lived entrants. Columns (6)–(7) split firms by sales orientation: *Poor Market* (*Rich Market*) indicates firms whose average buyer income percentile at entry is below (above) the sample median, where the buyer income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. Because this orientation is fixed at the firm's entry year, the split is predetermined with respect to subsequent innovation and inequality shocks, ruling out reverse causality from a firm's innovative activity to its customer composition. All specifications include firm fixed effects and year fixed effects. The estimating sample requires $t > t_0$, ensuring that the initial sales shares used to construct exposures are measured strictly before the outcome period. All continuous variables are winsorized at the 1st and 99th percentiles within each year. Standard errors (in parentheses) are clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A.10: Inequality Exposure and Firm Innovation: Generalized Entropy (GE) Index

	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel A. Dependent variable: Patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0032***	-0.0041***	-0.0032***	-0.0037**	-0.0031***	-0.0041**	-0.0014
	(0.0011)	(0.0013)	(0.0011)	(0.0015)	(0.0011)	(0.0016)	(0.0014)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	243,889	184,352	243,485	110,313	236,300	118,388	125,498
Adjusted R^2	0.546	0.548	0.547	0.531	0.538	0.520	0.571
	Baseline and Robustness					Heterogeneity	
Dep var. N_{ft}^{Pat}	(1) Full Sample	(2) 2004 Cohort	(3) Poor and Rich	(4) Small Firms	(5) Surviving Firms	(6) Poor Market	(7) Rich Market
Panel B. Dependent variable: Claim-weighted patent per thousand dollars of expenditure							
$Shock_{ft}^{ineq}$	-0.0281***	-0.0328***	-0.0282***	-0.0364***	-0.0265***	-0.0345***	-0.0138
	(0.0092)	(0.0105)	(0.0092)	(0.0125)	(0.0092)	(0.0133)	(0.0123)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	243,889	184,352	243,485	110,313	236,300	118,388	125,498
Adjusted R^2	0.548	0.554	0.548	0.524	0.541	0.529	0.565

Notes. This table reports firm-year regressions of innovation intensity on exposure to income-inequality shocks, where inequality is measured by the generalized entropy (GE) index with parameter two, which places greater weight on the upper tail of the income distribution. In Panel A, the dependent variable is the number of ultimately granted patent applications filed by firm f in year t (PatentsView/USPTO), scaled by the firm's total consumer expenditure in thousands of dollars. In Panel B, the dependent variable is the claim-weighted analogue, where each patent is weighted by its number of claims to capture technological scope. The main regressor, $Shock_{ft}^{ineq}$, is the firm-specific inequality exposure: a weighted average of predicted state-year GE indices, with weights given by the firm's complete initial sales shares across U.S. states in its entry year t_0. Predicted state-level inequality is constructed via the leave-one-out national-trend instrumental-variable procedure. The set of controls (denoted "Controls" above) comprises two classes. The first class consists of shift-share exposure-level controls aggregated with the same entry-year state shares: predicted log average household income, predicted log per-capita GDP, and predicted log population, each constructed via the same leave-one-out procedure as predicted inequality. These absorb firm exposure to first-moment and scale characteristics of state-level economic conditions that may co-move with inequality. The second class consists of firm-level covariates: the log of the number of product modules supplied (product scope), the firm's expenditure share (market-power proxy), and the log of firm age (years since first entry). Columns (1)–(5) report baseline and robustness samples. Column (1) is the full sample. Column (2) restricts to firms whose entry year is 2004. Column (3) restricts to firms with positive sales to both low- and high-income consumers during the sample period (low- and high-income households are defined in the NielsenIQ data description). Column (4) restricts to firms with below-median total consumer expenditure over the sample period (small firms). Column (5) restricts to firms observed in at least four sample years (surviving firms), thereby excluding short-lived entrants. Columns (6)–(7) split firms by sales orientation: *Poor Market* (*Rich Market*) indicates firms whose average buyer income percentile at entry is below (above) the sample median, where the buyer income percentile is computed as the expenditure-weighted average of household income percentiles among the firm's consumers. Because this orientation is fixed at the firm's entry year, the split is predetermined with respect to subsequent innovation and inequality shocks, ruling out reverse causality from a firm's innovative activity to its customer composition. All specifications include firm fixed effects and year fixed effects. The estimating sample requires $t > t_0$, ensuring that the initial sales shares used to construct exposures are measured strictly before the outcome period. All continuous variables are winsorized at the 1st and 99th percentiles within each year. Standard errors (in parentheses) are clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

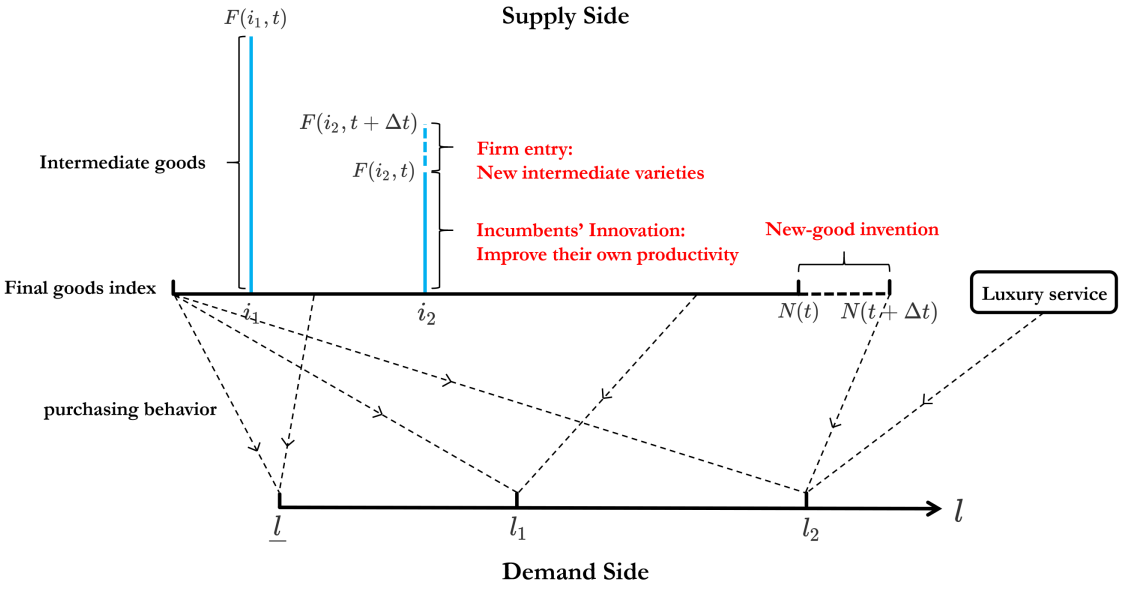
B Appendix: Baseline Model

In this appendix, we provide additional materials that clarify the overall structure of the model and present the detailed derivations. The model environment is the same as in the main text.

B.1 Summary of the Model Structure

Figure B.1 summarizes the structure of the model. On the demand side, a continuum of heterogeneous households, ordered by their endowment l, consume a hierarchy of goods whose affordability rises with income. Their non-homothetic preferences generate an endogenous demand ladder: richer households purchase goods with higher index i, and only the top-income group consumes the luxury service at the upper end of the hierarchy.

Figure B.1: Summary of the Model Structure



On the supply side, each final good is assembled by a final-goods producer that aggregates a continuum of intermediate varieties. Economic growth arises from multiple layers of innovation. At the frontier, inventors introduce new categories of final consumption goods. Within each sector, incumbent firms conduct R&D to improve productivity, while potential entrants decide whether to enter based on expected firm value. Productivity improvements and firm entry jointly expand sectoral output, lower prices, and extend market access to lower-income households, gradually transforming luxury goods into necessities.

In equilibrium, rising productivity and wages reshape the demand distribution across goods, guiding the direction of the next wave of innovation. The interaction between hierarchical demand, firm dynamics, and innovation thus endogenously generates a BGP in which goods sequentially diffuse down the income distribution, linking inequality, innovation, and long-run economic growth.

B.2 Derivations and Proofs for Section 3

B.2.1 Derivations for Section 3.2

We derive the profit function of intermediate-good producers, $\Pi_i(j,t)$, and then establish the expressions for sectoral output $Y(i,t)$ and labor demand $L(i,t)$.

First, we solve the optimization problem of the representative final-good producer:

$$\Pi(i,t) = \text{argmax}_{\{M_i(j,t)\}} \left\{ \frac{P(i,t)}{1-\beta} \int_0^{F(i,t)} M_i(j,t)^{1-\beta} dj - \int_0^{F(i,t)} Q_i(j,t) M_i(j,t) dj \right\},$$

the first-order condition implies the inverse demand for each intermediate variety is

$$P(i,t) M_i(j,t)^{-\beta} = Q_i(j,t).$$

Each intermediate-good producer operates under monopolistic competition and chooses $M_i(j,t)$ to maximize profits

$$\Pi_i(j,t) = \text{argmax}_{M_i(j,t)} \left\{ P(i,t) M_i(j,t)^{1-\beta} - W(t) \cdot \frac{M_i(j,t)}{\tilde{A}_i(j,t)^{\theta} \bar{A}(t)^{1-\theta}} \right\}.$$

The first-order condition yields

$$M_i(j,t) = \left[\frac{W(t)}{(1-\beta) P(i,t) \tilde{A}_i(j,t)^{\theta} \bar{A}(t)^{1-\theta}} \right]^{-\frac{1}{\beta}},$$

and

$$Q_i(j,t) = \frac{W(t)}{(1-\beta) \tilde{A}_i(j,t)^{\theta} \bar{A}(t)^{1-\theta}},$$

Substituting the expression of $M_i(j,t)$ into the profit function of the intermediate-good producer gives

$$\begin{aligned} \Pi_i(j,t) &= \frac{\beta}{1-\beta} \frac{W(t)}{\tilde{A}_i(j,t)^{\theta} \bar{A}(t)^{1-\theta}} \left[\frac{W(t)}{(1-\beta) P(i,t) \tilde{A}_i(j,t)^{\theta} \bar{A}(t)^{1-\theta}} \right]^{-\frac{1}{\beta}} \\ &= \beta (1-\beta)^{\frac{1-\beta}{\beta}} \left[\frac{P(i,t)^{\frac{1}{1-\beta}}}{W(t)} \bar{A}(t)^{1-\theta} \tilde{A}_i(j,t)^{\theta} \right]^{\frac{1-\beta}{\beta}}, \end{aligned} \tag{B.1}$$

which corresponds to equation (9) in the main text. Also, we obtain the profit of the final-good producer

$$\Pi(i,t) = \frac{P(i,t)}{1-\beta} \int_0^{F(i,t)} M_i(j,t)^{1-\beta} dj - \int_0^{F(i,t)} Q_i(j,t) M_i(j,t) dj$$

$$
\begin{aligned}
&= \frac{P(i,t)}{1-\beta}\int_0^{F(i,t)}\left[\frac{W(t)}{(1-\beta)P(i,t)\tilde{A}_i(j,t)^{\theta}\bar{A}(t)^{1-\theta}}\right]^{-\frac{1-\beta}{\beta}}dj \\
&\quad -\int_0^{F(i,t)}\frac{W(t)}{(1-\beta)\tilde{A}_i(j,t)^{\theta}\bar{A}(t)^{1-\theta}}\left[\frac{W(t)}{(1-\beta)P(i,t)\tilde{A}_i(j,t)^{\theta}\bar{A}(t)^{1-\theta}}\right]^{-\frac{1}{\beta}}dj \\
&= \frac{\beta}{1-\beta}\int_0^{F(i,t)}(1-\beta)^{\frac{1-\beta}{\beta}}\left[\frac{P(i,t)^{\frac{1}{1-\beta}}}{W(t)}\bar{A}(t)^{1-\theta}\tilde{A}_i(j,t)^{\theta}\right]^{\frac{1-\beta}{\beta}}dj \\
&= \frac{1}{1-\beta}\int_0^{F(i,t)}\Pi_i(j,t)dj. \qquad \text{(B.2)}
\end{aligned}
$$

The last equation follows from equation (B.1), and it states that the profit of the final-good producer equals the total profit of the intermediate-good producer times $\frac{1}{1-\beta}$.

Next, aggregate across all $j \in [0, F(i,t)]$ to derive the sectoral output

$$
\begin{aligned}
Y(i,t) &= \frac{1}{1-\beta}\int_0^{F(i,t)}M_i(j,t)^{1-\beta}dj \\
&= (1-\beta)^{\frac{1-2\beta}{\beta}}\left[\frac{W(t)}{P(i,t)}\right]^{-\frac{1-\beta}{\beta}}\bar{A}(t)^{(1-\theta)\frac{1-\beta}{\beta}}\int_0^{F(i,t)}\tilde{A}_i(j,t)^{\theta\frac{1-\beta}{\beta}}dj \\
&= (1-\beta)^{\frac{1-2\beta}{\beta}}\left[\frac{W(t)}{P(i,t)}\right]^{-\frac{1-\beta}{\beta}}B(i,t),
\end{aligned}
$$

where the last equality follows from the definition of $B(i,t)$ in equation (10).

Finally, total labor demand in sector i is

$$
\begin{aligned}
L(i,t) &= \int_0^{F(i,t)}L_i(j,t)dj \\
&= \left[\frac{W(t)}{(1-\beta)P(i,t)}\right]^{-\frac{1}{\beta}}\bar{A}(t)^{(1-\theta)\frac{1-\beta}{\beta}}\int_0^{F(i,t)}\tilde{A}_i(j,t)^{\theta\frac{1-\beta}{\beta}}dj \\
&= \left[\frac{W(t)}{(1-\beta)P(i,t)}\right]^{-\frac{1}{\beta}}B(i,t),
\end{aligned}
$$

which corresponds to equations (12) in the main text.

At last, we discuss the profit allocation in the production sector. Substitute the definition of $B(i,t)$ in equation (B.2)

$$
\begin{aligned}
\Pi(i,t) &= \frac{\beta}{1-\beta}\int_0^{F(i,t)}(1-\beta)^{\frac{1-\beta}{\beta}}\left[\frac{P(i,t)^{\frac{1}{1-\beta}}}{W(t)}\bar{A}(t)^{1-\theta}\tilde{A}_i(j,t)^{\theta}\right]^{\frac{1-\beta}{\beta}}dj \\
&= \beta(1-\beta)^{\frac{1-2\beta}{\beta}}\left[\frac{W(t)}{P(i,t)^{\frac{1}{1-\beta}}}\right]^{-\frac{1-\beta}{\beta}}B(i,t) \\
&= \beta P(i,t)Y(i,t).
\end{aligned}
$$

This relation shows that a share β of output in each sector accrues to the final-good producer. From equation (B.2), we also know that among the remaining share $1-\beta$, a fraction β accrues to intermediate-good producers, i.e.

$$\frac{\Pi(i,t)}{P(i,t)Y(i,t)} = \beta \quad \text{and} \quad \frac{\int_0^{F(i,t)} \Pi_i(j,t)dj}{P(i,t)Y(i,t)} = \beta(1-\beta). \tag{B.3}$$

B.2.2 Derivations and Proofs for Section 3.3

The main purpose of this appendix is to introduce Lemma B.1. Lemma B.1 shows that the value function is separable in firm-specific productivity and aggregate variables, implying that heterogeneity among firms within a sector can be summarized by the common sectoral component $V_i(t)$. As a result, the R&D intensity $z_i(t)$ is identical across firms in the same sector and depends solely on sector-level fundamentals. This property substantially simplifies the characterization of equilibrium innovation dynamics.

Lemma B.1 *Under Assumption 1, the value function $\tilde{V}_i(A,t)$ is linear on $A^{\theta\frac{1-\beta}{\beta}}$ and can be expressed as*

$$\tilde{V}_i(A,t) = A^{\theta\frac{1-\beta}{\beta}} V_i(t), \tag{B.4}$$

where $V_i(t)$ only depends on sector-level or aggregate-level variable. The associated R&D intensity is given by

$$z_i(j,t) = \psi \left[\frac{W(t)}{\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) V_i(t) \cdot \psi\alpha \bar{A}(t)^{1+\theta\frac{1-\beta}{\beta}}} \right]^{\frac{\alpha}{\alpha-1}} \equiv z_i(t). \tag{B.5}$$

Proof of Lemma B.1:

To solve the HJB equation (14), we adopt the standard guess-and-verify approach. Conjecture that the value function takes the form:

$$\tilde{V}_i(A,t) = A^{\theta\frac{1-\beta}{\beta}} V_i(t). \tag{B.6}$$

Substituting equation (B.6) into the incumbent firm's HJB equation (14) yields

$$\begin{aligned} rA^{\theta\frac{1-\beta}{\beta}} V_i(t) - A^{\theta\frac{1-\beta}{\beta}} \dot{V}_i(t) = \max_{L_i^I(j,t)} \beta(1-\beta)^{\frac{1-\beta}{\beta}} \left[\frac{P(i,t)^{\frac{1}{1-\beta}}}{W(t)} \bar{A}(t)^{1-\theta} A^{\theta} \right]^{\frac{1-\beta}{\beta}} \\ - W(t) L_i^I(j,t) + z_i(j,t) \left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) A^{\theta\frac{1-\beta}{\beta}} V_i(t) - \delta A^{\theta\frac{1-\beta}{\beta}} V_i(t). \end{aligned} \tag{B.7}$$

Using $z_i(j,t) = \psi \bar{A}(t)^{\xi} \tilde{A}_i(j,t)^{-\iota} \left(L_i^I(j,t)\right)^{\alpha}$, where we have denoted

$$\xi = \alpha\left(1 + \theta\frac{1-\beta}{\beta}\right) \tag{B.8}$$

to simplify expression, together with Assumption 1 that $\iota = \alpha\theta\frac{1-\beta}{\beta}$, the first-order condition with respect to $L_i^I(t)$ yields

$$L_i^I(j,t) = A^{\theta\frac{1-\beta}{\beta}} \left[\frac{W(t)}{\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) V_i(t)\psi\alpha\bar{A}(t)^{\xi}} \right]^{\frac{1}{\alpha-1}}. \tag{B.9}$$

Substituting into $z_i(j,t)$ gives the optimal innovation effort

$$z_i(j,t) = \psi \left[\frac{W(t)}{\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) V_i(t)\psi\alpha\bar{A}(t)^{\frac{\xi}{\alpha}}} \right]^{\frac{\alpha}{\alpha-1}}.$$

Plugging $L_i^I(j,t)$ and $z_i(j,t)$ into equation (B.7), and dividing both sides by $A^{\theta\frac{1-\beta}{\beta}}$, we obtain

$$\begin{aligned} rV_i(t) - \dot{V}_i(t) = {} & \beta(1-\beta)^{\frac{1-\beta}{\beta}} \left[\frac{P(i,t)^{\frac{1}{1-\beta}}}{W(t)} \bar{A}(t)^{1-\theta} \right]^{\frac{1-\beta}{\beta}} \\ & - W(t) \left[\frac{W(t)}{\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) V_i(t)\psi\alpha\bar{A}(t)^{\xi}} \right]^{\frac{1}{\alpha-1}} \\ & + \psi \left[\frac{W(t)}{\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) V_i(t)\psi\alpha\bar{A}(t)^{\xi/\alpha}} \right]^{\frac{\alpha}{\alpha-1}} \left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) V_i(t) - \delta V_i(t). \end{aligned} \tag{B.10}$$

Crucially, the resulting equation is independent of firm-level productivity A, confirming the conjecture in equation (B.6). Hence, the value function $\tilde{V}_i(A,t)$ is linear in $A^{\theta\frac{1-\beta}{\beta}}$, and $V_i(t)$ depends only on sectoral and aggregate variables such as r, $P(i,t)$, $W(t)$, and $\bar{A}(t)$. The associated R&D intensity is given by equation (B.5). This completes the proof. ■

B.2.3 Labor for Firm Entry

In this appendix, we discuss a technical issue related to the labor demand for firm entry. We begin by characterizing the dynamics of the number of firms in sector i, denoted by $F(i,t)$. In Lemma B.2, we show that $F(i,t)$ is a càdlàg process and may exist discontinuity only when $i = N(t)$. The proof of Lemma B.2 is somewhat technical and relies on several results established later for the BGP. Readers may therefore find it helpful to first review the statement and main intuition of the lemma, and then return to the detailed proof after reading the subsequent results.

Lemma B.2 *Fix a sector k, and let t^* denote the time such that $k = N(t^*)$. In the stationary equilibrium, the firm mass $F(k,t)$ satisfies the following properties: (i) for $t < t^*$, $F(k,t) = 0$; (ii) at $t = t^*$, $F(k,t) = F_1 > 0$, where F_1 is a constant; (iii) for $t > t^*$, $F(k,t)$ is continuous and*

differentiable in t.

Proof of Lemma B.2:

Since $N(t)$ is strictly increasing, sector k has not yet been invented for all $t < t^*$. Therefore,

$$F(k,t) = 0 \quad \text{for all } t < t^*.$$

At time $t = t^*$, Lemma B.5 implies that the demand for final good k is strictly positive and given by $1 - \mathcal{G}(\bar{l}) > 0$. Using equation (11), sectoral output satisfies

$$\begin{aligned} 1 - \mathcal{G}(\bar{l}) &= (1-\beta)^{\frac{1-2\beta}{\beta}} \left[\frac{W(t^*)}{P(k,t^*)}\right]^{-\frac{1-\beta}{\beta}} B(k,t^*) \\ &= (1-\beta)^{\frac{1-2\beta}{\beta}} \left[\frac{W(t^*)}{P(k,t^*)}\right]^{-\frac{1-\beta}{\beta}} \bar{A}(t^*)^{(1-\theta)\frac{1-\beta}{\beta}} \int_0^{F(k,t^*)} \tilde{A}_k(j,t^*)^{\theta\frac{1-\beta}{\beta}} dj \\ &= (1-\beta)^{\frac{1-2\beta}{\beta}} \left[\frac{W(t^*)}{\bar{A}(t^*)P(k,t^*)}\right]^{-\frac{1-\beta}{\beta}} \zeta^{\theta\frac{1-\beta}{\beta}} F(k,t^*). \end{aligned}$$

Since the left-hand side is strictly positive, it follows that

$$F(k,t^*) > 0.$$

Furthermore, by the time-translation invariance property established in Lemma B.10, firm mass depends only on the normalized index $\tilde{i} = i/N(t)$. At time t^*, we have $\tilde{i} = k/N(t^*) = 1$ by construction. Therefore,

$$F(k,t^*) = F_1,$$

where F_1 is a constant.

Finally, for $t > t^*$, entrants solve

$$\max_{L_k^E(t)} \left\{ e(k,t) \cdot \mathbb{E}_t \left[A^{\theta\frac{1-\beta}{\beta}} V_k(t) \right] - W(t) L_k^E(t) \right\},$$

where $e(k,t) = \psi_e \bar{A}(t) L_k^E(t)$.The first-order condition implies

$$\begin{aligned} W(t) &= V_k(t)\psi_e \bar{A}(t) \mathbb{E}_t \left[A^{\theta\frac{1-\beta}{\beta}} \right] \\ &= V_k(t)\psi_e \bar{A}(t) \frac{B(k,t)}{\bar{A}(t)^{(1-\theta)\frac{1-\beta}{\beta}} F(k,t)}. \end{aligned} \tag{B.11}$$

Rearranging yields

$$F(k,t) = V_k(t)\psi_e \bar{A}(t) \frac{B(k,t)}{\bar{A}(t)^{(1-\theta)\frac{1-\beta}{\beta}} W(t)}. \tag{B.12}$$

Since $W(t)$ and $\bar{A}(t)$ grow at constant rates, they are continuous and differentiable in t. Equation (B.32) implies that $V_k(t)$ is continuous and differentiable, which in turn implies from equa-

tion (B.33) that $P(k,t)$ is continuous and differentiable. Equation (11), together with Lemma B.8, implies that both $Y(k,t)$ and $B(k,t)$ are continuous and differentiable. Therefore, equation (B.12) implies that $F(k,t)$ is continuous and differentiable in t for all $t > t^*$. ■

To accommodate the possible discontinuities in firm creation characterized in Lemma B.2, we define the cumulative entry–labor process $L^{AE}(i,t)$ as the total amount of labor devoted to firm entry in sector i up to time t,

$$L^{AE}(i,t) = \int_0^t L^E(i,\tau)\, d\tau.$$

This cumulative formulation allows entry to occur both through continuous flows and through instantaneous jumps, while preserving bounded variation. Accordingly, the law of motion for firm mass can be written in the general differential form

$$dF(i,t) = \psi_e \bar{A}(t)\, dL^{AE}(i,t) - \delta F(i,t)\, dt, \tag{B.13}$$

where differentials with respect to L^{AE} are interpreted in the Lebesgue–Stieltjes sense. When entry is continuous, this expression reduces to the flow representation $\dot{F}(i,t) = e(i,t) - \delta F(i,t)$ used in the main text. Aggregating over sectors, define the total mass of intermediate-good firms at time t as

$$\bar{F}(t) \equiv \int_0^{N(t)} F(i,t)\, di.$$

We also define aggregate cumulative entry labor as

$$\bar{L}^{AE}(t) \equiv \int_0^{N(t)} L^{AE}(i,t)\, di. \tag{B.14}$$

Aggregating sector-level firm dynamics and accounting for the time variation in the frontier $N(t)$, the evolution of total firm mass satisfies

$$d\bar{F}(t) = \psi_e \bar{A}(t)\, d\bar{L}^{AE}(t) - \delta \bar{F}(t)\, dt.$$

On the other hand, by construction of the entry process, total firm mass also evolves according to

$$d\bar{F}(t) = \left(\int_0^{N(t)} e(i,t)\, di + \dot{N}(t) F_1 \right) dt - \delta \bar{F}(t)\, dt.$$

The first term inside the parentheses captures firm creation through entry within existing sectors, obtained by aggregating the Poisson arrival rates $e(i,t)$ across all interior sectors $i \le N(t)$. The second term, $\dot{N}(t)F_1$, reflects instantaneous firm creation associated with frontier expansion: when the technological frontier advances at rate $\dot{N}(t)$, a mass F_1 of firms is created in each newly opened

sector. Equating the two expressions for $d\bar{F}(t)$ yields

$$d\bar{L}^{AE}(t) = \frac{\int_0^{N(t)} e(i,t)\,di + \dot{N}(t)F_1}{\psi_e \bar{A}(t)}\,dt.$$

Since all terms on the right-hand side are locally bounded, $\bar{L}^{AE}(t)$ is absolutely continuous and therefore differentiable almost everywhere, with derivative given by

$$\dot{\bar{L}}^{AE}(t) = \frac{\int_0^{N(t)} e(i,t)\,di + \dot{N}(t)F_1}{\psi_e \bar{A}(t)}.$$

Finally, define aggregate entry labor at time t as $L^E(t) \equiv \dot{\bar{L}}^{AE}(t)$. Hence,

$$L^E(t) = \frac{\int_0^{N(t)} e(i,t)\,di + \dot{N}(t)F_1}{\psi_e \bar{A}(t)}. \tag{B.15}$$

B.3 Derivations and Proofs for Section 4.1

We first solve the consumer's problem and derive key properties of the function $\mathcal{L}$ in Appendix B.3.1, which summarizes the demand-side information. We then discuss the properties of the BGP in Appendix B.3.2. Finally, we detrend the entire dynamic system in Appendix B.3.3.

B.3.1 Properties of $\mathcal{L}$

We denote by $P_l(t)$ the price index of household l's final consumption. Consider the household's problem along a BGP with constant r:

$$\begin{aligned}
\max_{\{C_i(j,t)\}} \quad & \int_0^\infty e^{-\rho t}\frac{C_l(t)^{1-\sigma}}{1-\sigma}\,dt \\
\text{s.t.} \quad & C_l(t) = \int_0^{N(t)} i^{-\gamma} C_l(i,t)\,di + \min_{i\in(0,N_t]}\{C_l(i,t)\}\cdot X_l(t), \\
& P_l(t)C_l(t) = \int_0^{N(t)} P(i,t)C_l(i,t)\,di + P^X(t)X_l(t), \\
& \int_0^\infty e^{-rt}P_l(t)C_l(t)dt \le \int_0^\infty W(t)l\,e^{-rt}dt + \mathbb{V}(0)l.
\end{aligned}$$

We make the following conjecture, which will be verified in the proof of Proposition 1 in Appendix B.4.

Conjecture 1 *For any household with endowment l, the equilibrium price index $P_l(t)$ is constant over time t.*

Suppose that Conjecture 1 holds. Then the Euler equation for the household problem becomes

$$r = \rho + \sigma\frac{\dot{C}_l}{C_l}. \tag{B.16}$$

Let μ_l denote the Lagrange multiplier associated with household l's budget constraint. The household's demand for good i is determined by the following reservation price condition:

$$C_l(i,t) = \begin{cases} 1 & \text{if } P(i,t) \le \kappa_l(i,t), \\ 0 & \text{if } P(i,t) > \kappa_l(i,t). \end{cases} \tag{B.17}$$

Solving the household's optimization problem, we have

$$\kappa_l(i,t) = \frac{i^{-\gamma} e^{-\rho t}}{\mu_l C_l(t)^{\sigma}} \cdot e^{rt}. \tag{B.18}$$

From equation (B.18), the household's willingness to pay $\kappa_l(i,t)$ is strictly decreasing in i. Therefore, if the product price $P(i,t)$ is strictly increasing in i, the household demand rule in equation (B.17) implies a cut-off structure: each household purchases all goods with index below some threshold and does not purchase goods with index above it.

At this stage, we also cannot establish the monotonicity of $P(i,t)$ from the demand side alone. We therefore proceed by conjecturing that the equilibrium price schedule is strictly increasing in i, and conduct the analysis under this conjectured property. We will later show that the equilibrium constructed from the supply side indeed satisfies this property, thereby validating the conjecture and closing all the arguments in this appendix; see the end of Proposition 2 in Appendix B.5.

Conjecture 2 *At any time t, the equilibrium price $P(i,t)$ is strictly increasing in i.*

Lemma B.3 *Suppose that Conjecture 2 holds. At time t, for household with endowment l, there exists a positive number $\bar{i}(l,t)$, satisfying:*

$$C_l(i,t) = 1 \quad \textit{if and only if} \quad i \le \bar{i}(l,t),$$

Proof of Lemma B.3:

Fix a household with endowment l. Under Conjecture 2, the price schedule $P(i,t)$ is strictly increasing in i. From equation (B.18), the reservation price $\kappa_l(i,t)$ is strictly decreasing in i, with $\kappa_l(i,t) \to \infty$ as $i \to 0$. Hence, there are two cases.

Case 1: $\kappa_l(N(t),t) > P(N(t),t)$. Since $\kappa_l(i,t)$ is decreasing in i and $P(i,t)$ is increasing in i, for all $0 < i \le N(t)$ we have

$$\kappa_l(i,t) \ge \kappa_l(N(t),t) > P(N(t),t) \ge P(i,t).$$

Therefore, equation (B.17) implies that the household purchases all available goods. In this case, we set

$$\bar{i}(l,t) = N(t),$$

so that

$$C_l(i,t) = 1 \quad \text{if and only if} \quad i \le \bar{i}(l,t).$$

Case 2: $\kappa_l(N(t),t) \leq P(N(t),t)$. Since $\kappa_l(i,t) \to \infty$ as $i \to 0$, while $P(i,t)$ is finite for any given $i > 0$, and since $\kappa_l(i,t)$ is strictly decreasing in i whereas $P(i,t)$ is strictly increasing in i, there exists a unique cutoff $\bar{i}(l,t) \in (0, N(t)]$ such that

$$\kappa_l(\bar{i}(l,t),t) = P(\bar{i}(l,t),t).$$

Moreover, for all $i < \bar{i}(l,t)$, $\kappa_l(i,t) > P(i,t)$, and for all $i > \bar{i}(l,t)$, $\kappa_l(i,t) < P(i,t)$. It then follows from equation (B.17) that

$$C_l(i,t) = 1 \quad \text{if and only if} \quad i \leq \bar{i}(l,t).$$

In either case, the household's consumption bundle is characterized by a cut-off rule. This completes the proof. ■

In what follows, it is convenient to characterize demand for good i by defining

$$\ell(i,t) = \inf\{l \in [\underline{l}, \infty) : P(i,t) \leq \kappa_l(i,t)\} \tag{B.19}$$

to be the minimum labor endowment required for a household to purchase good i. By construction, when $l = \ell(i,t)$, the household is indifferent between purchasing and not purchasing good i, implying $P(i,t) = \kappa_l(i,t)$.

Lemma B.4 *Suppose Conjecture 2 holds. Then, for any $\Delta t > 0$ along the BGP, if two final goods i_1 and i_2 satisfy*

$$i_2 = e^{g_n \Delta t} i_1,$$

then

$$\ell(i_1,t) = \ell(i_2, t+\Delta t) \quad \text{and} \quad P(i_1,t) = e^{\gamma g_n \Delta t} P(i_2, t+\Delta t).$$

Proof of Lemma B.4:

First, we show that $\bar{i}(l,t)$ defined in Lemma B.3 must grows at the same rate as $N(t)$. Consider a household with endowment l at time t', who either have $X_l(t') = 0$ or $X_l(t') > 0$. Suppose $X_l(t') > 0$. Since $X_l(t)$ grows at a constant rate along the BGP, we have $X_l(t) > 0$ for all t. By equation (3), this household must have consumed all the available goods, i.e. $\bar{i}(l,t) = N(t)$, so $\bar{i}(l,t)$ grows at the rate g_n. In this case, the household's aggregate consumption is

$$C_l(t) = \int_0^{N(t)} i^{-\gamma} di + X_l(t) = \frac{N(t)^{1-\gamma}}{1-\gamma} + X_l(t).$$

Since $N(t)^{1-\gamma}$ grows at the rate $(1-\gamma)g_n$, to ensure $C_l(t)$ and $X_l(t)$ have constant growth rate, their growth rate must both be $(1-\gamma)g_n$.

Suppose $X_l(t') = 0$. Then we have $X_l(t) = 0$ for all t along the BGP. Thus, the household's

aggregate consumption is given by

$$C_l(t) = \int_0^{\bar{i}(l,t)} i^{-\gamma} di = \frac{\bar{i}(l,t)^{1-\gamma}}{1-\gamma}.$$

Since $C_l(t)$ grows at a constant rate, $\bar{i}(l,t)$ must also grow at some constant rate. We claim that this rate is g_n. Given that the distribution of l is unbounded above, there must exist a positive measure of households that can afford all available goods and have $X_l(t) > 0$. As we have shown, the consumption of these households will grow at the rate $(1-\gamma)g_n$. By the definition of the aggregate final consumption $C(t)$ and the requirement that $C(t)$ grows at a constant rate on BGP, it must be that all households' consumption grows at the same constant rate $(1-\gamma)g_n$. This implies that $\bar{i}(l,t)$ grows at the rate g_n.

In either case above, $\bar{i}(l,t)$ grows at a constant rate g_n. Then, we show that $\ell(i_1,t) = \ell(i_2, t+\Delta t)$. Let $l = \ell(i_1,t)$, it follows that

$$\bar{i}(l, t+\Delta t) = e^{g_n \Delta t}\bar{i}(l,t) = e^{g_n \Delta t} i_1 = i_2.$$

By definition, $\ell(i_2, t+\Delta t)$ is the poorest household who is indifferent between consuming and not consuming good i_2 at time $t+\Delta t$. We claim that this household has endowment l. Suppose, for contradiction, that there exists $l' < l$ such that a household with endowment l' consumes good i_2 at time $t+\Delta t$. This implies $\bar{i}(l', t+\Delta t) \geq i_2$, and then $\bar{i}(l',t) \geq i_1$. Thus, at time t, the household with endowment $l' < l$ would consume good i_1, contradicting the definition of $l = \ell(i_1,t)$. Therefore,

$$\ell(i_1,t) = \ell(i_2, t+\Delta t).$$

At last, we show that $P(i_1,t) = e^{\gamma g_n \Delta t} P(i_2, t+\Delta t)$. To see this, note that

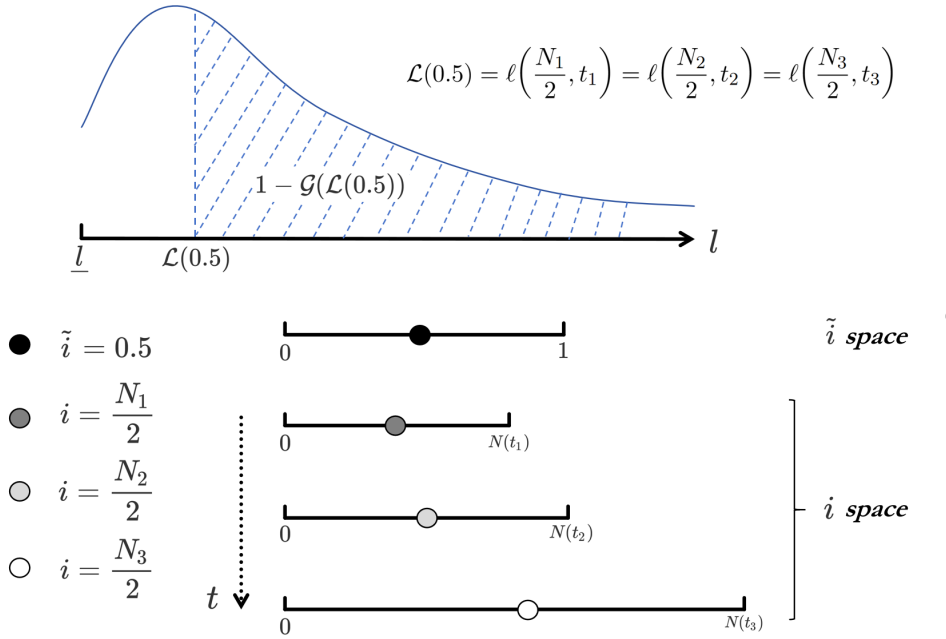
$$\begin{aligned}
P(i_2, t+\Delta t) &= \kappa_{\ell(i_2,t+\Delta t)}(i_2, t+\Delta t) \\
&= \frac{{i_2}^{-\gamma} e^{-\rho(t+\Delta t)}}{\mu_{\ell(i_2,t+\Delta t)} C_{\ell(i_2,t+\Delta t)}(t+\Delta t)^{\sigma}} \cdot e^{r(t+\Delta t)} \\
&= \frac{(e^{g_n \Delta t} i_1)^{-\gamma} e^{-\rho t}}{\mu_{\ell(i_1,t)} C_{\ell(i_1,t)}(t)^{\sigma}} \cdot e^{rt} \\
&= e^{-\gamma g_n \Delta t} P(i_1,t),
\end{aligned}$$

which complete our proof. ■

By Lemma B.4, if $i_1/N(t_1) = i_2/N(t_2)$, then $\ell(i_1,t_1) = \ell(i_2,t_2)$. Thus, we can define functions $\mathcal{L}(\tilde{i}(t)) = \ell(i,t)$ and $p(\tilde{i}(t)) = e^{\gamma g_n t} P(i,t)$, where $\tilde{i}(t) = i/N(t)$. For simplicity, we write these function as $\mathcal{L}(\tilde{i})$ and $p(\tilde{i})$, and then present several useful lemmas on the properties of $\mathcal{L}$. First, Lemma B.5 determines the domain and codomain of $\mathcal{L}$, and Lemma B.6 shows the monotonicity of $\mathcal{L}$. Second, Lemma B.7 shows that, given $\mathcal{L}$, each household's consumption decision follows a

simple threshold rule: if $l = \mathcal{L}(\tilde{i})$, the household consumes good j if and only if $j \leq \tilde{i}N(t)$. Finally, Lemma B.8 characterizes the relationship between $\mathcal{L}$ and the equilibrium price $p(\tilde{i})$.

Figure B.2: Time Translation Invariance of ℓ



Notes. The upper panel plots the cumulative distribution of labor endowments across households. For a good with normalized index $\tilde{i} = 0.5$, the marginal household that purchases this good has a labor endowment level $\mathcal{L}(0.5)$. The lower panel provides a graphical representation of this translation: each horizontal line corresponds to the product space at a particular time t_j, ranging from 0 to $N(t_j)$. A fixed normalized index $\tilde{i} = 0.5$ (black dot) maps to different absolute product indices $i = N(t_j)/2$ (gray or white dots) over time.

Figure B.2 illustrates the *time translation invariance* on the demand side. The upper panel plots the cumulative distribution of labor endowments across households. For a good with normalized index $\tilde{i} = 0.5$, the marginal household that purchases this good has a labor endowment level $\mathcal{L}(0.5)$. The total demand for this good, $1 - \mathcal{G}(\mathcal{L}(0.5))$, is represented by the shaded area to the right of $\mathcal{L}(0.5)$ under the distribution curve. The equality

$$\mathcal{L}(0.5) = \ell\left(\frac{N_1}{2}, t_1\right) = \ell\left(\frac{N_2}{2}, t_2\right) = \ell\left(\frac{N_3}{2}, t_3\right)$$

shows that, along the BGP, the same normalized product index $\tilde{i} = 0.5$ corresponds to goods located at different absolute positions in the product space ($i = N_t/2$) at different times. As the number of varieties $N(t)$ expands over time, moving forward in time by Δt is equivalent to shifting along the product space by a factor of $e^{g_n \Delta t}$.

The lower panel provides a graphical representation of this translation: each horizontal line corresponds to the product space at a particular time t_j, ranging from 0 to $N(t_j)$. A fixed normalized index $\tilde{i} = 0.5$ (black dot) maps to different absolute product indices $i = N(t_j)/2$ (gray or white dots) over time. Thus, the figure demonstrates that the demand for a given normalized product type remains invariant over time once expressed in terms of $\tilde{i}$ rather than the absolute index i.

Lemma B.5 *Suppose that Conjecture 2 holds. Then there exist constants $\eta > 0$ and $\bar{l} < \infty$ such that:*

$$\mathcal{L}(\tilde{i}) = \underline{l} \quad \textit{if and only if} \quad 0 < \tilde{i} \leq \eta,$$

and $\mathcal{L}(1) = \bar{l}$.

Proof of Lemma B.5:

We first establish the existence of $\bar{l}$. To do so, note that the set of households who purchase goods $N(t)$ is nonempty. This follows from the fact that endowment l is unbounded, whereas $N(t)$ is finite. Hence, by the household budget constraint, there exists a finite endowment level above which households purchase all goods, including goods $N(t)$. We therefore can define

$$\bar{l} \equiv \inf \{l : C_l(N(t), t) = 1\}. \tag{B.20}$$

By the definition of $\mathcal{L}(\cdot)$, it follows that $\mathcal{L}(1) = \bar{l}$.

Next, we demonstrate the existence of η. Define $\underline{\kappa}(i,t) = \inf_l \kappa_l(i,t)$, so that product i is consumed by all households if and only if $P(i,t) \leq \underline{\kappa}(i,t)$, as equation (B.17) implies. Under Conjecture 2, and $\underline{\kappa}(i,t)$ is decreasing in i with $\lim_{i \to 0} \underline{\kappa}(i,t) = \infty$, there exists a unique value $i = \tilde{\eta}(t)$ at each time such that

$$P(i,t) = \underline{\kappa}(i,t)$$

and

$$P(i,t) \leq \underline{\kappa}(i,t) \quad \text{if and only if} \quad 0 < i \leq \tilde{\eta}(t).$$

Thus, we have

$$\mathcal{L}\left(\frac{i}{N(t)}\right) = \underline{l} \quad \text{if and only if} \quad 0 < i \leq \tilde{\eta}(t).$$

As we have shown in Lemma B.4, $\mathcal{L}\left(\frac{i}{N(t)}\right)$ should be independent of time t, which implies $\tilde{\eta}(t)$ must grow at rate g_n. Therefore, we define $\eta = \frac{\tilde{\eta}(t)}{N(t)}$, which implies:

$$\mathcal{L}(\tilde{i}) = \underline{l} \quad \text{if and only if} \quad 0 < \tilde{i} \leq \eta.$$

This completes the proof. ■

In Lemma B.5, we establish the existence of a constant fraction η such that, for all $i \leq \eta N(t)$, every household purchases good i. Hence, subsequent analysis can primarily focus on the domain $\tilde{i} \in [\eta, 1]$.

Lemma B.6 *Suppose that Conjecture 2 holds. Then function $\mathcal{L}(\tilde{i})$ is strictly increasing on $[\eta, 1]$.*

Proof of Lemma B.6:

Take any $\eta \leq \tilde{i}_1 < \tilde{i}_2 \leq 1$. Suppose for contradiction, that $\mathcal{L}(\tilde{i}_1) \geq \mathcal{L}(\tilde{i}_2)$. By definition of $\mathcal{L}$,

this implies

$$\ell_1 \equiv \ell(\tilde{i}_1 N(t), t) \geq \ell(\tilde{i}_2 N(t), t) \equiv \ell_2.$$

By construction, a household with endowment ℓ_1 is indifferent at good $\tilde{i}_1 N(t)$, i.e.,$P(\tilde{i}_1 N(t), t) = \kappa_{\ell_1}(\tilde{i}_1 N(t), t)$. Under Conjecture 2, this household will only consume goods with $0 < i \leq \tilde{i}_1 N(t)$. Similarly, household with endowment ℓ_2 will only consume goods with $0 < i \leq \tilde{i}_2 N(t)$. Because $\tilde{i}_1 < \tilde{i}_2$, the consumption set of household ℓ_1 is a strict subset of that of household ℓ_2. Hence, the bundle chosen by household ℓ_2 is strictly preferred by household ℓ_1 under the same preference. Moreover, since $\ell_1 \geq \ell_2$, household ℓ_1 can afford the bundle chosen by household ℓ_2 whenever household ℓ_2 can. Therefore, the bundle associated with cutoff $\tilde{i}_1 N(t)$ cannot be optimal for household ℓ_1, a contradiction. Therefore, $\mathcal{L}(\tilde{i})$ must be strictly increasing on $[\eta, 1]$. ■

A direct implication of Lemma B.6 is that, for any given i, the willingness to pay $\kappa_l(i,t)$ is strictly increasing in l over $[\underline{l}, \bar{l}]$. Moreover, Lemma B.6 implies that $\mathcal{L}(\tilde{i})$ is invertible on $[\eta, 1]$. We therefore define its inverse function $\mathcal{L}^{-1} : [\underline{l}, \bar{l}] \to [\eta, 1]$ by

$$\mathcal{L}^{-1}(l) = \tilde{i} \quad \text{if and only if} \quad \mathcal{L}(\tilde{i}) = l. \tag{B.21}$$

In the next lemma, we show that the household consumption bundle is characterized by $\mathcal{L}^{-1}(l)$.

Lemma B.7 *Suppose that Conjectures 1 and 2 hold. Then, for a household with endowment $\underline{l} < l < \bar{l}$, its optimal consumption rule is given by*

$$C_l(i,t) = \begin{cases} 1 & \text{if } i \leq \mathcal{L}^{-1}(l) N(t), \\ 0 & \text{if } i > \mathcal{L}^{-1}(l) N(t), \end{cases}$$

and its consumption is given by

$$X_l(t) = 0.$$

For a household with endowment $l \geq \bar{l}$, its optimal consumption rule is given by

$$C_l(i,t) = 1 \quad \text{for all } i \leq N(t),$$

and its luxury service is given by

$$X_l(t) = \frac{Y(t)(l - \bar{l})}{P^X(t)}. \tag{B.22}$$

Proof of Lemma B.7:

Step 1. For a household with endowment $\underline{l} < l < \bar{l}$, the definition of $\mathcal{L}^{-1}(l)$ implies that:

$$P(\mathcal{L}^{-1}(l) N(t), t) = \kappa_l(\mathcal{L}^{-1}(l) N(t), t).$$

Under Conjecture 2, and the fact that $\kappa_l(i,t)$ is strictly decreasing in i, it follows that:

$$P(i,t) < \kappa_l(i,t) \quad \text{for all } i < \mathcal{L}^{-1}(l)N(t),$$

$$P(i,t) > \kappa_l(i,t) \quad \text{for all } i > \mathcal{L}^{-1}(l)N(t).$$

By equation (B.17), the household therefore consumes all goods with $i \leq \mathcal{L}^{-1}(l)N(t)$ and consumes no goods with $i > \mathcal{L}^{-1}(l)N(t)$. Finally, it is straightforward to see that $X_l(t) = 0$, since household l consumes no goods with $i > \mathcal{L}^{-1}(l)N(t)$, we have

$$\min_{i\in(0,N(t)]} C_l(i,t) = 0.$$

That is, luxury service consumption yields no utility at the margin.

Step 2. For a household with endowment $l \geq \bar{l}$, Lemma B.5 implies that the household consumes the good indexed by $N(t)$. By the same logic as in **Step 1**, it then follows that the household must consume all goods with $i \leq N(t)$.

To derive their luxury service consumption, we first rewrite the household budget constraint in per-period form. The present-value budget constraint for a household with endowment l is

$$\int_0^\infty e^{-rt} P_l C_l(t) dt = \int_0^\infty W(t) l \cdot e^{-rt} dt + \mathbb{V}(0) l.$$

Under Conjecture 1, P_l is the (constant) price index of household l's final consumption. Since $\mathbb{V}(0)$ denotes the aggregate market value of all firms at $t = 0$, we have

$$\mathbb{V}(0) \equiv \int_0^\infty e^{-rt} \left\{ \int_0^{N(t)} \left[\Pi(i,t) + \int_0^{F(i,t)} \Pi_i(j,t) dj \right] di - W(t) L^{RD}(t) \right\} dt, \tag{B.23}$$

where the total labor employed in R&D is given by

$$L^{RD}(t) = \int_0^{N(t)} \int_0^{F(i,t)} L_i^I(j,t) dj \, di + L^E(t) + L^N(t).$$

Substituting $\mathbb{V}(0)$ into the present-value constraint yields

$$\int_0^\infty e^{-rt} P_l C_l(t) dt = \int_0^\infty e^{-rt} \left\{ W(t) + \int_0^{N(t)} \left[\Pi(i,t) + \int_0^{F(i,t)} \Pi_i(j,t) dj \right] di - W(t) L^{RD}(t) \right\} l dt.$$

Since labor is the only input in the economy, the resource constraint implies that the total net profit flow across all firms must be equal to aggregate output minus labor income

$$\int_0^{N(t)} \left[\Pi(i,t) + \int_0^{F(i,t)} \Pi_i(j,t) dj \right] di - W(t) L^{RD}(t) = Y(t) - W(t).$$

So the present-value constraint simplifies to

$$\int_0^\infty e^{-rt} P_l C_l(t)\, dt = \int_0^\infty e^{-rt} Y(t)\, l\, dt. \tag{B.24}$$

Let $a_l(t)$ denote the household's net saving with $a_l(0) = 0$. The corresponding flow budget constraint is

$$\dot{a}_l(t) + P_l C_l(t) = r a_l(t) + Y(t) l. \tag{B.25}$$

We now show that $a_l(t) \equiv 0$ for all t and all l along the BGP. Under Conjecture 1, the Euler equation (B.16) implies all households' consumption grows at the common rate

$$g_c = \frac{\dot{C}_l(t)}{C_l(t)} = \frac{r - \rho}{\sigma}.$$

Along the BGP, aggregate income grows at rate g_y, i.e. $Y(t) = Y(0)e^{g_y t}$, and consumption expenditure satisfies $P_l C_l(t) = P_l C_l(0) e^{g_c t}$. Substituting these paths into (B.24) yields

$$\frac{P_l C_l(0)}{r - g_c} = \frac{Y(0) l}{r - g_y}. \tag{B.26}$$

Suppose $g_c > g_y$. Then (B.26) implies $P_l C_l(0) < Y(0) l$ for all l, and evaluating (B.25) at $t = 0$ gives

$$\dot{a}_l(0) > 0 \qquad \text{for all } l.$$

Therefore $\int \dot{a}_l(0)\, d\mathcal{G}(l) > 0$, which contradicts the credit-market clearing condition that the traded asset is in zero net supply:

$$\int a_l(t)\, d\mathcal{G}(l) = 0 \quad \text{for all } t.$$

An analogous contradiction arises if $g_c < g_y$. Hence we must have $g_c = g_y$. Then (B.26) implies $P_l C_l(0) = Y(0) l$, and since both sides grow at the same rate, it follows that

$$P_l C_l(t) = Y(t) l \qquad \text{for all } t \text{ and } l. \tag{B.27}$$

Substituting (B.27) into (B.25) yields $\dot{a}_l(t) = r a_l(t)$. Together with $a_l(0) = 0$, we obtain $a_l(t) \equiv 0$ for all t and l, and the per-period budget constraint simplifies to

$$\int_0^{N(t)} P(i,t) C_l(i,t)\, di + P^X(t) X_l(t) = Y(t) l. \tag{B.28}$$

From equation (B.27), for a household with endowment $l \geq \bar{l}$, we have

$$\int_0^{N(t)} P(i,t)\, di + P^X(t) X_l(t) = Y(t) l.$$

and for the household with endowment $\bar{l}$, we have

$$\int_0^{N(t)} P(i,t)\,di = Y(t)\bar{l}.$$

Combining the above two equations yields equation (B.22). ■

Lemma B.8 *Suppose that Conjectures 1 and 2 hold. Then the equilibrium price is related to the marginal value of $\mathcal{L}$:*

$$p(\tilde{i}) = y\mathcal{L}'(\tilde{i}) \quad \textit{for} \quad \tilde{i} \in [\eta, 1],$$

where the detrended price is defined as $p(\tilde{i}) = e^{\gamma g_n t} P(i,t)$.

Proof of Lemma B.8:

We solve the detrended price level $p(\tilde{i})$ using equation (B.28). Under Conjecture 1, combing equation (B.18) with the Euler equation (B.16) yields

$$\kappa_l(i,t) = \frac{i^{-\gamma}}{\tilde{\mu}_l}, \quad \text{with} \quad \tilde{\mu}_l = \mu_l C_l(0)^{\sigma}.$$

which implies $\kappa_l(i,t)$ is independent of time t. For $\tilde{i} \geq \eta$, we have

$$P(i,t) = \frac{i^{-\gamma}}{\tilde{\mu}_l} \quad \text{for} \quad l = \ell(i,t).$$

A household with endowment $\underline{l} \leq l \leq \mathcal{L}(1)$ consumes up to $\mathcal{L}^{-1}(l)N(t)$ and has zero luxury service $X_l(t) = 0$, where $\mathcal{L}^{-1}(l)$ is defined in equation (B.21). So equation (B.28) simplifies to

$$\int_0^{\mathcal{L}^{-1}(l)N(t)} P(i,t)di = Y(t)l \quad \text{for} \quad l \in [\underline{l}, \mathcal{L}(1)],$$

and detrended yields

$$\int_0^{\eta} p(\tilde{i})d\tilde{i} + \int_{\eta}^{\mathcal{L}^{-1}(l)} \frac{\tilde{i}^{-\gamma}}{\tilde{\mu}_{\mathcal{L}(\tilde{i})}} d\tilde{i} = yl \quad \text{for} \quad l \in [\underline{l}, \mathcal{L}(1)].$$

Differentiating with respect to l gives

$$\tilde{\mu}_l = \frac{\left[\mathcal{L}^{-1}(l)\right]^{-\gamma}}{y\,\mathcal{L}'(\mathcal{L}^{-1}(l))} \quad \text{for} \quad l \in [\underline{l}, \mathcal{L}(1)],$$

or equivalently,

$$\tilde{\mu}_{\mathcal{L}(\tilde{i})} = \frac{\tilde{i}^{-\gamma}}{y\mathcal{L}'(\tilde{i})} \quad \text{for} \quad \tilde{i} \in [\eta, 1].$$

Using the detrended expression of $p(\tilde{i}) = \frac{\tilde{i}^{-\gamma}}{\tilde{\mu}_{\mathcal{L}(\tilde{i})}}$, we obtain

$$p(\tilde{i}) = y\mathcal{L}'(\tilde{i}) \quad \text{for} \quad \tilde{i} \in [\eta, 1].$$

■

B.3.2 Properties of the BGE

In this appendix, we derive properties of the BGE. Let $\{g_c, g_x, g_y, g_w, g_n\}$ denote the growth rates of $\{C, X, Y, w, N\}$. The next lemma pins down their relationship on the BGP.

Lemma B.9 *Along the BGP, all aggregate quantities grow at constant rates. In particular,*

$$g_c = g_x = g_y = g_w = (1-\gamma)g_n.$$

Moreover, the luxury service price P^X satisfies

$$P^X(t) = \frac{W(t)}{\bar{A}^{1-\gamma}}, \tag{B.29}$$

which is constant along the BGP.

Proof of Lemma B.9:

We have shown that the growth rate of $C_l(t)$ is $(1-\gamma)g_n$ in Lemma B.4, and also equal to g_y in Lemma B.7. By the definition of $C(t)$, the growth rate of $C_l(t)$ is equal to the growth rate of $C(t)$. Therefore, we have established that $g_c = g_y = (1-\gamma)g_n$.

Next we show that $g_y = g_w$. It suffices to show that the aggregate labor share is constant. In the luxury service sector, the resource constraint implies that all output is distributed to labor, yielding a constant labor share of 1. In the production sector, the aggregate output is divided into three components: (i) profits from final-goods production, (ii) profits from intermediate-goods production, and (iii) labor compensation. As shown in equation (B.3), the labor share in the production sector across all final goods is constant and equal to $(1-\beta)^2$. Furthermore, Lemma B.7 implies the relative size of the luxury service sector is constant along the stationary equilibrium, so the aggregate labor share is time-invariant. With fixed labor supply, it follows that $g_y = g_w$.

Finally, the first-order condition for the luxury service producer problem gives equation (B.29). Because $g_w = (1-\gamma)g_n$, $P^X(t)$ is constant along the BGP. As shown in Lemma B.7, each household with $l < \bar{l}$ has $X_l(t) = 0$, and each household with $l \geq \bar{l}$ has $X_l(t) > 0$, which grows at rate $(1-\gamma)g_n$ by equation (B.22). Thus $X(t) = \int_{\bar{l}}^{\infty} X_l(t) d\mathcal{G}(l)$ grows at rate $(1-\gamma)g_n$. ■

To further simplify the analysis, it is convenient to exploit the symmetry property of the BGP, which allows the product space to be normalized from $(0, N(t)]$ to $(0, 1]$. In the following lemma, we show an important property of our BGP system: time-translation invariance, which implies a

simple invariance: moving forward in time by Δt is equivalent to moving along the product ladder by the factor $e^{g_n \Delta t}$.

Lemma B.10 *Along the BGP, for any $\Delta t > 0$, if two categories of final goods i_1 and i_2 satisfies*

$$i_2 = e^{g_n \Delta t} i_1,$$

then the following relationships hold:

$$\begin{aligned} Y(i_1,t) &= Y(i_2, t+\Delta t), & z_{i_1}(t) &= z_{i_2}(t+\Delta t), \\ F(i_1,t) &= F(i_2, t+\Delta t), & L(i_1,t) &= e^{g_n \Delta t} L(i_2, t+\Delta t), \\ B(i_1,t) &= e^{-\frac{1-\beta}{\beta} g_n \Delta t} B(i_2, t+\Delta t). \end{aligned}$$

Proof of Lemma B.10:

We will complete the proof in three steps.

Step 1. Derive the growth rate of sectoral productivity $B(i,t)$ in equation (19).

Let

$$Z(i,t) \equiv \int_0^{F(i,t)} \tilde{A}_i(j,t)^{\theta \frac{1-\beta}{\beta}} \, dj.$$

Considering a small time interval dt, we partition firms in sector i into two groups: incumbent firms indexed by $j \in [0, F(i,t)]$, and newly entering firms indexed by $j \in (F(i,t), F(i,t+dt)]$. For an incumbent firm $j \in [0, F(i,t)]$, productivity evolves according to an independent Poisson process $N_i^z(j,t)$, which denotes successful innovation. For an incumbent firm that remains active, we intuitively have

$$d\left(\tilde{A}_i(j,t)^{\theta \frac{1-\beta}{\beta}}\right) = \left(\lambda^{\theta \frac{1-\beta}{\beta}} - 1\right) \tilde{A}_i(j,t)^{\theta \frac{1-\beta}{\beta}} \, dN_i^z(j,t).$$

Taking conditional expectations, we obtain

$$\mathbb{E}_t\left[d\left(\tilde{A}_i(j,t)^{\theta \frac{1-\beta}{\beta}}\right)\right] = z_i(t)\left(\lambda^{\theta \frac{1-\beta}{\beta}} - 1\right) \tilde{A}_i(j,t)^{\theta \frac{1-\beta}{\beta}} \, dt.$$

Given that incumbent firms are subject to independent idiosyncratic shocks, by an exact law of large numbers, the aggregate realized flow equals its conditional expectation almost surely

$$\begin{aligned} \int_0^{F(i,t)} d\Big(\tilde{A}_i(j,t)^{\theta \frac{1-\beta}{\beta}}\Big) \, dj &= \int_0^{F(i,t)} \mathbb{E}_t\Big[d\Big(\tilde{A}_i(j,t)^{\theta \frac{1-\beta}{\beta}}\Big)\Big] \, dj. \\ &= z_i(t)\big(\lambda^{\theta \frac{1-\beta}{\beta}} - 1\big) Z(i,t) \, dt. \end{aligned}$$

For newly entering firms, we assume that entrants draw their productivity from the current

distribution of incumbents. Therefore, for any $j \in (F(i,t), F(i,t+dt)]$, we have

$$\mathbb{E}_t\Big[\tilde{A}_i(j,t)^{\theta\frac{1-\beta}{\beta}}\Big] = \frac{1}{F(i,t)}\int_0^{F(i,t)} \tilde{A}_i(j',t)^{\theta\frac{1-\beta}{\beta}}\, dj' = \frac{Z(i,t)}{F(i,t)}.$$

Since the measure of entrants over dt is $e(i,t)\,dt$, the total contribution of entry is

$$e(i,t)\,dt \cdot \frac{Z(i,t)}{F(i,t)}.$$

Since exit shocks are independent of firm productivity, exiting firms constitute a random sample of incumbents. Hence, by the law of large numbers, the expected productivity mass lost to exit is

$$\delta F(i,t) \cdot \frac{Z(i,t)}{F(i,t)} dt.$$

By Ito's Lemma, we obtain

$$\begin{aligned} dZ(i,t) &= dF(i,t)\cdot\frac{Z(i,t)}{F(i,t)} + \int_0^{F(i,t)} d\Big(\tilde{A}_i(j,t)^{\theta\frac{1-\beta}{\beta}}\Big)\, dj \\ &= (e(i,t) - \delta F(i,t))\frac{Z(i,t)}{F(i,t)}dt + z_i(t)\big(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\big)Z(i,t)dt. \end{aligned} \tag{B.30}$$

Dividing both sides by $Z(i,t)$, and substituting the resulting equation into Ito's formula applied to

$$B(i,t) = \bar{A}(t)^{(1-\theta)\frac{1-\beta}{\beta}} Z(i,t),$$

we obtain equation (19).

Step 2. Show that the firm value V satisfies a time translation invariance condition:

$$V_{i_2}(t+\Delta t) = e^{-\left(\gamma+\theta\frac{1-\beta}{\beta}\right)g_n\Delta t}\, V_{i_1}(t).$$

The entry condition for intermediate firms implies

$$\begin{aligned} W(t) &= \bar{A}(t)\psi_e\left[\frac{\int_0^{F(i,t)}\tilde{A}_i(j,t)^{\theta\frac{1-\beta}{\beta}}dj}{F(i,t)}\right]V_i(t) \\ &= \bar{A}(t)^{1-(1-\theta)\frac{1-\beta}{\beta}}\,\psi_e\,\frac{B(i,t)}{F(i,t)}V_i(t), \end{aligned}$$

where it follows from a law of large numbers that

$$\frac{\int_0^{F(i,t)}\tilde{A}_i(j,t)^{\theta\frac{1-\beta}{\beta}}dj}{F(i,t)} = \mathbb{E}[\tilde{A}_i(j,t)^{\theta\frac{1-\beta}{\beta}}]$$

represents the expected productivity drawn by an entrant from the distribution of incumbent firms'

productivities in sector i. By Ito's Lemma,

$$\frac{\dot{W}(t)}{W(t)} = \left(1-(1-\theta)\frac{1-\beta}{\beta}\right)\frac{\dot{\bar{A}}(t)}{\bar{A}(t)} + \frac{\dot{B}(i,t)}{B(i,t)} - \frac{\dot{F}(i,t)}{F(i,t)} + \frac{\dot{V}_i(t)}{V_i(t)}. \tag{B.31}$$

Combining with equation (19) and (B.31), we obtain

$$\frac{\dot{V}_i(t)}{V_i(t)} = \frac{\dot{W}(t)}{W(t)} - \frac{\dot{\bar{A}}(t)}{\bar{A}(t)} - z_i(j,t)\cdot\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right), \tag{B.32}$$

where $z_i(j,t)$ is given by equation (B.5).

From the firm's HJB equation in the proof of Lemma B.1, $V_i(t)$ satisfies

$$\begin{aligned} rV_i(t) - \dot{V}_i(t) &= \beta(1-\beta)^{\frac{1-\beta}{\beta}}\left[\frac{P(i,t)^{\frac{1}{1-\beta}}}{W(t)}\bar{A}(t)^{1-\theta}\right]^{\frac{1-\beta}{\beta}} \\ &\quad -W(t)\cdot\left[\frac{W(t)}{(\lambda^{\theta\frac{1-\beta}{\beta}}-1)V_i(t)\psi\alpha\bar{A}(t)^{\xi}}\right]^{\frac{1}{\alpha-1}} + z_i(j,t)(\lambda^{\theta\frac{1-\beta}{\beta}}-1)V_i(t) - \delta V_i(t), \end{aligned}$$

where ξ is a parameter defined in equation (B.8). Substituting the expression for $\dot{V}_i(t)$ in equation (B.32) and simplifying, we obtain

$$\begin{aligned} \left[\rho+\delta+(\sigma(1-\gamma)+\gamma)g_n\right]V_i(t) &= \beta(1-\beta)^{\frac{1-\beta}{\beta}}\left[\frac{P(i,t)^{\frac{1}{1-\beta}}}{W(t)}\bar{A}(t)^{1-\theta}\right]^{\frac{1-\beta}{\beta}} \\ &\quad -W(t)\cdot\left[\frac{W(t)}{(\lambda^{\theta\frac{1-\beta}{\beta}}-1)V_i(t)\psi\alpha\bar{A}(t)^{\xi}}\right]^{\frac{1}{\alpha-1}}, \end{aligned} \tag{B.33}$$

where we have used Lemma B.9 and the Euler equation $r=\rho+\sigma(1-\gamma)g_n$ along the BGP.

We now show that V satisfies a time translation invariance condition based on equation (B.33). Consider the index translation $i_2 = e^{g_n\Delta t}i_1$. For ease of notation, we ignore the constant multiplier $[\rho+\delta+(\sigma(1-\gamma)+\gamma)g_n]$. Then $V_{i_1}(t)$ and $V_{i_2}(t)$ are given by

$$\begin{aligned} V_{i_1}(t) &= \beta(1-\beta)^{\frac{1-\beta}{\beta}}\left[\frac{P(i_1,t)^{\frac{1}{1-\beta}}}{W(t)}\bar{A}(t)^{(1-\theta)}\right]^{\frac{1-\beta}{\beta}} \\ &\quad -W(t)\cdot\left[\frac{W(t)}{\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right)V_{i_1}(t)\cdot\psi\alpha\bar{A}(t)^{\xi}}\right]^{\frac{1}{\alpha-1}}, \\ V_{i_2}(t+\Delta t) &= \beta(1-\beta)^{\frac{1-\beta}{\beta}}\left[\frac{P(i_2,t+\Delta t)^{\frac{1}{1-\beta}}}{W(t+\Delta t)}\bar{A}(t+\Delta t)^{(1-\theta)}\right]^{\frac{1-\beta}{\beta}} \end{aligned}$$

$$-W(t+\Delta t)\cdot\left[\frac{W(t+\Delta t)}{\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right)V_{i_2}(t+\Delta t)\cdot\psi\alpha\bar{A}(t+\Delta t)^{\xi}}\right]^{\frac{1}{\alpha-1}}.$$

To show the relationship between $V_{i_1}(t)$ and $V_{i_2}(t+\Delta t)$, observe that Lemma B.4 implies $P(i_2,t+\Delta t)=e^{-\gamma g_n\Delta t}P(i_1,t)$, and the aggregate variables evolve according to $W(t+\Delta t)=e^{(1-\gamma)g_n\Delta t}W(t)$, $\bar{A}(t+\Delta t)=e^{g_n\Delta t}\bar{A}(t)$. Substituting these expressions into $V_{i_2}(t)$, we have

$$\begin{aligned}V_{i_2}(t+\Delta t) &= \beta(1-\beta)^{\frac{1-\beta}{\beta}}e^{-\left(\gamma+\theta\frac{1-\beta}{\beta}\right)g_n\Delta t}\left[\frac{P(i_1,t)^{\frac{1}{1-\beta}}}{W(t)}\bar{A}(t)^{(1-\theta)}\right]^{\frac{1-\beta}{\beta}}\\ &\quad -e^{(1-\gamma)g_n\Delta t}W(t)\cdot\left[\frac{e^{(1-\gamma)g_n\Delta t}W(t)}{\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right)V_{i_2}(t+\Delta t)\cdot\psi\alpha e^{\xi g_n\Delta t}\bar{A}(t)^{\xi}}\right]^{\frac{1}{\alpha-1}}.\end{aligned}$$

Multiply each side by $e^{\left(\gamma+\theta\frac{1-\beta}{\beta}\right)g_n\Delta t}$ to derive

$$\begin{aligned}e^{\left(\gamma+\theta\frac{1-\beta}{\beta}\right)g_n\Delta t}V_{i_2}(t+\Delta t) &= \beta(1-\beta)^{\frac{1-\beta}{\beta}}\left[\frac{P(i_1,t)^{\frac{1}{1-\beta}}}{W(t)}\bar{A}(t)^{(1-\theta)}\right]^{\frac{1-\beta}{\beta}}\\ &\quad -W(t)\cdot\left[\frac{e^{\left[(1-\gamma)+(\alpha-1)\left(\theta\frac{1-\beta}{\beta}+1\right)-\xi\right]g_n\Delta t}W(t)}{\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right)V_{i_2}(t+\Delta t)\cdot\psi\alpha\bar{A}(t)^{\xi}}\right]^{\frac{1}{\alpha-1}}\\ &= \beta(1-\beta)^{\frac{1-\beta}{\beta}}\left[\frac{P(i_1,t)^{\frac{1}{1-\beta}}}{W(t)}\bar{A}(t)^{(1-\theta)}\right]^{\frac{1-\beta}{\beta}}\\ &\quad -W(t)\cdot\left[\frac{W(t)}{\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right)e^{\left(\gamma+\theta\frac{1-\beta}{\beta}\right)g_n\Delta t}V_{i_2}(t+\Delta t)\cdot\psi\alpha\bar{A}(t)^{\xi}}\right]^{\frac{1}{\alpha-1}}.\end{aligned}$$

Thus, we have

$$V_{i_2}(t+\Delta t)=e^{-\left(\gamma+\theta\frac{1-\beta}{\beta}\right)g_n\Delta t}V_{i_1}(t).$$

Step 3. Show the temporal translation conditions for $z_i(j,t),Y(i,t),B(i,t),F(i,t)$.

1. Innovation intensity $z_i(j,t)$. Using the definition of $z_i(j,t)$ and the temporal translation of W, V, and $\bar{A}$, we have

$$\begin{aligned}z_{i_2}(j,t+\Delta t) &= \psi\left[\frac{W(t+\Delta t)}{(\lambda^{\theta\frac{1-\beta}{\beta}}-1)V_{i_2}(t+\Delta t)\cdot\psi\alpha\bar{A}(t+\Delta t)^{\frac{\xi}{\alpha}}}\right]^{\frac{\alpha}{\alpha-1}}\\ &= \psi\left[\frac{e^{(1-\gamma)g_n\Delta t}W(t)}{(\lambda^{\theta\frac{1-\beta}{\beta}}-1)e^{-(\gamma+\theta\frac{1-\beta}{\beta})g_n\Delta t}V_{i_1}(t)\cdot\psi\alpha e^{\frac{\xi}{\alpha}g_n\Delta t}\bar{A}(t)^{\frac{\xi}{\alpha}}}\right]^{\frac{\alpha}{\alpha-1}}\end{aligned}$$

$$= \psi \left[\frac{W(t)}{(\lambda^{\theta \frac{1-\beta}{\beta}} - 1) V_{i_1}(t) \cdot \psi \alpha \bar{A}(t)^{\frac{\xi}{\alpha}}} \right]^{\frac{\alpha}{\alpha-1}} = z_{i_1}(j,t).$$

2. Sectoral output $Y(i,t)$. Using $Y(i,t) = 1 - \mathcal{G}(\mathcal{L}(\tilde{i}))$ and the fact that $\tilde{i}_1 = \tilde{i}_2$, we get

$$Y(i_2, t+\Delta t) = 1 - \mathcal{G}(\mathcal{L}(\tilde{i}_2)) = 1 - \mathcal{G}(\mathcal{L}(\tilde{i}_1)) = Y(i_1, t).$$

3. Sectoral productivity $B(i,t)$. By equation (11),

$$\begin{aligned} B(i_2, t+\Delta t) &= \frac{Y(i_2, t+\Delta t) \left[\frac{W(t+\Delta t)}{P(i_2, t+\Delta t)} \right]^{\frac{1-\beta}{\beta}}}{(1-\beta)^{\frac{1-2\beta}{\beta}}} = \frac{Y(i_1, t) \left[\frac{e^{g_n \Delta t} W(t)}{P(i_1, t)} \right]^{\frac{1-\beta}{\beta}}}{(1-\beta)^{\frac{1-2\beta}{\beta}}} \\ &= e^{\frac{1-\beta}{\beta} g_n \Delta t} B(i_1, t). \end{aligned}$$

4. Sectoral production labor demand $L(i,t)$. By equation (12),

$$\begin{aligned} L(i_2, t+\Delta t) &= \left[\frac{W(t+\Delta t)}{(1-\beta) P(i_2, t+\Delta t)} \right]^{-\frac{1}{\beta}} B(i_2, t+\Delta t) \\ &= \left[\frac{e^{(1-\gamma)\Delta t} W(t)}{e^{-\gamma \Delta t}(1-\beta) P(i_1, t)} \right]^{-\frac{1}{\beta}} e^{\frac{1-\beta}{\beta} g_n \Delta t} B(i_1, t) \\ &= \left[\frac{W(t)}{(1-\beta) P(i_1, t)} \right]^{-\frac{1}{\beta}} e^{-g_n \Delta t} B(i_1, t) \\ &= e^{-g_n \Delta t} L(i_1, t). \end{aligned}$$

5. Firm number $F(i,t)$. Using the entry condition,

$$\begin{aligned} F(i_2, t+\Delta t) &= \bar{A}(t+\Delta t)^{1-(1-\theta)\frac{1-\beta}{\beta}} \psi_e \frac{B(i_2, t+\Delta t)}{W(t+\Delta t)} V_{i_2}(t+\Delta t) \\ &= e^{[1-(1-\theta)\frac{1-\beta}{\beta}] g_n \Delta t} \bar{A}(t)^{1-(1-\theta)\frac{1-\beta}{\beta}} \psi_e \frac{e^{\frac{1-\beta}{\beta} g_n \Delta t} B(i_1, t)}{e^{(1-\gamma) g_n \Delta t} W(t)} e^{-(\gamma + \theta \frac{1-\beta}{\beta}) g_n \Delta t} V_{i_1}(t) \\ &= F(i_1, t). \end{aligned}$$

Hence, we have verified that $z_i(j,t)$, $Y(i,t)$, $B(i,t)$, $L(i,t)$, and $F(i,t)$ all satisfy the time translation invariance conditions. ■

To better illustrate this lemma, define the normalized product index $\tilde{i}(t) \equiv \frac{i}{N(t)}$, which measures the relative luxury level of sector i at time t. Lemma B.10 implies that two sectors with the same $\tilde{i}$ at different points in time will share identical (possibly detrended) values for output, innovation rates, firm numbers, and other sector-specific variables. This property can be interpreted as a *time translation invariance*: shifting forward in time by Δt is equivalent to moving along the product space by a factor of $e^{g_n \Delta t}$. Based on that, we can translate all the dynamics of a specific sector into

a set of differential equations across sectors. Building on this property, we proceed by detrending all sector-level variables by their respective asymptotic growth rates in the next subsection.

B.3.3 Full Detrending Procedure

To characterize the BGE, we eliminate the common exponential growth components of all variables and transform the system into a stationary representation. The detrending procedure proceeds in three steps.

Step 1. Identifying asymptotic growth rates. We first compute the asymptotic growth rates of all aggregate- and sector-level variables. For example, the measure of varieties $N(t)$ grows at rate g_n, sectoral productivity $B(i,t)$ grows at rate $\frac{1-\beta}{\beta}g_n$, and wages $W(t)$ grow at rate $(1-\gamma)g_n$. From these primitives, we can infer the growth trend of any endogenous variable in the system.

Step 2. Applying time translation invariance. Lemma B.10 shows that along the BGP, shifting the good index by a factor of $e^{g_n \Delta t}$ while moving time forward by Δt preserves the structure of the economy. This invariance ensures that the system admits a representation in which all real allocations are invariant under such translations and differ only by deterministic exponential factors.

Step 3. Constructing detrended variables. Using the identified growth rates and translation invariance, we define stationary counterparts of all variables by removing their exponential trends.

For aggregate variables, we set detrended variables as

$$y = e^{-(1-\gamma)g_n t}Y(t), \quad w = e^{-(1-\gamma)g_n t}W(t), \quad 1 = e^{-g_n t}N(t),$$

where we normalize the detrended $N(t)$ to 1.

For sector-level variables, we normalize indices by the measure of variety, $\tilde{i} = i/N(t)$, and define detrended functions as

$$v(\tilde{i}) = e^{\left(\gamma+\theta\frac{1-\beta}{\beta}\right)g_n t}V_i(t), \quad z(\tilde{i}) = z_i(t), \quad e(\tilde{i}) = e(i,t), \quad f(\tilde{i}) = F(i,t),$$

$$p(\tilde{i}) = e^{\gamma g_n t}P(i,t), \quad b(\tilde{i}) = e^{-\frac{1-\beta}{\beta}g_n t}B(i,t), \quad v^f(\tilde{i}) = e^{\gamma g_n t}V^F(i,t).$$

After this normalization, the system becomes stationary, which provides the foundation for characterizing the equilibrium. In what follows, we derive explicit expressions for the seven sector-level variables defined above.

Incumbent's decision $z(\tilde{i})$ and $v(\tilde{i})$. It follows from equation (B.5) that

$$z(\tilde{i}) = \psi\left[\frac{w}{(\lambda^{\theta\frac{1-\beta}{\beta}}-1)\,v(\tilde{i})\,\psi\alpha}\right]^{\frac{\alpha}{\alpha-1}}. \tag{B.34}$$

Since $\tilde{i} = i/N(t)$ and $\dot{N}/N = g_n$, we have the relation $\frac{d\tilde{i}}{dt} = -\tilde{i}\,g_n$. Using the HJB equa-

tion (B.10), we can derive

$$\begin{aligned}
&\left[\rho + ((\sigma-1)(1-\gamma)+1+\theta\tfrac{1-\beta}{\beta})g_n + \delta\right] v(\tilde{i}) \\
= \quad & \beta(1-\beta)^{\frac{1-\beta}{\beta}} \cdot p(\tilde{i})^{\frac{1}{\beta}} \cdot w^{-\frac{1-\beta}{\beta}} \\
& -w\left(\tfrac{z(\tilde{i})}{\psi}\right)^{\frac{1}{\alpha}} + z(\tilde{i})\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) \cdot v(\tilde{i}) - \tilde{i} g_n v'(\tilde{i}),
\end{aligned} \tag{B.35}$$

where we have used the Euler equation $r = \rho + \sigma(1-\gamma)g_n$ along the BGP.

Variety of intermediate input $e(\tilde{i})$ and $f(\tilde{i})$. We next characterize the entry dynamics of intermediate firms. For a new entrant, it follows from equation (10) that the expected firm value is given by

$$\begin{aligned}
\mathbb{E}_t[A^{\theta\frac{1-\beta}{\beta}} V_i(t)] &= \left[\frac{\int_0^{F(i,t)} \tilde{A}_i(j,t)^{\theta\frac{1-\beta}{\beta}} dj}{F(i,t)}\right] V_i(t) \\
&= \left[\frac{B(i,t)}{\bar{A}(t)^{(1-\theta)\frac{1-\beta}{\beta}} F(i,t)}\right] V_i(t).
\end{aligned}$$

Define the detrended R&D input for entry as $L_i^E(j,t) = l^E(\tilde{i}) \cdot e^{-g_n t}$, then the innovation production function can be written as

$$e(\tilde{i}) = \psi_e l^E(\tilde{i}),$$

and the entrant's decision problem becomes

$$\max_{l^E(\tilde{i})} \quad \psi_e l^E(\tilde{i}) \cdot \left(\frac{b(\tilde{i})}{f(\tilde{i})}\right) v(\tilde{i}) - w \cdot l^E(\tilde{i}).$$

The corresponding first-order condition with respect to $l^E(\tilde{i})$ is

$$w = v(\tilde{i}) \cdot \psi_e \cdot \left(\frac{b(\tilde{i})}{f(\tilde{i})}\right). \tag{B.36}$$

Finally, the law of motion for the measure of firms $F(i,t)$ is

$$\begin{aligned}
\frac{\partial F(i,t)}{\partial t} &= e(i,t) - \delta F(i,t), \\
\Rightarrow -\tilde{i} g_n \frac{df(\tilde{i})}{d\tilde{i}} &= e(\tilde{i}) - \delta f(\tilde{i}).
\end{aligned} \tag{B.37}$$

Household's decision $p(\tilde{i})$. As we have shown in Lemma B.8, for $\tilde{i} \geq \eta$, $p(\tilde{i}) = y\mathcal{L}'(\tilde{i})$. The equilibrium pricing condition holds for all $\tilde{i} \in (0,1]$:

$$Y(i,t) = (1-\beta)^{\frac{1-2\beta}{\beta}} \left[\frac{W(t)}{P(i,t)}\right]^{-\frac{1-\beta}{\beta}} B(i,t)$$

$$\Rightarrow 1-\mathcal{G}(\mathcal{L}(\tilde{i})) \quad = \quad (1-\beta)^{\frac{1-2\beta}{\beta}}\left[\frac{w}{p(\tilde{i})}\right]^{-\frac{1-\beta}{\beta}} b(\tilde{i}). \tag{B.38}$$

Since $\mathcal{G}(\underline{l})=0$ and $\mathcal{L}(\tilde{i})=\underline{l}$ when $0<\tilde{i}<\eta$, it follows that

$$1 \quad = \quad (1-\beta)^{\frac{1-2\beta}{\beta}}\left[\frac{w}{p(\tilde{i})}\right]^{-\frac{1-\beta}{\beta}} b(\tilde{i}) \qquad \text{for} \quad \tilde{i}\in(0,\eta). \tag{B.39}$$

Productivity dynamics $b(\tilde{i})$.

We use equation (19) to derive $b(\tilde{i})$. As shown in Lemma B.10, $B(i,t)$ admits the representation $B(i,t)=e^{\frac{1-\beta}{\beta}g_n t}b(\tilde{i})$. Fix any i, since $N(t)$ grows at rate g_n, we have $\tilde{i}=i/N(t)\to 0$ as $t\to\infty$. Denote $b_0=\lim_{\tilde{i}\to 0} b(\tilde{i})$, then

$$B(i,t)\sim e^{\frac{1-\beta}{\beta}g_n t}b_0 \quad \text{as} \quad t\to\infty.$$

Therefore, the asymptotic growth rate of $B(i,t)$ is $\lim_{t\to\infty}\frac{\dot{B}(i,t)}{B(i,t)}=\frac{1-\beta}{\beta}g_n$. Moreover, the firm-number dynamics in equation (B.37) imply that $\lim_{t\to\infty} e(i,t)=\delta F(i,t)$. Hence, taking limits as $t\to\infty$ in equation (19) gives

$$g_n=\lim_{t\to\infty}\frac{1}{\theta}\frac{\beta}{1-\beta}z_i(t)\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right)=\frac{1}{\theta}\frac{\beta}{1-\beta}z_0\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right), \tag{B.40}$$

where $z_0\equiv\lim_{\tilde{i}\to 0}z(\tilde{i})$. Then using equation (19) in detrended terms, the law of motion for $b(\tilde{i})$ becomes

$$-\tilde{i}g_n\frac{db(\tilde{i})}{d\tilde{i}}=\left\{\left[z(\tilde{i})-z_0\right]\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right)+\frac{e(\tilde{i})}{f(\tilde{i})}-\delta\right\}b(\tilde{i}). \tag{B.41}$$

Final goods firm $v^f(\tilde{i})$. The aggregate profit of the intermediate producers is:

$$\int_0^{F(i,t)}\Pi_i(j,t)dj=\beta(1-\beta)^{\frac{1-\beta}{\beta}}\cdot P(i,t)^{\frac{1}{\beta}}\cdot W(t)^{-\frac{1-\beta}{\beta}}\cdot B(i,t).$$

The profit flow of the final goods producer is $\frac{1}{1-\beta}$ times the above profit:

$$\Pi(i,t)=\beta(1-\beta)^{\frac{1-2\beta}{\beta}}\cdot P(i,t)^{\frac{1}{\beta}}\cdot W(t)^{-\frac{1-\beta}{\beta}}\cdot B(i,t).$$

So the value of a final-goods firm can be represented by:

$$\begin{aligned}V^F(i,t)&=\int_t^\infty e^{-r(\tau-t)}\,\beta(1-\beta)^{\frac{1-2\beta}{\beta}}\,P(i,\tau)^{\frac{1}{\beta}}\,W(\tau)^{-\frac{1-\beta}{\beta}}\,B(i,\tau)\,d\tau\\&=e^{-\gamma g_n t}\int_t^\infty e^{-(r+\gamma g_n)(\tau-t)}\,\beta(1-\beta)^{\frac{1-2\beta}{\beta}}\,p(i,\tau)^{\frac{1}{\beta}}\,w^{-\frac{1-\beta}{\beta}}\,b(i,\tau)\,d\tau.\end{aligned}$$

Let $V^F(i,t) = e^{-\gamma g_n t} v^f(\tilde{i})$ and $x = i/N(\tau)$. Since $x = \tilde{i}\, e^{-g_n(\tau - t)}$, we have $\tau = \frac{\ln \tilde{i} - \ln x}{g_n} + t$, and

$$\begin{aligned} v^f(\tilde{i}) &= \int_{\tilde{i}}^{0} e^{-(r+\gamma g_n)\frac{\ln \tilde{i} - \ln x}{g_n}} \beta(1-\beta)^{\frac{1-2\beta}{\beta}} p(x)^{\frac{1}{\beta}} w^{-\frac{1-\beta}{\beta}} b(x)\, d\left(\frac{\ln \tilde{i} - \ln x}{g_n} + t\right) \\ &= \int_{0}^{\tilde{i}} \frac{e^{-(\frac{r}{g_n}+\gamma)(\ln \tilde{i} - \ln x)}}{x g_n} \beta(1-\beta)^{\frac{1-2\beta}{\beta}} p(x)^{\frac{1}{\beta}} w^{-\frac{1-\beta}{\beta}} b(x)\, dx \\ &= \int_{0}^{\tilde{i}} \frac{x^{(\frac{r}{g_n}+\gamma-1)}}{\tilde{i}^{(\frac{r}{g_n}+\gamma)} g_n} \beta(1-\beta)^{\frac{1-2\beta}{\beta}} p(x)^{\frac{1}{\beta}} w^{-\frac{1-\beta}{\beta}} b(x)\, dx \\ &= \int_{0}^{\tilde{i}} \frac{x^{\left[\frac{\rho}{g_n}+\sigma(1-\gamma)+\gamma-1\right]}}{\tilde{i}^{\left[\frac{\rho}{g_n}+\sigma(1-\gamma)+\gamma\right]} g_n} \beta(1-\beta)^{\frac{1-2\beta}{\beta}} p(x)^{\frac{1}{\beta}} w^{-\frac{1-\beta}{\beta}} b(x)\, dx. \end{aligned} \tag{B.42}$$

B.4 Proofs for Proposition 1

Proof of Proposition 1:

To prove Proposition 1, we show that a BGE can be characterized by a pair of functions $\{v(\tilde{i}), \mathcal{L}(\tilde{i})\}_{\tilde{i}\in(0,1]}$ and a set of parameters $\{w, \eta, y, g_n\}$. The argument proceeds by verifying that these objects jointly satisfy the following system of conditions:

(i) The ordinary differential equation for $v(\tilde{i})$ together with its boundary condition;

(ii) The ordinary differential equation for $\mathcal{L}(\tilde{i})$ together with its boundary condition;

(iii) The labor market clearing condition;

(iv) The aggregate resource constraint;

(v) The entry condition of final producers;

(vi) The price normalization.

Establishing that $\{v(\tilde{i}), \mathcal{L}(\tilde{i}), w, \eta, y, g_n\}$ satisfy all six conditions completes the proof of the proposition.

Step 1. Derivation of the ODE for $v(\tilde{i})$ and its boundary condition.

We begin by deriving the differential equation satisfied by $v(\tilde{i})$. From the detrended entry condition in equation (B.36),

$$w = v(\tilde{i}) \cdot \psi_e \cdot \frac{b(\tilde{i})}{f(\tilde{i})},$$

taking logs and differentiating with respect to $\tilde{i}$ yields

$$\frac{v'(\tilde{i})}{v(\tilde{i})} = \frac{f'(\tilde{i})}{f(\tilde{i})} - \frac{b'(\tilde{i})}{b(\tilde{i})}.$$

Substituting $\frac{f'(\tilde{i})}{f(\tilde{i})}$ from equation (B.37) and $\frac{b'(\tilde{i})}{b(\tilde{i})}$ from equation (B.41) gives

$$v'(\tilde{i}) = \left[z(\tilde{i}) - z_0\right] \cdot \left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) \frac{v(\tilde{i})}{\tilde{i} g_n}. \tag{B.43}$$

Using equation (B.40), we can rewrite the above equation as

$$v'(\tilde{i}) = \theta \frac{1-\beta}{\beta} \cdot \left[\tfrac{z(\tilde{i})}{z_0} - 1\right] \cdot \frac{v(\tilde{i})}{\tilde{i}}. \tag{B.44}$$

Since $z(\tilde{i})$ satisfies equation (B.34), by denoting $v_0 = \lim_{\tilde{i}\to 0} v(\tilde{i})$, we can derive $\frac{z(\tilde{i})}{z_0} = \left(\frac{v(\tilde{i})}{v_0}\right)^{\frac{\alpha}{1-\alpha}}$. Substituting this equation into equation (B.44), we arrive at

$$v'(\tilde{i}) = \theta \frac{1-\beta}{\beta} \cdot \left[\left(\tfrac{v(\tilde{i})}{v_0}\right)^{\frac{\alpha}{1-\alpha}} - 1\right] \frac{v(\tilde{i})}{\tilde{i}}. \tag{B.45}$$

Next, equations (B.40) and equation (B.34) allow us to eliminate z_0 and express v_0 as a function of g_n:

$$v_0 = h(g_n) = \frac{w\left(\theta\frac{1-\beta}{\beta}\right)^{\frac{1-\alpha}{\alpha}}}{\left[\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right)\psi\right]^{1/\alpha}\alpha} g_n^{\frac{1-\alpha}{\alpha}}.$$

Substituting this expression into equation (B.45) yields

$$v'(\tilde{i}) = \theta \frac{1-\beta}{\beta} \cdot \left[\left(\tfrac{v(\tilde{i})}{h(g_n)}\right)^{\frac{\alpha}{1-\alpha}} - 1\right] \frac{v(\tilde{i})}{\tilde{i}},$$

which corresponds exactly to equation (22) in the proposition.

Finally, we derive the boundary condition. We define $A(i,t)$ as the average firm productivity in sector i at time t:

$$A(i,t) = \left[\frac{\int_0^{F(i,t)} \tilde{A}_i(j,t)^{\theta\frac{1-\beta}{\beta}} dj}{F(i,t)}\right]^{\frac{1}{\theta}\frac{\beta}{1-\beta}}. \tag{B.46}$$

By assumption, every intermediate-good producers in the newest variety $N(t)$ has productivity $\tilde{A}_{N(t)}(j,t) = \zeta\bar{A}(t)$. Substituting this equation into equation (B.46) yields $A(N(t),t) = \zeta\bar{A}(t)$. Defining the detrended average firm productivity in a given sector as $a(\tilde{i}) = e^{-g_n t}A(i,t)$, we obtain $a(1) = \zeta$. It follows from equation (10) that

$$A(i,t)^{\theta\frac{1-\beta}{\beta}} = \frac{B(i,t)}{\bar{A}(t)^{(1-\theta)\frac{1-\beta}{\beta}} F(i,t)},$$

which implies $a(\tilde{i})^{\theta\frac{1-\beta}{\beta}} = \frac{b(\tilde{i})}{f(\tilde{i})}$. Substituting this into equation (17), we obtain

$$w = v(\tilde{i}) \cdot \psi_e \cdot a(\tilde{i})^{\theta\frac{1-\beta}{\beta}}, \tag{B.47}$$

and using $a(1) = \zeta$, it follows that

$$v(1) = \frac{w}{\psi_e \zeta^{\theta\frac{1-\beta}{\beta}}}.$$

This completes the derivation of equation (22) and its boundary condition.

Step 2. Derivation of the ODE for $\mathcal{L}(\tilde{i})$ and its boundary condition.
The detrended HJB equation (B.35) for an incumbent is

$$\begin{aligned}
&\big[\rho + (\sigma-1)(1-\gamma)g_n + g_n + \theta\frac{1-\beta}{\beta}g_n + \delta\big]v(\tilde{i}) \\
&= \beta(1-\beta)^{\frac{1-\beta}{\beta}} \cdot p(\tilde{i})^{\frac{1}{\beta}} \cdot w^{-\frac{1-\beta}{\beta}} - w\left[\frac{w}{\big(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\big)\, v(\tilde{i})\, \psi\alpha}\right]^{\frac{1}{\alpha-1}} \\
&\quad + z(\tilde{i})\big(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\big)\, v(\tilde{i}) - \tilde{i} g_n v'(\tilde{i}).
\end{aligned}$$

Using equation (B.43) from **Step 1** to substitute $v'(\tilde{i})$ and using equation (B.40) to eliminate z_0, the above HJB equation is simplified to

$$\begin{aligned}
\big[\rho + \delta + \big((\sigma-1)(1-\gamma)+1\big)g_n\big]v(\tilde{i}) &= \beta(1-\beta)^{\frac{1-\beta}{\beta}} \cdot p(\tilde{i})^{\frac{1}{\beta}} \cdot w^{-\frac{1-\beta}{\beta}} \\
&\quad - w\left[\frac{w}{\big(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\big)\, v(\tilde{i})\, \psi\alpha}\right]^{\frac{1}{\alpha-1}}.
\end{aligned}$$

Rearranging gives the closed-form pricing function

$$\begin{aligned}
p(\tilde{i}) &= \frac{w^{1-\beta}}{\beta^\beta(1-\beta)^{(1-\beta)}} \cdot \Bigg\{\Big[\rho + \delta + \Big((\sigma-1)(1-\gamma)+1\Big)g_n\Big]v(\tilde{i}) \\
&\quad + w \cdot \left[\frac{w}{\big(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\big)v(\tilde{i}) \cdot \psi\alpha}\right]^{\frac{1}{\alpha-1}}\Bigg\}^{\beta}.
\end{aligned} \tag{B.48}$$

From Lemma B.5, the demand system implies $p(\tilde{i}) = y\mathcal{L}'(\tilde{i})$ for $\tilde{i} \in [\eta, 1]$, so that the expression above directly characterizes the ODE for $\mathcal{L}(\tilde{i})$ in terms of $v(\tilde{i})$ and the model primitives:

$$\begin{aligned}
\mathcal{L}'(\tilde{i}) &= \frac{w^{1-\beta}}{y\beta^\beta(1-\beta)^{(1-\beta)}} \cdot \Bigg\{\Big[\rho + \delta + \Big((\sigma-1)(1-\gamma)+1\Big)g_n\Big]v(\tilde{i})
\end{aligned}$$

$$+w\cdot\left[\frac{w}{\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right)v(\tilde{i})\cdot\psi\alpha}\right]^{\frac{1}{\alpha-1}}\Bigg\}^{\beta}. \tag{B.49}$$

At last, from Lemma B.8, the boundary condition is $\mathcal{L}(\eta)=\underline{l}$. And if $\tilde{i}<\eta$, $\mathcal{L}(\tilde{i})=\underline{l}$, which pines down $\mathcal{L}(\tilde{i})$.

Having solved for $v(\tilde{i})$ and $\mathcal{L}(\tilde{i})$ in **Step 1** and **Step 2**, we can then recover $p(\tilde{i})$ from equation (B.48), $b(\tilde{i})$ from equation (B.38), $v^f(\tilde{i})$ from equation (B.42), $f(\tilde{i})$ from equation (B.36), and $e(\tilde{i})$ from equation (B.37), taking $\{w,\eta,y,g_n\}$ as given. These detrended objects are then directly used in the subsequent steps.

Step 3. Labor market clearing condition.

Labor is demanded by production ($\int_0^{N(t)}\int_0^{F(i,t)}L_i(j,t)djdi$), by incumbent R&D ($\int_0^{N(t)}\int_0^{F(i,t)}L_i^I djdi$), by entry R&D ($L^E(t)$), by luxury service ($L^X(t)$), and by idea production ($L^X(t)$). So the labor market clearing condition is

$$1=\int_0^{N(t)}\left[\int_0^{F(i,t)}L_i(j,t)+L_i^I(j,t)dj\right]di+L^E(t)+L^X(t)+L^N(t).$$

Specifically, along the BGP, this condition can be re-expressed as

$$1\;=\;\int_0^1\Big[l(\tilde{i})+l^I(\tilde{i})\Big]d\tilde{i}\;+\;L^E\;+\;L^X\;+\;L^N, \tag{B.50}$$

where:

(i) $l(\tilde{i})\equiv e^{g_n t}L(i,t)$ is defined as the detrended labor demand in final-good sector $\tilde{i}$. The expression for $L(i,t)=\int_0^{F(i,t)}L_i(j,t)dj$ is given in equation (12), with a detrended version

$$l(\tilde{i})=\left[\frac{w}{(1-\beta)p(\tilde{i})}\right]^{-\frac{1}{\beta}}b(\tilde{i}). \tag{B.51}$$

(ii) $l^I(\tilde{i})\equiv e^{g_n t}\int_0^{F(i,t)}L_i^I(j,t)dj$ is defined as the incumbents' R&D demand in final-good sector $\tilde{i}$. The expression for $L_i^I(j,t)$ is given in equation (B.9), which leads to

$$l^I(\tilde{i})=\left[\frac{w}{\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right)v(\tilde{i})\psi\alpha}\right]^{\frac{1}{\alpha-1}}\cdot b(\tilde{i}). \tag{B.52}$$

(iii) L^E is the entrants' R&D labor demand defined in equation (B.15). Expressing it in detrended terms yields

$$L^E=\int_0^1\frac{e(\tilde{i})}{\psi_e}d\tilde{i}+\frac{f(1)g_n}{\psi_e}. \tag{B.53}$$

(iv) L^X denotes labor demand in the luxury service sector, which is pinned down by the luxury service market clearing condition. As we have shown in Lemma B.7, the aggregate demand for luxury service is

$$\frac{\int_{\mathcal{L}(1)}^{\infty}\big(l-\mathcal{L}(1)\big)Y(t)d\mathcal{G}(l)}{P^X(t)},$$

so we have

$$\begin{aligned} L^X &= \frac{\int_{\mathcal{L}(1)}^{\infty}\big(l-\mathcal{L}(1)\big)Y(t)d\mathcal{G}(l)}{\bar{A}(t)^{1-\gamma}P^X(t)} \\ &= \frac{\int_{\mathcal{L}(1)}^{\infty}\big(l-\mathcal{L}(1)\big)Y(t)d\mathcal{G}(l)}{W(t)} \\ &= \frac{y\int_{\mathcal{L}(1)}^{\infty}\big(l-\mathcal{L}(1)\big)d\mathcal{G}(l)}{w}, \end{aligned} \tag{B.54}$$

where the second equality follows from the expression for $P^X(t)$ in equation (B.29).

(v) L^N is the labor demand in the idea production sector. From equation (20), we have

$$L^N = \frac{g_n}{\phi}. \tag{B.55}$$

In equilibrium, equation (B.50) pins down the wage w: the left–hand side is total labor supply (normalized to one), while the right–hand side is total labor demand expressed in detrended variables.

Step 4. Aggregate resource constraint.

The aggregate resource constraint is

$$\underbrace{Y(t)}_{\text{aggregate output}} = W(t)\underbrace{\left[\int_0^{N(t)}L(i,t)di+L^X(t)\right]}_{\text{total production labor}}+\underbrace{\int_0^{N(t)}\left[\Pi(i,t)+\int_0^{F(i,t)}\Pi_i(j,t)dj\right]di}_{\text{total production profit}}, \tag{B.56}$$

which states that aggregate output must equal the sum of labor income from production and profits. From the equilibrium condition of the luxury service producer, we have

$$X(t)P^X(t)=W(t)L^X(t).$$

Then equation (B.56) becomes

$$\int_0^{N(t)}P(i,t)Y(i,t)di=W(t)\int_0^{N(t)}L(i,t)di+\int_0^{N(t)}\left[\Pi(i,t)+\int_0^{F(i,t)}\Pi_i(j,t)dj\right]di. \tag{B.57}$$

In Appendix B.2.1, we have shown that

$$\frac{\Pi(i,t)}{P(i,t)Y(i,t)} = \beta \quad \text{and} \quad \frac{\int_0^{F(i,t)} \Pi_i(j,t)dj}{P(i,t)Y(i,t)} = \beta(1-\beta).$$

Using these two equations in (B.57), we derive

$$(1-\beta)^2 \int_0^{N(t)} P(i,t)Y(i,t)di = W(t) \int_0^{N(t)} L(i,t)di. \tag{B.58}$$

Using the market-clearing condition of the final-goods producer, we derive

$$\begin{aligned}
\int_0^{N(t)} P(i,t)Y(i,t)di &= \int_{\underline{l}}^{\infty} \left[\int_0^{N(t)} P(i,t)C_l(i,t)\, di \right] d\mathcal{G}(l). \\
&= \int_{\underline{l}}^{\bar{l}} \int_0^{N(t)\mathcal{L}^{-1}(l)} P(i,t)didG(l) + \int_{\bar{l}}^{\infty} \int_0^{N(t)} P(i,t)didG(l) \\
&= \int_{\underline{l}}^{\mathcal{L}(1)} Y(t)ld\mathcal{G}(l) + \int_{\mathcal{L}(1)}^{\infty} Y(t)\mathcal{L}(1)d\mathcal{G}(l) \\
&= Y(t) \left[1 - \int_{\mathcal{L}(1)}^{\infty} \big(l - \mathcal{L}(1)\big)\, d\mathcal{G}(l) \right],
\end{aligned}$$

where we have used Lemma B.7 and equation (B.28). Substituting the last expression into equation (B.58), we have the detrended version of the resource constraint:

$$\begin{aligned}
(1-\beta)^2 Y(t) \left[1 - \int_{\mathcal{L}(1)}^{\infty} \big(l - \mathcal{L}(1)\big)\, d\mathcal{G}(l) \right] &= W(t) \int_0^{N(t)} L(i,t)di \\
\Rightarrow y(1-\beta)^2 \left[1 - \int_{\mathcal{L}(1)}^{\infty} \big(l - \mathcal{L}(1)\big)\, d\mathcal{G}(l) \right] &= w \int_0^1 l(\tilde{i})\, d\tilde{i}.
\end{aligned} \tag{B.59}$$

Step 5. Entry condition of intermediate goods producers.

The inventor's optimization problem in equation (21) implies

$$w = v^f(1)\phi, \tag{B.60}$$

where $v^f(\tilde{i})$ denotes the detrended value of the newest final–good firm, defined in equation (B.42).

Step 6. Price normalization.

First, we show that the price normalization implies $C(t) = Y(t)$. We define the price of final consumption bundle as $P^C(t)$, which satisfies

$$P^C(t)C(t) = \int_{\underline{l}}^{\infty} \left[\int_0^{N(t)} P(i,t)C_l(i,t)\, di + P^X(t)X_l(t) \right] d\mathcal{G}(l). \tag{B.61}$$

Since this price is normalized to one, i.e. $P^C(t) = 1$, we have

$$\begin{aligned} C(t) &= \int_{\underline{l}}^{\infty} \left[\int_0^{N(t)} P(i,t) C_l(i,t)\, di + P^X(t) X_l(t) \right] d\mathcal{G}(l) \\ &= \int_{\underline{l}}^{\infty} \left[\int_0^{N(t)} P(i,t) C_l(i,t)\, di \right] d\mathcal{G}(l) + \int_{\bar{l}}^{\infty} P^X(t) X_l(t)\, d\mathcal{G}(l). \end{aligned} \tag{B.62}$$

The market-clearing condition for the luxury services implies

$$\int_{\bar{l}}^{\infty} P^X(t) X_l(t)\, d\mathcal{G}(l) = P^X(t) X(t),$$

and the market-clearing condition for the final goods implies

$$\int_0^{N(t)} P(i,t) Y(i,t) di = \int_{\underline{l}}^{\infty} \left[\int_0^{N(t)} P(i,t) C_l(i,t)\, di \right] d\mathcal{G}(l). \tag{B.63}$$

So equation (B.62) simplifies to $C(t) = \int_0^{N(t)} P(i,t) Y(i,t) di + P^X(t) X(t) = Y(t)$.

Next, we use $C(t) = Y(t)$ to rewrite $Y(t)$ as

$$\begin{aligned} Y(t) &= C(t) = \int_{\underline{l}}^{\infty} C_l(t) d\mathcal{G}(l) \\ &= \int_{\underline{l}}^{\infty} \left[\int_0^{N(t)} i^{-\gamma} C_l(i,t) di + \min_{\{i \in (0, N_t]\}} \{(C_l(i,t))\} \cdot X_l(t) \right] d\mathcal{G}(l). \end{aligned} \tag{B.64}$$

The market-clearing condition for the luxury service implies that

$$\int_{\underline{l}}^{\infty} \min_{\{i \in (0, N_t]\}} \{(C_l(i,t))\} \cdot X_l(t) d\mathcal{G}(l) = X(t) = \bar{A}(t)^{1-\gamma} L^X(t),$$

and it follows from Lemma B.7 that

$$\begin{aligned} \int_{\underline{l}}^{\infty} \left[\int_0^{N(t)} i^{-\gamma} C_l(i,t) di \right] d\mathcal{G}(l) &= \int_{\underline{l}}^{\mathcal{L}(1)} \left[\int_0^{\mathcal{L}^{-1}(l) N(t)} i^{-\gamma} di \right] d\mathcal{G}(l) + \int_{\mathcal{L}(1)}^{\infty} \left[\int_0^{N(t)} i^{-\gamma} di \right] d\mathcal{G}(l) \\ &= \int_{\underline{l}}^{\mathcal{L}(1)} \frac{[\mathcal{L}^{-1}(l) N(t)]^{1-\gamma}}{1-\gamma} d\mathcal{G}(l) + \int_{\mathcal{L}(1)}^{\infty} \frac{N(t)^{1-\gamma}}{1-\gamma} d\mathcal{G}(l). \end{aligned}$$

Substituting these two relations into equation (B.64) yields

$$Y(t) = \int_{\underline{l}}^{\mathcal{L}(1)} \frac{[\mathcal{L}^{-1}(l) N(t)]^{1-\gamma}}{1-\gamma} d\mathcal{G}(l) + \int_{\mathcal{L}(1)}^{\infty} \frac{N(t)^{1-\gamma}}{1-\gamma} d\mathcal{G}(l) + \bar{A}(t)^{1-\gamma} L^X(t). \tag{B.65}$$

The detrended version of equation (B.65) is

$$y = \int_{\underline{l}}^{\mathcal{L}(1)} \frac{[\mathcal{L}^{-1}(l)]^{1-\gamma}}{1-\gamma} d\mathcal{G}(l) + \int_{\mathcal{L}(1)}^{\infty} \frac{1}{1-\gamma} d\mathcal{G}(l) + L^X. \tag{B.66}$$

Finally, we verify Conjecture 1 proposed in Appendix B.3.1, which states that $P_l(t)$ is constant for each time t and for all households. For households with $l < \mathcal{L}(1) = \bar{l}$, we have

$$P_l(t) = \frac{\int_0^{N(t)} P(i,t)C_l(i,t)\,di}{\int_0^{N(t)} i^{-\gamma}C_l(i,t)\,di} = \frac{\int_0^{\mathcal{L}^{-1}(l)} p(\tilde{i})\,d\tilde{i}}{\int_0^{\mathcal{L}^{-1}(l)} \tilde{i}^{-\gamma}\,d\tilde{i}}. \tag{B.67}$$

For households with $l > \mathcal{L}(1) = \bar{l}$, we have

$$P_l(t) = \frac{\int_0^{N(t)} P(i,t)C_l(i,t)\,di + P^X(t)X_l(t)}{\int_0^{N(t)} i^{-\gamma}C_l(i,t)\,di + X_l(t)} = \frac{\int_0^1 p(\tilde{i})\,d\tilde{i} + y(l-\bar{l})}{\int_0^1 \tilde{i}^{-\gamma}\,d\tilde{i} + \frac{y(l-\bar{l})}{w}}, \tag{B.68}$$

where the second equality follows from the expression for $X_l(t)$ in equation (B.22) and the expression for $P^X(t)$ in equation (B.29). ■

B.5 Proofs for Proposition 2

Before formally proving the proposition, we first establish the condition of ζ, which is summarized in the following assumption and lemma.

Assumption 2 *We assume ζ is relatively small, such as*

$$\zeta < \left\{ \frac{\left[\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right)\psi\right]^{1/\alpha}\alpha}{\psi_e\left(\theta\frac{1-\beta}{\beta}\right)^{\frac{1-\alpha}{\alpha}} g_n^{\frac{1-\alpha}{\alpha}}} \right\}^{\frac{1}{\theta}\frac{\beta}{1-\beta}}. \tag{B.69}$$

Lemma B.11 *Under the Assumption 2, the average firm productivity in a given sector $a(\tilde{i})$ satisfies*

$$a(1) < a(0), \text{ and } v(1) > v_0 \equiv \lim_{\tilde{i}\to 0} v(\tilde{i}),$$

where $a(\tilde{i})$ is defined as the detrended version of equation (B.46), while $a(0) \equiv \lim_{\tilde{i}\to 0} a(\tilde{i})$.

Proof of Lemma B.11:

Using the expressions for $v(1)$ and v_0 provided in Proposition 1, we obtain

$$\frac{v(1)}{v_0} = \frac{\left[\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right)\psi\right]^{1/\alpha}\alpha}{\psi_e\left(\theta\frac{1-\beta}{\beta}\right)^{\frac{1-\alpha}{\alpha}} g_n^{\frac{1-\alpha}{\alpha}} \zeta^{\theta\frac{1-\beta}{\beta}}} > 1.$$

Then, from equation (B.47),

$$\frac{a(1)}{a(0)} = \left[\frac{v_0}{v(1)}\right]^{\frac{1}{\theta}\frac{\beta}{1-\beta}} < 1.$$

■

Proof of Proposition 2:

We begin by establishing the monotonicity of $v(\tilde{i})$, then we show the monotonicity of $z(\tilde{i})$, $p(\tilde{i})$, and $b(\tilde{i})$, respectively.

Step 1. Monotonicity of $v(\tilde{i})$.

We first show that $\lim_{\tilde{i}\to 0} v'(\tilde{i}) = 0$, and then establish that $v'(\tilde{i}) > 0$ for all $\tilde{i} \in (0,1]$. From Proposition 1, $v(\tilde{i})$ satisfies the ODE

$$v'(\tilde{i}) = \theta \frac{1-\beta}{\beta} \left[\left(\frac{v(\tilde{i})}{v_0} \right)^{\frac{\alpha}{1-\alpha}} - 1 \right] \frac{v(\tilde{i})}{\tilde{i}}, \qquad \tilde{i} \in (0,1]. \tag{B.70}$$

To compute $\lim_{\tilde{i}\to 0} v'(\tilde{i})$, we apply L'Hospital's rule:

$$\begin{aligned} \lim_{\tilde{i}\to 0} v'(\tilde{i}) &= \theta \frac{1-\beta}{\beta} \cdot \lim_{\tilde{i}\to 0} \frac{\alpha}{1-\alpha} \left(\frac{v(\tilde{i})}{v_0} \right)^{\frac{\alpha}{1-\alpha}} v'(\tilde{i}) \\ &= \theta \frac{1-\beta}{\beta} \cdot \frac{\alpha}{1-\alpha} \lim_{\tilde{i}\to 0} v'(\tilde{i}). \end{aligned}$$

Since $\theta \frac{1-\beta}{\beta} \frac{\alpha}{1-\alpha} \neq 1$ in general, it must be that $\lim_{\tilde{i}\to 0} v'(\tilde{i}) = 0$.

Given this condition, the right-hand side of equation (B.70) is bounded and satisfies Lipschitz condition; therefore, the solution to the ODE is unique. Now we prove v is monotonically increasing by contradiction. Suppose there exists $k \in (0,1]$ such that $v'(k) \leq 0$. Two cases arise:

- If $v'(k) = 0$, then $v(\tilde{i}) = v_0$ solves equation (B.70). By uniqueness of solutions, this would imply $v(1) = v_0$, which is contradicted with Lemma B.11.
- If $v'(k) < 0$, note that $v'(\tilde{i}) > 0$ if and only if $v(\tilde{i}) > v_0$. Thus $v(k) < v_0$. Since $v(1) > v_0$ and $v(\tilde{i})$ is continuous, there must exist some $k' \in (k,1)$ with $v(k') = v_0$, which implies $v'(k') = 0$, still contradicted.

Therefore $v'(\tilde{i}) > 0$ for all $\tilde{i} \in (0,1]$, i.e. $v(\tilde{i})$ is strictly increasing.

Step 2. Monotonicity of other variables.

- *Innovation intensity $z(\tilde{i})$ and price $p(\tilde{i})$:* Equation (B.34) and equation (B.48)

$$\begin{aligned} z(\tilde{i}) &= \psi \left[\frac{w}{\left(\lambda^{\theta \frac{1-\beta}{\beta}} - 1 \right) v(\tilde{i}) \cdot \psi \alpha} \right]^{\frac{\alpha}{\alpha-1}} \\ p(\tilde{i}) &= \frac{w^{1-\beta}}{\beta^{\beta}(1-\beta)^{(1-\beta)}} \cdot \left\{ \left[\rho + \delta + \Big((\sigma-1)(1-\gamma) + 1 \Big) g_n \right] v(\tilde{i}) \right. \end{aligned}$$

$$+w \cdot \left[\frac{w}{\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) v(\tilde{i}) \cdot \psi\alpha} \right]^{\frac{1}{\alpha-1}} \Bigg\}^{\beta}.$$

implies that $z(\tilde{i})$ and $p(\tilde{i})$ increase with $v(\tilde{i})$. Given $v(\tilde{i})$ is strictly increasing, so are the two functions.

- *Sectoral productivity* $b(\tilde{i})$*:* For $\tilde{i} \in [\eta, 1]$, from equation (B.38),

$$\begin{aligned} b(\tilde{i}) &= \frac{w^{\frac{1-\beta}{\beta}} p(\tilde{i})^{-\frac{1-\beta}{\beta}} (1 - \mathcal{G}(\mathcal{L}(\tilde{i})))}{(1-\beta)^{\frac{1-2\beta}{\beta}}} \\ \Rightarrow b'(\tilde{i}) &= \frac{w^{\frac{1-\beta}{\beta}}}{(1-\beta)^{\frac{1-2\beta}{\beta}}} \cdot \left[-\frac{1-\beta}{\beta} p(\tilde{i})^{-\frac{1}{\beta}} p'(\tilde{i})(1 - \mathcal{G}(\mathcal{L}(\tilde{i}))) - \mathcal{G}'(\mathcal{L}(\tilde{i}))\mathcal{L}'(\tilde{i}) \right] < 0. \end{aligned}$$

The last inequality comes from the facts that $\mathcal{L}' > 0$. For $\tilde{i} \in (0, \eta)$, from equation (B.39),

$$b(\tilde{i}) = \frac{w^{\frac{1-\beta}{\beta}} p(\tilde{i})^{-\frac{1-\beta}{\beta}}}{(1-\beta)^{\frac{1-2\beta}{\beta}}},$$

which implies that $b(\tilde{i})$ is strictly decreasing give that $p(\tilde{i})$ is strictly increasing.

■

Finally, we verify Conjecture 2 proposed in Appendix B.3.1. In particular, along the BGP, the equilibrium price can be written as

$$P(i,t) = e^{-\gamma g_n t} p(\tilde{i}), \qquad \tilde{i} = \frac{i}{N(t)}.$$

Fix any date t. Since $p(\tilde{i})$ is strictly increasing in $\tilde{i}$, it follows immediately that $P(i,t)$ is strictly increasing in i. Hence, Conjecture 2 holds in equilibrium.

B.6 Proofs for Proposition 3

Proof of Proposition 3:

For notational convenience, we use superscripts 1 and 2 to denote variables corresponding to the equilibrium outcomes under the income distributions $\mathcal{G}^1$ and $\mathcal{G}^2$, respectively.

Step 1. Show that $\frac{z^2(\tilde{i})}{z^1(\tilde{i})}$ is strictly increasing in $\tilde{i}$.

From equation (B.34), we can express

$$\frac{z^2(\tilde{i})}{z^1(\tilde{i})} = \left[\frac{v^2(\tilde{i})}{v^1(\tilde{i})} \right]^{\frac{\alpha}{1-\alpha}} \cdot \left(\frac{w^1}{w^2} \right)^{\frac{\alpha}{1-\alpha}}$$

$$= \left[\frac{\hat{v}^2(\tilde{i})}{\hat{v}^1(\tilde{i})}\right]^{\frac{\alpha}{1-\alpha}} \cdot \left(\frac{v_0^2}{v_0^1}\right)^{\frac{\alpha}{1-\alpha}} \cdot \left(\frac{w^1}{w^2}\right)^{\frac{\alpha}{1-\alpha}},$$

where we use $\hat{v}(\tilde{i}) = v(\tilde{i})/v_0$ to represent the value of $v(\tilde{i})$ relative to v_0. Hence, $\frac{z^2(\tilde{i})}{z^1(\tilde{i})}$ is strictly increasing in $\tilde{i}$ if and only if $\frac{\hat{v}^2(\tilde{i})}{\hat{v}^1(\tilde{i})}$ is strictly increasing in $\tilde{i}$.

According to equation (B.70), $\hat{v}(\tilde{i})$ satisfies the ODE

$$\hat{v}'(\tilde{i}) = \theta\frac{1-\beta}{\beta} \cdot \frac{\left[\hat{v}(\tilde{i})^{\frac{\alpha}{1-\alpha}} - 1\right]\hat{v}(\tilde{i})}{\tilde{i}}, \tag{B.71}$$

with boundary condition

$$\hat{v}(1) = \frac{v(1)}{v_0} = \frac{\left[(\lambda^{\theta\frac{1-\beta}{\beta}} - 1)\,\psi\right]^{1/\alpha}\alpha}{\psi_e\left(\theta\frac{1-\beta}{\beta}\right)^{\frac{1-\alpha}{\alpha}} g_n^{\frac{1-\alpha}{\alpha}} \zeta^{\theta\frac{1-\beta}{\beta}}}.$$

Since $\hat{v}(1)$ is decreasing in g_n and $g_n^1 > g_n^2$, it follows that $\hat{v}^2(1) > \hat{v}^1(1)$. Moreover, by uniqueness and continuity of the ODE solution, and $\lim_{\tilde{i}\to 0}\hat{v}^1(\tilde{i}) = \lim_{\tilde{i}\to 0}\hat{v}^2(\tilde{i}) = 1$, we obtain

$$\hat{v}^2(\tilde{i}) > \hat{v}^1(\tilde{i}) > 1 \qquad \text{for all } 0 < \tilde{i} \leq 1.$$

Finally, from equation (B.71),

$$\frac{\hat{v}^{2\prime}(\tilde{i})}{\hat{v}^{1\prime}(\tilde{i})} = \frac{\left[\hat{v}^2(\tilde{i})^{\frac{\alpha}{1-\alpha}} - 1\right]\hat{v}^2(\tilde{i})}{\left[\hat{v}^1(\tilde{i})^{\frac{\alpha}{1-\alpha}} - 1\right]\hat{v}^1(\tilde{i})} > \frac{\hat{v}^2(\tilde{i})}{\hat{v}^1(\tilde{i})},$$

which implies

$$\frac{\mathrm{d}}{\mathrm{d}\tilde{i}}\left(\frac{\hat{v}^2(\tilde{i})}{\hat{v}^1(\tilde{i})}\right) = \frac{\hat{v}^{2\prime}(\tilde{i})\hat{v}^1(\tilde{i}) - \hat{v}^{1\prime}(\tilde{i})\hat{v}^2(\tilde{i})}{\left[\hat{v}^1(\tilde{i})\right]^2} > 0.$$

Step 2. Show that $\frac{z^2(\tilde{i})}{z^1(\tilde{i})} \leq 1$.

Observe that

$$z(1) = \psi\left[\frac{w}{(\lambda^{\theta\frac{1-\beta}{\beta}} - 1)v(1)\cdot\psi\alpha}\right]^{\frac{\alpha}{\alpha-1}} = \psi\left[\frac{\psi_e\,\zeta^{\theta\frac{1-\beta}{\beta}}}{(\lambda^{\theta\frac{1-\beta}{\beta}} - 1)\psi\alpha}\right]^{\frac{\alpha}{\alpha-1}},$$

which is independent of $\mathcal{G}$. Hence $z^2(1) = z^1(1)$. Since $\frac{z^2(\tilde{i})}{z^1(\tilde{i})}$ is strictly increasing in $\tilde{i}$, it follows that

$$\frac{z^2(\tilde{i})}{z^1(\tilde{i})} \leq 1 \quad \text{for all } \tilde{i}.$$

Combining Steps 1 and 2, we conclude that if $g_n^1 > g_n^2$, then

$$\log(z^1(\tilde{i})) - \log(z^2(\tilde{i})) \geq 0 \quad \text{for all } \tilde{i},$$

and this difference is strictly decreasing in $\tilde{i}$. ■

B.7 Proofs for Proposition 4

Proof of Proposition 4:

The result follows from Appendix B.3.3 and the proof of Lemma B.10. Here, we provide an alternative derivation from the perspective of sectoral labor demand.

The key observation is that growth in $N(t)$ reduces the amount of labor allocated to each sector, because the expansion of the product space spreads the aggregate labor force across an increasing number of activities. Combining equations (11) and (12) yields the labor demand

$$L(i,t) = \frac{(1-\beta)^2 Y(i,t) P(i,t)}{W(t)}.$$

For any fixed i, as $t \to \infty$, sector i becomes an basic necessity sector, and its output converges to 1. Moreover, Lemma B.4 shows that, for any fixed i, the asymptotic growth rate of $P(i,t)$ is $-\gamma g_n$. Taking into account the corresponding growth rate of the $W(t)$, it follows that $L(i,t)$ declines asymptotically at a rate g_n.

The decline in production labor implies that sectoral productivity $B(i,t)$ must grow sufficiently rapidly to maintain the constant output level of necessity goods. Indeed, equations (11) and (12) imply

$$Y(i,t) = \frac{B(i,t)^\beta}{1-\beta} L(i,t)^{1-\beta}.$$

Since $Y(i,t) \to 1$ and $L(i,t)$ declines at rate g_n, $B(i,t)$ must grow asymptotically at rate $\frac{1-\beta}{\beta} g_n$, which is the same asymptotic relationship established in Lemma B.10.

By equation (19), and because the number of firms $F(i,t)$ is asymptotically stationary, we therefore obtain

$$g_n = \lim_{t\to\infty} \frac{1}{\theta} \frac{\beta}{1-\beta} z_i(t) \left(\lambda^{\theta \frac{1-\beta}{\beta}} - 1 \right).$$

Finally, since $\lim_{t\to\infty} z_i(t) = \lim_{\tilde{i}\to 0} z(\tilde{i})$, it follows that

$$g_n = \lim_{\tilde{i}\to 0} \frac{1}{\theta} \frac{\beta}{1-\beta} z(\tilde{i}) \left(\lambda^{\theta \frac{1-\beta}{\beta}} - 1 \right).$$

■

To illustrate the demand mechanism depicted in Figure 6, we consider a static partial equilibrium at a fixed date t, holding $W(t)$, and $Y(t)$ fixed. We then examine how a productivity change in a given sector affects output and prices throughout the product hierarchy. This exercise provides the demand-side intuition behind Proposition 4.

Define cumulative expenditure relative to income $E(i,t) \equiv \frac{1}{Y(t)} \int_0^i P(k,t)\, dk$. Combining sequential consumption rules and the sectoral supply from equation (11), we have a single differential equation for $E(i,t)$ that pins down the static partial equilibrium:

$$\frac{\partial E(i,t)}{\partial i} = \frac{W(t)}{Y(t)} \left[\frac{1 - \mathcal{G}(\max\{\underline{l}, E(i,t)\})}{(1-\beta)^{\frac{1-2\beta}{\beta}} B(i,t)} \right]^{\frac{\beta}{1-\beta}}, \qquad E(0,t) = 0. \tag{B.72}$$

We now study how this static equilibrium responds to a sectoral productivity improvement. For an arbitrary perturbation $h(\cdot)$, consider $\log B(j,t) \mapsto \log B(j,t) + \varepsilon h(j)$, and let $E(i,t;\varepsilon)$ denote the corresponding solution to equation (B.72). We define the functional derivative by

$$\left.\frac{dE(i,t;\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} = \int_0^{N(t)} \frac{\delta E(i,t)}{\delta \log B(s,t)} h(s)\, ds. \tag{B.73}$$

From equation (B.72), we have

$$\begin{aligned} \left.\frac{dE(i,t;\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} = & -\frac{\beta}{1-\beta} \frac{W(t)}{Y(t)} (1-\beta)^{-\frac{1-2\beta}{1-\beta}} \\ & \times [1 - \mathcal{G}(\max\{\underline{l}, E(i,t)\})]^{\frac{\beta}{1-\beta}} \int_0^i B(s,t)^{-\frac{\beta}{1-\beta}} h(s)\, ds. \end{aligned} \tag{B.74}$$

Comparing equations (B.73) and (B.74) yields

$$\frac{\delta E(i,t)}{\delta \log B(s,t)} = \begin{cases} 0, & i < s, \\ -\frac{\beta}{1-\beta} \frac{W(t)}{Y(t)} (1-\beta)^{-\frac{1-2\beta}{1-\beta}} B(s,t)^{-\frac{\beta}{1-\beta}} [1 - \mathcal{G}(\max\{\underline{l}, E(i,t)\})]^{\frac{\beta}{1-\beta}}, & i > s. \end{cases} \tag{B.75}$$

Thus, higher productivity in sector s leaves the expenditure thresholds of lower-ranked products unchanged, but lowers the thresholds of all higher-ranked products.

For a higher-ranked product with responsive demand, $E(i,t) > \underline{l}$, equation (B.75) implies

$$\frac{\delta Y(i,t)}{\delta \log B(s,t)} = -\mathcal{G}'(E(i,t)) \frac{\delta E(i,t)}{\delta \log B(s,t)} > 0, \qquad i > s. \tag{B.76}$$

Its equilibrium price also rises:

$$\frac{\delta \log P(i,t)}{\delta \log B(s,t)} = -\frac{\beta}{1-\beta} \frac{\mathcal{G}'(E(i,t))}{1 - \mathcal{G}(E(i,t))} \frac{\delta E(i,t)}{\delta \log B(s,t)} > 0, \qquad i > s. \tag{B.77}$$

For products ranked below the shocked sector,

$$\frac{\delta Y(i,t)}{\delta \log B(s,t)} = \frac{\delta P(i,t)}{\delta \log B(s,t)} = 0, \qquad i < s. \tag{B.78}$$

The productivity improvement therefore propagates only upward along the product hierarchy. A necessity-sector improvement affects a broad range of higher-ranked products, whereas a luxury-sector improvement affects only the smaller set of products ranked above that luxury. This is the asymmetric demand spillover illustrated in Figure 6.

B.8 Proofs for Proposition 5

Proof of Proposition 5:

The proof proceeds in four steps. First, we derive closed-form solutions for $v(\tilde{i})$, $z(\tilde{i})$, and $p(\tilde{i})$. Second, we derive the two conditions that determine the equilibrium. Third, at $\rho = 0$, we reduce these conditions to a single equation in g_n and show that a strict mean-preserving spread lowers g_n. Finally, we extend the result to sufficiently small positive values of ρ.

Step 1. Solve for $v(\tilde{i})$, $z(\tilde{i})$ and $p(\tilde{i})$.

Under equation (25), equation (B.71) reduces to $\tilde{i}\hat{v}'(\tilde{i}) = \hat{v}(\tilde{i})[\hat{v}(\tilde{i})-1]$. Its solution is $\hat{v}(\tilde{i}) = \frac{1}{1-D\tilde{i}}$ for some constant D. To solve for D, we use the boundary condition at $\hat{v}(1)$. Equations (B.5) and (B.40) imply $v_0 = \frac{2wg_n}{(\lambda-1)^2\psi^2}$, and Proposition 1 gives $v(1) = w/(\psi_e\zeta)$, hence $\hat{v}(1) = \frac{v(1)}{v_0} = \frac{(\lambda-1)^2\psi^2}{2\psi_e\zeta g_n} = \frac{\bar{g}}{g_n}$. The restriction of ζ in Assumption 2 (see equation (B.69)) guarantees $0 < g_n < \bar{g}$, which leads to $D = 1 - \frac{g_n}{\bar{g}} > 0$. Then, we have the following closed-form solutions in Section 4.4:

$$v(\tilde{i}) = \frac{2wg_n}{(\lambda-1)^2\psi^2}\frac{1}{1-(1-g_n/\bar{g})\,\tilde{i}}, \qquad z(\tilde{i}) = \frac{g_n}{\lambda-1}\frac{1}{1-(1-g_n/\bar{g})\,\tilde{i}}.$$

Substituting these results into the incumbent HJB equation (B.35) gives the price schedule

$$\frac{p(\tilde{i})}{w} = \left\{\frac{2g_n}{\beta(1-\beta)^{\frac{1-\beta}{\beta}}(\lambda-1)^2\psi^2}\left[(\rho+\delta+g_n)\hat{v}(\tilde{i}) + \frac{g_n}{2}\hat{v}(\tilde{i})^2\right]\right\}^{\beta}. \tag{B.79}$$

Step 2. Derive the equilibrium conditions.

Let $\mu \equiv y/w$. The equilibrium values of g_n and μ are determined by two conditions: the no-arbitrage condition for new-product invention and aggregate labor-market clearing.

New-product invention. Sequential consumption rule gives $Y(\tilde{i}) = 1 - \mathcal{G}\left(\frac{1}{\mu}\int_0^{\tilde{i}}\frac{p(s)}{w}\,ds\right)$. Under $\sigma = 1$, evaluating equation (B.42) at $\tilde{i} = 1$ and using equation (B.38) gives $\frac{v^f(1)}{w} = \frac{\beta}{g_n}\int_0^1 \tilde{i}^{\rho/g_n}\frac{p(\tilde{i})}{w}Y(\tilde{i})d\tilde{i}$. The inventor's condition (B.60) becomes

$$\mathcal{F}(g_n,\mu) \equiv g_n - \phi\beta\int_0^1 \tilde{i}^{\rho/g_n}\frac{p(\tilde{i})}{w}\left\{1-\mathcal{G}\left[\frac{1}{\mu}\int_0^{\tilde{i}}\frac{p(s)}{w}\,ds\right]\right\}d\tilde{i} = 0. \tag{B.80}$$

Labor-market clearing condition. Define the wage-normalized price of the complete product bundle as $P(g_n,\rho)\equiv\int_0^1\frac{p(\tilde{i})}{w}\,d\tilde{i}$. Sequential consumption rule implies that aggregate expenditure on final goods, relative to the wage, is

$$E(g_n,\mu)\equiv\int_0^1\frac{p(\tilde{i})}{w}Y(\tilde{i})\,d\tilde{i}=\mu M\left(\frac{P(g_n,\rho)}{\mu}\right), \tag{B.81}$$

where $M(x)\equiv\int_0^\infty\min\{l,x\}d\mathcal{G}(l)=\int_0^x[1-\mathcal{G}(l)]dl$. Total labor demand consists of production labor, incumbent R&D labor, entry labor, luxury-service labor, and new-product invention labor. We derive these five components separately. First, equations (B.51) and (B.38) imply $l(\tilde{i})=(1-\beta)^2\frac{p(\tilde{i})}{w}Y(\tilde{i})$. Hence aggregate production labor is

$$L^P=(1-\beta)^2E(g_n,\mu). \tag{B.82}$$

Second, under equation (25), equation (B.52) gives $l^I(\tilde{i})=[z(\tilde{i})/\psi]^2b(\tilde{i})$. Moreover, equations (B.43) and (B.40) imply $\tilde{i}g_nv'(\tilde{i})=z(\tilde{i})(\lambda-1)v(\tilde{i})-g_nv(\tilde{i})$. Substituting these two expressions into the incumbent HJB equation (B.35), and using equation (B.38), yields $l^I(\tilde{i})+(\rho+\delta+g_n)\frac{v(\tilde{i})}{w}b(\tilde{i})=\beta(1-\beta)\frac{p(\tilde{i})}{w}Y(\tilde{i})$. Integrating over sectors gives

$$L^I=\beta(1-\beta)E(g_n,\mu)-(\rho+\delta+g_n)\int_0^1\frac{v(\tilde{i})}{w}b(\tilde{i})\,d\tilde{i}. \tag{B.83}$$

Third, equations (B.53) and (B.37) imply

$$L^E=\frac{1}{\psi_e}\int_0^1\left[-\tilde{i}g_nf'(\tilde{i})+\delta f(\tilde{i})\right]d\tilde{i}+\frac{g_nf(1)}{\psi_e}=\frac{g_n+\delta}{\psi_e}\int_0^1f(\tilde{i})\,d\tilde{i}=(g_n+\delta)\int_0^1\frac{v(\tilde{i})}{w}b(\tilde{i})\,d\tilde{i}, \tag{B.84}$$

where the second equality follows from integration by parts and the third uses the free-entry condition (B.36). Finally, the cutoff income for purchasing the complete product bundle is P/μ. Equation (B.54) and the normalization $\int_0^\infty l\,d\mathcal{G}(l)=1$ therefore give

$$L^X=\mu-E(g_n,\mu), \tag{B.85}$$

while equation (B.55) gives

$$L^N=\frac{g_n}{\phi}. \tag{B.86}$$

Substituting equations (B.82), (B.83), (B.84), (B.85), (B.86) into $1=L^P+L^I+L^E+L^X+L^N$ gives

$$\mathcal{H}(g_n,\mu)\equiv\mu-1-\beta E(g_n,\mu)+\frac{g_n}{\phi}-\rho\int_0^1\frac{v(\tilde{i})}{w}b(\tilde{i})\,d\tilde{i}=0. \tag{B.87}$$

Equations (B.80) and (B.87) jointly determine (g_n,μ).

Step 3. The limiting case at $\rho=0$.

At $\rho = 0$, equation (B.80) becomes $g_n = \phi\beta E(g_n, \mu)$. Substituting this equality into equation (B.87) gives $\mu = 1$ and $E(g_n) = M(P(g_n))$, the equilibrium conditions reduce to

$$\frac{M(P(g_n))}{g_n} = \frac{1}{\phi\beta}. \tag{B.88}$$

It can be shown that the left-hand side of equation (B.88) is strictly decreasing in g_n. We first calculate the response of $P(g_n)$ to g_n. At $\rho = 0$, equation (B.79) implies

$$\frac{\partial \log[p(\tilde{i})/w]}{\partial \log g_n} = \beta \left\{ 1 + \frac{g_n \frac{\partial}{\partial g_n} \left[(\delta + g_n)\hat{v} + \frac{g_n}{2}\hat{v}^2 \right]}{(\delta + g_n)\hat{v} + \frac{g_n}{2}\hat{v}^2} \right\}.$$

The closed-form solution in **Step 1** gives $\hat{v}_{g_n}(\tilde{i}) = -\frac{\tilde{i}/\bar{g}}{[1-(1-g_n/\bar{g})\tilde{i}]^2} \leq 0$. Therefore,

$$\frac{\partial}{\partial g_n}\left[(\delta + g_n)\hat{v} + \frac{g_n}{2}\hat{v}^2 \right] = \hat{v} + \frac{1}{2}\hat{v}^2 + \left[\delta + g_n(1+\hat{v})\right]\hat{v}_{g_n} \leq \hat{v} + \frac{1}{2}\hat{v}^2.$$

It follows that

$$\frac{\partial \log[p(\tilde{i})/w]}{\partial \log g_n} \leq 2\beta = \frac{2\theta}{1+\theta} < 1 \qquad \text{for every } \tilde{i} > 0.$$

Since $P(g_n) = \int_0^1 [p(\tilde{i})/w] d\tilde{i}$, we have:

$$\frac{g_n P'(g_n)}{P(g_n)} = \int_0^1 \frac{p(\tilde{i})/w}{P(g_n)} \frac{\partial \log[p(\tilde{i})/w]}{\partial \log g_n} d\tilde{i} < 1. \tag{B.89}$$

We can now differentiate the left-hand side of equation (B.88). Since $M'(P) = 1 - \mathcal{G}(P)$, we obtain

$$\frac{d}{d \log g_n} \log\left[\frac{M(P(g_n))}{g_n} \right] = \frac{P[1-\mathcal{G}(P)]}{M(P)} \frac{g_n P'(g_n)}{P} - 1 < 0. \tag{B.90}$$

To see the last inequality, note that $1 - \mathcal{G}(l)$ is decreasing in l, so $M(P) \geq P[1 - \mathcal{G}(P)]$. Thus, the left-hand side of equation (B.88) is decreasing in g_n.

For $j \in \{1, 2\}$, superscript j denotes the corresponding object under $\mathcal{G}^j$. The strict mean-preserving-spread condition implies $M^2(P) - M^1(P) = -\int_0^P [\mathcal{G}^2(l) - \mathcal{G}^1(l)]\, dl < 0$. Evaluating this equation at $P(g_n^1(0))$ gives $\frac{M^2(P(g_n^1(0)))}{g_n^1(0)} < \frac{M^1(P(g_n^1(0)))}{g_n^1(0)} = \frac{1}{\phi\beta}$. For distribution $\mathcal{G}^2$, the same ratio is strictly decreasing in g_n and equals $1/(\phi\beta)$ at $g_n^2(0)$. It follows that

$$g_n^2(0) < g_n^1(0). \tag{B.91}$$

Step 4. Extend the result to sufficiently small ρ.

For each $j \in \{1, 2\}$, let $\mathcal{F}^j$ and $\mathcal{H}^j$ denote the two equilibrium conditions (B.80) and (B.87). At the limiting equilibrium, direct differentiation gives $\det \frac{\partial(\mathcal{F}^j, \mathcal{H}^j)}{\partial(g_n, \mu)} = 1 - \phi\beta E_{g_n}^j > 0$. The inequality follows from the strict monotonicity established in equation (B.90). The equilibrium mapping is continuously differentiable around $\rho = 0$. The implicit function theorem therefore gives continuous

local equilibrium paths $g_n^j(\rho)$ and $\mu^j(\rho)$ for both distributions. By equation (B.91), their growth-rate difference is strictly negative at $\rho = 0$. Continuity then implies that there exists $\bar{\rho} > 0$ such that $g_n^2(\rho) < g_n^1(\rho)$ for every $0 \leq \rho < \bar{\rho}$. Finally, Proposition 3 implies that $\log z^1(\tilde{i}) - \log z^2(\tilde{i})$ is positive for every $\tilde{i} < 1$ and strictly decreasing in $\tilde{i}$.

■

C Appendix: Alternative Model Specification

This section examines the robustness of the main results under different model specifications. While we modify several assumptions of the baseline framework, the core mechanisms of the model remain unchanged. The main qualitative results continue to hold under these alternative settings.

C.1 Extension: Skilled and Unskilled Labor

In the baseline model, we assume that each worker supplies one unit of homogeneous labor, which can be employed either in production or in innovation. In this section, we extend the framework by introducing two types of labor: *skilled workers*, who are specialized in R&D activities, and *unskilled workers*, who are specialized in production.

Each household is endowed with labor l, drawn from a continuous distribution $\mathcal{G}(l)$ with mean normalized to unity, i.e., $\mathbb{E}[l] = 1$. A fraction Ω of this labor endowment is *skilled*, while the remaining fraction $(1-\Omega)$ is *unskilled*. Households supply both types of labor inelastically at market wages $W^s(t)$ for skilled labor and $W^u(t)$ for unskilled labor. Similar to the baseline model, the present value budget constraint for a household with labor endowment l is modified to:

$$\int_0^\infty e^{-R(t,0)} \left[\int_0^{N(t)} P(i,t)C(i,t)di + P^X(t)X(t) \right] dt \leq \quad (1-\Omega) \int_0^\infty W^u(t) l \cdot e^{-R(t,0)} dt + \Omega \int_0^\infty W^s(t) l \cdot e^{-R(t,0)} dt + \mathbb{V}(0) l.$$

The supply side is the same as in the baseline model, except that only *unskilled workers* are employed in production—that is, in the production of intermediate goods and luxury services. By contrast, all three innovation activities use *skilled workers* as inputs: incumbent R&D, firm entry, and idea production. We define detrended wages as

$$w^u = e^{-(1-\gamma)g_n t} W^u(t), \quad w^s = e^{-(1-\gamma)g_n t} W^s(t).$$

Under this alternative model with two types of labor, we obtain a counterpart of Proposition 1, which characterizes the mathematical properties of the system.

Proposition C.1 *The BGE can be characterized by two functions,* $\{v(\tilde{i}), \mathcal{L}(\tilde{i})\}$, $\tilde{i} \in (0,1]$, *and five undetermined parameters,* $\{w^s, w^u, \eta, y, g_n\}$, *which jointly satisfy the following system:*

(i) The functions $v(\tilde{i})$ *solves the same ODE as in Proposition 1, subject to the boundary conditions*

$$v(1) = \frac{w^s}{\psi_e \zeta^{\theta \frac{1-\beta}{\beta}}}.$$

Here, $h(\cdot)$ is given by

$$h(g_n) = \frac{w^s \left(\theta \frac{1-\beta}{\beta}\right)^{\frac{1-\alpha}{\alpha}}}{\left[\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right)\psi\right]^{\frac{1}{\alpha}} \alpha} g_n^{\frac{1-\alpha}{\alpha}}.$$

(ii) There exists a constant $\eta > 0$ denoting the share of product varieties universally consumed across all households. If $\tilde{i} \geq \eta$, $\mathcal{L}(\tilde{i})$ satisfies

$$\begin{aligned}\mathcal{L}'(\tilde{i}) \;=\; & \frac{w^{u1-\beta}}{y\beta^\beta(1-\beta)^{1-\beta}} \Big\{ \big[\rho + \delta + \big((\sigma-1)(1-\gamma)+1\big) g_n\big]\, v(\tilde{i}) \\ & + w^s \left[\frac{w^s}{\left(\lambda^{\theta\frac{1-\beta}{\beta}} - 1\right) v(\tilde{i}) \cdot \psi\alpha}\right]^{\frac{1}{\alpha-1}} \Big\}^{\beta},\end{aligned}$$

subject to the same boundary conditions as in Proposition 1.

(iii) w^u is determined by the unskilled labor market clearing condition; w^s is determined by the skilled labor market clearing condition; other parameters are determined in the same way as in Proposition 1.

The proof follows almost identically to that of Proposition 1, with the main difference arising from the segmented labor market. For *unskilled workers*, the market clearing condition is

$$1 - \Omega \;=\; \int_0^1 l(\tilde{i})\, d\tilde{i} \;+\; L^X.$$

For *skilled workers*, the market clearing condition is

$$\Omega \;=\; \int_0^1 l^I(\tilde{i}) d\tilde{i} \;+\; L^E \;+\; L^N.$$

Moreover, it is straightforward to verify that the conclusions of Proposition 2, Proposition 3, and Proposition 4 remain valid under this alternative model setup.

With separate markets for skilled and unskilled labor, it is generally difficult to derive a sharp counterpart to Proposition 5 under an unrestricted income distribution $\mathcal{G}(l)$. To obtain a transparent analytical result, we therefore specialize the income distribution to a Pareto distribution. This specification allows us to characterize how growth and the allocation of innovation respond to inequality when the income distribution is sufficiently unequal. Suppose that income follows a Pareto distribution with exponent $\epsilon > 1$. As in our quantitative exercise, we set its lower bound to $\underline{l} = (\epsilon - 1)/\epsilon$, which normalizes mean income to 1. Its CDF is

$$\mathcal{G}(l) = \begin{cases} 0, & l < \underline{l}, \\ 1 - \left(\frac{\underline{l}}{l}\right)^{\epsilon}, & l \geq \underline{l}. \end{cases}$$

The Gini coefficient of the Pareto distribution is $1/(2\epsilon-1)$, so a lower ϵ represents greater inequality. The following Proposition C.2 shows that, when the initial level of inequality is sufficiently high, greater inequality will always reduce growth and redirect innovation away from necessities.

Proposition C.2 *Suppose equation (25) holds. There exists $\bar{\epsilon} > 1$ such that, for every $\epsilon \in (1, \bar{\epsilon})$, there exists $\bar{\rho}(\epsilon) > 0$ satisfying $\frac{\partial g_n}{\partial \epsilon} > 0$ for every $\rho \in (0, \bar{\rho}(\epsilon))$. Thus, a further increase in already high income inequality lowers growth and shifts innovation away from necessity sectors.*

Proposition C.2 establishes that, when income inequality is already sufficiently high, a further increase in inequality reduces long-run growth. More importantly, the proposition highlights the allocation of R&D resources as a key channel through which inequality affects growth. In the present setting, the aggregate supply of skilled labor—and hence the total amount of resources devoted to innovation—is fixed. If incumbent R&D and firm entry were absent, so that skilled labor were used exclusively for the invention of new products as in Foellmi and Zweimüller (2006), the skilled-labor market-clearing condition would imply $g_n = \phi\Omega$, independently of the income distribution. Inequality affects growth here precisely because it reallocates R&D resources across innovation activities and sectors. In particular, as innovation shifts away from necessity sectors and toward luxury sectors, the rate at which new products are introduced into the economy declines, thereby reducing aggregate growth.

Proof of Proposition C.2:

The proof proceeds in four steps. First, we reduce the equilibrium to two conditions. Second, we solve the economy at $\rho = 0$. Third, we extend the system to a small positive discount rate. Finally, we sign the response to inequality and derive the implications for innovation allocation.

Step 1. Derive the two equilibrium conditions.

As in the proof of Proposition 5, let $v_0 \equiv \lim_{\tilde{i}\downarrow 0} v(\tilde{i})$ and $\hat{v}(\tilde{i}) \equiv v(\tilde{i})/v_0$. Then $\hat{v}(\tilde{i}) = \frac{1}{1-(1-g_n/\bar{g})\tilde{i}}$, and it follows that

$$v(\tilde{i}) = \frac{2w^s g_n}{(\lambda-1)^2\psi^2}\hat{v}(\tilde{i}), \qquad z(\tilde{i}) = \frac{g_n}{\lambda-1}\hat{v}(\tilde{i}).$$

The incumbent HJB implies

$$p(\tilde{i}) = P_0 Q(\tilde{i})^\beta, \tag{C.1}$$

where

$$P_0 \equiv \left[\frac{v_0}{\beta(1-\beta)^{\frac{1-\beta}{\beta}}}\right]^\beta (w^u)^{1-\beta}, \qquad Q(\tilde{i}) \equiv (\rho+\delta+g_n)\hat{v}(\tilde{i}) + \frac{g_n}{2}\hat{v}(\tilde{i})^2.$$

Define $A(\tilde{i}) \equiv \int_0^{\tilde{i}} Q(x)^\beta dx$, so $\mathcal{L}(1) = \frac{P_0 A(1)}{y} \equiv c$. The income required to purchase products up to $\tilde{i}$ is $cA(\tilde{i})/A(1)$. Hence sectoral demand, which equals sectoral output in equilibrium, is $Y(\tilde{i}) = 1 - \mathcal{G}\left(c\frac{A(\tilde{i})}{A(1)}\right)$. As in the proof of Proposition 5, the share of aggregate income spent on manufactured goods is $M(c) = \int_0^\infty \min\{l, c\}\, d\mathcal{G}(l)$. Also, production labor is $(1-\beta)^2 y M(c)/w^u$,

and luxury service labor is $y[1-M(c)]/w^u$. Production-labor clearing condition gives

$$\frac{P_0}{w^u} = \frac{c(1-\Omega)}{A(1)[1-\beta(2-\beta)M(c)]}. \tag{C.2}$$

The value of a newly invented product is $v^f(1) = \frac{\beta P_0}{g_n}\mathcal{D}(c)$, where $\mathcal{D}(c) \equiv \int_0^1 Y(\tilde{i})\tilde{i}^{\rho/g_n}Q(\tilde{i})^\beta d\tilde{i}$. Combining inventor free entry, $w^s = \phi v^f(1)$, with the definition of P_0 yields

$$\frac{P_0}{w^u} = \frac{1}{1-\beta}\left[\frac{2\phi\mathcal{D}(c)}{(\lambda-1)^2\psi^2}\right]^{\frac{\beta}{1-\beta}}. \tag{C.3}$$

Equations (C.2) and (C.3) give the first equilibrium condition.

The skilled labor market gives the second equilibrium condition. Combining product-market clearing with equation (C.3) gives $b(\tilde{i}) = \frac{(1-\beta)(\lambda-1)^2\psi^2}{2\phi\,\mathcal{D}(c)}Y(\tilde{i})Q(\tilde{i})^{-(1-\beta)}$. Thus, incumbent R&D and entry together use

$$L^I + L^E = \frac{(1-\beta)g_n^2}{2\phi}\frac{\mathcal{R}(c)}{\mathcal{D}(c)},$$

where $\mathcal{R}(c) \equiv \int_0^1 Y(\tilde{i})\frac{\hat{v}(\tilde{i})^2+2(1+\delta/g_n)\hat{v}(\tilde{i})}{Q(\tilde{i})^{1-\beta}}d\tilde{i}$. Since $L^N = g_n/\phi$, skilled labor market clearing becomes

$$\Omega = \frac{g_n}{\phi} + \frac{(1-\beta)g_n^2}{2\phi}\frac{\mathcal{R}(c)}{\mathcal{D}(c)}. \tag{C.4}$$

Step 2. Solve the limiting economy at $\rho = 0$.

At $\rho = 0$, $Q(\tilde{i}) = g_n\left[\left(1+\frac{\delta}{g_n}\right)\hat{v}(\tilde{i}) + \frac{1}{2}\hat{v}(\tilde{i})^2\right]$, which gives us $\frac{\mathcal{R}(c)}{\mathcal{D}(c)} = \frac{2}{g_n}$. Equation (C.4) therefore implies $g_n = g_0 = \frac{\phi\Omega}{2-\beta}$. Then, we use the first equilibrium condition to solve for c. When $\rho = 0$, $\mathcal{D}(c) = \frac{A(1)M(c)}{c}$. Substituting this expression into equation (C.3) and equating the result with equation (C.2) gives $c = KM(c)^\beta[1-\beta(2-\beta)M(c)]^{1-\beta}$, where $K \equiv \frac{A(1)}{[(1-\Omega)(1-\beta)]^{1-\beta}}\left[\frac{2\phi}{(\lambda-1)^2\psi^2}\right]^\beta > 0$. Here $A(1)$ is evaluated at $(g_n,\rho) = (g_0,0)$, so K does not depend on ϵ. To simplify the expressions below, define $L \equiv \log\left(\frac{c}{l}\right)$. Substituting $c = (\epsilon-1)e^L/\epsilon$ gives the equilibrium condition

$$\frac{\epsilon-1}{\epsilon}e^L = KM^\beta[1-\beta(2-\beta)M]^{1-\beta}, \tag{C.5}$$

where $M = M(\epsilon,L) = 1 - \frac{e^{-(\epsilon-1)L}}{\epsilon}$.

To establish the existence and uniqueness of L (equivalently c), define the log difference between the two sides of equation (C.5) as

$$\Phi(\epsilon,L) \equiv \log\frac{\epsilon-1}{\epsilon} + L - \log K - \beta\log M - (1-\beta)\log[1-\beta(2-\beta)M].$$

Because $M_L = (\epsilon-1)(1-M)$ and $M_L/M \leq 1$, we have $\Phi_L > 0$. At $L = 0$, $\Phi(\epsilon,0) = (1-\beta)\log\left[\frac{(\epsilon-1)/\epsilon}{1-\beta(2-\beta)(\epsilon-1)/\epsilon}\right] - \log K$. Define $\bar{\epsilon}^1 \equiv \left[1 - \min\left\{\frac{1}{2}, \frac{K^{1/(1-\beta)}}{2[1+\beta(2-\beta)K^{1/(1-\beta)}]}\right\}\right]^{-1} > 1$. Then, for every $\epsilon \in (1,\bar{\epsilon}^1)$, we have $\Phi(\epsilon,0) < 0$. Moreover, for every fixed $\epsilon > 1$, $\Phi(\epsilon,L) \to \infty$ as $L \to \infty$. Since

$\Phi(\epsilon, L)$ is continuous and strictly increasing in L, it follows that, for every $\epsilon \in (1, \bar{\epsilon}^1)$, equation (C.5) has a unique solution $L(\epsilon) > 0$ (equivalently $c(\epsilon) > \underline{l}$).

We next characterize some properties of $L(\epsilon)$ when $\epsilon \downarrow 1$, which will be useful in **Step 4**. Since $0 < M \leq 1$, equation (C.5) implies $0 < L \leq \log\left(\frac{K\epsilon}{\epsilon-1}\right)$, and hence $(\epsilon-1)L \to 0$. Expanding the Pareto truncated mean gives $M = (\epsilon-1)(1+L)[1+o(1)]$. If L were bounded, the two sides of equation (C.5) would be of orders $\epsilon - 1$ and $(\epsilon-1)^\beta$, respectively, which is impossible when $\beta < 1$. Thus, $L \to \infty$, $M \sim (\epsilon-1)L \to 0$ and $(\epsilon-1)L(\epsilon)^2 \to 0$. Differentiating $\Phi(\epsilon, L) = 0$ to get

$$L'(\epsilon) = -\frac{\Phi_\epsilon}{\Phi_L} = -\frac{1-\beta}{\epsilon-1}[1+o(1)] \qquad (\epsilon \downarrow 1). \tag{C.6}$$

Step 3. Jointly linearize the two equilibrium conditions around $\rho = 0$.

Fix $\epsilon \in (1, \bar{\epsilon}^1)$ and write $c_0(\epsilon) \equiv \underline{l} e^{L(\epsilon)}$. In this step, we make the dependence of A, $\mathcal{D}$, and $\mathcal{R}$ on (g_n, ρ) explicit. The equilibrium values (c, g_n) solve

$$F_1(c, g_n, \rho; \epsilon) \equiv \frac{c(1-\Omega)}{A(1; g_n, \rho)[1-\beta(2-\beta)M(c)]} - \frac{1}{1-\beta}\left[\frac{2\phi\mathcal{D}(c; g_n, \rho)}{(\lambda-1)^2\psi^2}\right]^{\frac{\beta}{1-\beta}} = 0. \tag{C.7}$$

$$F_2(c, g_n, \rho; \epsilon) \equiv \frac{g_n}{\phi} + \frac{(1-\beta)g_n^2}{2\phi}\frac{\mathcal{R}(c; g_n, \rho)}{\mathcal{D}(c; g_n, \rho)} - \Omega = 0. \tag{C.8}$$

At $\rho = 0$, **Step 2** establishes the reference equilibrium $(c_0(\epsilon), g_0)$. It's easy to verify that the Jacobian of (F_1, F_2) with respect to (c, g_n) is nonsingular at $(c_0(\epsilon), g_0, 0; \epsilon)$. The implicit function theorem therefore supplies a number $\bar{\rho}^1(\epsilon) > 0$ and a unique smooth local branch $(c(\epsilon, \rho), g_n(\epsilon, \rho))$ for $|\rho| < \bar{\rho}^1(\epsilon)$. Let $\Delta c \equiv c(\epsilon, \rho) - c_0(\epsilon)$ and $\Delta g_n \equiv g_n(\epsilon, \rho) - g_0$. A first-order Taylor expansion of $F_1 = F_2 = 0$ around $(c_0(\epsilon), g_0, 0; \epsilon)$ gives

$$\begin{pmatrix} F_{1c} & F_{1g_n} \\ 0 & \dfrac{2-\beta}{\phi} \end{pmatrix} \begin{pmatrix} \Delta c \\ \Delta g_n \end{pmatrix} + \rho \begin{pmatrix} F_{1\rho} \\ F_{2\rho} \end{pmatrix} = O_\epsilon(\rho^2). \tag{C.9}$$

All derivatives are evaluated at $(c, g_n, \rho; \epsilon) = (c_0(\epsilon), g_0, 0; \epsilon)$. The second row of equation (C.9) is

$$\frac{2-\beta}{\phi}\Delta g_n + \rho F_{2\rho} = O_\epsilon(\rho^2).$$

To derive Δg_n, we need the partial derivative of $\mathcal{R}/\mathcal{D}$ with respect to ρ. Recall from the definition in **Step 1** that

$$\frac{\mathcal{R}}{\mathcal{D}} = \frac{\displaystyle\int_0^1 Y(\tilde{i})\frac{\hat{v}(\tilde{i})^2 + 2(1+\delta/g_n)\hat{v}(\tilde{i})}{Q(\tilde{i})^{1-\beta}}d\tilde{i}}{\displaystyle\int_0^1 Y(\tilde{i})\tilde{i}^{\rho/g_n}Q(\tilde{i})^\beta d\tilde{i}}.$$

For fixed (c, g_n), ρ affects this ratio through three objects: the market-size schedule $Y(\tilde{i})$, the

inventor's discount factor $\tilde{i}^{\rho/g_n}$, and $Q(\tilde{i})$. At $\rho = 0$, the numerator integrand is pointwise $2/g_0$ times the denominator integrand, so the first-order changes in $Y(\tilde{i})$ cancel in the quotient rule. Differentiating the remaining terms gives

$$\frac{\partial}{\partial \rho}\left(\frac{\mathcal{R}}{\mathcal{D}}\right)\Big|_{(c,g_n,\rho)=(c_0(\epsilon),g_0,0)} = -\frac{2}{g_0^2}\overline{q}(\epsilon), \tag{C.10}$$

where

$$\overline{q}(\epsilon) \equiv \frac{\int_0^1 Y(\tilde{i})A'(\tilde{i})q(\tilde{i})d\tilde{i}}{\int_0^1 Y(\tilde{i})A'(\tilde{i})d\tilde{i}}, \qquad q(\tilde{i}) \equiv \log \tilde{i} + \frac{2}{\hat{v}(\tilde{i}) + 2(1+\delta/g_0)}.$$

Here Y, A', and $\hat{v}$ are evaluated at $(c, g_n, \rho) = (c_0(\epsilon), g_0, 0)$. Thus q is a fixed function of $\tilde{i}$, while $\overline{q}(\epsilon)$ varies with ϵ through the reference-equilibrium market-size weights $Y(\tilde{i})$. Substituting this result into the second row of equation (C.9) yields

$$g_n(\epsilon, \rho) = g_0 + \frac{1-\beta}{2-\beta}\rho\,\overline{q}(\epsilon) + O_\epsilon(\rho^2). \tag{C.11}$$

Step 4. Sign the growth and innovation responses.

Equation (C.11) shows that it remains to determine the sign of $\overline{q}'(\epsilon)$. The difficulty comes from the term $\log \tilde{i}$ in $q(\tilde{i})$, which diverges as $\tilde{i} \downarrow 0$. As shown below, $A(\tilde{i})$ is locally proportional to $\tilde{i}$, so $\log A(\tilde{i})$ and $\log \tilde{i}$ have the same logarithmic behavior near zero. We therefore use $\log A(\tilde{i})$ to capture this logarithmic part. This choice also allows its weighted average to be calculated explicitly, since $\frac{d}{d\tilde{i}}\left[A(\tilde{i})\log A(\tilde{i}) - A(\tilde{i})\right] = A'(\tilde{i})\log A(\tilde{i})$. We write

$$\overline{q}(\epsilon) = \overline{\log A(\tilde{i})} + \overline{q(\tilde{i}) - \log A(\tilde{i})}.$$

We will show that the first term can be calculated explicitly, and the second term has a lower-order derivative.

First, we derive $\overline{\log A(\tilde{i})}$. Let $\eta(\epsilon)$ denote the universal-consumption cutoff satisfying $A(\eta(\epsilon)) = e^{-L(\epsilon)}A(1)$. The market-size schedule is therefore: $Y(\tilde{i}) = 1$ if $0 < \tilde{i} \le \eta(\epsilon)$ and $Y(\tilde{i}) = \left[\frac{A(\eta(\epsilon))}{A(\tilde{i})}\right]^\epsilon$ if $\eta(\epsilon) < \tilde{i} \le 1$. After splitting the numerator and denominator at $\eta(\epsilon)$ and simplifying the resulting integrals, we obtain the following exact expression:

$$\overline{\log A(\tilde{i})} = \frac{\int_0^1 Y(\tilde{i})A'(\tilde{i})\log A(\tilde{i})d\tilde{i}}{\int_0^1 Y(\tilde{i})A'(\tilde{i})d\tilde{i}} = \log A(1) - L(\epsilon) + \frac{\int_0^{L(\epsilon)} xe^{-(\epsilon-1)x}dx - 1}{1 + \int_0^{L(\epsilon)} e^{-(\epsilon-1)x}dx}.$$

A change in ϵ affects the final fraction in two ways: it changes the exponential weights directly, and it changes the upper limit $L(\epsilon)$. Because $(\epsilon-1)L(\epsilon) \to 0$, the exponential weight is uniformly close to one over $[0, L(\epsilon)]$. The final fraction therefore behaves like the average of x over this interval,

which is $L(\epsilon)/2$. Formally, we can differentiate to get

$$\frac{d}{d\epsilon}\frac{\int_0^{L(\epsilon)} xe^{-(\epsilon-1)x}dx - 1}{1+\int_0^{L(\epsilon)} e^{-(\epsilon-1)x}dx} = \frac{1}{2}L'(\epsilon) + O\left(L(\epsilon) + \frac{1}{(\epsilon-1)L(\epsilon)} + L(\epsilon)^2\right).$$

The expression for $L'(\epsilon)$ in **Step 2** implies $L(\epsilon) = O(\log(1/(\epsilon-1)))$, so the remainder is $o(1/(\epsilon-1))$. Since $A(1)$ does not depend on ϵ, it follows that

$$\frac{d}{d\epsilon}\overline{\log A(\tilde{i})} = -\frac{1}{2}L'(\epsilon) + o\left(\frac{1}{\epsilon-1}\right).$$

Next, we show that the second term in the decomposition does not affect the leading sign. Since $A(\tilde{i}) = A'(0)\tilde{i} + O(\tilde{i}^2)$ and $A'(0) > 0$, the function $q(\tilde{i}) - \log A(\tilde{i})$ extends to a continuously differentiable function at $\tilde{i} = 0$. It does not itself depend on ϵ; its weighted average changes only through the market-size weights. Formally, we can differentiate to get

$$\frac{d}{d\epsilon}\overline{q(\tilde{i}) - \log A(\tilde{i})} = O\left(L(\epsilon) + \frac{|L'(\epsilon)|}{L(\epsilon)}\right) = o\left(\frac{1}{\epsilon-1}\right).$$

Here the first term comes from the direct change in the Pareto weights and the second from movement of the cutoff. The last equality follows from $(\epsilon-1)L(\epsilon) \to 0$, $L(\epsilon) \to \infty$, and $|L'(\epsilon)| = O(1/(\epsilon-1))$. Combining the above two parts yields

$$\overline{q}'(\epsilon) = \frac{1-\beta}{2(\epsilon-1)}[1+o(1)] \qquad (\epsilon \downarrow 1). \tag{C.12}$$

By the definition of the $o(1)$ term in equation (C.12), there exists $1 < \bar{\epsilon} < \bar{\epsilon}$[1] such that, for every $\epsilon \in (1, \bar{\epsilon})$, $\left|\frac{2(\epsilon-1)}{1-\beta}\overline{q}'(\epsilon) - 1\right| < \frac{1}{2}$. Consequently,

$$\overline{q}'(\epsilon) > \frac{1-\beta}{4(\epsilon-1)} > 0.$$

At last, it remains to construct the discount-rate bound. Fix any $\epsilon \in (1, \bar{\epsilon})$, we have $\frac{\partial g_n}{\partial \epsilon}\Big|_{\rho=0} = 0$ and $\frac{\partial^2 g_n}{\partial \epsilon \partial \rho}\Big|_{\rho=0} = \frac{1-\beta}{2-\beta}\overline{q}'(\epsilon) > 0$. Consequently, $\lim_{\rho \downarrow 0} \frac{1}{\rho}\frac{\partial g_n(\epsilon,\rho)}{\partial \epsilon} = \frac{1-\beta}{2-\beta}\overline{q}'(\epsilon) > 0$. By the definition of this limit, there exists $\bar{\rho}(\epsilon) > 0$, such that

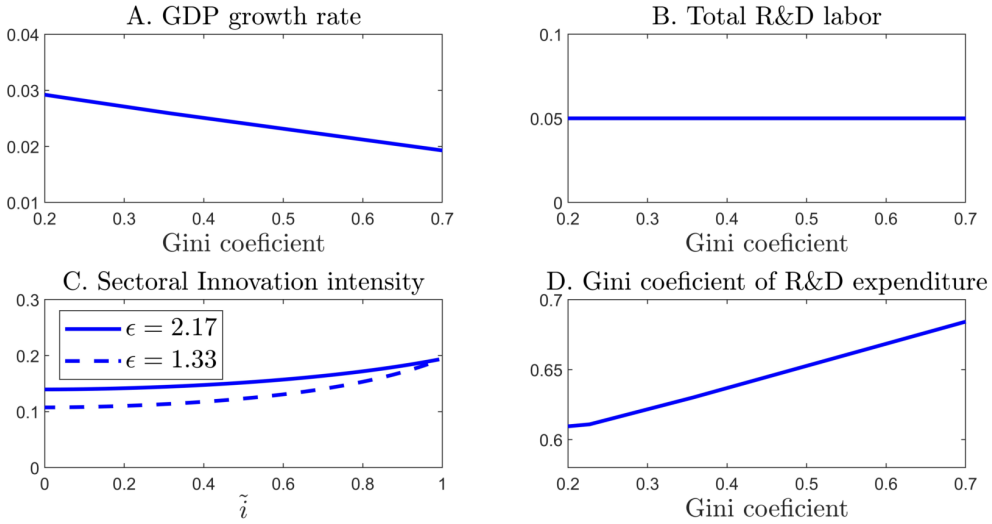
$$\frac{\partial g_n(\epsilon,\rho)}{\partial \epsilon} > \frac{1-\beta}{2(2-\beta)}\overline{\rho q}'(\epsilon) > 0$$

for every $0 < \rho < \bar{\rho}(\epsilon)$. The closed-form expression for $z(\tilde{i})$ then shows that, when inequality increases, innovation falls for every $\tilde{i} < 1$, with a larger decline in more necessary sectors. ■

Although Proposition C.2 imposes a restriction on the degree of inequality, we find quantitatively that this conclusion is relatively robust across different values of ϵ. Following the main text, we

provide a numerical example to illustrate how income inequality affects innovation. The results are shown in Figure C.1. Since the total supply of *skilled workers* is fixed at $\Omega = 0.05$, the total R&D labor is also fixed at 0.05, as displayed in Panel B. The impact of income inequality on R&D allocation and growth remains qualitatively the same as in the baseline model.

Figure C.1: Inequality and Growth in Alternative Model Setup



Notes. The parameters used in this example are: $\Omega = 0.05, \rho = 0.02, \sigma = 2, \gamma = 0.1, \beta = 0.1, \theta = 0.2, \delta = 0.06, \alpha = 0.5, \xi = 1.4, \iota = 0.9, \psi_e = 1, \psi = 1, \lambda = 1.2, \phi = 2, \zeta = 1$. We vary the Pareto exponent ϵ to generate different levels of inequality, targeting a Gini coefficient in the range 0.2-0.7. In Figure C, the solid line corresponds to $\epsilon = 2.17$ (Gini coefficient 0.3), while the dashed line corresponds to $\epsilon = 1.33$ (Gini coefficient 0.6).

C.2 Extension: Constant Returns to Scale in Final-Good Production

In the baseline model, the production of final goods exhibits decreasing returns with respect to inputs. In this section, following Akcigit and Kerr (2018), we consider an alternative specification in which the final-good sector operates under constant returns to scale technology. Specifically, the representative final-good producer has the following production dunction:

$$Y(i,t) = \frac{\left(\bar{A}(t)L^f(i,t)\right)^{\beta}}{1-\beta} \int_0^{F(i,t)} M_i(j,t)^{1-\beta} dj.$$

Following a similar derivation in Appendix B.2.1, the profit of an intermediate-good producer is given by

$$\Pi_i(j,t) = \beta(1-\beta)^{\frac{1-\beta}{\beta}} \left[\frac{P(i,t)^{\frac{1}{1-\beta}} L^f(i,t)^{\frac{\beta}{1-\beta}}}{W(t)} \bar{A}(t)^{1-\theta} \tilde{A}_i(j,t)^{\theta} \right]^{\frac{1-\beta}{\beta}}, \tag{C.13}$$

and under constant returns to scale, the final-good producer earns zero profits in equilibrium, i.e., $\Pi(i,t) = 0$. To maintain incentives for idea creation, we assume that the idea producer receives a fraction Ψ of the profits generated by all intermediate-good producers. The value of an idea can

therefore be written as

$$V^F(i,t) = \Psi \int_t^{\infty} e^{-r(\tau-t)}\, \beta(1-\beta)^{\frac{1-2\beta}{\beta}}\, P(i,\tau)^{\frac{1}{\beta}}\, L^f(i,\tau)\, W(\tau)^{-\frac{1-\beta}{\beta}}\, B(i,\tau)\, d\tau.$$

Apart from these modifications, the rest of the environment remains the same as in the baseline model. Under this alternative specification, we obtain a counterpart of Proposition 1 that characterizes the BGP of the economy.

Proposition C.3 *The BGP can be characterized by two functions,* $\{v(\tilde{i}), \mathcal{L}(\tilde{i})\}$, $\tilde{i} \in (0,1]$, *and four undetermined parameters,* $\{w, \eta, y, g_n\}$, *which jointly satisfy the following system:*

(i) The function $v(\tilde{i})$ *solves the same ODE subject to the same boundary conditions as in Proposition 1.*

(ii) There exists a constant $\eta > 0$ *denoting the share of product varieties universally consumed across all households. If* $\tilde{i} \geq \eta$, $\mathcal{L}(\tilde{i})$ *satisfies*

$$\begin{aligned}\mathcal{L}'(\tilde{i}) \;=\; & \frac{w^{\frac{1}{1+\beta}}}{y\beta^{\frac{2\beta}{1+\beta}}(1-\beta)^{\frac{1-\beta}{1+\beta}}\left[1-\mathcal{G}(\mathcal{L}(\tilde{i}))\right]^{\frac{\beta}{1+\beta}}} \cdot \Bigg\{ \Big[\rho+\delta+\Big((\sigma-1)(1-\gamma)+1\Big)g_n\Big] v(\tilde{i}) \\ & + w \cdot \left[\frac{w}{\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right) v(\tilde{i}) \cdot \psi\alpha}\right]^{\frac{1}{\alpha-1}} \Bigg\}^{\frac{\beta}{1+\beta}}. \end{aligned} \tag{C.14}$$

subject to the same boundary conditions as in Proposition 1.

(iii) The parameters are determined in the same way as in Proposition 1.

The proof follows closely that of Proposition 1, with the main difference arising in the derivation of equation (C.14) rather than equation (23). This difference stems from the modified profit function for intermediate-good producers, that is, equation (C.13) instead of equation (B.1). To derive equation (C.14), note that the labor demand of the final-good producer satisfies

$$L^f(i,t) = \frac{\beta P(i,t) Y(i,t)}{W(t)}.$$

Substituting this expression into equation (C.13) yields a detrended HJB equation analogous to (B.35):

$$\begin{aligned}\left[\rho + \left((\sigma-1)(1-\gamma)+1+\theta\frac{1-\beta}{\beta}\right)g_n + \delta\right] v(\tilde{i}) \;=\; & \beta(1-\beta)^{\frac{1-\beta}{\beta}} \cdot p(\tilde{i})^{\frac{1}{\beta}} \cdot \frac{\beta p(\tilde{i})\left[1-\mathcal{G}(\mathcal{L}(\tilde{i}))\right]}{w} \cdot w^{-\frac{1-\beta}{\beta}} \\ & - w\left(\frac{z(\tilde{i})}{\psi}\right)^{\frac{1}{\alpha}} + z(\tilde{i})\left(\lambda^{\theta\frac{1-\beta}{\beta}}-1\right) v(\tilde{i}) \\ & - \tilde{i} g_n v'(\tilde{i}). \end{aligned} \tag{C.15}$$

Equation (C.14) then follows from a sequence of derivations analogous to those in Appendix B.3.3. Moreover, it is straightforward to verify that the conclusions of Proposition 2, Proposition 3 and Proposition 4 remain valid under this alternative specification.

D Appendix: Numerical Algorithm

We solve for the balanced growth equilibrium by exploiting the structural properties characterized in Proposition 1. In particular, Proposition 1 implies that the BGE can be summarized by two functions, $\{v(\tilde{i}), \mathcal{L}(\tilde{i})\}$ for $\tilde{i} \in (0,1]$, together with four endogenous variables, $\{w, \eta, y, g_n\}$. Numerically, this characterization turns the equilibrium problem into a shooting problem. In general, one needs to guess the initial conditions for the two differential equations at the point $\tilde{i} = 1$, together with the four aggregate variables $\{w, \eta, y, g_n\}$, and then solve the implied system so that the two ODEs, their boundary conditions, and the four equilibrium conditions in Proposition 1 are jointly satisfied. This yields a six-dimensional nonlinear system.

In practice, the problem can be reduced to a four-dimensional nonlinear system. First, the boundary value $v(1)$ is pinned down directly by the boundary condition in Proposition 1, and therefore does not need to be guessed in the shooting procedure. Second, the equilibrium system exhibits a homogeneity property with respect to y: some variables are homogeneous of degree one in y, while others are homogeneous of degree zero. In particular, if $(\{v(\tilde{i}), \mathcal{L}(\tilde{i})\}_{\tilde{i}\in(0,1]}, w, \eta, y, g_n)$ satisfies all the conditions in Proposition 1, then for any scalar $c > 0$,

$$(\{cv(\tilde{i}), \mathcal{L}(\tilde{i})\}_{\tilde{i}\in(0,1]}, cw, \eta, cy, g_n)$$

also satisfies all the conditions in Proposition 1, except for equation (B.66), which pins down the level of y. We therefore normalize $y = 1$ and solve for $\{v(\tilde{i}), \mathcal{L}(\tilde{i})\}_{\tilde{i}\in(0,1]}$ together with the remaining aggregate unknowns, leaving equation (B.66) to be imposed afterward. Once the normalized solution is obtained, we recover the equilibrium value y^* from equation (B.66), and rescale the value function and the wage according to

$$v^*(\tilde{i}) = y^* v(\tilde{i}), \qquad w^* = y^* w,$$

while leaving $\mathcal{L}(\tilde{i})$, η, and g_n unchanged.

We now describe the numerical algorithm used to solve for the equilibrium.

Step 1: Set $y = y^0 = 1$.

Step 2: Construct a four-dimensional nonlinear system in $(\bar{l}, w, \eta, g_n)$.

For any candidate vector $(\bar{l}, w, \eta, g_n)$, we define the mapping

$$\mathbf{F}(\bar{l}, w, \eta, g_n) = \begin{pmatrix} F_1(\bar{l}, w, \eta, g_n) \\ F_2(\bar{l}, w, \eta, g_n) \\ F_3(\bar{l}, w, \eta, g_n) \\ F_4(\bar{l}, w, \eta, g_n) \end{pmatrix},$$

where the four residuals are given by:

(i) *Boundary condition of* $\mathcal{L}(\tilde{i})$.

For any candidate vector $(\bar{l}, w, \eta, g_n)$, we first solve the ODE system characterized by equations (22) and (23) in Proposition 1, subject to the boundary conditions

$$v(1) = \frac{w}{\psi_e \zeta^{\theta \frac{1-\beta}{\beta}}}, \qquad \mathcal{L}(1) = \bar{l}.$$

This yields the function $v(\tilde{i})$ and $\mathcal{L}(\tilde{i})$. We then evaluate whether the implied value of $\mathcal{L}(\tilde{i})$ at $\tilde{i} = \eta$ satisfies the boundary condition in Proposition 1. Accordingly, the first residual is defined as

$$F_1(\bar{l}, w, \eta, g_n) = \mathcal{L}(\eta) - \underline{l}.$$

(ii) *Labor market clearing condition.*

Using $v(\tilde{i})$ and $\mathcal{L}(\tilde{i})$, we recover the remaining sectoral objects pointwise. In particular, we obtain $p(\tilde{i})$ from equation (B.48), $b(\tilde{i})$ from equation (B.38), $v^f(\tilde{i})$ from equation (B.42), $f(\tilde{i})$ from equation (B.36), and $e(\tilde{i})$ from equation (B.37). These objects in turn determine labor demand in production, incumbent R&D, entry, luxury services, and new-good invention, denoted by $l(\tilde{i})$, $l^I(\tilde{i})$, L^E, L^X, and L^N, respectively, as characterized by equations (B.51)–(B.55). The second residual is then defined by the labor market clearing condition:

$$F_2(\bar{l}, w, \eta, g_n) = \int_0^1 \left[l(\tilde{i}) + l^I(\tilde{i}) \right] d\tilde{i} + L^E + L^X + L^N - 1.$$

(iii) *Aggregate resource constraint.*

Using the detrended aggregate resource constraint in equation (B.59), we define the third residual as

$$F_3(\bar{l}, w, \eta, g_n) = y(1-\beta)^2 \left[1 - \int_{\mathcal{L}(1)}^{\infty} \left(l - \mathcal{L}(1) \right) d\mathcal{G}(l) \right] - w \int_0^1 l(\tilde{i})\, d\tilde{i}.$$

(iv) *Entry condition for final-good producers.*

Using the detrended entry condition for final-good producers in equation (B.60), we define the fourth residual as

$$F_4(\bar{l}, w, \eta, g_n) = w - \phi v^f(1).$$

Step 3: Solve the nonlinear system

$$\mathbf{F}(\bar{l}, w, \eta, g_n) = 0$$

using the robust Newton method. Denote the solution by

$$(\bar{l}^0, w^0, \eta^0, g_n^0),$$

and the corresponding functions by $\{v^0(\tilde{i}), \mathcal{L}^0(\tilde{i})\}_{\tilde{i}\in(0,1]}$. Together with the normalization $y^0 = 1$, these functions and variables satisfy all the conditions in Proposition 1, except for equation (B.66), which pins down the level of y through the price normalization condition.

Step 4: Use equation (B.66) to recover the equilibrium level of y and rescale the normalized solution obtained in **Step 3**.

Using equation (B.66), we update y from its normalized value $y^0 = 1$ to

$$y^1 = \int_{\underline{l}}^{\mathcal{L}^0(1)} \frac{\left[(\mathcal{L}^{0-1}(l))\right]^{1-\gamma}}{1-\gamma} d\mathcal{G}(l) + \int_{\mathcal{L}^0(1)}^{\infty} \frac{1}{1-\gamma} d\mathcal{G}(l) + \frac{g_n^0}{\phi}.$$

The final equilibrium solution is then given by

$$\begin{cases} v^1(\tilde{i}) = y^1 v^0(\tilde{i}), & \tilde{i} \in (0,1], \\ \mathcal{L}^1(\tilde{i}) = \mathcal{L}^0(\tilde{i}), & \tilde{i} \in (0,1], \\ w^1 = y^1 w^0, \\ \eta^1 = \eta^0, \\ g_n^1 = g_n^0, \\ y^1 \text{ as given above.} \end{cases}$$

E Appendix: Quantitative Exercise

This appendix provides additional materials for the quantitative exercise. We present several technical derivations and discuss further robustness checks to support the quantitative results reported in the main text.

E.1 Dynamics of Firm Distribution

We characterize the evolution of the within-sector firm distribution by tracking firms' positions on a discrete productivity ladder. When a product is newly created, all firms start from the same productivity level $\zeta \bar{A}(t)$, where $\bar{A}(t)$ denotes the aggregate productivity level. Thereafter, productivity improvements arise only through innovation. Each successful innovation raises firm productivity by a factor $\lambda > 1$. Therefore, a firm's productivity is fully determined by the number of successful innovations it has accumulated.

Let $j \in \{0, 1, \dots\}$ denote the number of successful innovations, which is truncated for numerical purpose. Then firm productivity at ladder step j is given by

$$\tilde{A}_i(j,t) = \zeta \bar{A}(t) \lambda^j.$$

Productivity is a state variable, and the cross-sectional distribution of firms over j fully characterizes the distribution of firm productivity. In turn, this distribution is sufficient to determine equilibrium objects such as firm size, and firm level R&D expenditure.

Let $M_i(j,t)$ denote the mass of firms that have experienced j successful innovations at time t in sector i. The law of motion for the distribution is given by the following system of differential equations. For the lowest productivity level $j = 0$,

$$\dot{M}_i(0,t) = \left(\frac{e(i,t)}{F(i,t)} - \delta \right) M_i(0,t) - z_i(t) M_i(j,t),$$

which reflects firm entry, firm exit, and innovation to higher ladders. For higher productivity levels $j = 1, 2, \dots,$

$$\dot{M}_i(j,t) = \left(\frac{e(i,t)}{F(i,t)} - \delta \right) M_i(j,t) + z_i(t) \left[(M_i(j-1,t) - M_i(j,t) \right].$$

The first term captures firm entry and exit, while the second term describes innovation-driven redistribution. Firms at level $j-1$ move to level j upon successful innovation, while firms at level j move further up the ladder. This system defines a linear transport equation over the productivity ladder, where innovation induces an upward flow of firms across grid points.

Numerical implementation. An important simplification is that the computation of the balanced growth equilibrium does not require firm-level information. In particular, the equilibrium objects that characterize the BGE—including the innovation intensity $z_i(t)$, firm entry $e(i,t)$, and the total mass of firms $F(i,t)$—are determined entirely at the sectoral level. We therefore solve for

the full equilibrium first, and then use the implied paths of $z_i(t)$, $e(i,t)$, and $F(i,t)$ as inputs to characterize the within-sector firm distribution.

Let $j \in \{0, 1, \ldots, N_{\text{grid}} - 1\}$ denote the number of successful innovations, where the state space is truncated at N_{grid} for numerical implementation. For a sector i^* invented at time t^*, the state vector

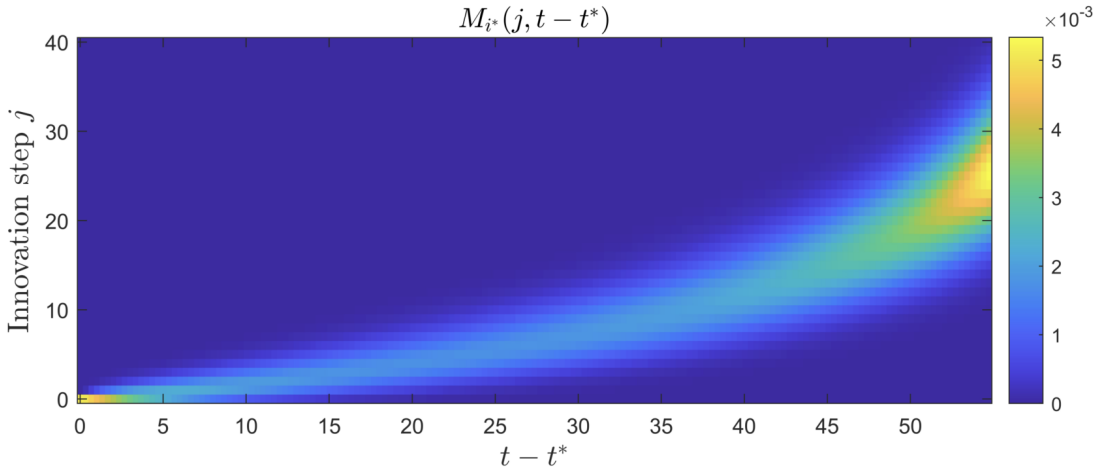
$$M_{i^*}(t) = \left[M_{i^*}(0,t), \ldots, M_{i^*}(N_{\text{grid}} - 1, t)\right]'$$

evolves according to the ODE system above. Conditional on the equilibrium paths of $z_{i^*}(t)$, $e(i^*,t)$, and $F(i^*,t)$, together with the boundary condition

$$M_{i^*}(t^*) = \left[F(i^*,t^*), 0, \ldots, 0\right]',$$

we solve this system using standard ODE solvers and obtain the implied dynamics of the firm distribution along the balanced growth path. As shown in Figure E.1, with firm entry and innovation accumulate over time, the distribution gradually shifts toward higher productivity ladders.

Figure E.1: Firm Distribution Dynamics



Notes. The figure illustrates the evolution of $M_{i^*}(t)$ under the calibrated parameter values.

E.2 Quantitative Result with Different γ

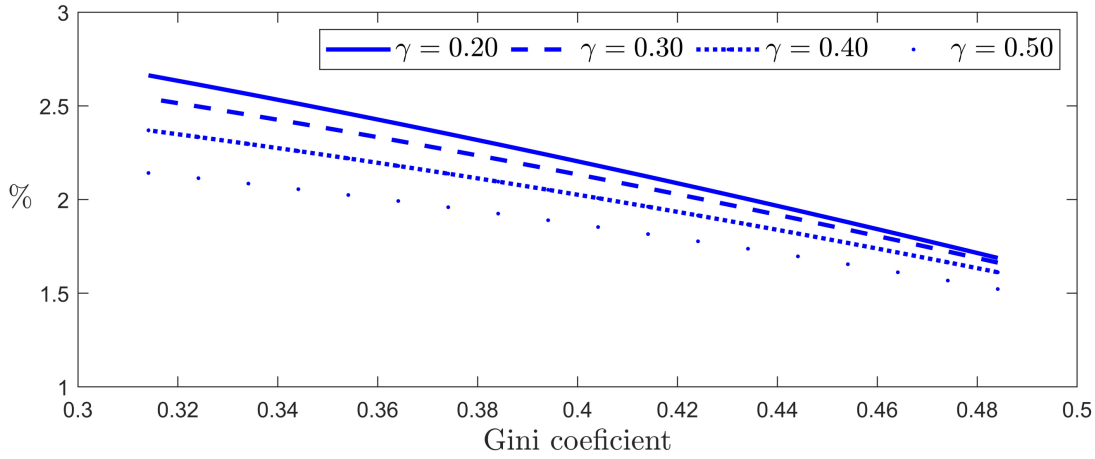
This appendix examines the sensitivity of our quantitative results to the preference curvature parameter γ, which governs the degree of non-homotheticity in household demand. In the baseline calibration, we set $\gamma = 0.343$ to match the observed average price growth rate of "necessities" in our sample period. To assess robustness, we recompute the main counterfactual exercise under alternative values of $\gamma \in \{0.20, 0.30, 0.40, 0.50\}$, holding all other parameters fixed.

Figure E.2 reports the implied per-capita growth rates with different inequality levels under each value of γ. Two patterns emerge. First, varying γ mainly affects the level of growth, reflecting differences in the strength of income-driven demand reallocation across goods. Second and more importantly, the relative decline in growth resulting from rising inequality is remarkably stable

across specifications. For all values of γ, the model continues to generate a sizable slowdown in aggregate growth when the income distribution shifts toward greater inequality.

Quantitatively, the relative contribution of rising inequality to the observed growth slowdown remains similar across specifications, indicating that our main quantitative conclusions are not driven by a particular choice of γ. Overall, these results show that the mechanism linking inequality, innovation incentives, and long-run growth is robust to plausible variation in the degree of demand non-homotheticity.

Figure E.2: Per-capita Growth with Different value of γ



Notes. This figure compares model-implied growth rates with different inequality levels under different value of γ.

E.3 Distribution of After-tax Income

Let pre-tax labor income be $l \geq \underline{l} > 0$ with Pareto density

$$f(l) = \frac{\epsilon\, \underline{l}^{\,\epsilon}}{l^{\epsilon+1}}, \qquad \epsilon > 0,$$

and consider the balanced-budget tax schedule

$$T(l) = l - \frac{l^{1-\upsilon}}{\displaystyle\int_{\underline{l}}^{\infty} l^{1-\upsilon}\, d\mathcal{G}(l)} = l - \frac{l^{1-\upsilon}}{\displaystyle\int_{\underline{l}}^{\infty} l^{1-\upsilon}\, \frac{\epsilon\, \underline{l}^{\,\epsilon}}{l^{\epsilon+1}}\, dl}, \qquad \upsilon \in [0,1], \tag{E.1}$$

where $\mathcal{G}$ is the CDF of l. The denominator equals

$$\int_{\underline{l}}^{\infty} l^{1-\upsilon}\, d\mathcal{G}(l) = \epsilon\, \underline{l}^{\,\epsilon} \int_{\underline{l}}^{\infty} l^{-(\epsilon+\upsilon)}\, dl = -\epsilon\, \underline{l}^{\,\epsilon} \left[\frac{l^{\,1-(\epsilon+\upsilon)}}{\epsilon+\upsilon-1} \right]_{l=\underline{l}}^{\infty} = \frac{\epsilon}{\epsilon+\upsilon-1}\, \underline{l}^{\,1-\upsilon}.$$

Hence after-tax income $Y \equiv l - T(l)$ is

$$Y = \frac{l^{1-\upsilon}}{\int l^{1-\upsilon} d\mathcal{G}(l)} = \underbrace{\frac{\epsilon + \upsilon - 1}{\epsilon}}_{C} \left(\frac{l}{\underline{l}}\right)^{1-\upsilon}, \tag{E.2}$$

with constant $C \equiv (\epsilon + \upsilon - 1)/\epsilon$.

For $\upsilon \in [0, 1)$, the mapping $l \mapsto Y$ in (E.2) is strictly increasing. The support of Y is $[C, \infty)$ since $Y = C$ at $l = \underline{l}$. Using a monotone transformation,

$$\Pr(Y \leq y) = \Pr\left(l \leq \underline{l} \left(\frac{y}{C}\right)^{\frac{1}{1-\upsilon}}\right) = 1 - \left(\frac{\underline{l}}{\underline{l}\left(y/C\right)^{1/(1-\upsilon)}}\right)^{\epsilon} = 1 - \left(\frac{C}{y}\right)^{\frac{\epsilon}{1-\upsilon}}.$$

Therefore, the CDF and PDF of Y are

$$F_Y(y) = 1 - \left(\frac{C}{y}\right)^{\frac{\epsilon}{1-\upsilon}}, \qquad f_Y(y) = \frac{\epsilon}{1-\upsilon} \frac{C^{\frac{\epsilon}{1-\upsilon}}}{y^{\frac{\epsilon}{1-\upsilon}+1}}. \tag{E.3}$$

i.e., $Y \sim \text{Pareto}\left(C, \frac{\epsilon}{1-\upsilon}\right)$. When $\upsilon = 0$ (no redistribution), $Y = C(l/\underline{l}) = l$ and the shape parameter is ϵ (the pre-tax shape), as expected. When $\upsilon \to 1$, $Y \to 1$ for all l, and the distribution degenerates at a constant (perfect equality). Moreover, the mean of Y exists and equals one:

$$\mathbb{E}[Y] = C \cdot \frac{\frac{\epsilon}{1-\upsilon}}{\frac{\epsilon}{1-\upsilon} - 1} = \frac{\epsilon + \upsilon - 1}{\epsilon} \cdot \frac{\epsilon}{\epsilon + \upsilon - 1} = 1,$$

confirming that the schedule (E.1) is balanced-budget for any $\upsilon \in [0, 1)$ with $\epsilon + \upsilon > 1$.

E.4 Redistribution with constant tax rate at the top

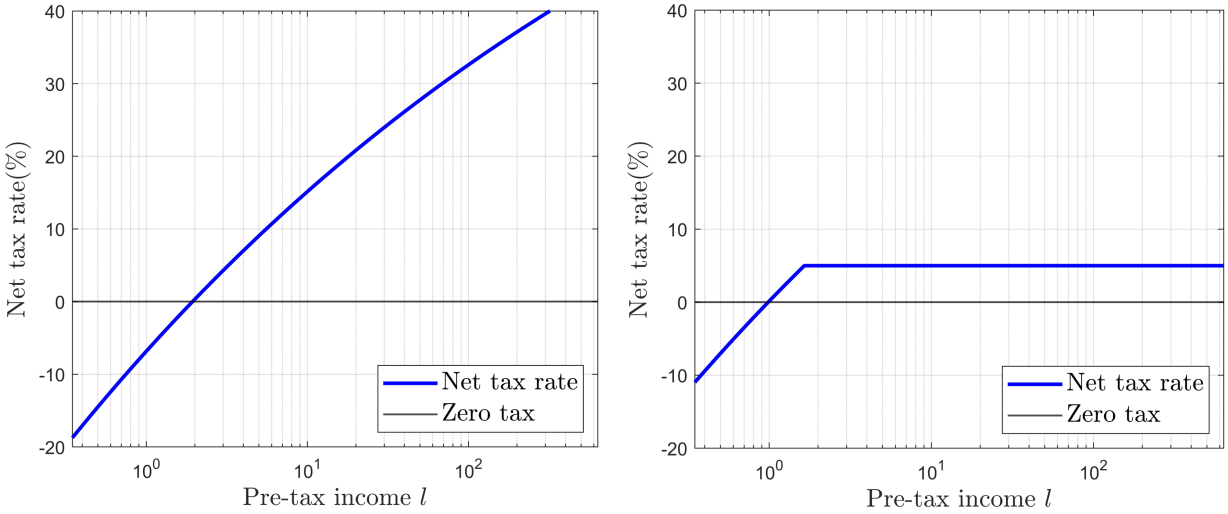
First, under the benchmark Pareto transformation, the tax rate continues to increase with income, implying relatively large welfare losses for households in the far upper tail. To limit these losses, we introduce a two-stage schedule. The tax endowment is:

$$T(l) = l - \max\left\{Al^{1-\upsilon}, (1-\bar{\tau})l\right\}, \tag{E.4}$$

where A is determined by the balanced-budget condition $\mathbb{E}[T(l)] = 0$. We define the endogenous threshold $l_{\text{cap}} = \left(\frac{A}{1-\bar{\tau}}\right)^{1/\upsilon}$. For a household with endowment $l < l_{\text{cap}}$, the tax schedule is $l - Al^{1-\upsilon}$, which is similar with the transfer rule in equation (28). For households with $l \geq l_{\text{cap}}$ face the constant tax rate $\bar{\tau}$. Figure E.3 compares the net tax rates implied by the two redistribution schemes. Under the transfer rule in equation (28), with $\upsilon = 0.1$, the net tax rate increases continuously with income. Under the two-stage rule in equation (E.4), with $\upsilon = 0.1$ and $\bar{\tau} = 0.05$, the tax rate follows a similar pattern at lower income levels but is capped at 5% in the upper tail. Thus, both schemes provide net transfers to low-income households, while the two-stage rule limits the tax burden on

the richest households.

Figure E.3: Net tax rate under different redistribution rules



(a) Net tax rate under equation (28)

(b) Net tax rate under equation (E.4)

Notes. The figure plots the net tax rate under different redistribution rules. Panel (a) reports the net tax rate under equation (28) with $\upsilon = 0.1$. Panel (b) reports the net tax rate under equation (E.4) with $\upsilon = 0.1$ and $\bar{\tau} = 0.05$.

Capping the tax rate substantially reduces welfare losses at the top. For a household at the 99.5th percentile of the endowment distribution, the benchmark redistribution rule in equation (28) implies a net tax rate of 15.96% and a welfare loss of 5.47%. Under the capped two-stage rule in equation (E.4), the corresponding net tax rate and welfare loss fall to 5.00% and 2.08%, respectively. This protection comes at the cost of a weaker growth response: the post-redistribution growth rate is 2.10% under the benchmark rule, compared with 1.76% under the capped rule.

E.5 Local redistribution with Pareto improvement

Second, we construct a more targeted local redistribution scheme to show the existence of Pareto improving redistribution. Consider the following schedule. Rich households with $l \geq l_{\text{cap}}$ pay a constant tax rate τ_R, and the resulting revenue is transferred uniformly to a group of households with $l_R \leq l \leq l_D$:

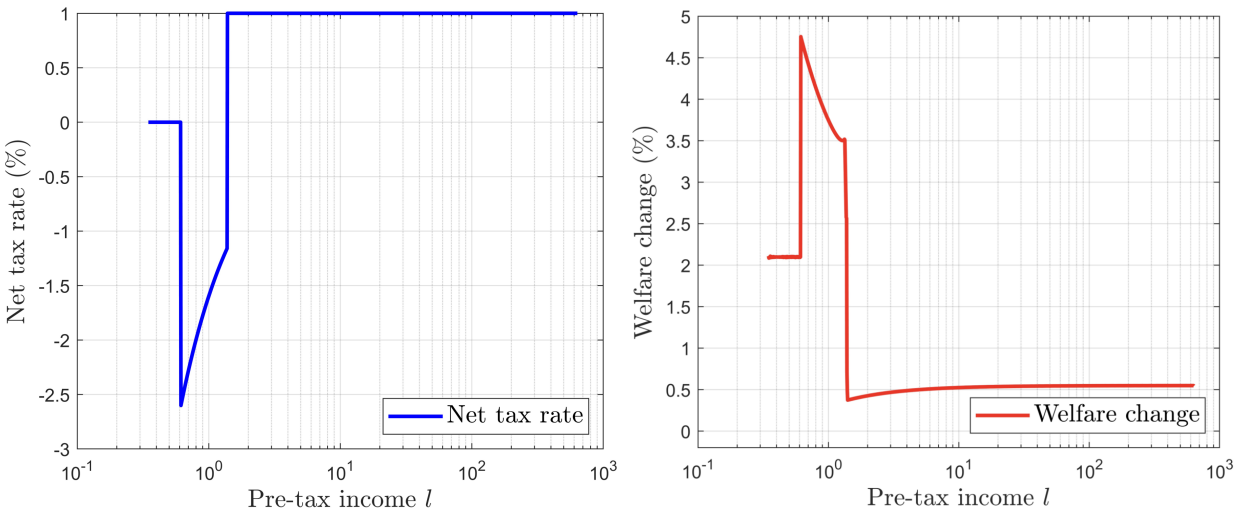
$$T(l) = \begin{cases} 0, & l < l_{\text{tran}}, \\ b, & l_{\text{tran}} \leq l < l_{\text{cap}}, \\ \tau_R l, & l \geq l_{\text{cap}}, \end{cases} \tag{E.5}$$

where b is determined by the balanced-budget condition $\mathbb{E}[T(l)] = 0$.

In our numerical exercise, we set $\tau_R = 0.01$ and transfer the resulting revenue uniformly to the 30% of households immediately below $l_{\text{cap}} = 1.39$. This implies a lower transfer cutoff of $l_{\text{tran}} = 0.61$ and a per-household transfer of $b = 0.016$. The net tax rates and welfare changes implied by this policy are plotted in Figure E.4. The policy raises the consumption growth rate from 1.65% to

1.70%. Welfare increases for every household type on the evaluated income grid, including the richest households, whose minimum welfare gain is approximately 0.37%. The targeted scheme therefore constitutes a numerical Pareto improvement without requiring any ex post compensation.

Figure E.4: Net tax rate and welfare change under redistribution rule equation (E.5)



(a) Net tax rate (b) Welfare change

Notes. The figure plots the net tax rate and welfare change under the redistribution rule in equation (E.5), with $\tau_R = 0.01$, $l_{\text{tran}} = 0.61$ and $l_{\text{cap}} = 1.39$. Panel (a) reports the net tax rate, and Panel (b) reports the welfare change.